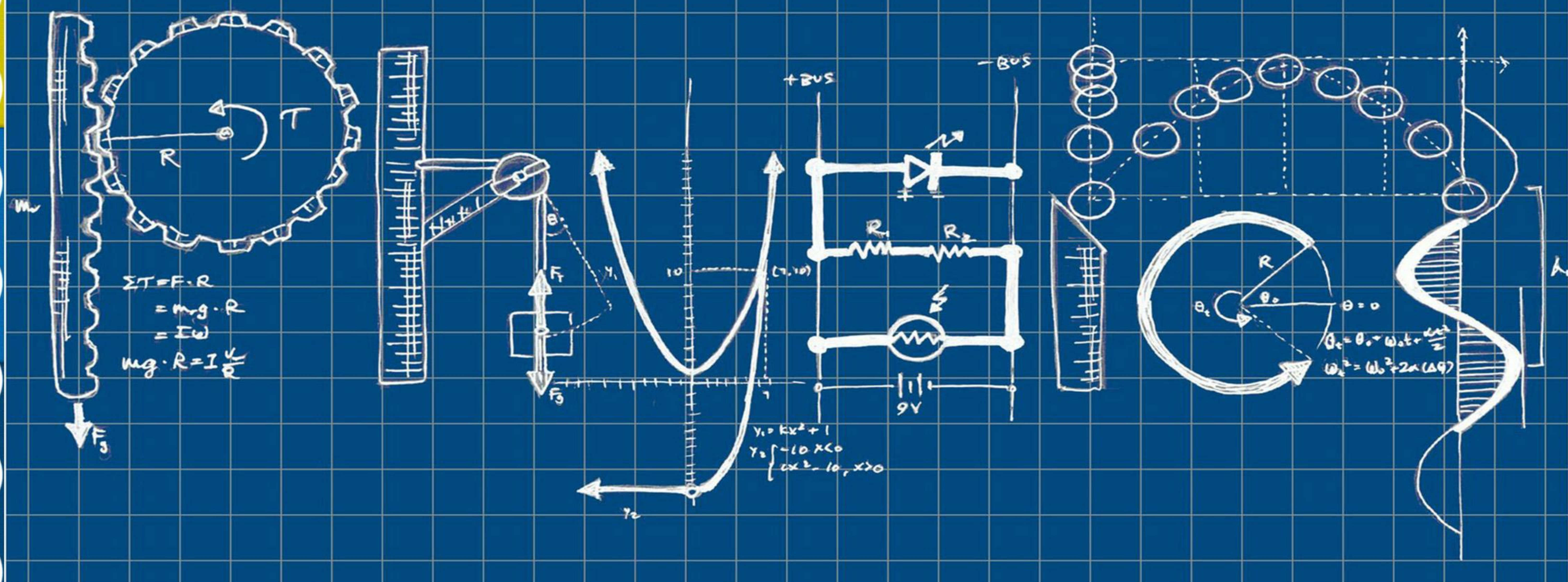


Physics I

notebook



Dr.
Ziad Abu Wa'ar

الموضوع: position, velocity and speed 2.1 التاريخ 17/9/08

"Position"

* Particale moving in one dimension.

"Position"

حركة: motion

* A particale position is the location of the particale with respect to a reference point.

$x = 0$ = reference point

$$\Delta x = x_f - x_i$$

Δx = displacement

x_i = initial position

x_f = Final position

displacements are a vector quantity [we need direction and magn]

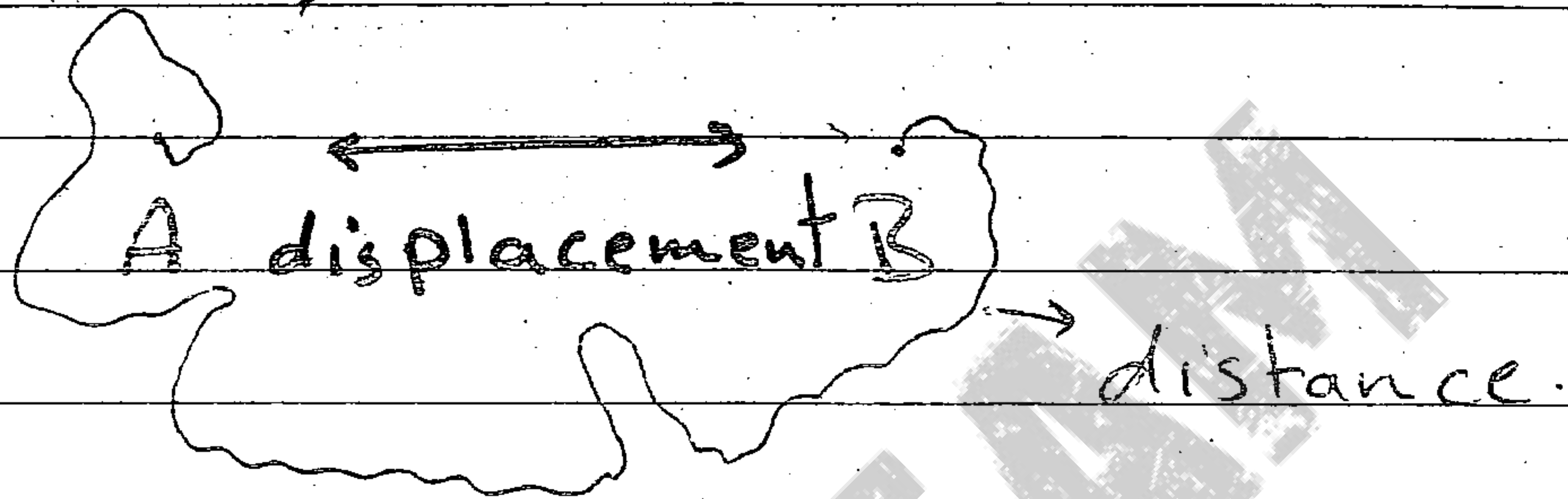
17/9/08 التاريخ follow

الموضوع:

- distance "d" :- the length of a path followed by a particle "position"
مسافة

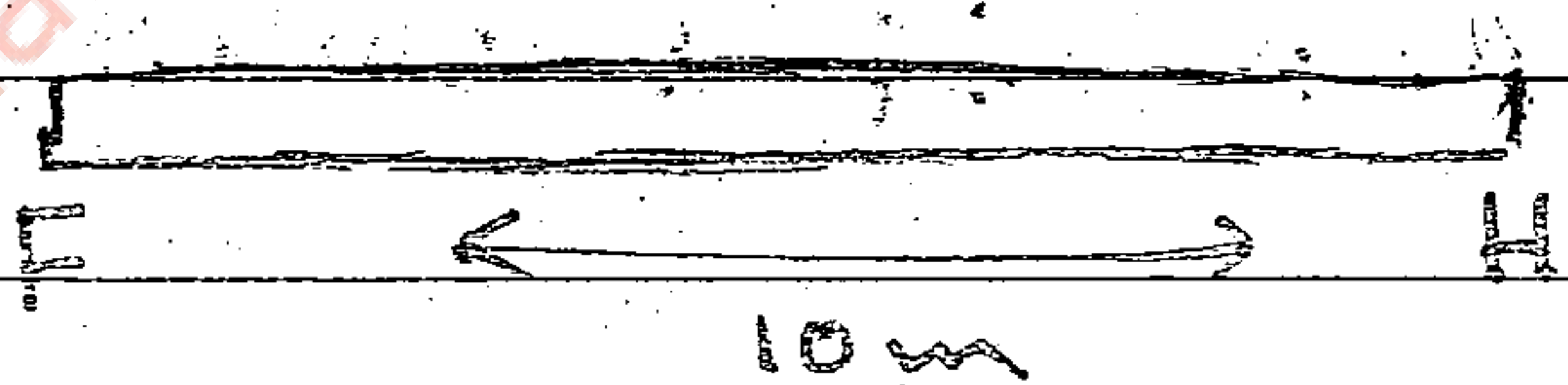
- distance :- + "always positive"

- displacement +, -



ex: A car moves from position H to position F and then returns to H.

- Find the displacement and distance of the car?



Solution:

$$x_i = H = x_F$$

displacement $\Delta x = x_F - x_i = 0$

distance :- $d = 10 + 10 = 20 \text{ m}$

الموضوع: "velocity" follow التاريخ 108/9/171

"velocity"
سرعة متجهة

$$V_{x, \text{avg}} = \frac{\Delta x}{\Delta t}$$

متجهة السرعة المتوسطة = $\frac{\Delta x}{\Delta t}$
↓
متجه الإزاحة
↓
متجه الزمن

"displacement vector"

Δx : displacement
 Δt : time

$$V_{\text{avg}} = \frac{d}{\Delta t}$$

↓
average speed

d : distance
 Δt : time

قياسية (scalar) تتسم speed "distance"

متجهة (vector) تتسم displacement, velocity

هدف chapter 1 هو إيجاد موقع و سرعة
جسم متحرك

17/9/107 follow "IS"

الموضوع:

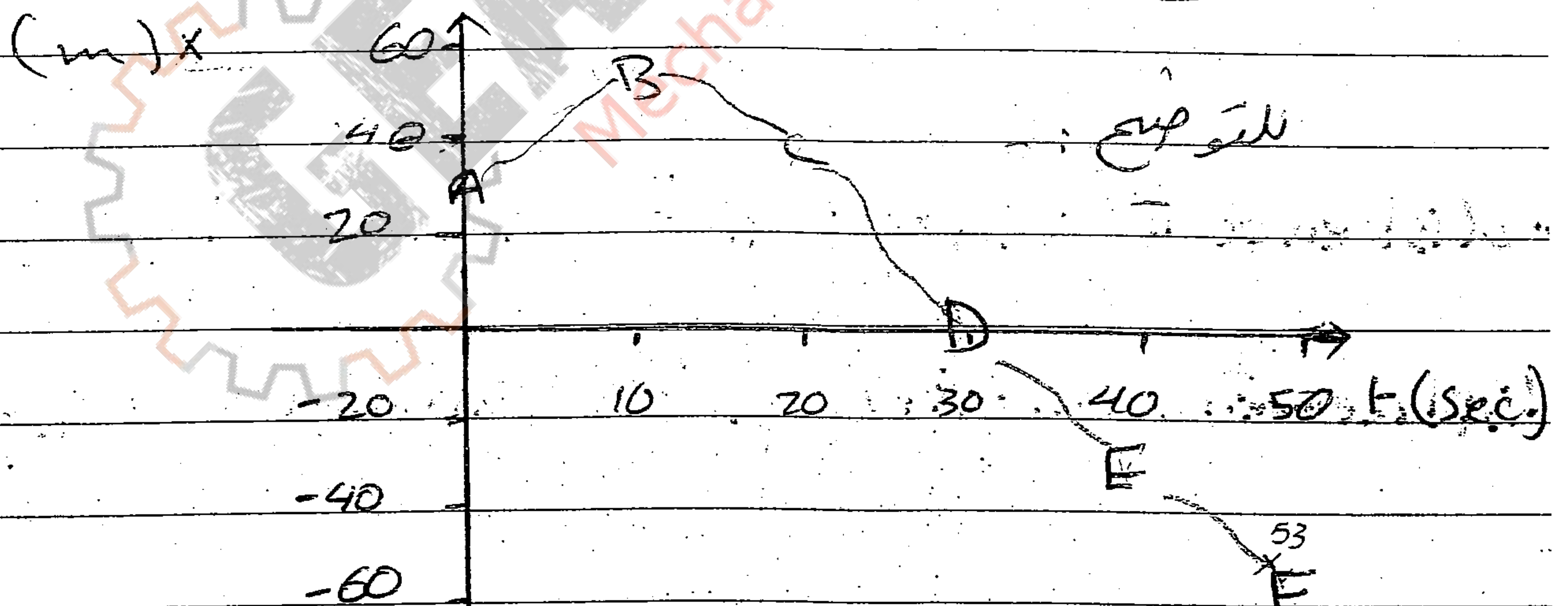
SI "International system" :-

$$[v] = \frac{m}{s}$$
 "متر في الثانية"

ex 2.1: Position

	t (s)	x (m)
A	0	30
B	10	52
C	20	38
D	30	0
E	40	-37
F	50	-53

Q: Find displacement, average velocity, average speed between positions A and E



التمثيل البياني
Graphical representation
(position - time graph)

تتبع 108 التاريخ 19 19

الموضوع:

Solution:

$$* \quad x_i = x_A = 30 \text{ m at } 0 \text{ sec! } t_A = 0$$

$$x_F = x_F = -53 \text{ m at } 50 \text{ sec } t_F = 50$$

$$\text{displacement} = \Delta x = x_F - x_i$$

$$\Rightarrow \Delta x = -53 - 30$$

$$= -83$$

لـ تغير في مكان الجسم إلى

$$* \quad v_{x, \text{avg}} = \frac{\Delta x}{\Delta t} \Rightarrow \frac{-83}{50} \Rightarrow -1.7 \frac{\text{m}}{\text{s}}$$

السرعة المتوسطة التي استقرت للوصول

لـ متوسط السرعة الكلية إلى F

$$* \quad v_{\text{avg}} = \frac{d}{\Delta t} \Rightarrow$$

$$d = 22 + 14 + 38 + 37 + 16 =$$

$$127 \text{ m}$$

لـ d

$$\therefore v_{\text{avg}} = \frac{d}{\Delta t} \Rightarrow \frac{127}{50} \Rightarrow \underline{2.54 \text{ m/s}}$$

note:- تكون الحركة منتظمة إذا تحرك الجسم بسرعة أو
بالتساوي. تكون الحركة غير منتظمة إذا تحرك الجسم
بالتسارع أو بالتأخر.

-note:- we can categorise motion in
three types:-

1. transitional motion حركة انتقالية
2. rotational motion حركة دورانية
3. vibrational motion حركة اهتزازية

الموضوع Instantaneous speed and velocity التاريخ 108 / 9 / 14

$$* V_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} \Rightarrow \frac{dx}{dt}$$

فئة السرعة $\frac{dx}{dt}$
 فئة الزمن Δt
 arg $\Delta t \rightarrow 0$ $\rightarrow$ صفر
 goto zero

a vector

تجهة لأننا نعلم اتجاهه displacement

The instantaneous speed equal to the magnitude of instantaneous velocity.

$$\vec{v}_{ins} = \vec{v}_{ins}$$

ins. speed = ins. velocity

exp 2.3 :- A particle moves along its position

$$x = 4t + 2t^2$$

a. Find the displacement in the time interval $t=0$ to $t=1$ sec

$t=0$ to $t=3$ sec

b. Find average velocity at the same time

c. Find instantaneous velocity at $t=2.5$

Solution :-

a. $t=0 \rightarrow t=1$

$$x_i(t=0) = 0 \Rightarrow x = 4t + 2t^2 \Rightarrow 4x(0) + 2(0)^2 = 0$$

$$x_F(t=1) \Rightarrow -4(1) + 2(1)^2 = -2 \text{ m}$$

$$\Delta x = x_F - x_i \Rightarrow -2 - 0 = -2 \text{ m}$$

$t=1 \rightarrow t=3$

$$x_i(t=1) = -2$$

$$x_F(t=3) = -4(3) + 2(9) = 6 \text{ m}$$

$$\Delta x = x_F - x_i \Rightarrow 6 \text{ m} - (-2) = 8 \text{ m}$$

b. $v_{x, \text{avg}} = \frac{\Delta x}{\Delta t}$

$$t=0 \rightarrow t=1 \Rightarrow v_{x, \text{avg}} = \frac{-2}{1} = -2 \text{ m/s}$$

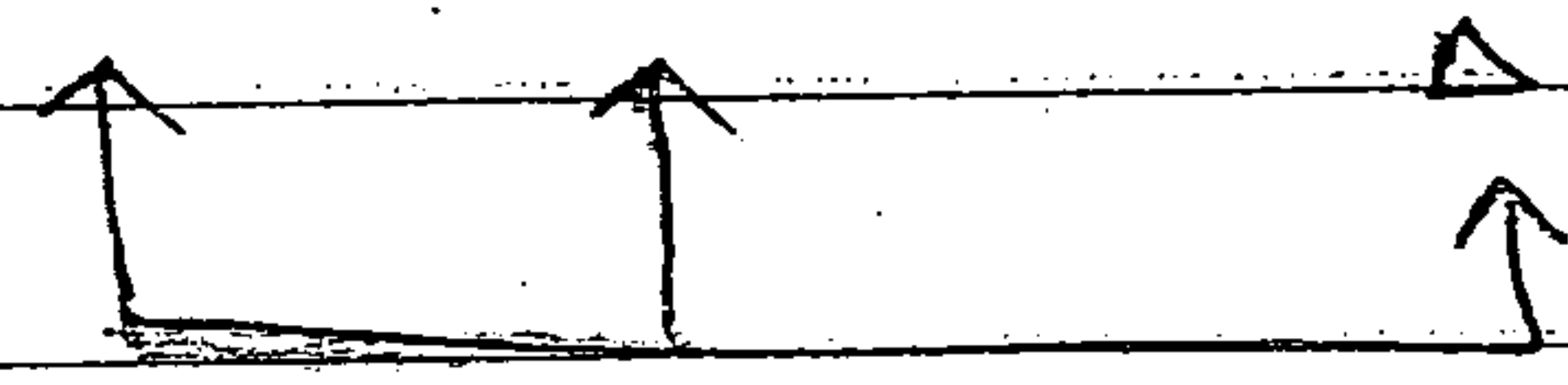
$$t=1 \rightarrow t=3 \Rightarrow v_{x, \text{avg}} = \frac{8}{2} = 4 \text{ m/s}$$

c. $v_x = \frac{dx}{dt} = 4 + 4t$

$$v_x(t=2.5) = 10 + 4 = 14 \text{ m/s}$$

2.3
الموضوع: The particle under constant velocity
التاريخ: 22/9/108

$-V_x = V_x$, any for constant velocity.
 $\vec{a} = \vec{0}$

$$\Rightarrow V_x = \frac{\Delta x}{\Delta t} = \frac{x_F - x_i}{\Delta t}$$


$$\Rightarrow x_F = x_i + V_x \Delta t$$

$$\Delta t = t_F - t_i, \text{ if } t_i = 0$$

$$x_F = x_i + V_x t \text{ for constant } V_x$$

$$v = \frac{d}{\Delta t} \text{ for constant speed}$$

exp: A particle moving at a constant speed in a circular path. If the speed is 5 m/s and the radius of the circle is 10 m.

calculate the time required to complete one trip around the circle.

تاريخ 10/8/2021

solution:-

$$v = \frac{d}{\Delta t}$$
 - استخرجنا speed
 - سرعة

$$\Delta t = \frac{d}{v} = \frac{2\pi r}{v} = \frac{2\pi (10)}{5}$$

$$= 12.6 \text{ sec}$$

r : radius

$$2\pi r = \text{محيط الدائرة}$$

$$2\pi r =$$

r : radius

note:- عند تحريك الجسم لليمين تكون الحركة موجبة

عند تحريك الجسم لليسار تكون الحركة سالبة



2.4 acceleration "تسارع" الموضوع:

التاريخ 22/9/108

$$a_{x, \text{avg}} = \frac{\Delta v_x}{\Delta t} = \frac{v_{x,F} - v_{x,i}}{t_F - t_i}$$

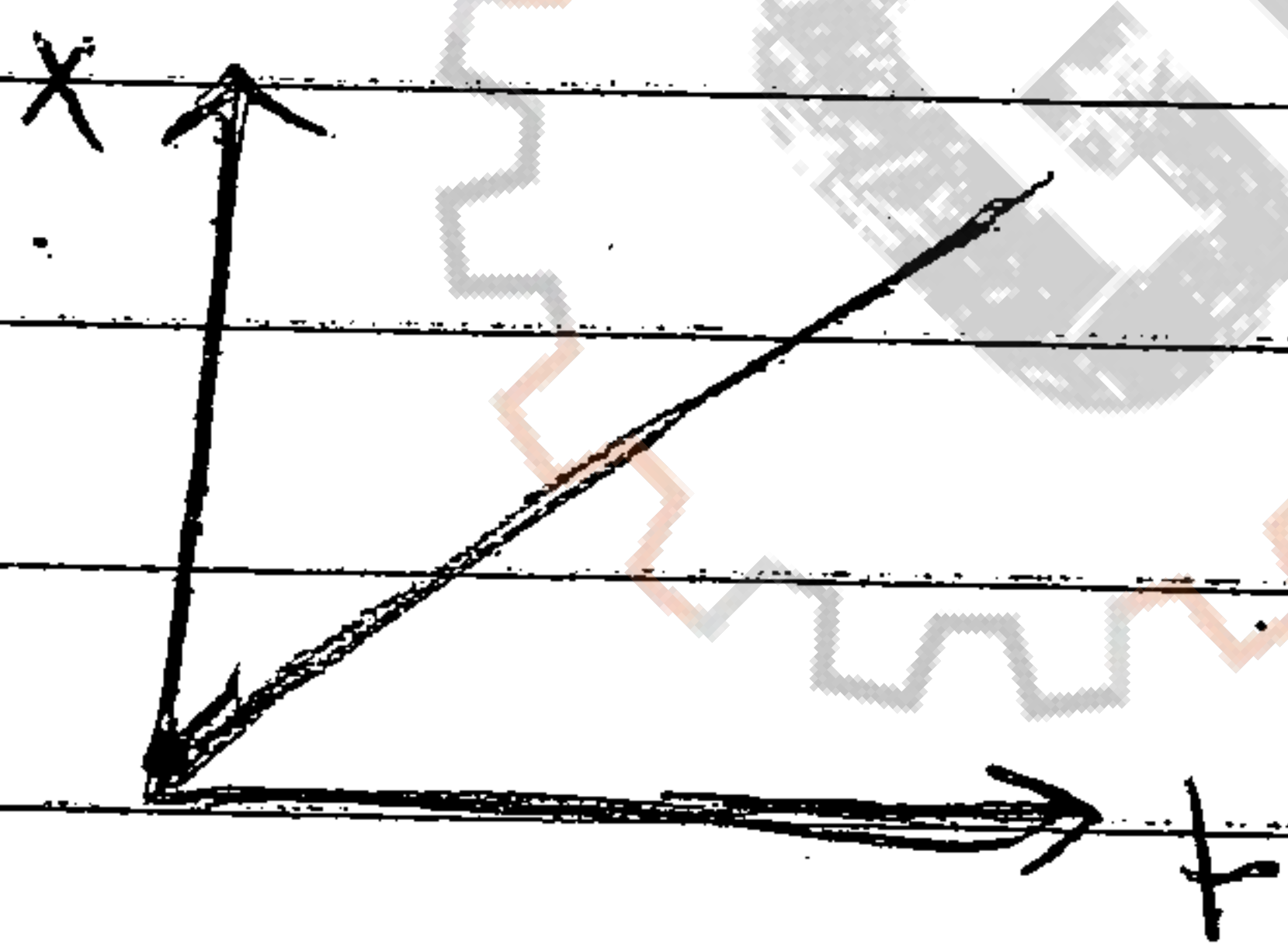
average acceleration
change in velocity
time interval

$$a_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta v_x}{\Delta t} = \frac{dv_x}{dt} = \frac{d^2 x}{dt^2} \quad \downarrow \text{How}$$

instantaneous acceleration

$$a_x = \frac{dv_x}{dt} = \frac{d(dx)}{dt(dt)} = \frac{d^2 x}{dt^2}$$

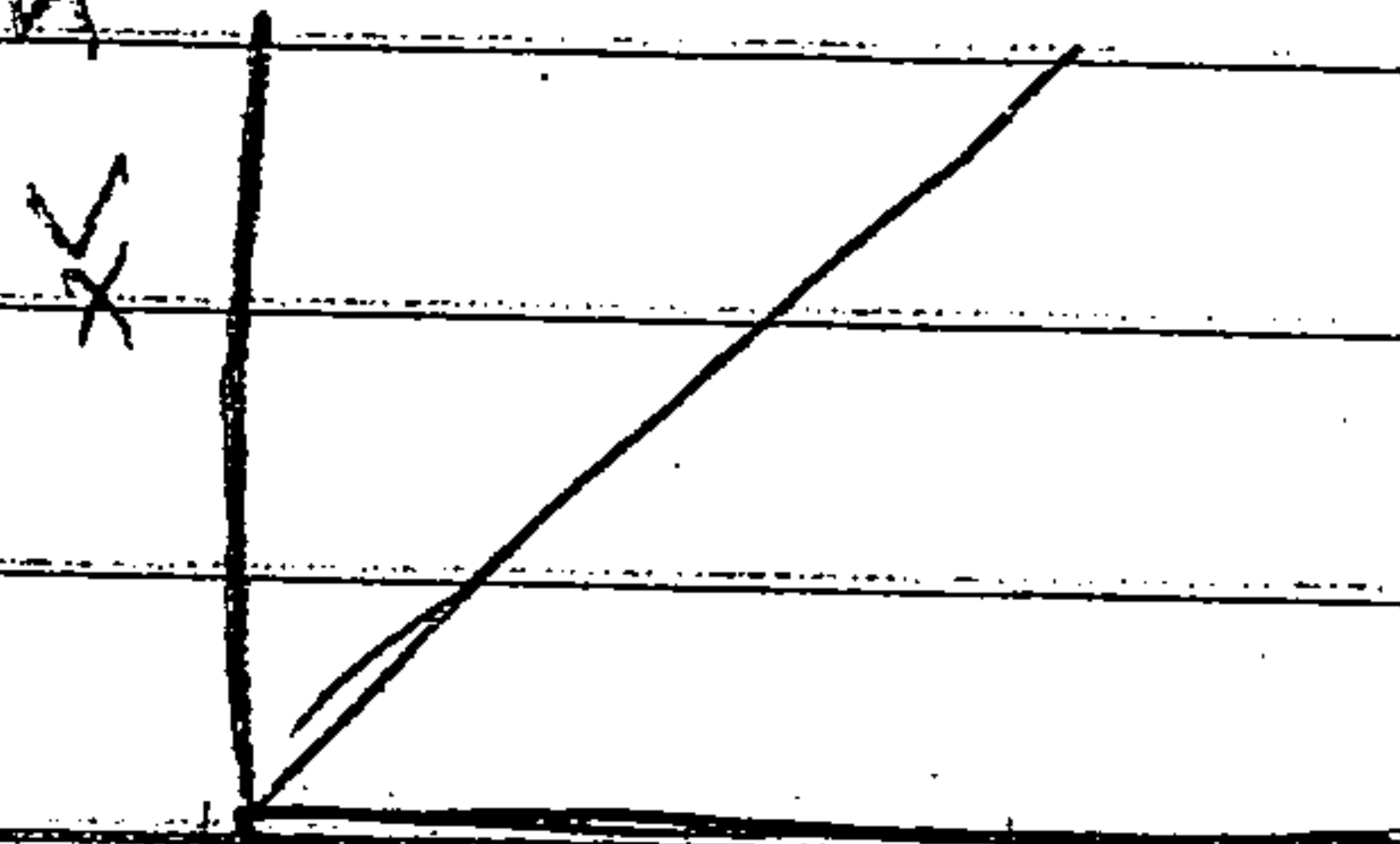
* Note: - x-t graph



velocity!
is the slope at some point.

لأنه سرعة المماس في
نقطة معينة

v_x -t graph



acceleration is
the slope at some point.

22/9/108 التاريخ Follow

الموضوع:

*note:- IF v_x and a_x in the same direction
the object speeding up.

*note:- IF v_x and a_x in opposite direction
the object slowing down.

exp 2.6:- x -axis $\vec{v_i}$ $\vec{a_x}$

$$v_x = (40 - 5t^2) \text{ m/s}$$

(a) Find average acceleration in the
time interval $t=0 \rightarrow t=2 \text{ sec}$.

(b) The acceleration at $t=2 \text{ sec}$.

solution:- $t_i = 0$, $t_f = 2$, $v_{xi} = 40 \text{ m/s}$

$$v_{xf} = 40 - 5(2)^2 = 20 \text{ m/s}$$

$$\begin{aligned} (a) a_{x \text{ avg}} &= \frac{v_{xf} - v_{xi}}{t_f - t_i} \Rightarrow \frac{20 - 40}{2 - 0} \\ &= -10 \text{ m/s} \end{aligned}$$

تاريخ 22/9/08

note:- $[a] = m/s^2$ unit.

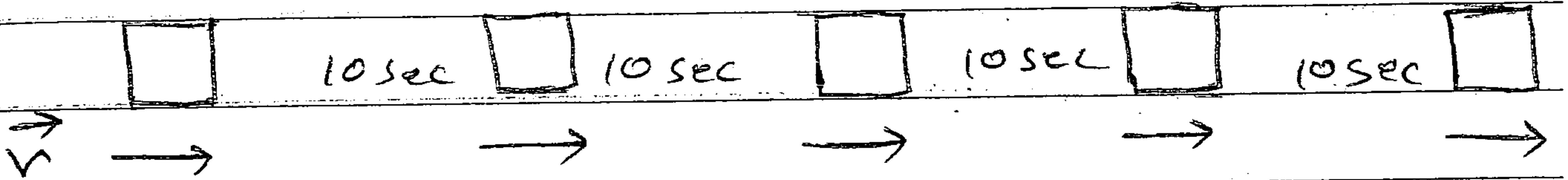
$$(b) a_x = \frac{dv_x}{dt} = -10t \rightarrow v_x = 40 - 5t^2$$

$$\frac{dv_x}{dt} = -10t$$

$$\Rightarrow a_x (t=2) = -20 m/s^2$$

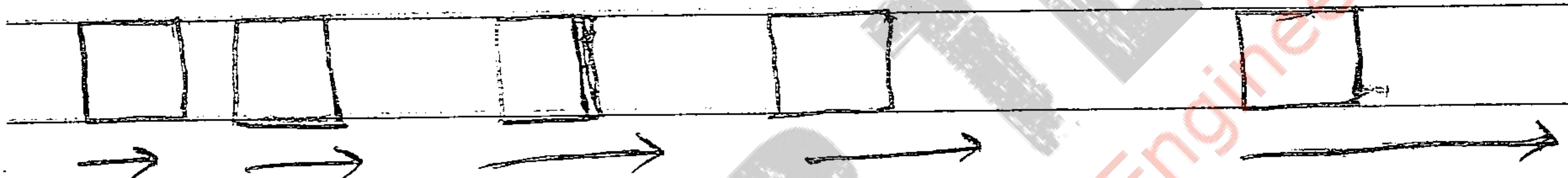
note:- لا تستعمل الاشتقاق إلا عند طلب السرعة
والتسارع اللحظي أو السرعة والتسارع
في لحظة معينة مسبقاً.

- car $\vec{v}$ constant, $a = 0$

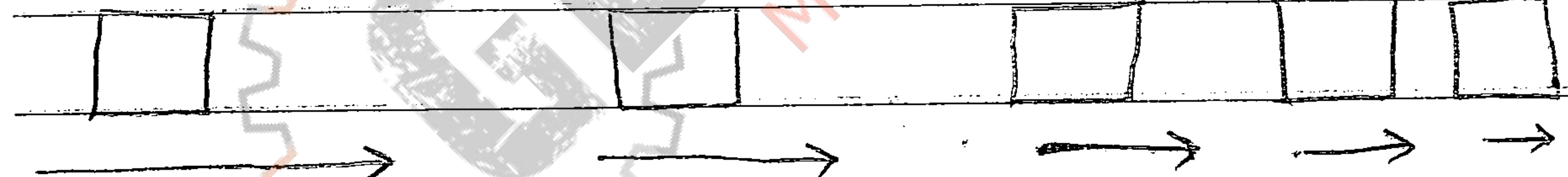


$\vec{v}$:- قسمة المسافة على الزمن

- car $\vec{v}$ increasing, a positive $a \rightarrow$



- car $\vec{v}$ decreasing, a negative $a \leftarrow$



هذه الدرس تشرح كيف نرسم وتفسر الحركة
والسريع، البطيء، وقفة لحظية

31/10/08 Follow التاريخ

الموضوع:

- we get a motion diagram by using a stroboscope which shows several images of the object taken.

- motion diagram:- we use it to describe the velocity and acceleration while an object is in motion.

الموضوع: Acceleration
 22/9/108 ^{2.6} The particle under constant

*IF the particle moves in a straight line with a constant acceleration a_x , its motion is described by the kinematic equations:

المعادلات الحركية

$$v_{x,f} = v_{x,i} + a_x t$$

$$v_{x,avg} = \frac{v_{x,i} + v_{x,f}}{2} \quad \text{= متوسط السرعة}$$

$$x_f = x_i + \frac{1}{2} (v_{x,i} + v_{x,f}) t$$

$$x_f = x_i + v_{x,i} t + \frac{1}{2} a_x t^2$$

$$v_{x,f}^2 = v_{x,i}^2 + 2a_x (x_f - x_i)$$

where:-

$v_{x,f}$:- Final velocity

$v_{x,i}$:- initial velocity

a_x :- acceleration

t :- time

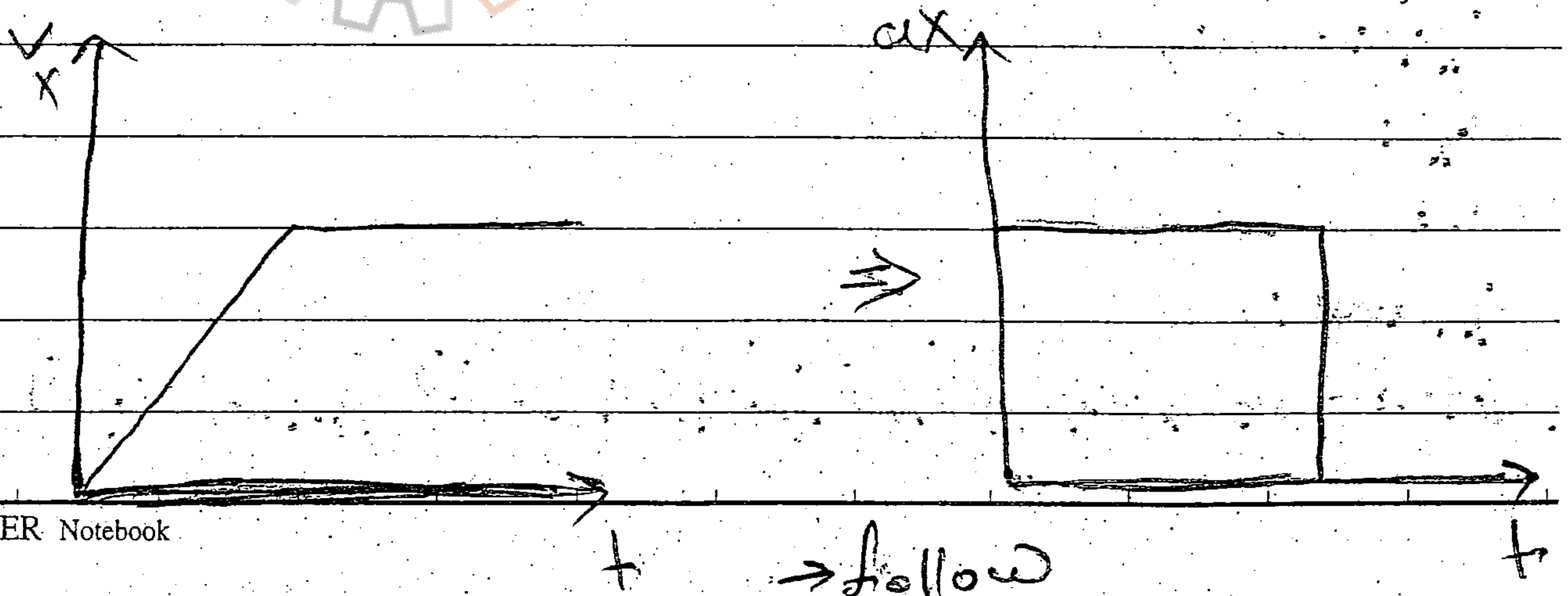
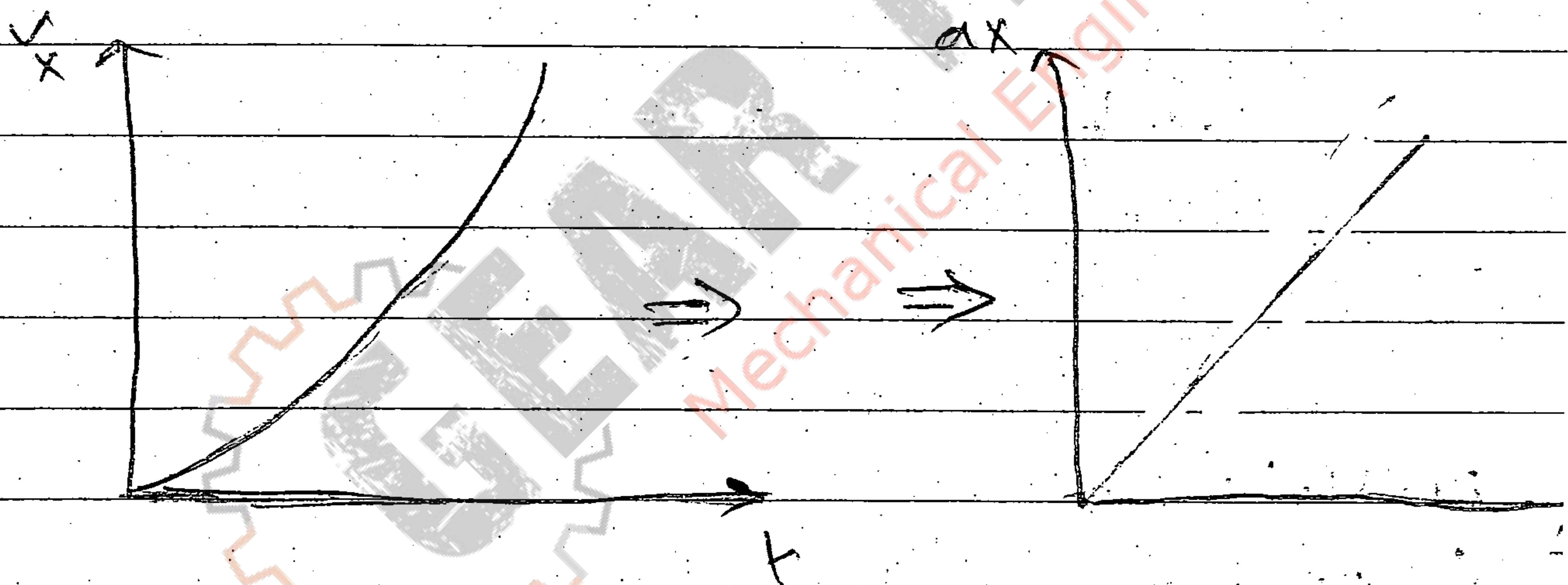
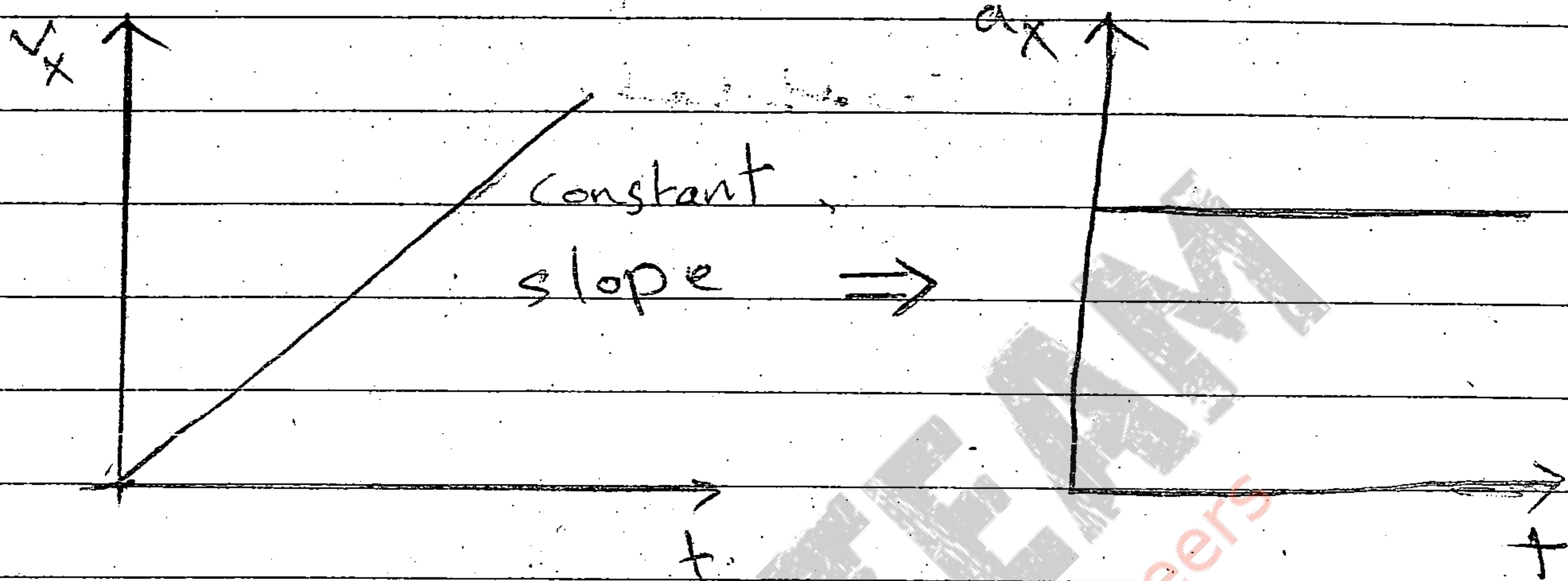
$v_{x,avg}$:- average velocity

*note:- في الحركة الحركية نفسه القوانين كما في الجاذبية

22 / 9 / 108 التاريخ Follow

الموضوع:

exp:- what is the possible acceleration as a function of time graph represents the following $v_x - t$ graph.



22/8/09 follow التاريخ

الموضوع:

ex 2.7:- A jet lands on an aircraft carrier at 63 m/s

(a) Find a_x if it stops in 2 seconds?

(b) IF the Jet touches down at position $x_i = 0$, what is final position?

solution:- $\rightarrow v_{xi} = 63 \text{ m/s}, v_{xf} = 0$
 $t = 2 \text{ sec}, x_i = 0$

$$(a) v_{xf} = v_{xi} + a_x t$$

$$a_x = \frac{v_{xf} - v_{xi}}{t} = \frac{0 - 63}{2} = -31.5 \text{ m/s}^2$$

$$(b) x_f = x_i + \frac{1}{2} (v_{xi} + v_{xf}) t$$

$$= 0 + \frac{1}{2} (63 + 0) (2) = 63 \text{ m}$$

$$(c) x_f = x_i + a_x t$$

$$x_f = 0 + 31.5 \times 2$$

$$x_f = 63 \text{ m}$$

* A Freely falling object is any object moving freely under the influence of gravity alone, regardless its initial motion.

الشیخ

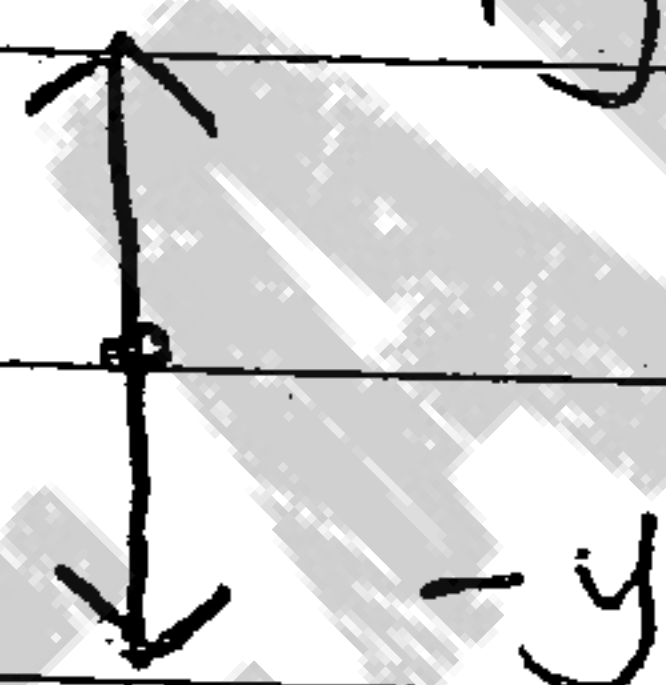
~~$a_x \rightarrow a_y$~~

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

5-1-1

كل شيء للأفضل

CVL a gini



*note:- في الحركة للأعلى والأسفل نضع $g = -9.8$
عند أول فصل أنه سقوط

* note :-
البرهان في توضيح قبيل الرحمن
- A ، - 4 ، - B

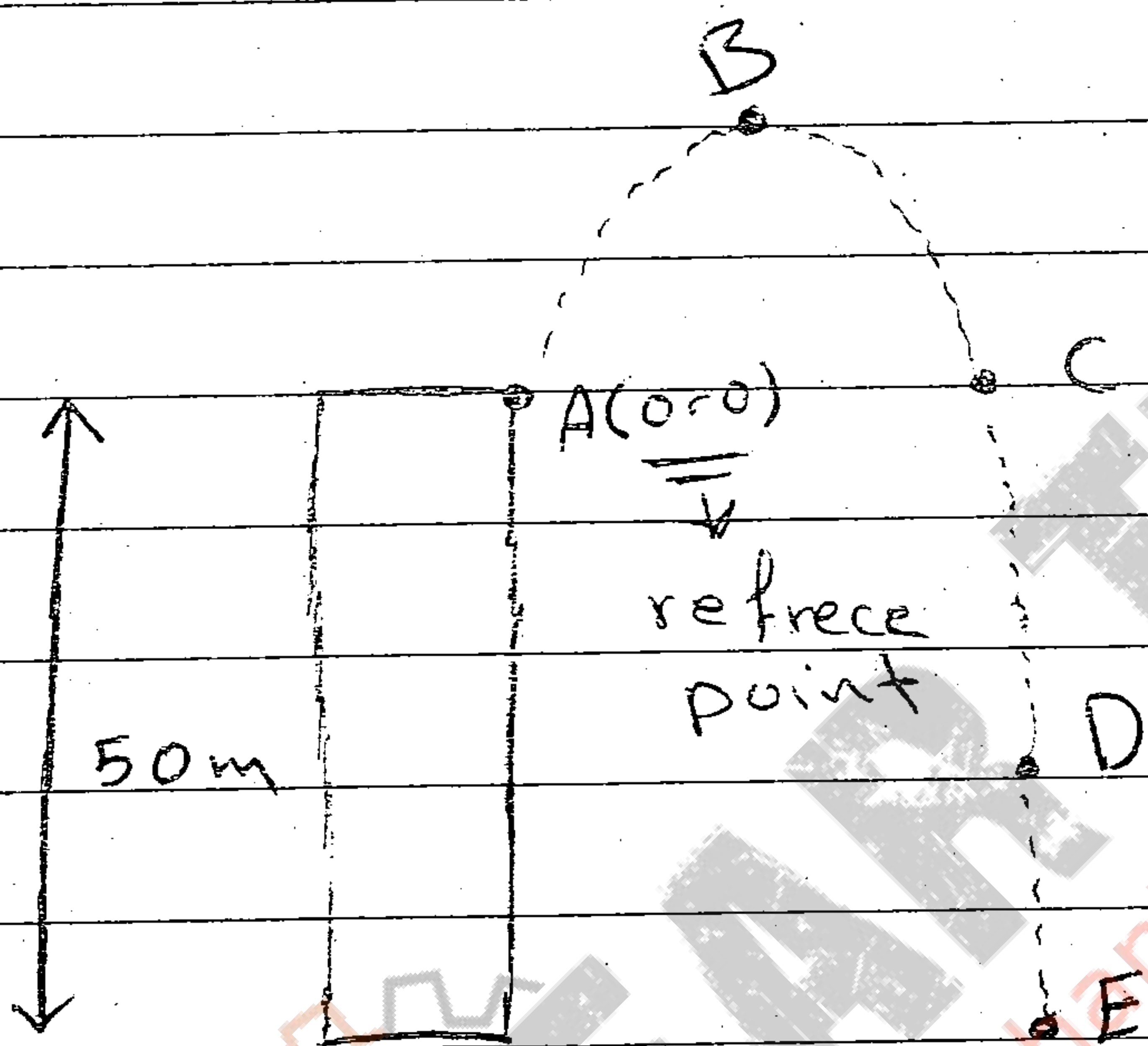
note:-
 1. $\frac{1}{x^2} = x^{-2}$
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 143.

Follow التاريخ 7/10/08

الموضوع:

ex 2.10:- A stone thrown with $V_i = V_A = 20$

- Find t at maximum height.
- Find the maximum height.
- Find velocity of the stone at position C.
- Find the velocity and position at $t = 5$ sec.



Solu:- $V_A = 20 \text{ m/s}$, $a = -g$

- السرعة عند أقصى ارتفاع تساوي صفر.
 - عندما يتغير الجسم اتجاه حركته: السرعة تساوي صفر.
 (a) $V_{yF} = V_{yi} + a_y t \Rightarrow t = t_b = \frac{V_{yF} - V_{yi}}{a_y}$

$$\frac{0 - 20}{-9.8} \Rightarrow 2.04 \text{ sec.}$$

تاريخ 10/10/2021

(b) reference point $y_A = 0$ "set point"

$$y_{B_{\max}} = y_A + V_{y_A} t + \frac{1}{2} a_y t^2$$

$$= 0 + 20 \times 2.04 + \frac{1}{2} (-9.8 \times (2.04)^2)$$

$$= 20.4 \text{ m} \quad \text{A is}$$

$$= 70.4 \text{ m} \quad \text{up, } \dot{y} > 0$$

$$(c) V_{y_c}^2 = V_{y_A}^2 + 2a_y (y_c - y_A)$$

$$= (20)^2 + 2 \times -9.8 (0 - 0)$$

$$= 400 \text{ m/s}$$

$$V_{y_c} = -20 \text{ m/s} \quad \text{لأنه في الأعلى}$$

$$(d) t = 5 \text{ sec.}$$

$$V_{y_D} = V_{y_A} + a_y t$$

$$= 20 + -9.8 \times 5$$

$$V_{y_D} = -29 \text{ m/s} \quad \text{لأنه في الأسفل}$$

$$y_D = y_A + V_{y_A} t + \frac{1}{2} a_y t^2$$

$$= 0 + 20 \times 5 + \frac{1}{2} \times -9.8 \times 25$$

$$= -22.5 \text{ m}$$

Kinematic Equations Derived

7/10/07 التاريخ 2.8 From calculus. Not Important الموضوع:

① $a_x = \frac{dv_x}{dt} \Rightarrow dv_x = a_x dt$

← سرعة

كل الرتبة من الثاني الى الابدائي

$$v_{xF} - v_{xi} = \int_0^t a_x dt = a_x \int_0^t dt$$

$$= a_x(t-0) = a_x t$$

$$\Rightarrow v_{xF} = v_{xi} + a_x t$$

في E! لا يمكن
شاهد

② $v_x = \frac{dx}{dt} \Rightarrow dx = v_x dt$

$$x_F - x_i = \int v_x dt = \int (v_{xi} + a_x t) dt$$

$$v_{xF} = v_{xi} + a_x t$$

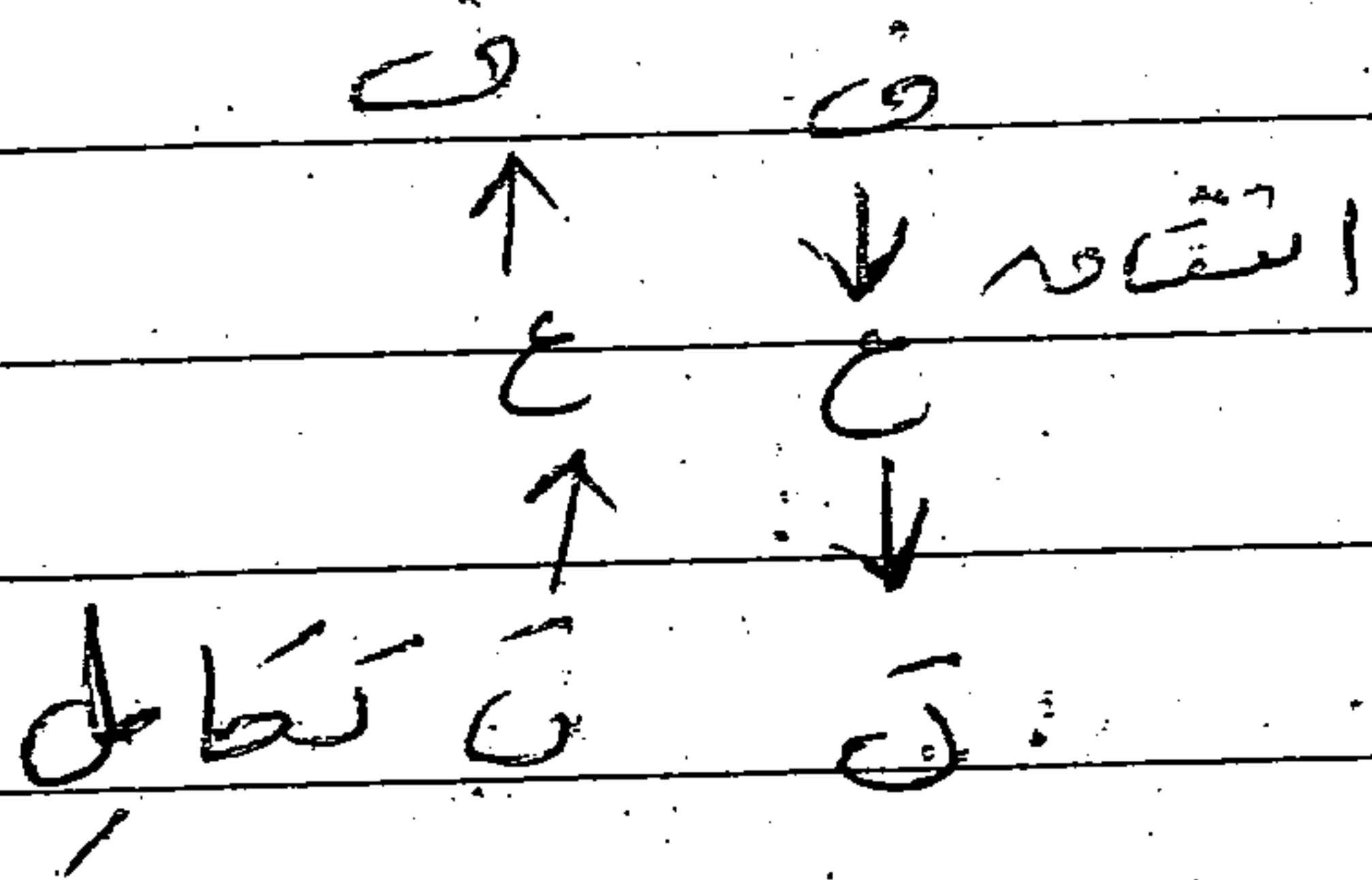
$$x_F - x_i = \int v_{xi} dt + \int a_x t dt$$

$$= v_{xi} t + a_x \frac{t^2}{2}$$

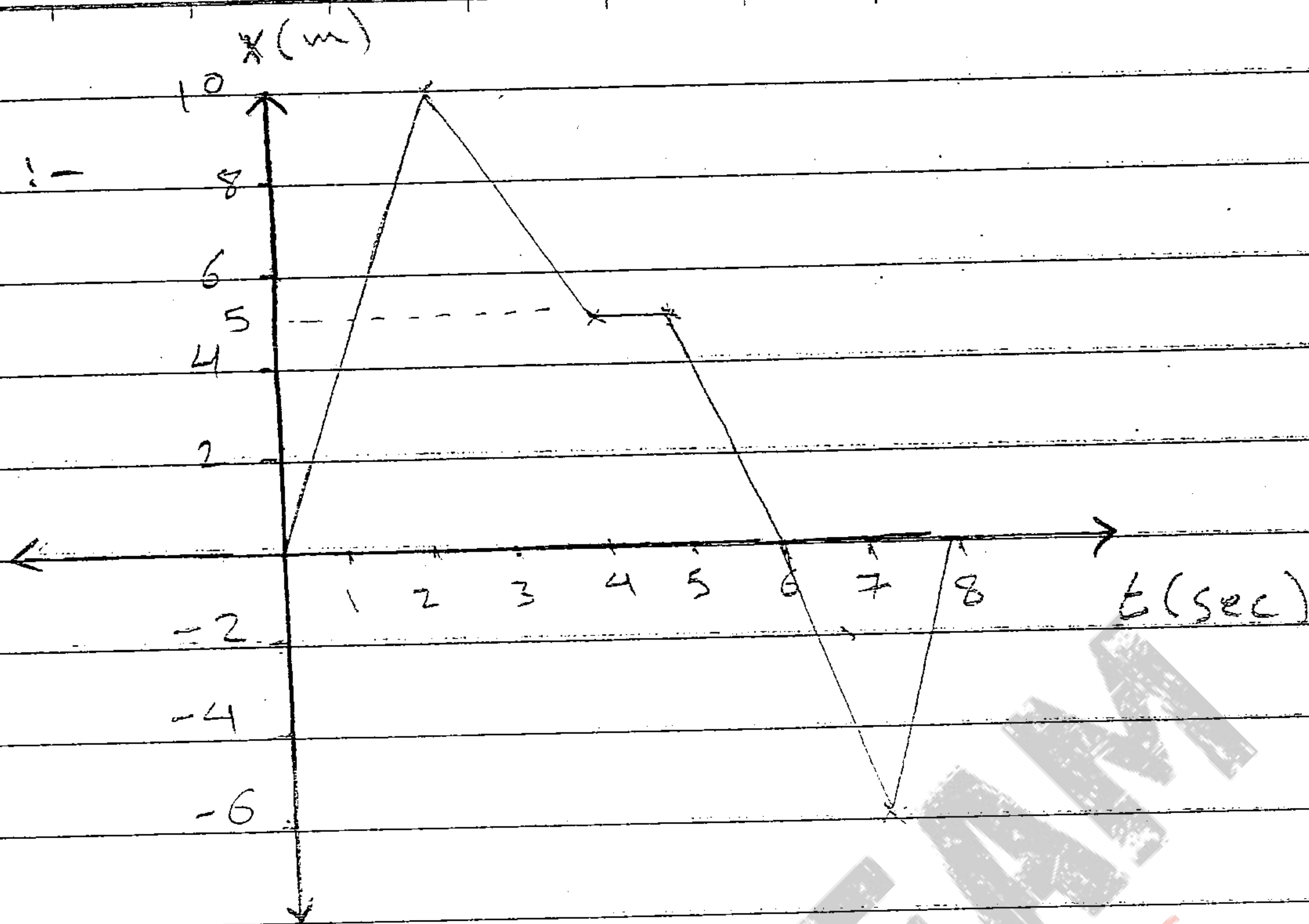
$$\Rightarrow x_F = x_i + v_{xi} t + \frac{1}{2} a_x t^2$$

تاريخ 10/10/108

التمثيل الجزيئي للدور في المجموع بالإنجليزية - note:



2.1 :-



Find $v_{x, \text{avg}}$ in the following intervals :-

(a) $0 \rightarrow 2 \text{ sec}$

(b) $0 \rightarrow 4 \text{ sec}$

(c) $2 \rightarrow 4 \text{ sec}$

(d) $4 \rightarrow 7 \text{ sec}$

(e) $0 \rightarrow 8 \text{ sec}$

Solu :-
$$v_{x, \text{avg}} = \frac{\Delta x}{\Delta t} = \frac{x_F - x_i}{t_F - t_i}$$

(a) $v = \frac{10 - 0}{2 - 0} = 5 \text{ m/s}$

(b) $v = \frac{5 - 0}{4 - 0} = 1.2 \text{ m/s}$

$$(c) v = \frac{5-10}{4-2} = -2.5 \text{ m/s}$$

$$(d) v = \frac{-5-5}{7-4} = -3.3 \text{ m/s}$$

$$(e) v = \frac{0-0}{8-0} = 0 \text{ m/s}$$

2.4:- A particle moves according to the equation $x = 10t^2$

where:- x is in m

(a) Find $v_{x, \text{avg}}$ from $2 \rightarrow 3$ sec.

(b) Find $v_{x, \text{avg}}$ from $2 \rightarrow 2.1$ sec

Sol:-

$$x = 10t^2$$

$$x(t=2) = 10 \times 4 = 40 \text{ m}$$

$$x(t=3) = 10 \times 9 = 90 \text{ m}$$

$$x(t=2.1) = 10 \times (2.1)^2 = 44.1 \text{ m}$$

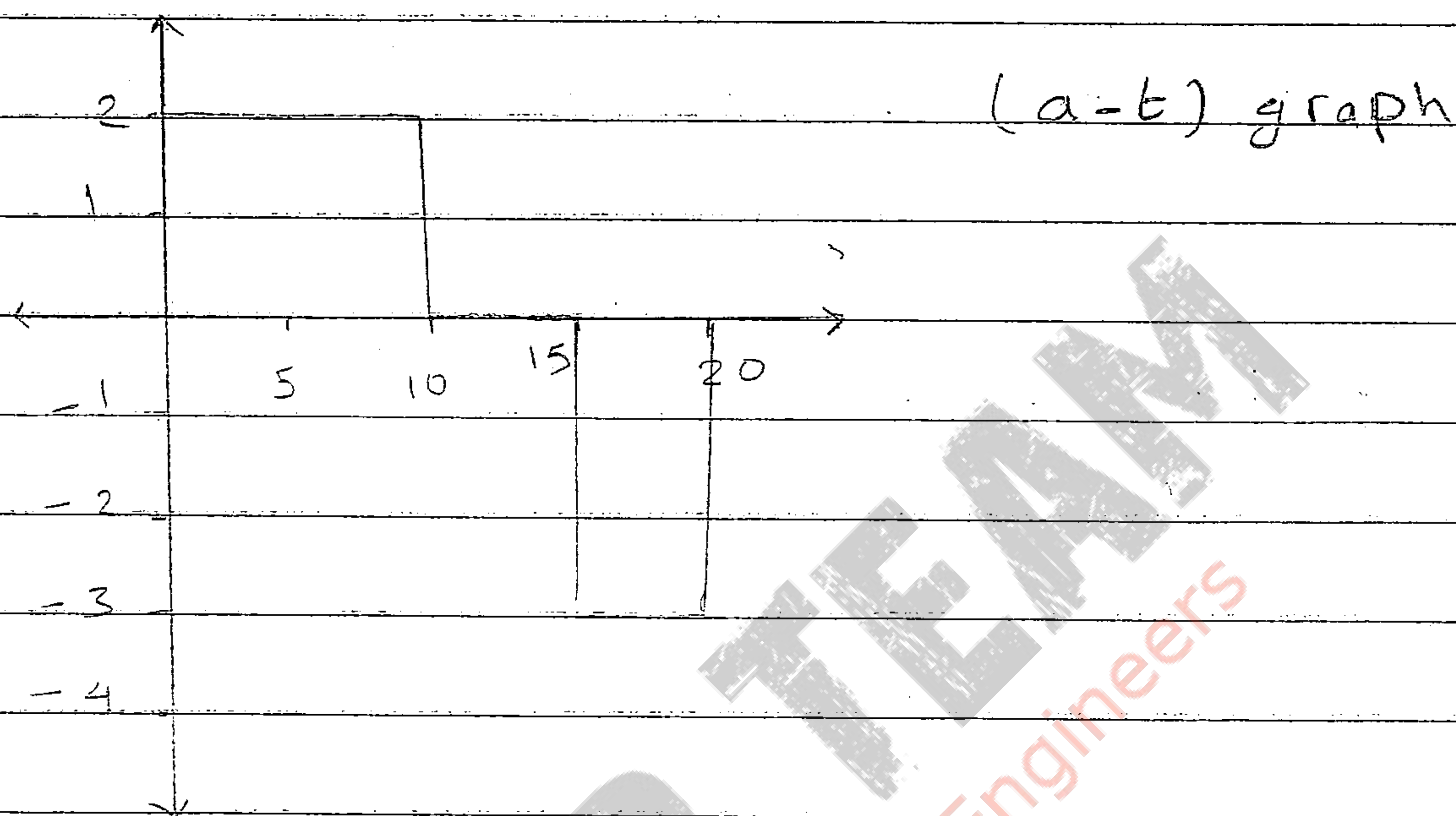
$$(a) v_{x, \text{avg}} = \frac{\Delta x}{\Delta t} \Rightarrow \frac{90-40}{3-2} \Rightarrow 50 \text{ m/s}$$

$$(b) v_{x, \text{avg}} = \frac{\Delta x}{\Delta t} \Rightarrow \frac{44.1-40}{2.1-2} \Rightarrow \frac{4.1}{0.1} = 41 \text{ m/s}$$

8/10/108 التاريخ Follow

الموضوع:

2.11 :- A particle starts from rest and accelerates as shown :-



Find (a) speed at $t=10$ sec and $t=20$ sec.
(b) distance traveled in the first 20 sec.

Solu:- $V_i = 0$

(a) From 0 to 10 sec $a=2$

$$V_f = V_i + a \times t \Rightarrow 0 + 2 \times 10$$

20 m/s

speed at $t=15$ sec is 20 m/s

why, $a=0$. V is constant

$$V_f = V_i + at$$

$$= 20 - 3(5) = 5 \text{ m/s}$$

تتبع التاريخ 8/10/108

الموضوع:

في تلك الفترة لأنه تلك ساعة
عالية.

$$\begin{aligned}x_F &= x_i + v_i t + \frac{1}{2} a t^2 \\&= 0 + 0 + \frac{1}{2} \times 2 \times 10^2 \\&= 100 \text{ m}\end{aligned}$$

over the next 5 sec.

$$\begin{aligned}x_F &= x_i + v_i t + \frac{1}{2} a t^2 \\&= 100 + 20 \times 5 + \frac{1}{2} \times 0 \times 5^2 \\&= 200 \text{ m}\end{aligned}$$

at $t = 20 \text{ sec}$.

$$\begin{aligned}x_F &= x_i + v_i t + \frac{1}{2} a t^2 \\&= 200 + 20 \times 5 + \frac{1}{2} \times 3 \times (5)^2\end{aligned}$$

$$x_F = 262 \text{ m}$$

2.13 :- A particle moves along the x-axis according to the equation $x = 2 + 3t - t^2$, where x is in meters and t is in sec. At $t = 3.00 \text{ s}$.

Find (a) the position of the particle
(b) its velocity. (c) its acceleration

8/10/08 التاريخ

الموضوع:

$$\begin{aligned} \Delta X \\ \text{(a)} \quad X_F - X_i &= \frac{1}{2} (v_i + v_F) t \\ 40 &= \frac{1}{2} (v_i + 2.8) 8.5 \\ v_i &= 6.61 \text{ m/s} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad v_F &= v_i + at \\ a &= \frac{v_F - v_i}{t} \end{aligned}$$

$$a = \frac{2.8 - 6.61}{8.5}$$

$$a = -0.448 \text{ m/s}^2$$

slowdown
↓
Low

$$2.24 \quad X = 2 + 3t - 4t^2$$

(a) Find its position when it changes direction.

(b) Find its velocity when it returns to the position it had at $t=0$

81/10/08 follow التاريخ

الموضوع:

Solu:-

$$x = 2 + 3t - t^2, t = 3 \text{ sec.}$$

$$\begin{aligned} (a) \quad x(t=3) &= 2 + 3t - t^2 \\ &= 2 + 9 - 9 \\ &= 2 \text{ m} \end{aligned}$$

$$(b) \quad x = 2 + 3t - t^2$$

$$v = 3 - 2t$$

$$= 3 - 6 = -3 \text{ m/s For left}$$

$$(c) \Rightarrow a = ?$$

$$\therefore a = 2 \text{ m/s}^2$$

من سرعة متساوية إلى سرعة متساوية

2.18:- A truck covers 40m in 8.5sec while smoothly slowing down to a final speed of 2.8 m/s:-

(a) Find its original speed.

(b) Find its acceleration.

$$\text{Solu:- } \Delta x = 40 \text{ m}, t = 8.5 \text{ sec}, v_f = 2.8 \text{ m/s}$$

تاریخ 8/10/08 follow

الموضوع:

Soln:-

$$x_f = x_i + v_i t + \frac{1}{2} a t^2$$

$$x = 2 + 3t - 4t^2$$

$$\frac{dx}{dt} = 3 - 8t$$

$$\Rightarrow x_i = 2, v_i = 3, a = -8$$

مع $a = -8$

$$(a) \quad v = \frac{dx}{dt} = 3 - 8t \Rightarrow 0 = 3 - 8t$$

وقت التوقف

$$t = \frac{3}{8} \text{ sec}$$

$$x = 2 + 3t - 4t^2$$

$$x = 2 + \frac{3 \times 3}{8} - 4 \times \frac{9}{64}$$

$$x = 2.56 \text{ m}$$

*note:- $ax^2 + bx + c = 0$

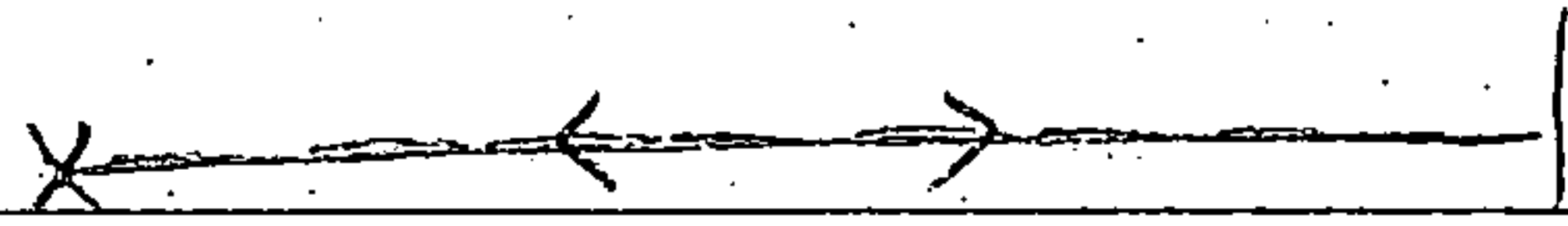
$$\Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

النتيجة

→ follow

تاريخ 29/9/08 follow

$$(b) t = 2 \times \frac{3}{8} = \frac{3}{4} \text{ sec.}$$



عند التوقف

2.56 m

دوران

نصف المسار

$$x_F = x_i$$

$$\Delta x = 0$$

$$v = 3 - 8t$$

$$v = 3 - 8 \times \frac{3}{4}$$

$$= -3 \text{ m/s}$$

solu [2] :-

$$x_F = x_i$$

$$x_F - x_i = v_i t + \frac{1}{2} a t^2$$

$$0 = v_i t + \frac{1}{2} a t^2$$

$$\Rightarrow t = \frac{-2v_i}{a} \Rightarrow t = \frac{-2 \times 3}{-8}$$

$$t = \frac{3}{4} \text{ sec.}$$

التاريخ 10/10/2020 Follow

الموضوع:

2.341 - A ball is thrown directly downward, with an initial speed 8 m/s, from a height 30 m. After what time does the ball strike the ground?

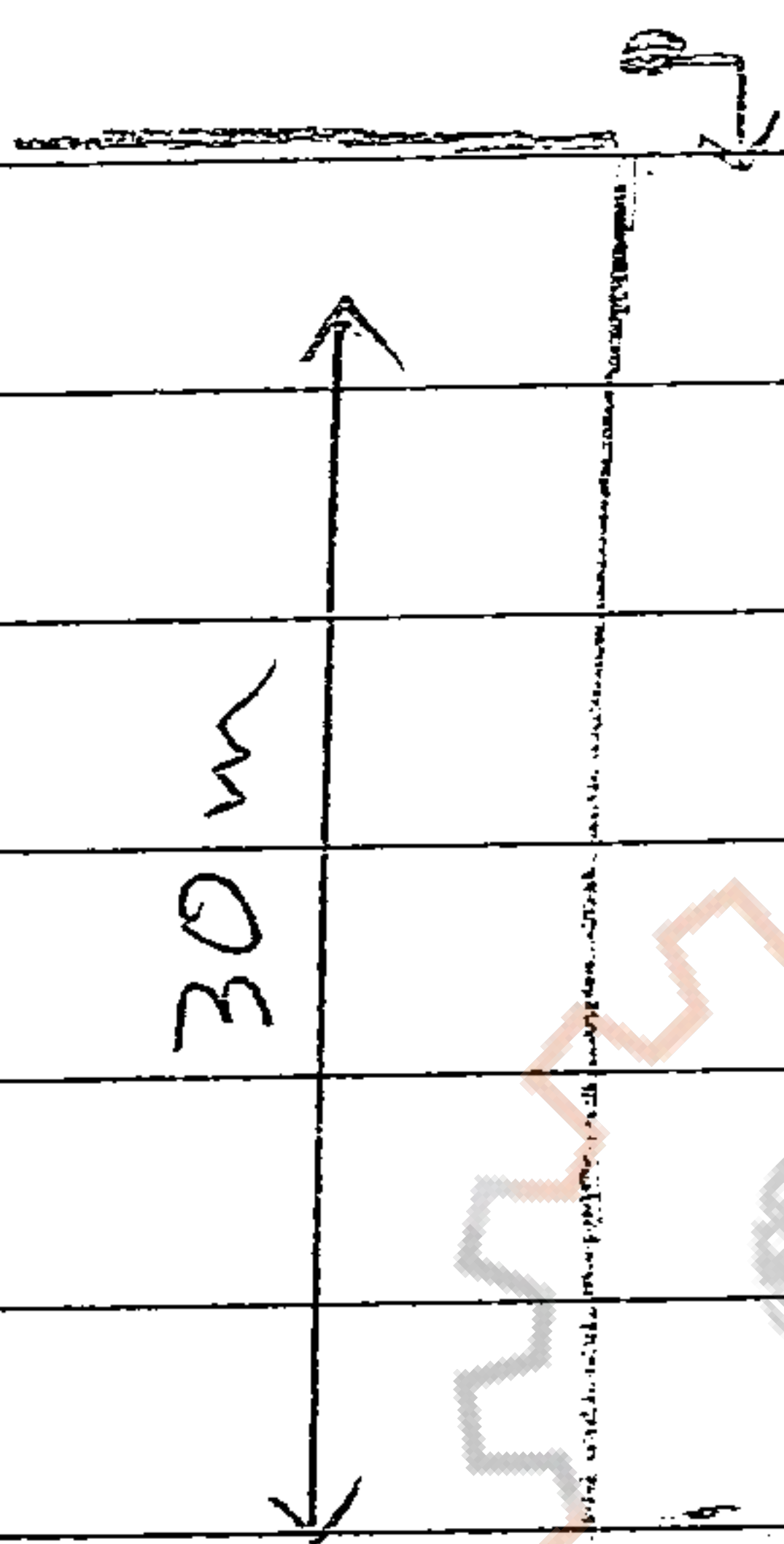
Solu: $V_i = -8 \text{ m/s}$, $\Delta x = 30 \text{ m}$

في البداية، $a = -9.8 \text{ m/s}^2$, $y_i = 30 \text{ m}$

في النهاية $y_f = 0 \text{ m}$ (في الأسفل دائماً)

إذاً إذا مضى، $v_f = ?$

سؤال



أي شيء سألنا في البداية؟

$$y_f = y_i + v_i t + \frac{1}{2} a t^2$$

$$0 = 30 - 8t - 4.9t^2$$

$$t = 1.7 \text{ sec}$$

10/10/08 التاريخ

الموضوع:

2.37

$$t = 3.00 \text{ sec}, V_{yF} = 0 \text{ m/s}, a_y = -9.8 \text{ m/s}^2$$

Find (a) the ball's initial velocity
(b) the height it reaches

Solu:-

(a) $V_{yi} = ?$

$$V_{yF} = V_{yi} + a_y t$$

$$0 = V_{yi} + (-9.8) \times 3$$

$$V_{yi} = +29.4$$

→ 29.4 m/s

(b) $\Delta y = ?$

$$V_{yF}^2 = V_{yi}^2 + 2a(y_F - y_i)$$

$$0 = (29.4)^2 + 2 \times (-9.8) \Delta y$$

$$797.97 = 19.6 \Delta y$$

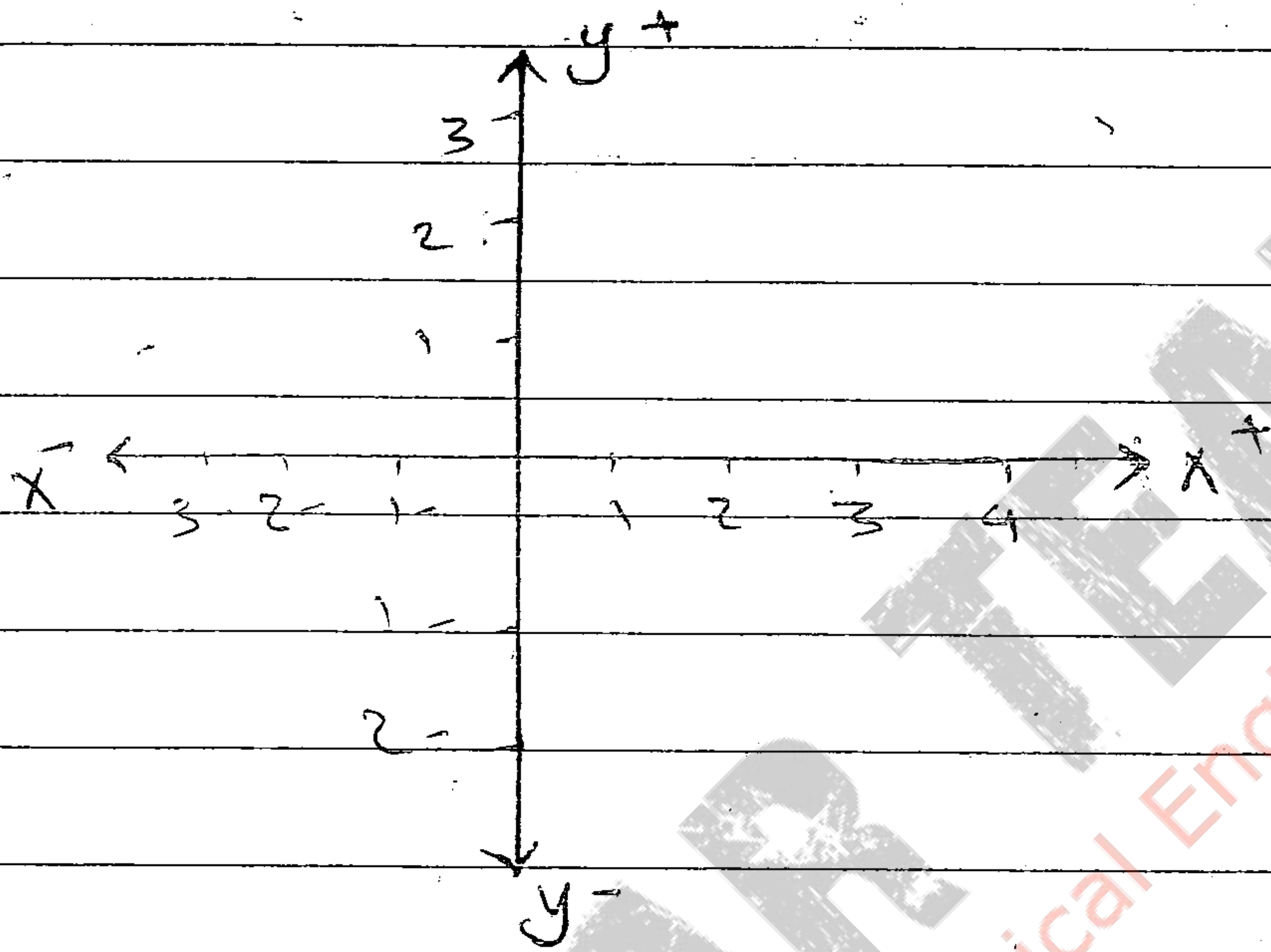
$$\therefore \Delta y = 35.56 \text{ m}$$

الموضوع: 3.1 coordinate systems التاريخ 10/10/08

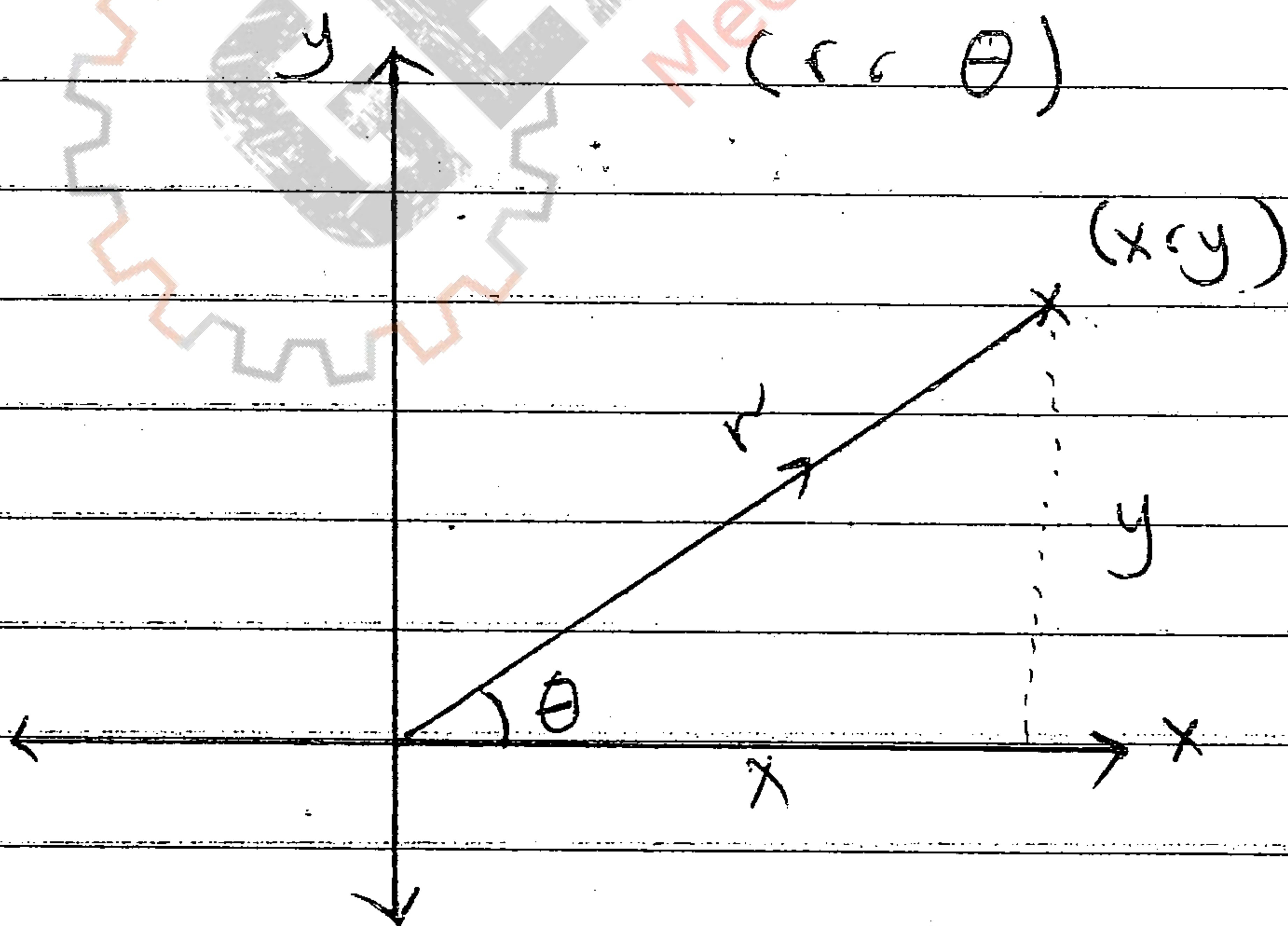
محاور

* cartesian coordinates (rectangular coordinate)

(x, y) (x, y, z)



* polar coordinates :-



r :- المسافة عن الـ origin للنقطة

θ :- الزاوية التي يصنعها المحاور

للنقطة في أي System آخر

polar

* conversion between cartesian and coordinate:-

$$\cos \theta = \frac{x}{r} \Rightarrow x = r \cos \theta$$

$$\sin \theta = \frac{y}{r} \Rightarrow y = r \sin \theta$$

$$r = \sqrt{x^2 + y^2}, \quad \theta = \tan^{-1} \frac{y}{x}$$

* exp 3.1:- page 54:- Four Quarter

$(x, y) = (-3.5, -2.5)$ Cartesian

Find its polar coordinates:-

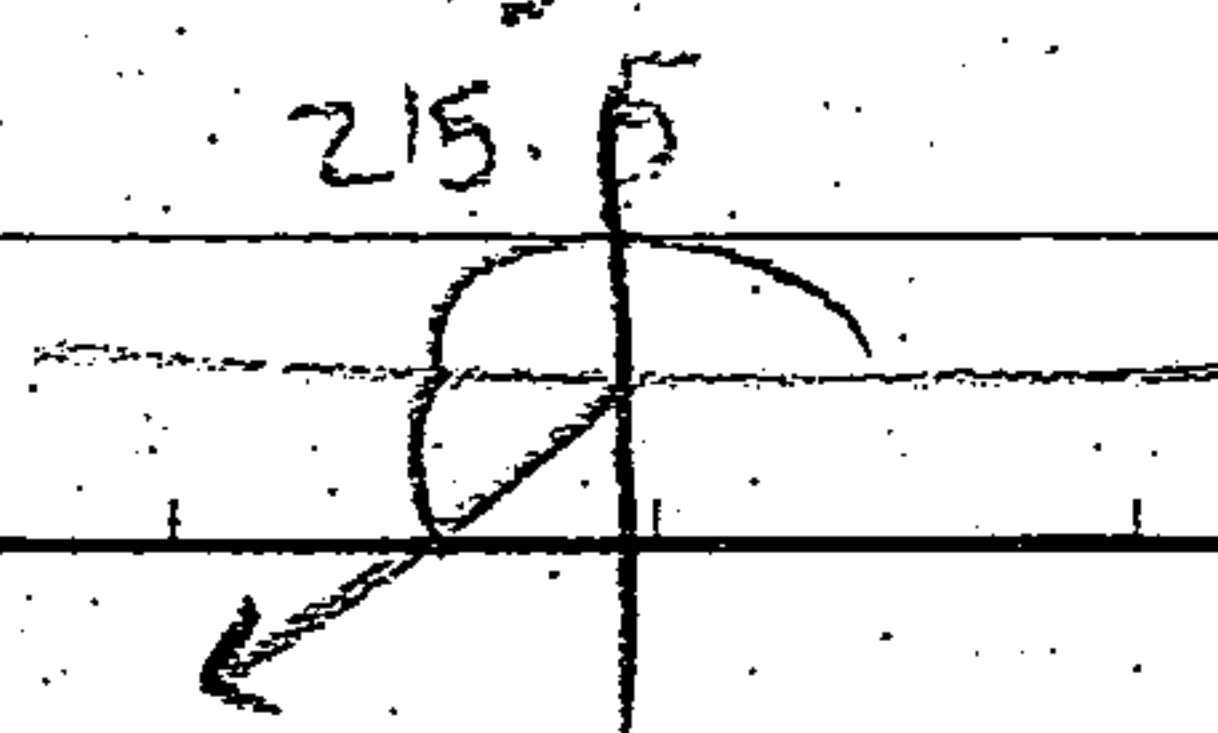
Solu:- r, θ up

$$r = \sqrt{x^2 + y^2} = \sqrt{(-3.5)^2 + (-2.5)^2} = 4.3$$

$$\theta = \tan^{-1} \frac{y}{x} \Rightarrow \tan^{-1} \left(\frac{-2.5}{-3.5} \right) = 35.5^\circ$$

* $(r, \theta) \Rightarrow (4.3, 35.5^\circ)$ النقطة في الربع الثالث

$(4.3, 215.5^\circ)$ الحل:-



الموضوع: vector and scalar quantities 3.2 التاريخ 10/10/101

3.3 some properties of vectors.

3.2 :-

* scalar :- is completely specified by a single magnitude with an appropriate unit without direction

exp $\Rightarrow$ speed, distance, mass, ...

* vector :- is completely specified by a number and appropriate unit with direction

exp $\Rightarrow$ velocity, displacement, force, ...

$\vec{A}$ vector

$|\vec{A}|$ magnitude

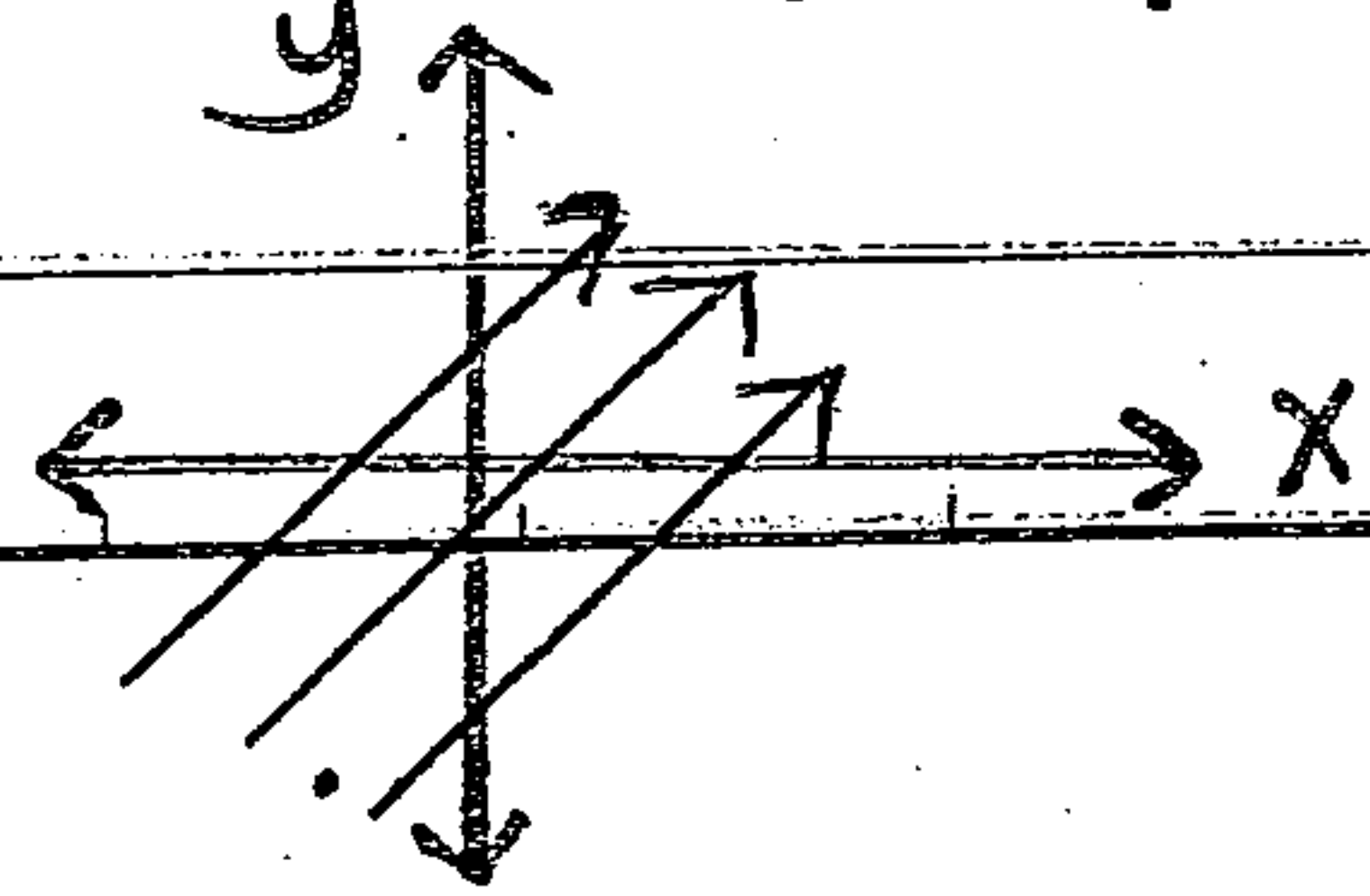
3.3 Some Properties of vectors :-

بعض خواص المتجهات

* Equality of two vectors :-

$\vec{A} = \vec{B}$ only if $|\vec{A}| = |\vec{B}|$ and

if $\vec{A}$ and $\vec{B}$ point in the same direction.

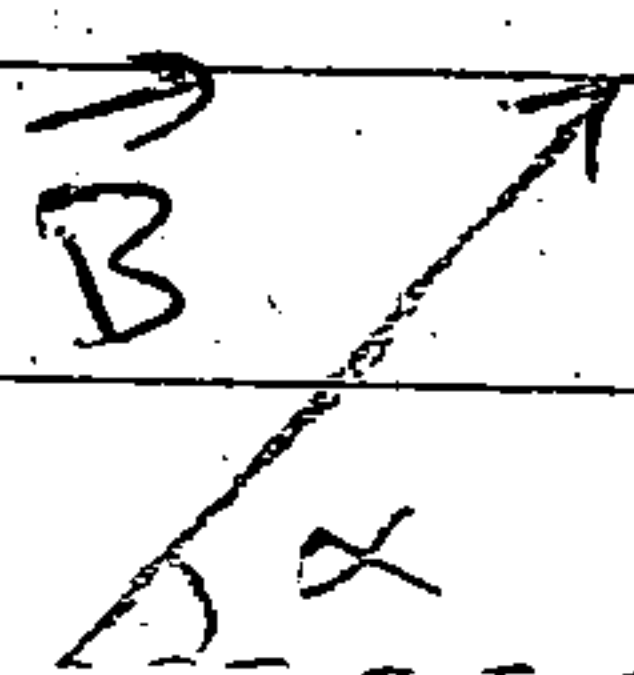


$\rightarrow$ Follow

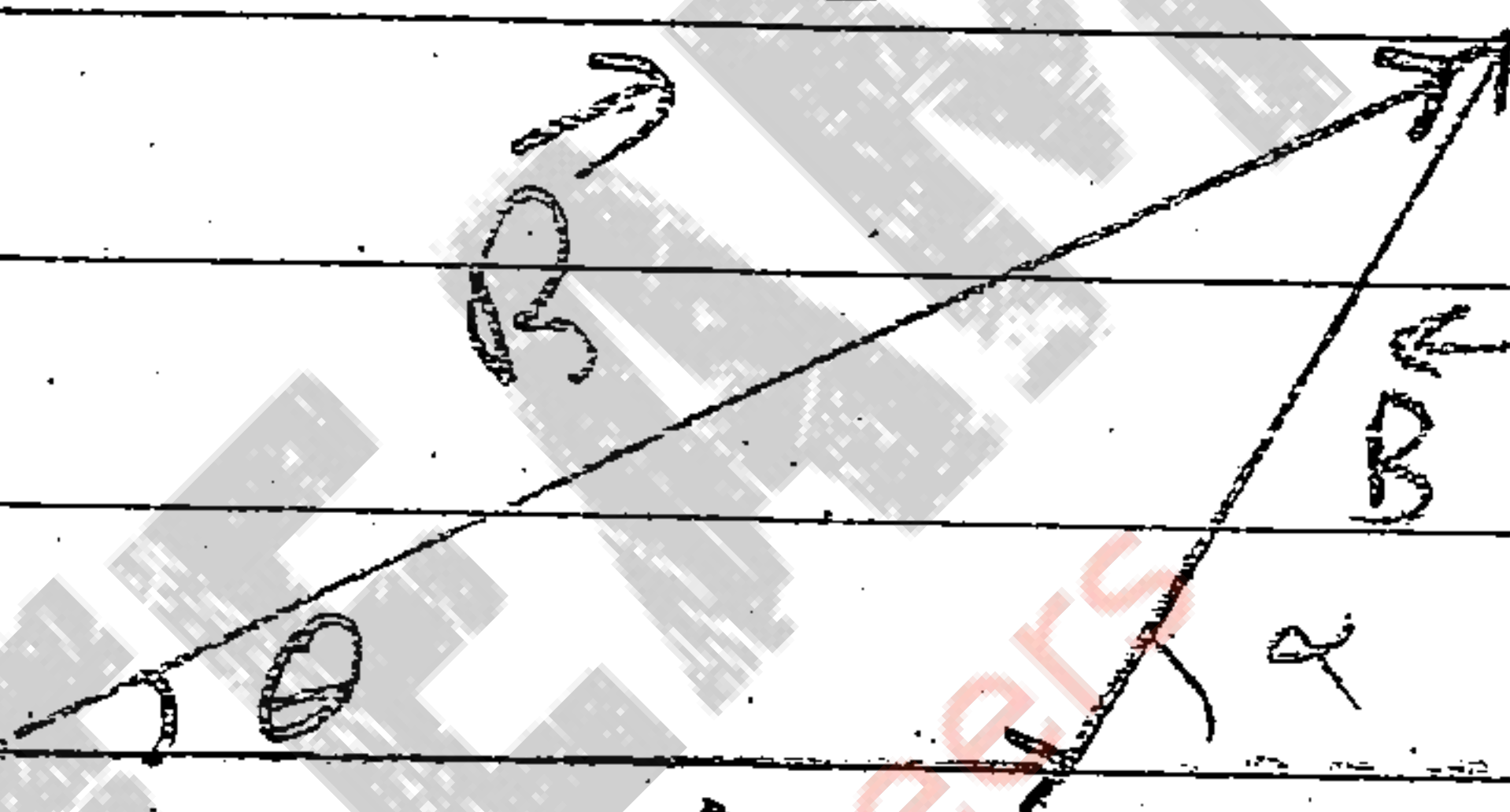
* Adding vectors :-

$$\vec{R} = \vec{A} + \vec{B}$$

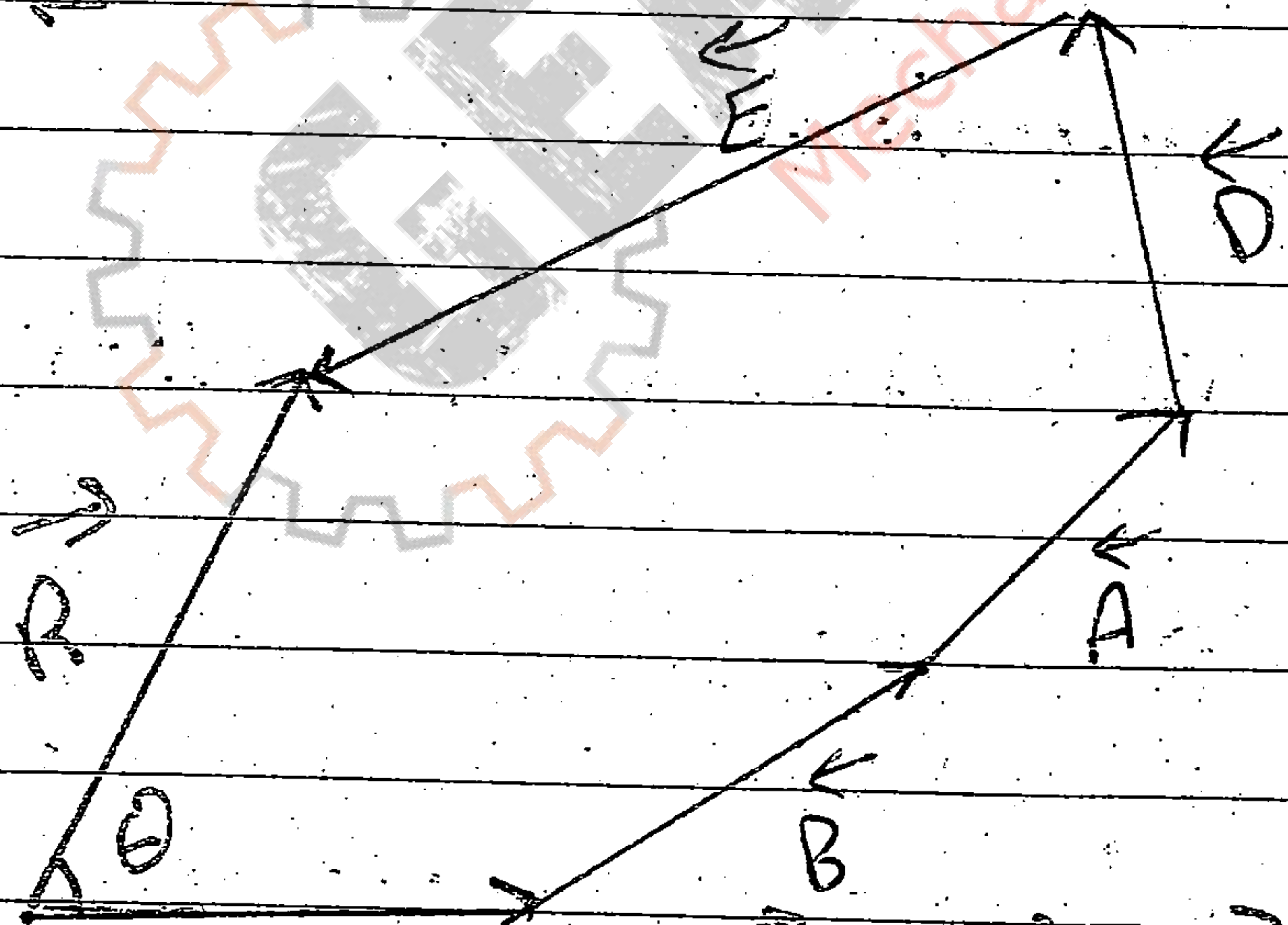
Resultant



$$\vec{R} = \vec{A} + \vec{B}$$



R drawn from the tail of A to the tip of B



$$\vec{R} = \vec{A} + \vec{B} + \vec{C} + \vec{D} + \vec{E}$$

تتبع التاريخ 10/10/108

$$\vec{A} + \vec{B} = \vec{B} + \vec{A} \quad \text{"commutative"}$$

قانون الجمع التبادلي "خاصية التبادلي"

$$\vec{A} + (\vec{B} + \vec{C}) = (\vec{A} + \vec{B}) + \vec{C}$$

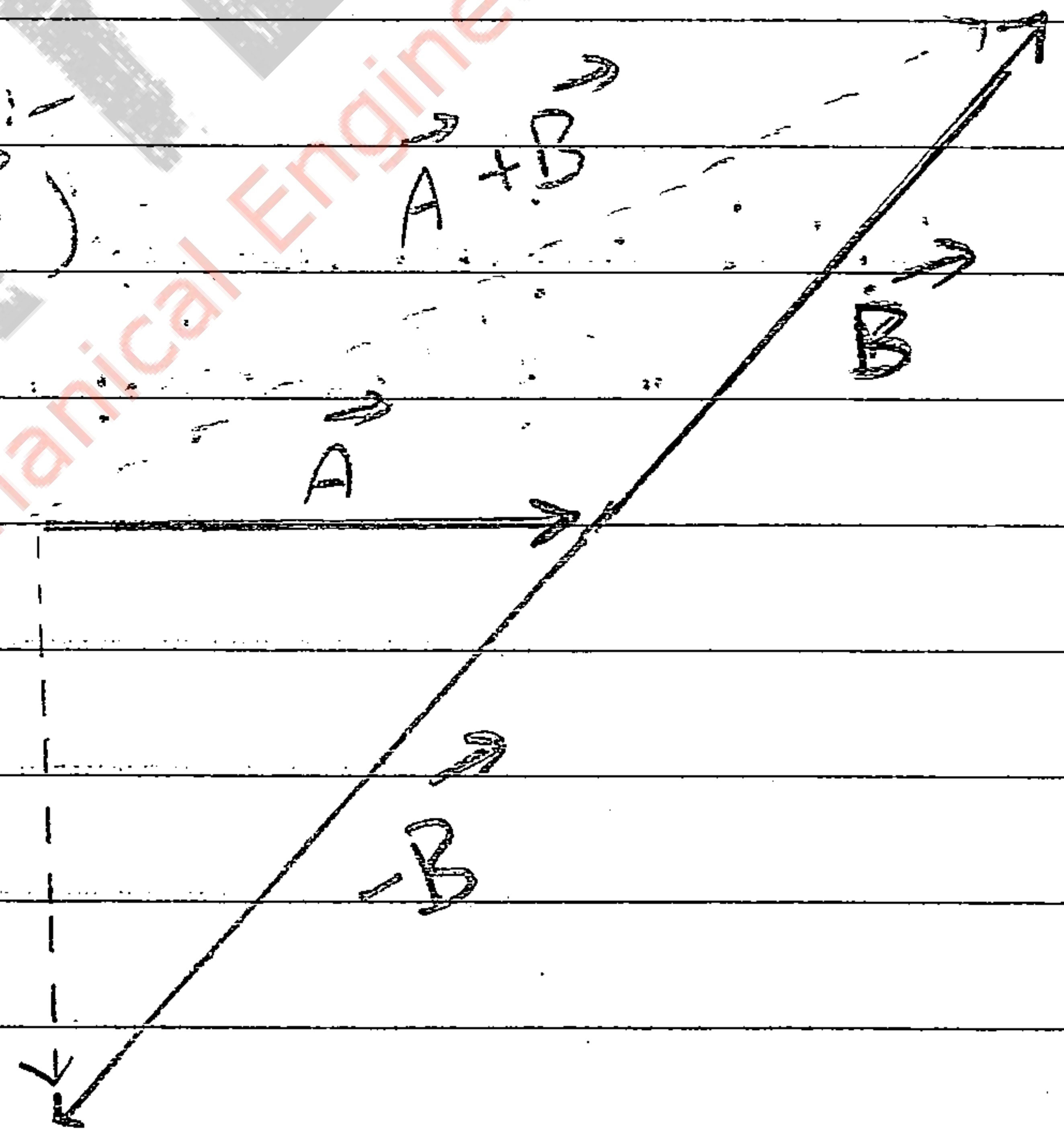
* negative of a vector:-

$-\vec{A}$ negative of $\vec{A}$

$\vec{A}$ and $-\vec{A}$ have the same magnitude but point in the opposite direction

* Subtracting vectors:-

$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$$



- Subtracting vectors means adding the negative of vectors.

* Multiplying a vector by a scalar :-

$\vec{A}, m \rightarrow m\vec{A}$ ← stays same direction but the magnitude change.

- If vector $\vec{A}$ is multiplied by a positive scalar quantity (m) the product $m\vec{A}$ is a vector that has the same direction as $\vec{A}$ and magnitude $m\vec{A}$. If vector $\vec{A}$ is multiplied by a negative scalar quantity $-m$, the product $-m\vec{A}$ is directed opposite $\vec{A}$.

« إذا ضربنا بوجه يتغير الاتجاه منه الاتجاه إذا
ضربنا بـ -1 يتغير الاتجاه والقيمة »

« سم الموجه يكون بين رأس النقطتين، ثابت »

« الفرق بين $\vec{A} + \vec{B}$ و $\vec{B} + \vec{A}$ هو أنه في الأول
نسب $\vec{B}$ واسم $\vec{A}$ »

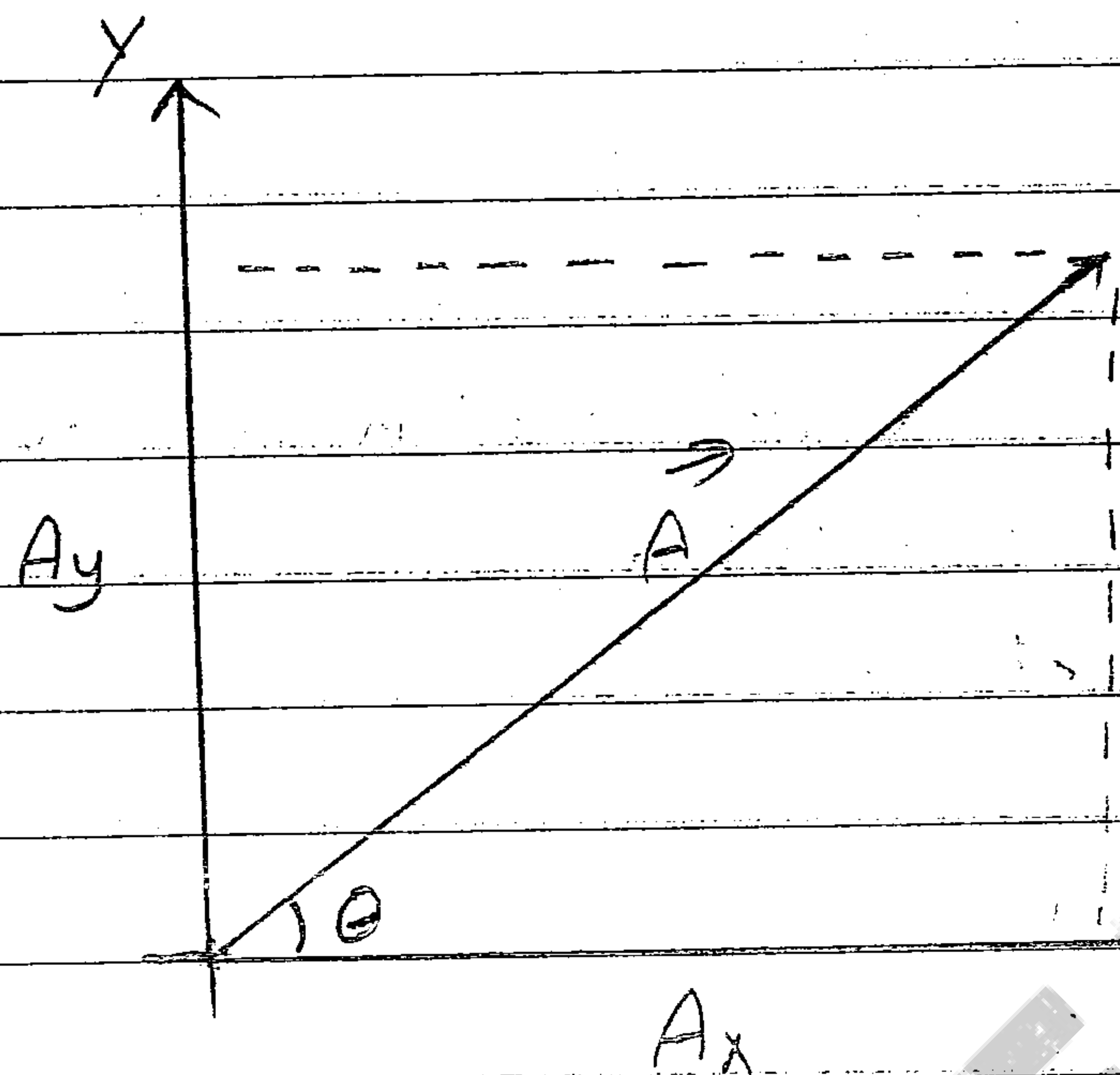
« موجه $\vec{A} + \vec{B}$ ليس موجه $\vec{A}$ »

« موجه $\vec{A} + \vec{B}$ ليس موجه $\vec{A}$ »

3.4 components of a vector and unit vectors

الموضوع:

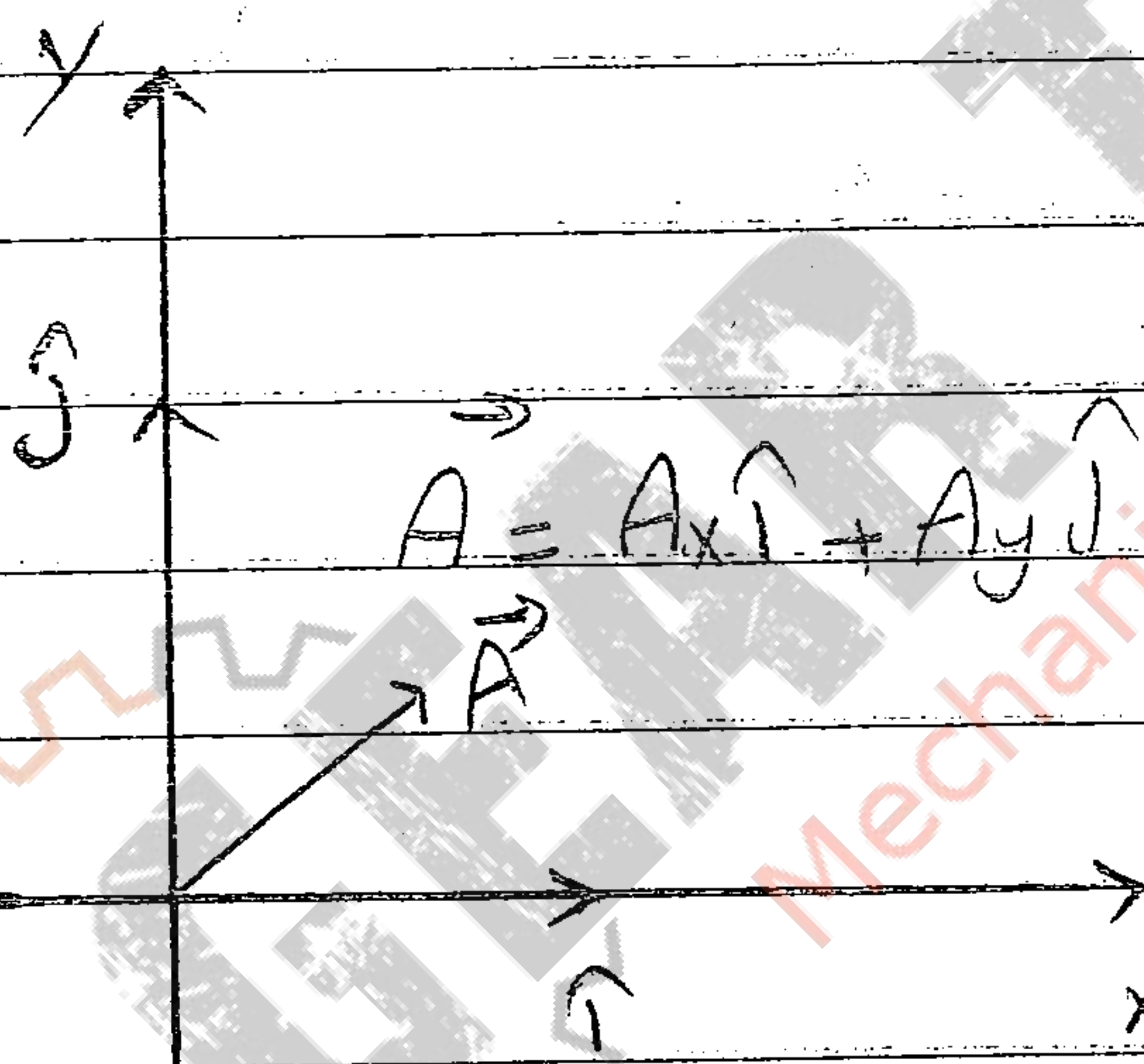
التاريخ 10/12/18



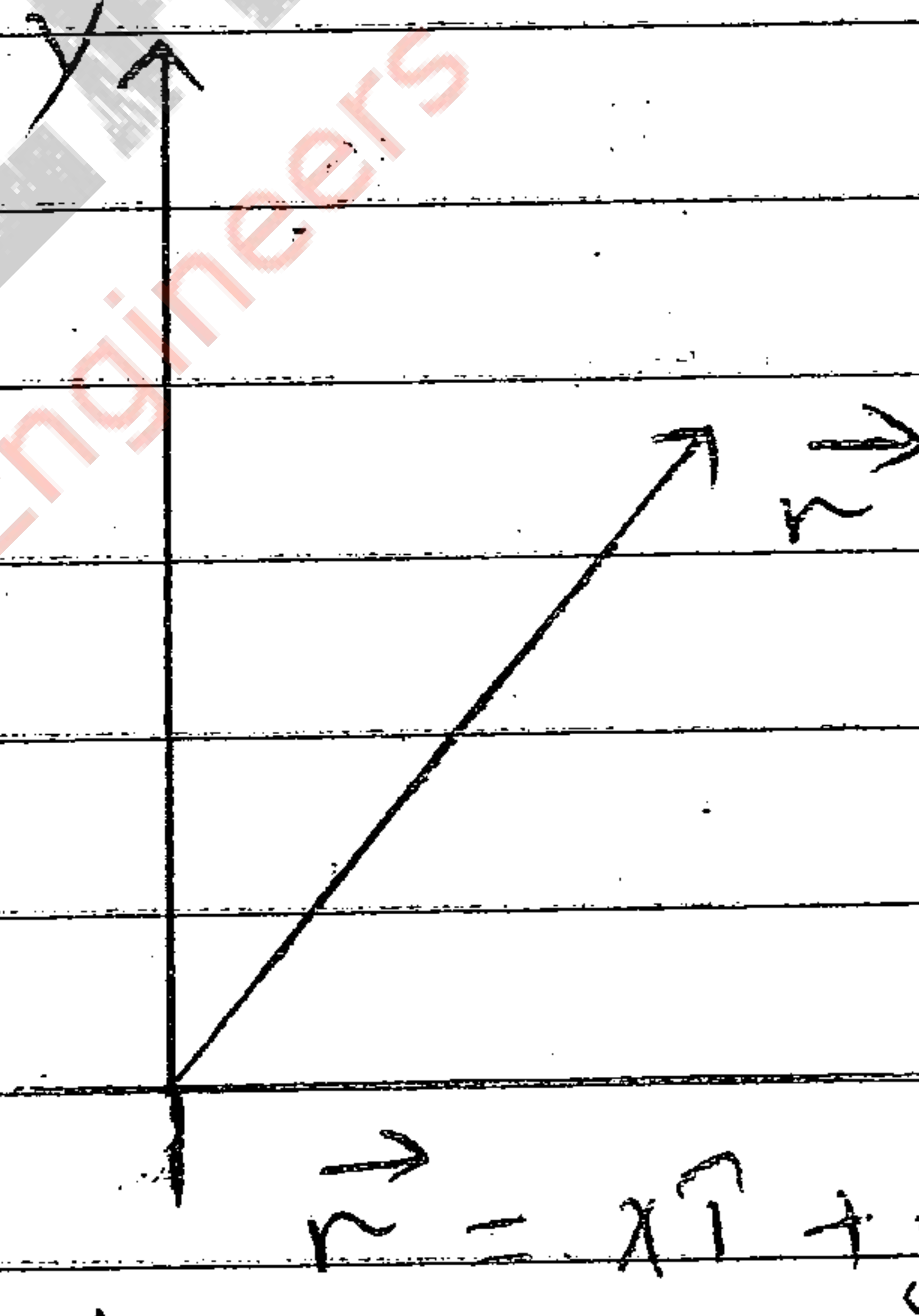
$$A_x = |\vec{A}| \cos \theta$$

$$A_y = |\vec{A}| \sin \theta$$

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2}$$



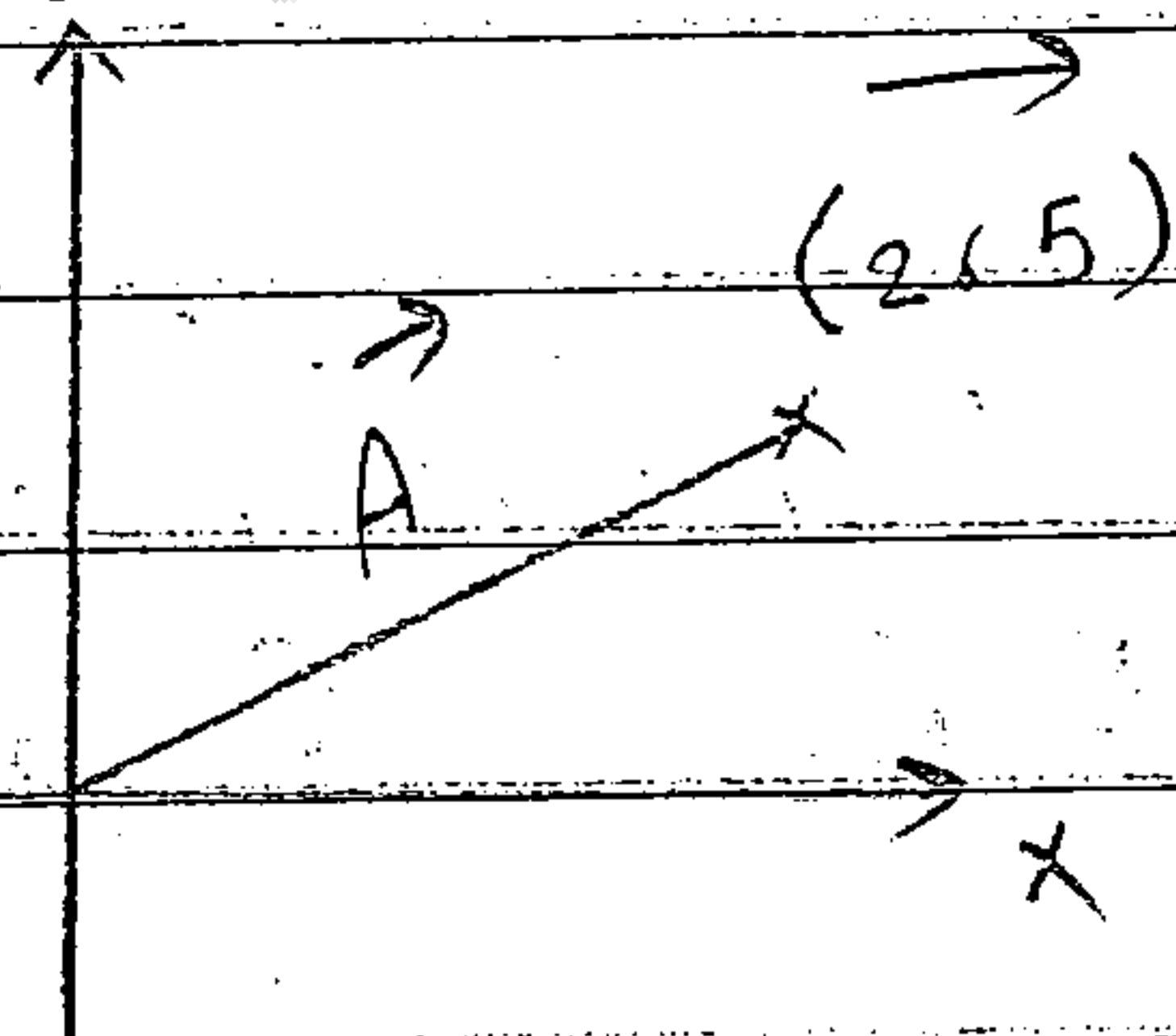
$$\vec{A} = A_x \hat{i} + A_y \hat{j}$$



$$\vec{r} = x \hat{i} + y \hat{j}$$

displacement

ex:-



$$\hat{i} = 2$$

$$\hat{j} = 5$$

note:- عندنا متجه اتحاد المحصلة $\vec{R}$ من 3 أشياء
 ① مركبات components ② اتجاه θ ③ مقدار $|\vec{R}|$ "محصلة"

* $\vec{A}$ Unit vector is a dimensionless vector having a magnitude of 1.

$$|\hat{i}| = |\hat{j}| = |\hat{k}| = 1$$

$$\vec{A} = A_x \hat{i} + A_y \hat{j}$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j}$$

$$\vec{R} = \vec{A} + \vec{B}$$

$$= (A_x + B_x) \hat{i} + (A_y + B_y) \hat{j}$$

$$= R_x \hat{i} + R_y \hat{j}$$

$$|\vec{R}| = \sqrt{R_x^2 + R_y^2}$$

$$\theta = \tan^{-1} \frac{R_y}{R_x}$$

(magnitude) R_x $\hat{i}$ $\hat{j}$ $\hat{k}$

* in 3-D (x, y, z)

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

$$\vec{R} = \vec{A} + \vec{B} = (A_x + B_x) \hat{i} + (A_y + B_y) \hat{j} + (A_z + B_z) \hat{k}$$

$$= R_x \hat{i} + R_y \hat{j} + R_z \hat{k}$$

$$|\vec{R}| = \sqrt{R_x^2 + R_y^2 + R_z^2}$$

$$\cos \theta = \frac{R_x}{|\vec{R}|}$$

الموضوع: Follow التاريخ 10/10/12

vector with units

example: - $\vec{A} = (2\hat{i} + 2\hat{j}) \text{ m}$

$\vec{B} = (2\hat{i} - 4\hat{j}) \text{ m}$

Find $\vec{A} + \vec{B} = \vec{R}$ components مركبات

solu: - components اجمع سينت
مع سينت والصوات مع الصوات

$$\vec{R} = \vec{A} + \vec{B}$$

$$= (2+2)\hat{i} + (2-4)\hat{j}$$
$$\vec{R} \Rightarrow 4\hat{i} - 2\hat{j}$$

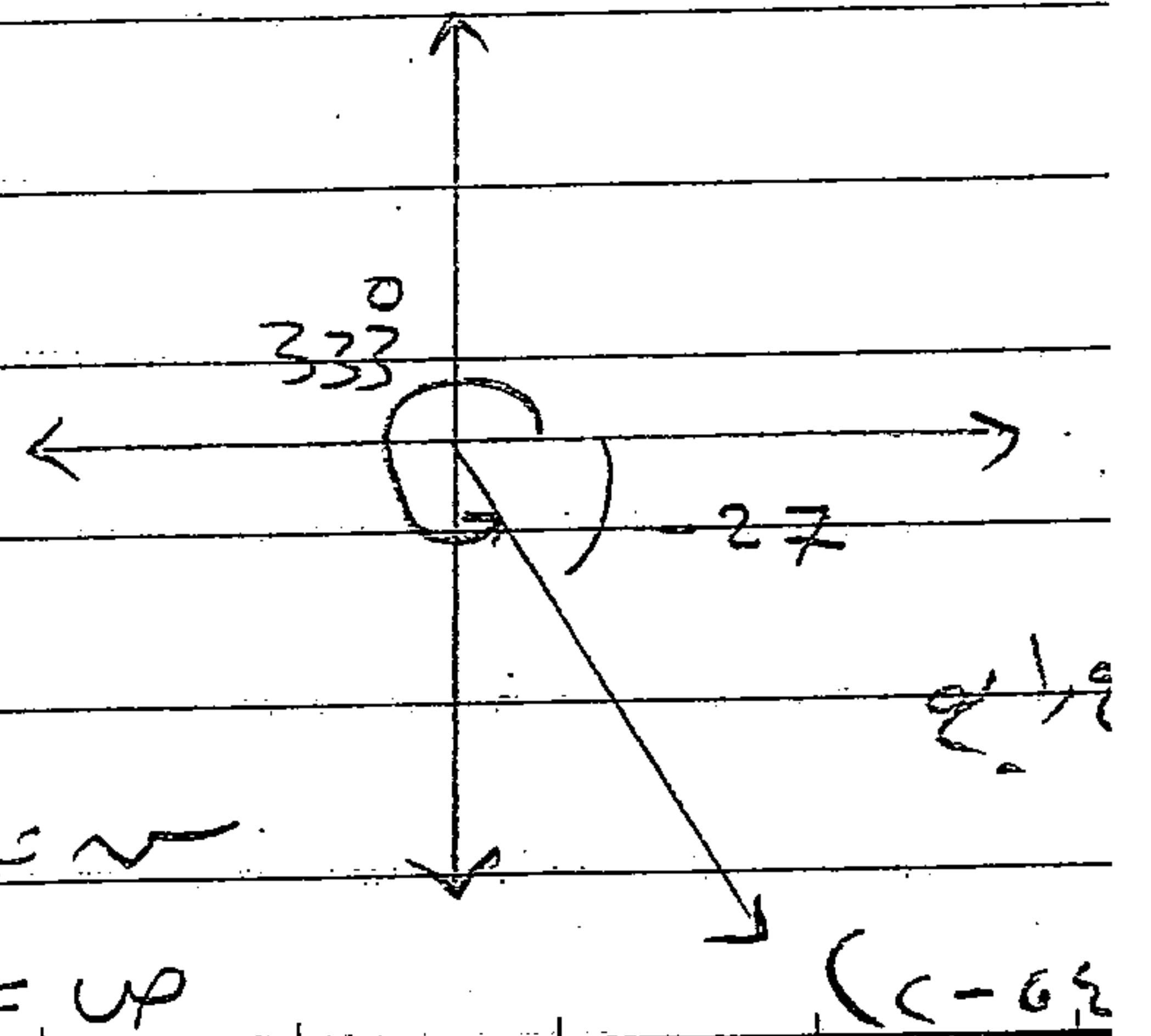
components $\vec{R}$

$$|\vec{R}| = \sqrt{16+4} = \sqrt{20} = 4.5 \text{ cm}$$

$$\theta = \tan^{-1} \frac{-2}{4} \Rightarrow -27^\circ \text{ or } 333^\circ$$

-note:- عندما كد الدقات θ

نقيم بالرسم لايجاد الزاوية
الصحيحة



ex 2:- A particle under goes three displacements :-

$$\vec{dr}_1 = (15\hat{i} + 30\hat{j} + 12\hat{k}) \text{ cm}$$

$$\vec{dr}_2 = (23\hat{i} - 14\hat{j} - 5\hat{k}) \text{ cm}$$

$$\vec{dr}_3 = (13\hat{i} + 15\hat{j} \quad \quad \quad) \text{ cm}$$

لا يوجد $\hat{k}$

Find the components of the resultant displacement and its magnitude ?

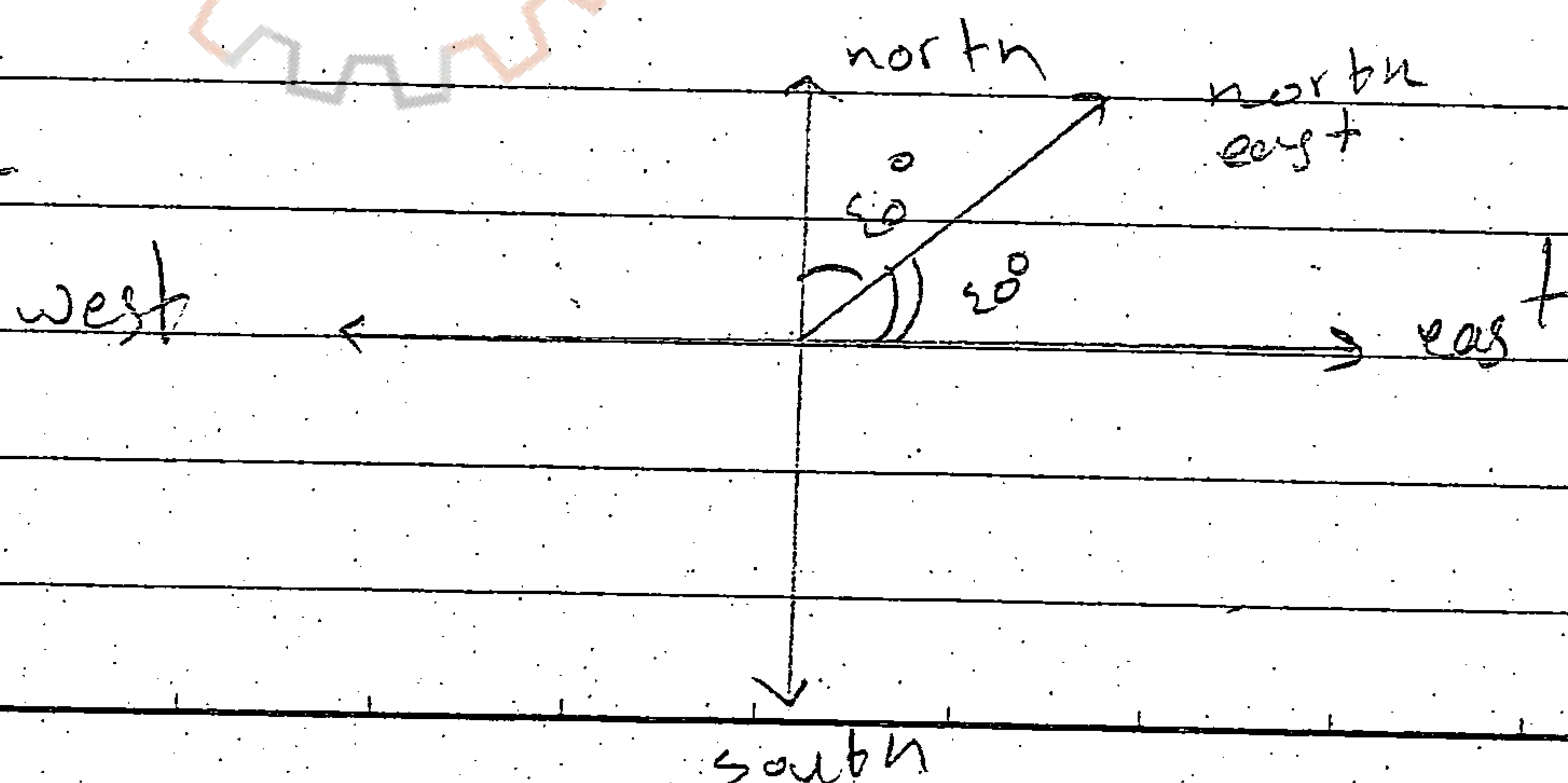
Solu:-

$$\begin{aligned} \vec{dr} &= \vec{dr}_1 + \vec{dr}_2 + \vec{dr}_3 \\ &= (15 + 23 - 13)\hat{i} + (30 - 14 + 15)\hat{j} \\ &\quad + (12 - 5 + 0)\hat{k} \end{aligned}$$

$$= (25\hat{i} + 31\hat{j} + 7\hat{k})$$

$$|\vec{dr}| = \sqrt{(25)^2 + (31)^2 + (7)^2} = 40 \text{ cm}$$

-note:-



7.3 the scalar product of two vectors

التاريخ 12/10/08

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

* Scalar or dot product $\vec{A} \cdot \vec{B}$

الجواب في الفيزياء، لنفعل كونه فقط جواب من الإجابات

$$\textcircled{1} \vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$$

θ angle between $\vec{A}$ and $\vec{B}$

$$|\vec{A}| = \sqrt{x^2 + y^2 + z^2}$$

$$\textcircled{2} \vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

سنستخدم طريقة (أ) لا يجب أن فقط

سنستخدم طريقة (ب) لأنه أسهل

* Note: $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1 \rightarrow \cos \theta = 1$

$$\hat{i} \cdot \hat{j} = \hat{i} \cdot \hat{k} = \hat{j} \cdot \hat{k} = 0 \rightarrow \cos 90^\circ = 0$$

$$\Rightarrow \vec{A} \cdot \vec{A} = A_x^2 + A_y^2 + A_z^2 = |\vec{A}|^2$$

ex:- $w = \vec{F} \cdot \vec{r} \leftarrow$ displacement

work $\leftarrow$ Force

أخذ

"قياس فقط للشد تقاسم" قوة

$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$

$$\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$$

Ex 1:-

$$\vec{A} = 2\hat{i} + 3\hat{j}, \quad \vec{B} = -1\hat{i} + 2\hat{j}$$

(a) Find $\vec{A} \cdot \vec{B}$

(b) Find the angle θ between $\vec{A}$ and $\vec{B}$

Solu:-

$$\begin{aligned} (a) \vec{A} \cdot \vec{B} &= A_x B_x + A_y B_y \\ &= 2(-1) + (3)(2) \\ &= -2 + 6 \Rightarrow 4 \end{aligned}$$

$$(b) \vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$$

$$4 = \sqrt{4+9} \sqrt{1+4} \cos \theta$$

$$4 = \sqrt{13} \sqrt{5} \cos \theta$$

$$\theta = \cos^{-1} \frac{4}{\sqrt{65}} \approx 60.3$$

تاريخ 12 / 10 / 08 Follow

الموضوع:

Ex 2

Ex 2 :- A particle moving in x-y plane undergoes a displacement given by

$$\vec{\Delta r} = (2\hat{i} + 3\hat{j}) \text{ m and Force}$$

$$\vec{F} = (5\hat{i} + 2\hat{j}) \text{ N}$$

(a) Find magnitude of the force and displacement.

(b) Find the work done by the force at the particle.

(c) Find the angle between Force and displacement.

Solu. -

$$(a) |\vec{F}| = \sqrt{25 + 4} = \sqrt{29} = 5.4 \text{ N}$$

$$|\vec{\Delta r}| = \sqrt{4 + 9} = \sqrt{13} = 3.6 \text{ m}$$

$$(b) W = \vec{F} \cdot \vec{\Delta r}$$

$$= (5)(2) + (2)(3)$$

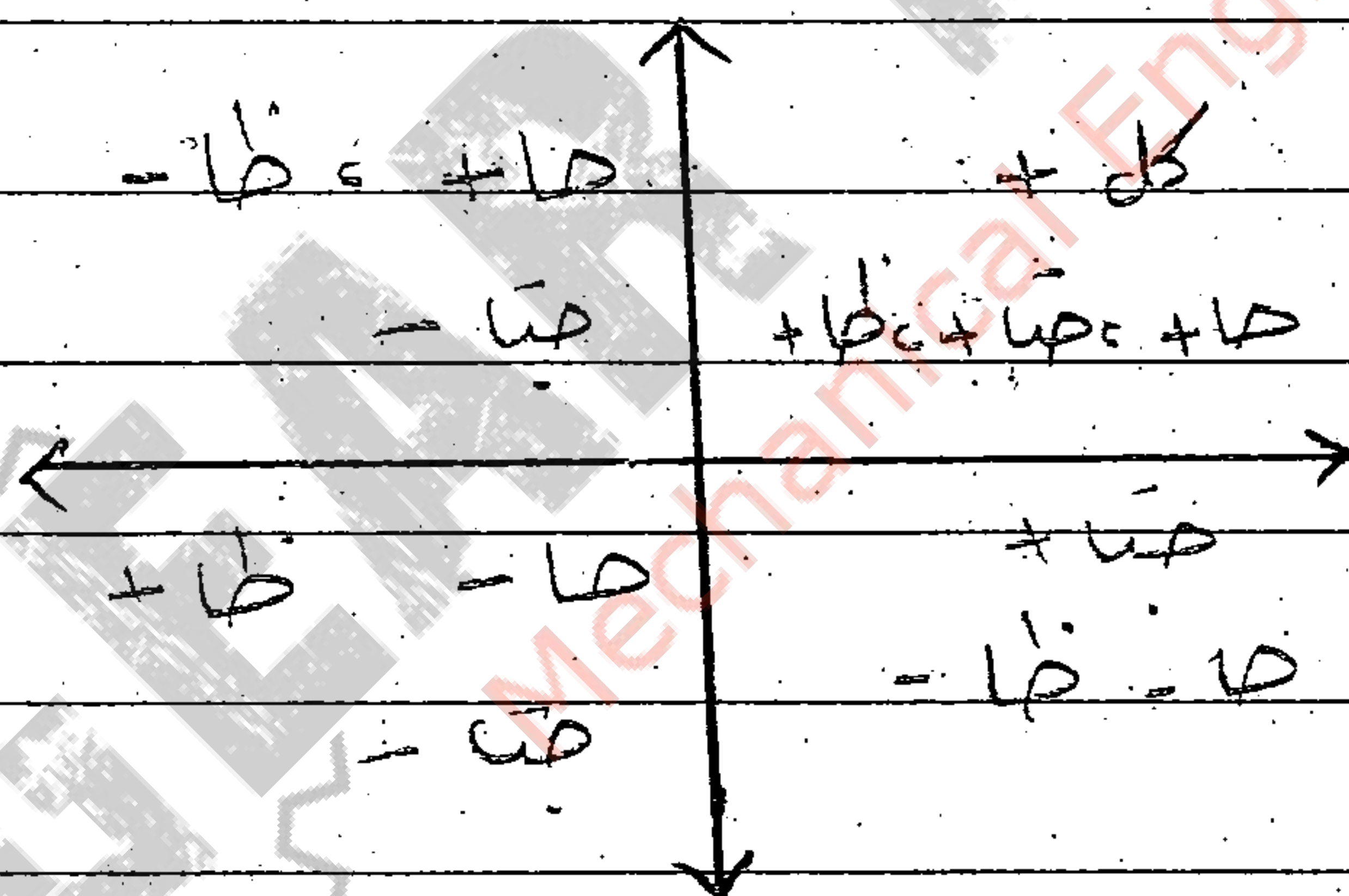
$$= 10 + 6 = 16 \text{ Joule (N.m)}$$

$$C) \cos \theta = \frac{|F|}{|A|} = \frac{16}{5.4}$$

$$\cos \theta = \frac{3.6}{5.4}$$

$$\theta = \cos^{-1} \left(\frac{3.6}{5.4} \right) = 34.38^\circ$$

أي قوة تنطوي على أحد الحمار تتعامل مع كل مباشر و... تحليل
 ولكن القوة التي تصنع زاوية مع محور السينات تحليل إلى
 مركبتين أفقية وأخرى رأسية الأولى المركبة الأفقية السبق
 وكما والأخرى المركبة الرأسية السبق.



note:- look at exercises at first
 secondary.

الموضوع: 11.3 The vector product and Torque التاريخ 12/10/08

$\vec{A}, \vec{B}$ two vectors.

* vector or cross product :-

$$\vec{A} \times \vec{B} = \vec{C}$$

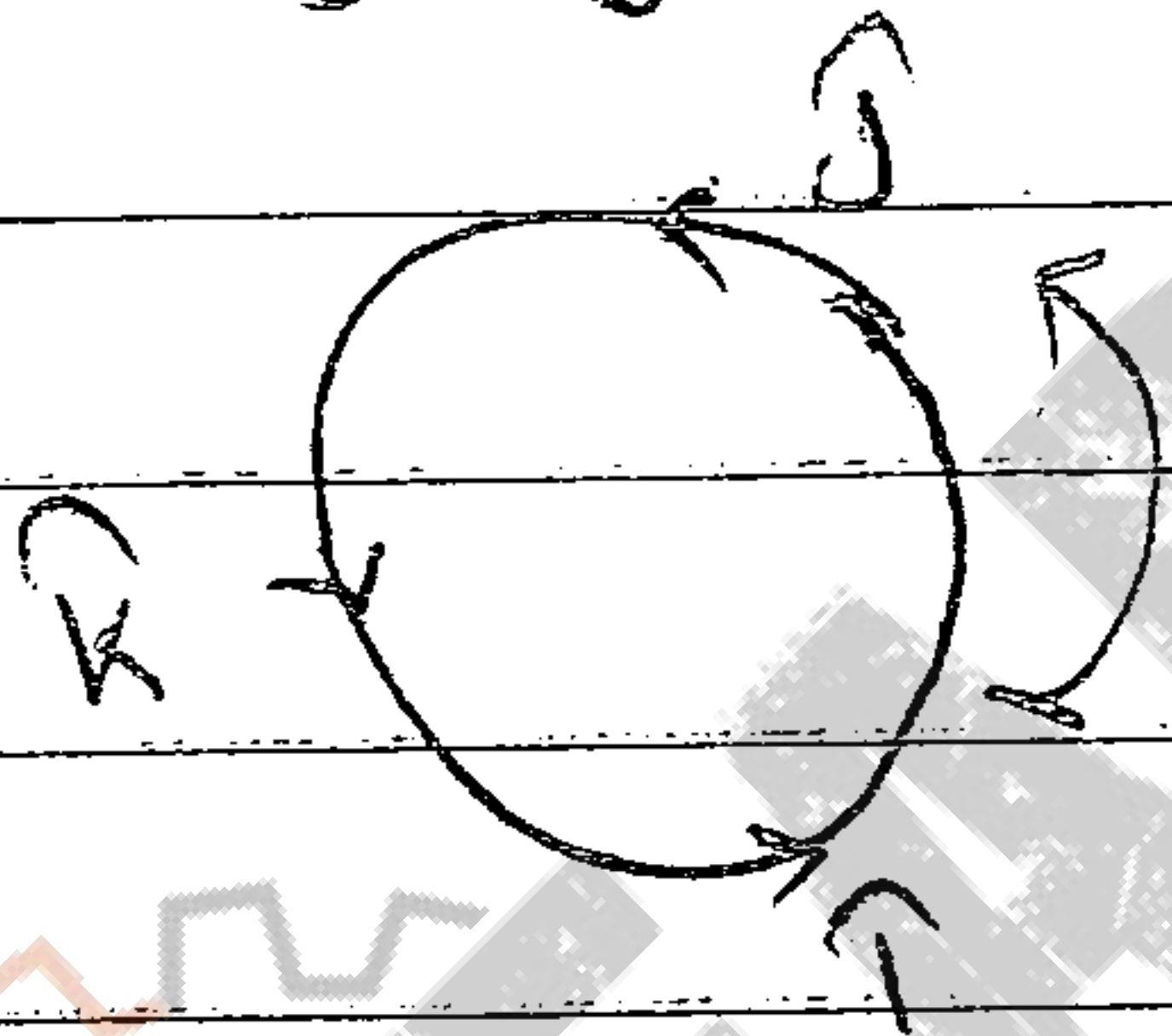
$$\vec{A} = \vec{C} \times \vec{P}$$

$$|\vec{C}| = |\vec{A}| |\vec{B}| \sin \theta$$

$$\vec{C} \text{ or } \vec{P} = \vec{C} \times \vec{P}$$

θ angle between $\vec{A}$ and $\vec{B}$

$$\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0$$



كان اتجاه المحاور متساوي

$$\hat{i} \times \hat{j} = \hat{k}$$

$$\hat{j} \times \hat{i} = -\hat{k}$$

$$\hat{j} \times \hat{k} = \hat{i}$$

$$\hat{k} \times \hat{j} = -\hat{i}$$

$$\hat{k} \times \hat{i} = \hat{j}$$

$$\hat{i} \times \hat{k} = -\hat{j}$$

أحيانا نكتب $\vec{P} \times \vec{r}$ ونسميها عزم الدوران $\vec{L}$ أو عزم الزخم $\vec{L}$

تاریخ 12/10/08 Follow

الموضوع:

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

نصف دائرة

ممكن ان يكون

$$= \hat{i} (A_y B_z - A_z B_y)$$

$$= -\hat{j} (A_x B_z - A_z B_x)$$

$$= +\hat{k} (A_x B_y - A_y B_x)$$

$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$$

$$\vec{A} \times \vec{A} = 0 \quad \text{"sin } 0 = 0 \text{"}$$

$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$$

$$\text{exp: } \vec{\tau} = \vec{r} \times \vec{F}$$

Torque ← position × Force

12/10/08 التاريخ

الموضوع:

exp 11.1 :-

$$\vec{A} = 2\hat{i} + 3\hat{j}$$

$$\vec{B} = -\hat{i} + 2\hat{j}$$

Find $\vec{A} \times \vec{B}$ and $\vec{B} \times \vec{A}$

Solu:-

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 0 \\ -1 & 2 & 0 \end{vmatrix}$$

$$= \hat{i}(0-0) - \hat{j}(0-0) + \hat{k}[(2)(2) - (3)(-1)]$$

$$= 7\hat{k}$$

$$\vec{A} \times \vec{B} = 7\hat{k}$$

$$\Rightarrow \vec{B} \times \vec{A} = -\vec{A} \times \vec{B} \\ = -7\hat{k}$$

exp 11.2:-

Force $\vec{F} = (2\hat{i} + 3\hat{j}) \text{ N}$

$\vec{r} = (4\hat{i} + 5\hat{j}) \text{ m}$

Find the Torque vector.

Solu:-

$\vec{T} = \vec{r} \times \vec{F}$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 5 & 0 \\ 2 & 3 & 0 \end{vmatrix}$$

دائياً سالب

$$= 0\hat{i} - 0\hat{j} + \hat{k} [(4)(3) - (5)(2)]$$

$$= 2\hat{k} \text{ N.m}$$

لـ عزم الازدواج.

دائماً موجب

الموضوع: "chapter 3" p65 $\Rightarrow$ p69
التاريخ 12/10/08 problems

3.1 :- The polar coordinates of a point are $r = 5.50$ and $\theta = 240^\circ$!! Find the cartesian coordinates !!!

solu :- $x = r \cos \theta$

$$x = 5.50 \cos 240$$

$$x = -2.75$$

$$y = r \sin \theta$$

$$y = 5.50 \sin 240$$

$$y = -4.76$$

240



$(-2.75, -4.76)$

$\Rightarrow$ solu $(-2.75, -4.76)$ في الربع الثالث

3.2 :- point one $(2.50 \text{ m}, 30.0^\circ)$

point two $(3.80 \text{ m}, 120.0^\circ)$

Determine :-

(a) the cartesian coordinates of these points

(b) the distance between them

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الموضوع:

Solu:-

(a) point 1 $(2.50 \text{ m}, 30.0^\circ)$

$$x = r \cos \theta$$

$$x = 2.50 \cos 30.0^\circ$$

$$x = 2.165$$

$$y = r \sin \theta$$

$$y = 2.50 \sin 30.0^\circ$$

$$y = 1.25$$

$$\Rightarrow (2.165, 1.25)$$

point 2 $(3.80 \text{ m}, 120.0^\circ)$

$$x = r \cos \theta$$

$$x = 3.80 \cos 120.0^\circ$$

$$x = -1.9$$

$$y = r \sin \theta$$

$$y = 3.80 \sin 120.0^\circ$$

$$y = 3.29$$

$$\Rightarrow (-1.9, 3.29)$$

$$(b) d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(2.165 + 1.9)^2 + (1.25 - 3.29)^2}$$

$$d = \sqrt{16.524 + 4.177} = 4.548 \text{ m}$$

→ follow

12/10/08 Follow التاريخ

الموضوع:

3.8 :- $\vec{F}_1$ 6.00 unites , the origin (0,0)

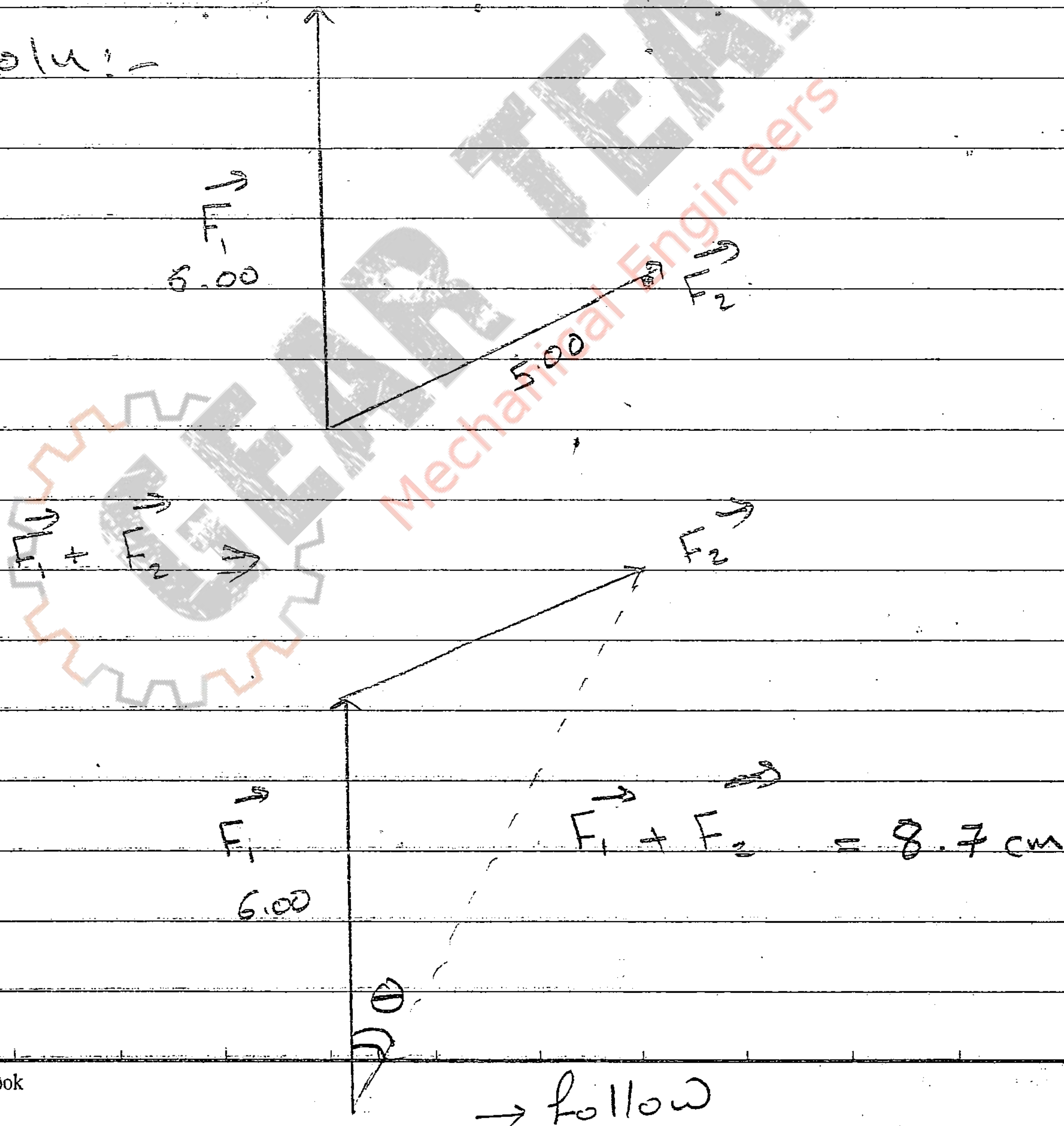
30.0° above the positive x-axis

$\vec{F}_2$ 5.00 unites the same objects

above the positive y-axis.

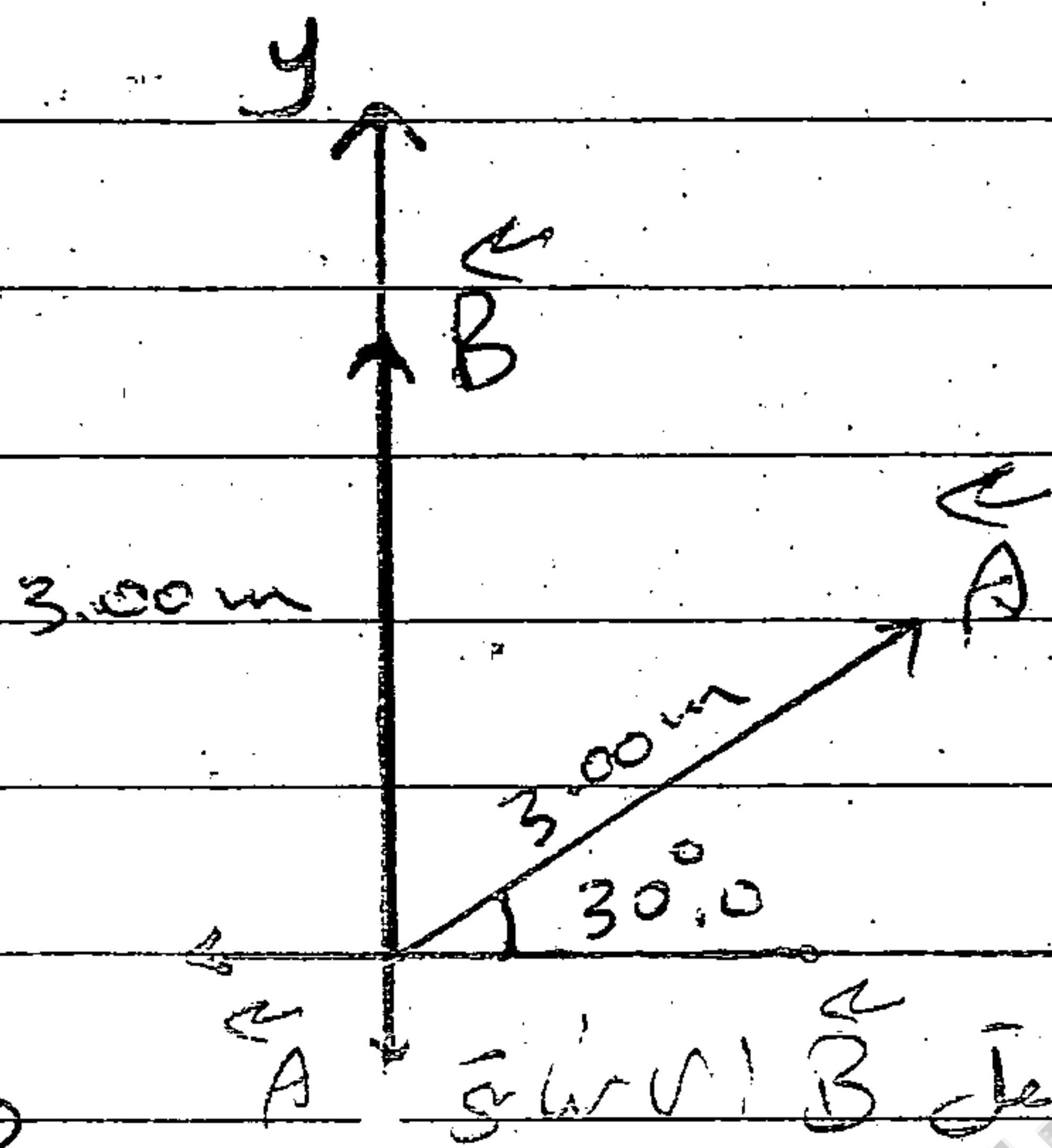
Find Graphically the magnitude and direction of the resultant $\vec{F}_1 + \vec{F}_2$.

solu:-

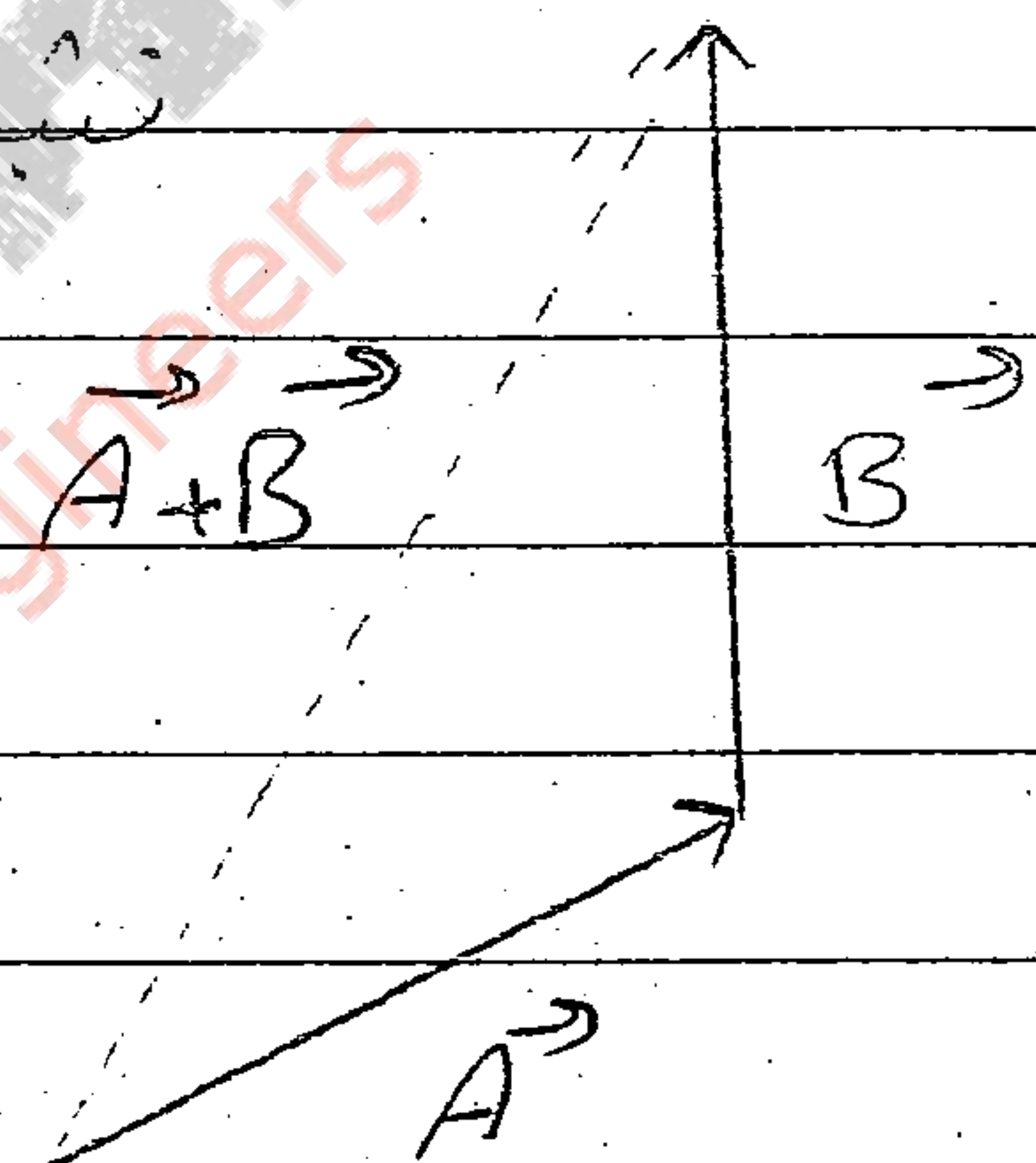


3.11 :- $|\vec{A}| = |\vec{B}| = 3.00 \text{ m}$

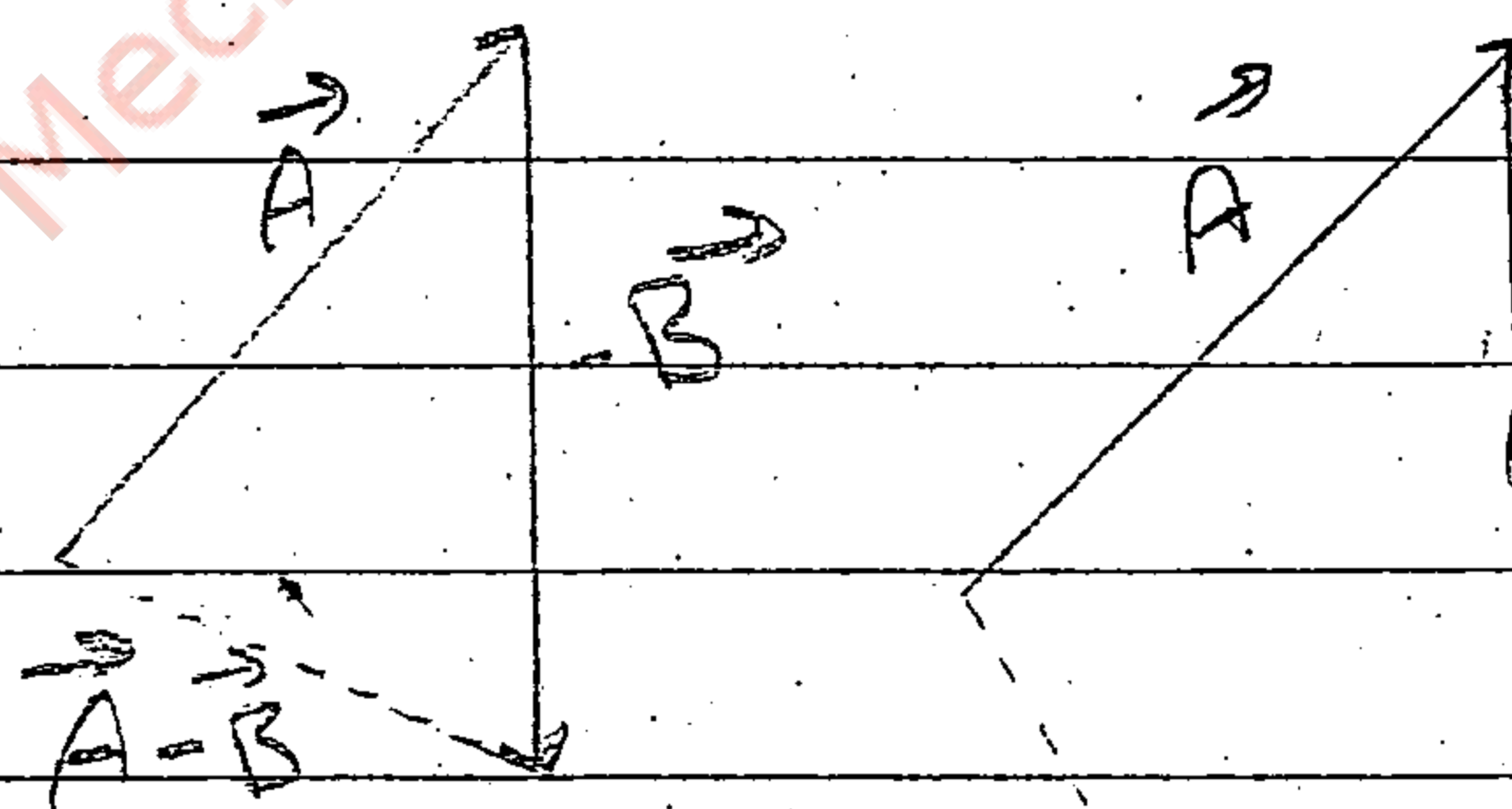
Graphically Find:-



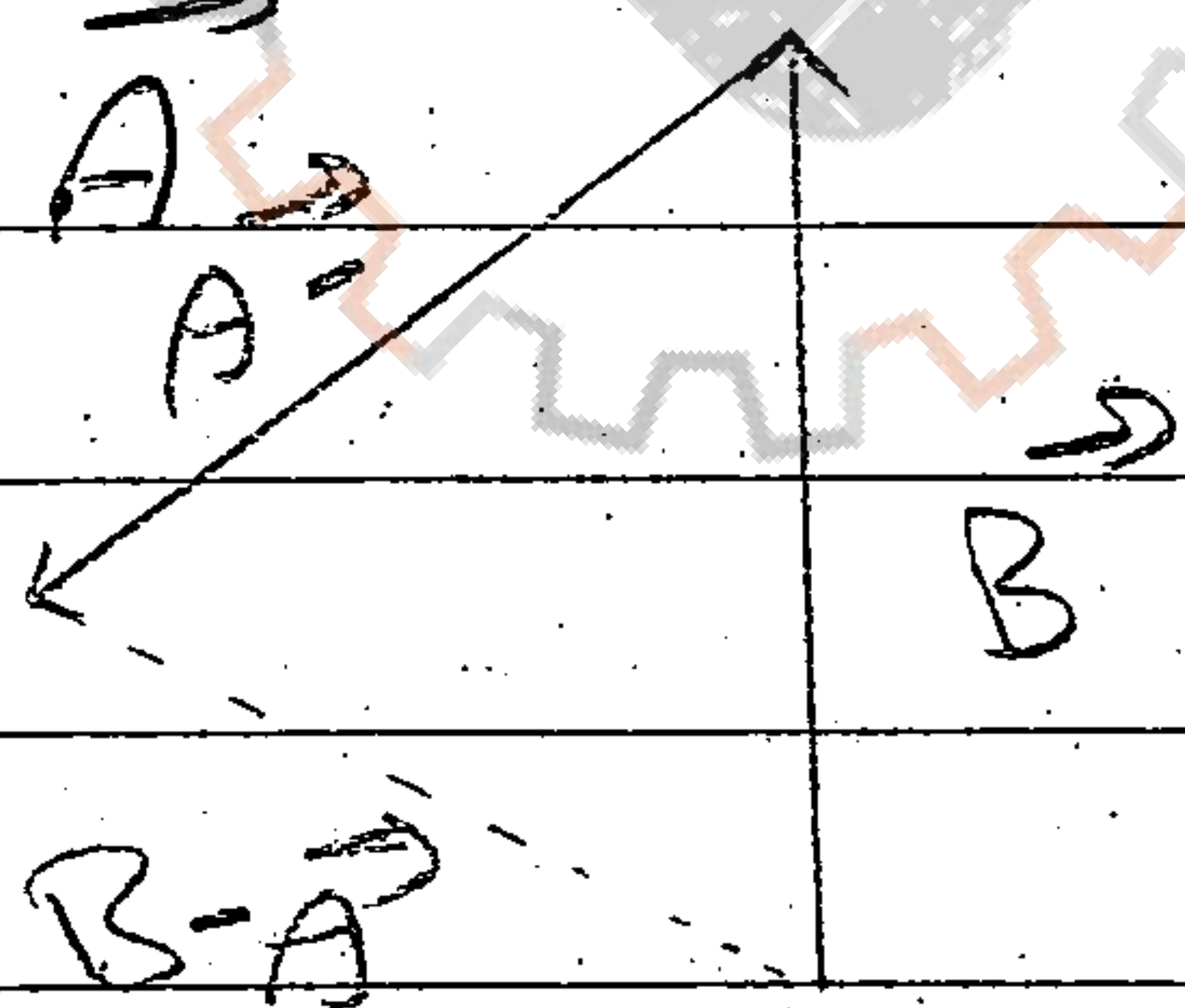
(a) $\vec{A} + \vec{B}$



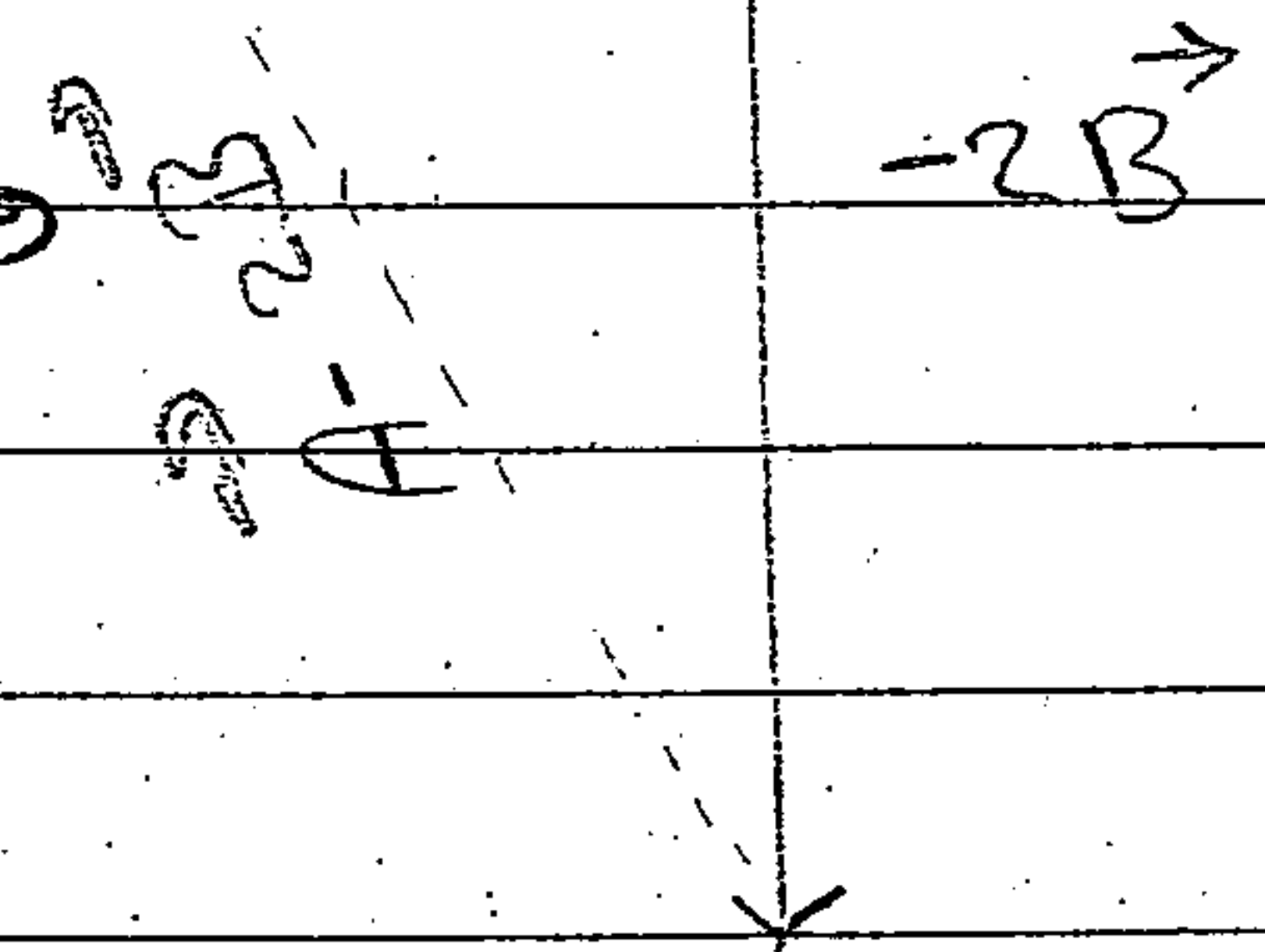
(b) $\vec{A} - \vec{B} \Rightarrow \vec{A} + (-\vec{B})$



(c) $\vec{B} - \vec{A}$



(d) $\vec{A} - 2\vec{B}$



الموضوع: 14/10/08 التاريخ Follow

3.21 :- A particle undergoes the following displacements :-

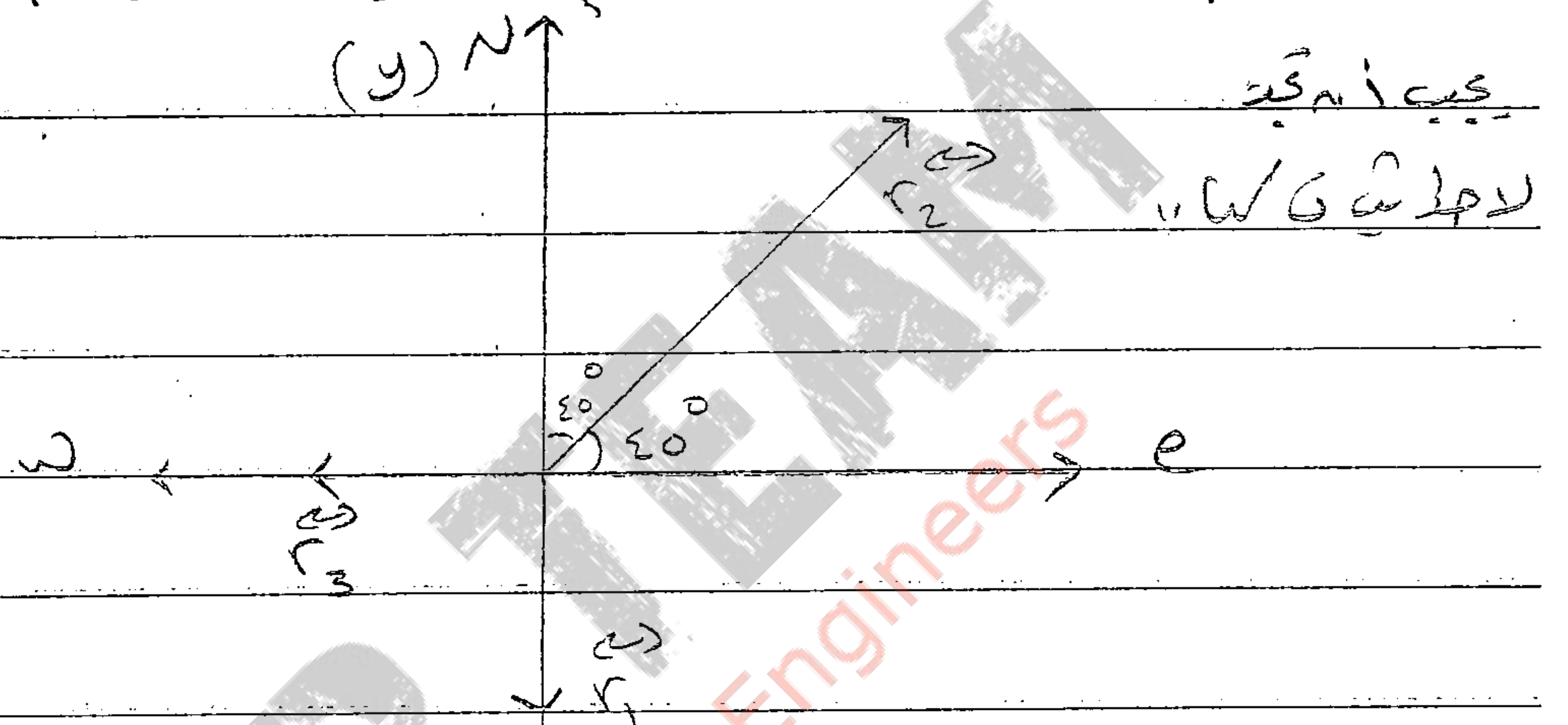
$$\Delta r_1 = 3.5 \text{ m South } \downarrow$$

$$\Delta r_2 = 8.2 \text{ m north east } 45^\circ \nearrow$$

$$\Delta r_3 = 15 \text{ m west } \leftarrow$$

what is the resultant of displacements?

Solu:-



$$r_1 = -3.5 \hat{j} \text{ (m)}$$

$$r_2 = 8.2 \cos 45^\circ \hat{i} + 8.2 \sin 45^\circ \hat{j}$$

$$= 5.8 \hat{i} + 5.8 \hat{j} \text{ (m)}$$

$$r_3 = -15 \hat{i}$$

$$\Delta r = r_1 + r_2 + r_3$$

$$\Delta r = (0 + 5.8 - 15) \hat{i} + (-3.5 + 5.8 + 0) \hat{j}$$

$$= (-9.2) \hat{i} + (2.3) \hat{j} \text{ m}$$

$$|\Delta r| = \sqrt{(-9.2)^2 + (2.3)^2} = 9.48 \text{ m}$$

$$\theta = \tan^{-1} \frac{2.3}{-9.2}$$

$$\therefore \theta = 166^\circ$$

3.24:- $\vec{A}$ has x and y components of -8.7 cm and 15 cm .

$\vec{B}$ has x and y components of 13.2 cm and -6.6 cm .

if $\vec{A} - \vec{B} + 3\vec{C} = 0$ what are the components of $\vec{C}$?

Solu:-

$$\vec{A} = (-8.7\hat{i} + 15\hat{j}) \text{ cm}$$

$$\vec{B} = (13.2\hat{i} - 6.6\hat{j}) \text{ cm}$$

$$\vec{A} - \vec{B} + 3\vec{C} = 0$$

$$3\vec{C} = \vec{B} - \vec{A}$$

$$= [13.2 - (-8.7)]\hat{i} + [-6.6 - 15]\hat{j}$$

$$3\vec{C} = 21.9\hat{i} - 21.6\hat{j}$$

$$\vec{C} = \frac{1}{3} (21.9\hat{i} - 21.6\hat{j}) \text{ cm}$$

$$\vec{C} = (7.3\hat{i} - 7.2\hat{j}) \text{ cm}$$

$$\Rightarrow C_x = 7.3 \text{ cm}$$

$$\Rightarrow C_y = -7.2 \text{ cm}$$

12/10/08 التاريخ Follow

الموضوع:

$$3.2a :- \vec{A} = (3\hat{i} - 4\hat{j} + 4\hat{k}) \text{ m}$$

$$\vec{B} = (2\hat{i} + 3\hat{j} - 7\hat{k}) \text{ m}$$

Find the magnitude of a vector:-

$$(a) \vec{C} = \vec{A} + \vec{B}$$

$$(b) \vec{D} = 2\vec{A} - \vec{B}$$

$$\text{solu :- } \vec{C} = \vec{A} + \vec{B}$$

$$\vec{C} = (3+2)\hat{i} + (-4+3)\hat{j} + (4-7)\hat{k}$$

$$\vec{C} = (5\hat{i} - 1\hat{j} - 3\hat{k})$$

$$\vec{D} = 2\vec{A} - \vec{B}$$

$$\Rightarrow \vec{A} \times 2 \Rightarrow \vec{A} (3\hat{i} - 4\hat{j} + 4\hat{k}) \times 2$$

$$2\vec{A} (6\hat{i} - 8\hat{j} + 8\hat{k})$$

$$\vec{D} = 2\vec{A} - \vec{B}$$

$$\vec{D} = (6\hat{i} - 8\hat{j} + 8\hat{k}) - (2\hat{i} + 3\hat{j} - 7\hat{k})$$

$$\vec{D} = (4\hat{i} - 11\hat{j} + 15\hat{k})$$

تاريخ 12/10/08 Follow

الموضوع:

$$7.7:- \vec{F} = (6\hat{i} - 2\hat{j}) \text{ N}$$

$$\vec{\Delta r} = (3\hat{i} + \hat{j}) \text{ m}$$

Find

- (a) the work done by the force on the particle
 (b) the angle between $\vec{F}$ and $\vec{\Delta r}$
- استنتجنا اننا لنستطيع ان نحل المسألة

Solu :-

الحل :-

$$(a) W = \vec{F} \cdot \vec{\Delta r}$$

$$W = F_x \Delta r_x + F_y \Delta r_y$$

$$W = 6 \times 3 + -2 \times 1$$

$$W = 16$$

$$(b) W = 16$$

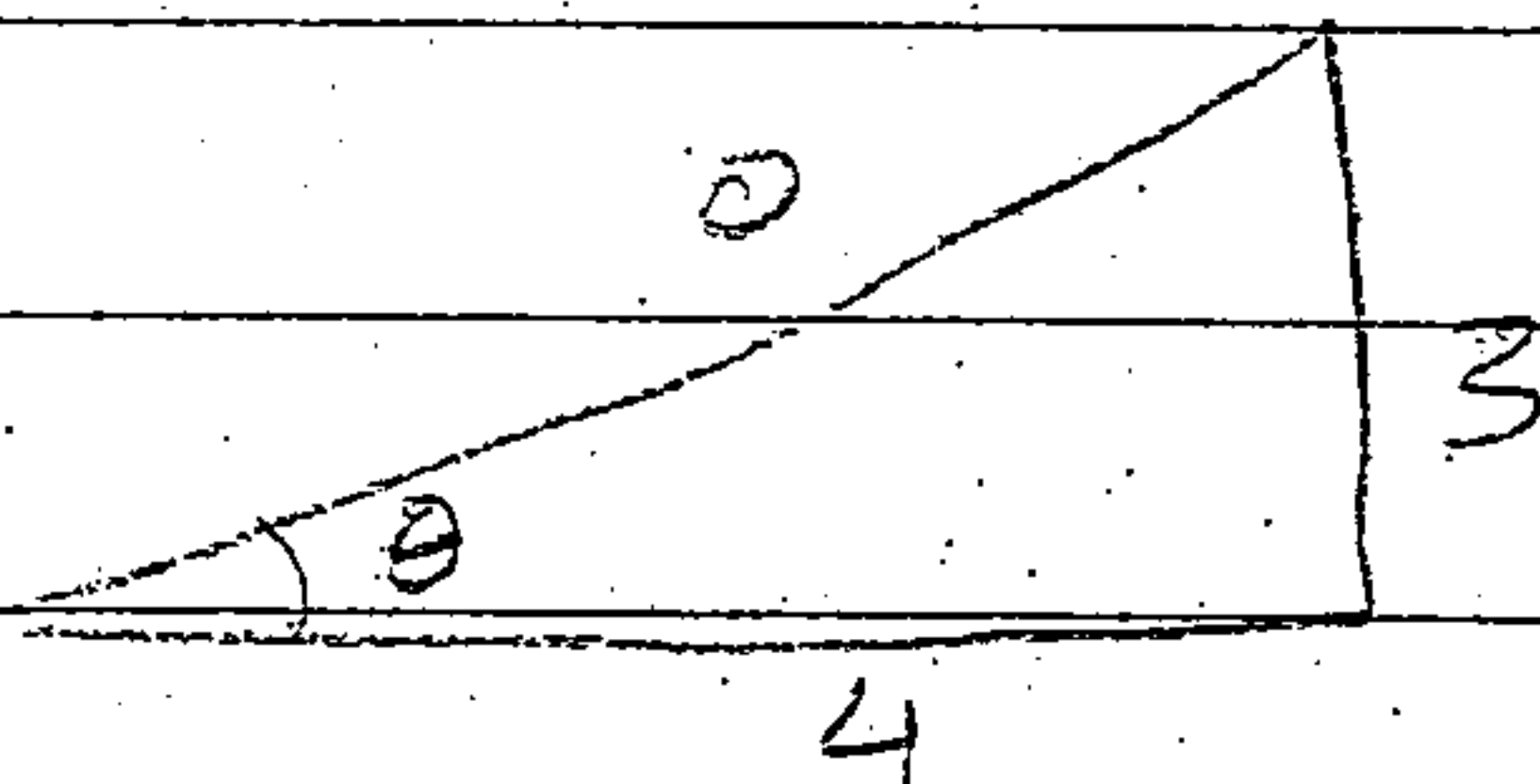
$$W = |\vec{A}| |\vec{B}| \cos \theta$$

$$16 = \sqrt{40} \sqrt{10} \cos \theta$$

$$16 = 2 \times 10 \cos \theta$$

$$\cos \theta = \frac{4}{5}$$

$$\theta = 36.87^\circ$$



18/10/08 التاريخ follow

الموضوع:

7.9:- Using the definition of scalar product to find the angle between $\vec{A}$ and $\vec{B}$ where:-

$$\vec{A} = 3\hat{i} - 2\hat{j}$$

$$\vec{B} = 4\hat{i} - 4\hat{j}$$

Solu:-

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y$$

$$\vec{A} \cdot \vec{B} = (3)(4) + (-2)(-4) = 20$$

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$$

but

$$|\vec{A}| = \sqrt{9 + 4} = \sqrt{13}$$

$$|\vec{B}| = \sqrt{16 + 16} = \sqrt{32}$$

$$20 = \sqrt{32} \sqrt{13} \cos \theta$$

$$\therefore \cos \theta = \frac{20}{\sqrt{32} \sqrt{13}}$$

$$\therefore \theta = \cos^{-1} \frac{20}{\sqrt{13} \sqrt{32}}$$

$$\therefore \theta = 11.3^\circ$$

7.11 :- A particle subject to force F_x .

Find the work done by the force on the particle

(a) from $x=0$ to $x=5$ m

(b) from $x=5$ to $x=10$ m

(c) from $x=0$ to $x=15$ m

(d) what is the total work done by the force over the distance $x=0$ to $x=15$ m?

Solu:- F_x is the force

$\theta = \cos \theta$ is the angle

(a) $x=0$ to $x=5$ m

$$\Delta x = 5 \text{ m}$$

$$W = \Delta x \times F \quad F = 3 \text{ N}$$

$$W = 15 \text{ Jol}$$

(b) $x=5$ m to $x=10$ m

$$F = 3 \text{ N}$$

$$W = 3 \times 5 = 15 \text{ Jol}$$

(c) $\Delta x = 5$ m

$$W = \Delta x \times F \Rightarrow 5 \times 3 = 15 \text{ Jol}$$

$$(d) 15 + 15 + 15 \Rightarrow 45 \text{ Jole}$$

12/10/08 التاريخ Follow

الموضوع:

11.1 :- Given $\vec{M} = 6\hat{i} + 2\hat{j} - \hat{k}$ and
 $\vec{N} = 2\hat{i} - \hat{j} - \hat{k}$, calculate the
vector product $\vec{M} \times \vec{N}$

Soln:-

$$\vec{M} \times \vec{N} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 6 & 2 & -1 \\ 2 & -1 & -1 \end{vmatrix}$$

$$= \hat{i} (2 \times -1 - (-1 \times -1)) \Rightarrow -3\hat{i}$$

$$= -\hat{j} (6 \times -1 - (-1 \times 2)) \Rightarrow 4\hat{j}$$

$$= \hat{k} (6 \times -1 - (2 \times 2)) \Rightarrow -10\hat{k}$$

$$\therefore \vec{M} \times \vec{N} = (-3\hat{i} + 4\hat{j} - 10\hat{k})$$

12/10/08 Follow التاريخ

الموضوع:

11.3 :- $\vec{A} = -3\hat{i} + 4\hat{j}$

$$\vec{B} = 2\hat{i} + 3\hat{j}$$

Find

(a) $\vec{A} \times \vec{B}$

(b) the angle between $\vec{A}$ and $\vec{B}$

Solu:-

$$(a) \vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & 4 & 0 \\ 2 & 3 & 0 \end{vmatrix}$$

$$= 0\hat{i} - 0\hat{j} + \hat{k}((-3)(3) - (4)(2))$$
$$\vec{A} \times \vec{B} = -17\hat{k}$$

$$(b) |\vec{A} \times \vec{B}| = |\vec{A}| |\vec{B}| \sin \theta$$

$$\text{but } |\vec{A} \times \vec{B}| = \sqrt{0^2 + 0^2 + (-17)^2}$$

$$= 17$$

$$\theta = \sin^{-1} \frac{17}{5 \cdot 5}$$

$$\Rightarrow \theta = 70.6^\circ$$

12/10/08 التاريخ Follow

الموضوع:

11.4:- vector $(2\hat{i} - 3\hat{j} + 4\hat{k}) \times \vec{A} =$
 $\uparrow (4\hat{i} + 3\hat{j} - \hat{k})$

Do you Believe that claim?!!

Solu:- $\vec{A} \times \vec{B} = \vec{C}$

$$\vec{B} \perp \vec{C} \quad \& \quad \vec{A} \perp \vec{C}$$

$$\vec{B} \times \vec{A} = 0$$

$$0 = \vec{B} \times \vec{A}$$

$$(2\hat{i} - 3\hat{j} + 4\hat{k}) \cdot (4\hat{i} + 3\hat{j} - \hat{k}) = 0$$

$$8 - 9 - 4 = 0$$

$$-5 \neq 0$$

! دعاه بالاذن

False him

تاريخ 12/10/08 Follow

الموضوع:

$$\text{ex: } \vec{A} = \hat{i} + \hat{j} - \hat{k}$$

$$\vec{B} = 2\hat{i} - 3\hat{j} + 4\hat{k}$$

$$A \times B = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 2 & -3 & 4 \end{vmatrix}$$

$$\Rightarrow \hat{i}(4-3) - \hat{j}(4+2) + \hat{k}(-3-2) \\ = \hat{i} - 6\hat{j} - 5\hat{k}$$

note 1:- في ضرب المتجهات والتقاطعي، راعي الترتيب
بين الأضلاع.

note 2:- ضرب المتجهات التقاطعي "dot"
والضرب المتقاطعي "cross".

note 3:-

$\hat{i}$	$\hat{j}$	$\hat{k}$	لا يحدد اتجاه
1	1	-1	جدول $\hat{i}$ والجدول
2	-3	4	الضرب

الموضوع: The position, velocity and Acceleration vectors 4.1 التاريخ 12/10/09

$$\vec{\Delta r} = \vec{r}_f - \vec{r}_i$$

↓
displacement

$$\vec{r}_f = x_f \hat{i} + y_f \hat{j}$$

$$\vec{r}_i = x_i \hat{i} + y_i \hat{j}$$

$$\vec{v}_{avg} = \frac{\vec{\Delta r}}{\Delta t} \rightarrow \text{displacement}$$

↓ $\Delta t \rightarrow \text{time intervals}$

average velocity

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\vec{\Delta r}}{\Delta t} = \frac{d\vec{r}}{dt}$$

↓
instantaneous velocity

$$\vec{a}_{avg} = \frac{\vec{\Delta v}}{\Delta t} = \frac{\vec{v}_f - \vec{v}_i}{t_f - t_i}$$

↓
average acceleration

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{d\vec{v}}{dt} \Rightarrow \vec{a} = a_x \hat{i} + a_y \hat{j}$$

↓
instantaneous acceleration

الموضوع: 4.2 two-dimensional motion with constant acceleration التاريخ 12/ 10 /08

$$\vec{r} = x\hat{i} + y\hat{j} \Rightarrow \text{position vector}$$

$$* \vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} \Rightarrow v_x\hat{i} + v_y\hat{j} \text{ velocity}$$

$$* \vec{a} = \frac{d\vec{v}}{dt} = \frac{dv_x}{dt}\hat{i} + \frac{dv_y}{dt}\hat{j} \Rightarrow a_x\hat{i} + a_y\hat{j} \text{ acceleration}$$

* equations in two dimension :-

$$① \vec{v}_f = \vec{v}_i + a\vec{t}$$

$$\vec{v}_f = (v_{xi} + a_x t)\hat{i} + (v_{yi} + a_y t)\hat{j}$$

$$② \vec{r}_f = \vec{r}_i + \vec{v}_i t + \frac{1}{2} \vec{a} t^2$$

$$\text{للمبدال } x = (x_i + v_{xi} t + \frac{1}{2} a_x t^2) + (y_i + v_{yi} t + \frac{1}{2} a_y t^2)$$

” معادلات (1)، (2)، (3) نفس الطريقة ”

note:- يمكن استخدام هذه المعادلات في الحركة ثنائية الأبعاد، حيث يكون التسارع ثابتاً في كل من المحاور x و y.

note:- هذه المعادلات تصبح معادلات x و معادلات y

26/10/108 follow التاريخ

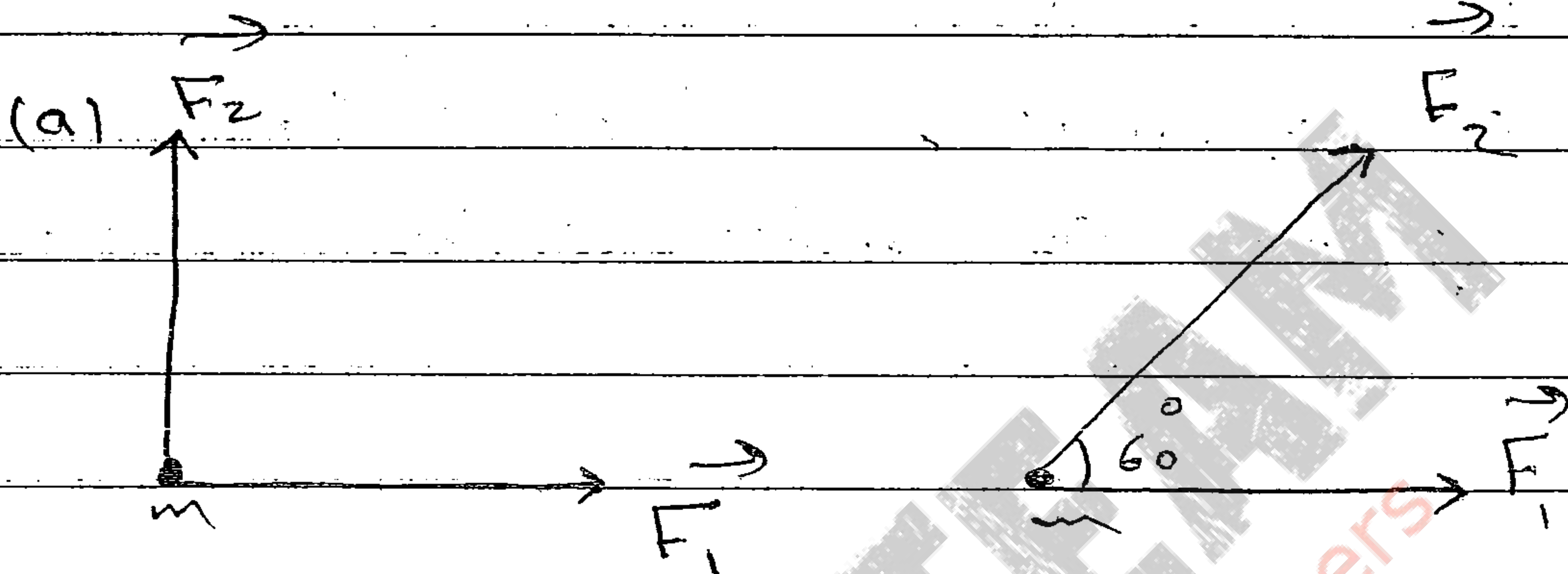
الموضوع:

5.9:-

$$m = 5 \text{ kg}$$

$$|\vec{F}_1| = 20 \text{ N}$$

$$|\vec{F}_2| = 15 \text{ N}$$



Find the acceleration in two situation of the mass.

$$(b) \vec{F}_1 = 20\hat{i} \quad \vec{F}_2 = 15\cos 60^\circ\hat{i} + 15\sin 60^\circ\hat{j}$$

solu:-

$$(a) \sum \vec{F} = m\vec{a}$$

$$20\hat{i} + 15\hat{j} = 5\vec{a}$$

$$4\hat{i} + 3\hat{j} = \vec{a}$$

$$|\vec{a}| = \sqrt{16+9} = 5 \text{ m/s}^2$$

$$F_{2x} = 15\cos 60^\circ = 7.5 \text{ N}$$

$$F_{2y} = 15\sin 60^\circ = 13 \text{ N}$$

$$\sum \vec{F} = m\vec{a}$$

$$(20 + 7.5)\hat{i} + 13\hat{j} = 5\vec{a}$$

$$27.5\hat{i} + 13\hat{j} = 5\vec{a}$$

$$\vec{a} = 5.5\hat{i} + 2.6\hat{j}$$

$$|\vec{a}| = \sqrt{(5.5)^2 + (2.6)^2} = 6.08 \text{ m/s}^2$$

$$\theta = \tan^{-1} \frac{2.6}{5.5} = 25.3^\circ$$

تتبع التاريخ 26/10/08

الموضوع:

Solu:-

$$(a) \vec{F} = m_1 a_1 = m_2 a_2$$

الكتلة نفس

القوة

$$m_1(3) = m_2(1)$$

$$\Rightarrow \frac{m_1}{m_2} = \frac{1}{3}$$

$$(b) m_2 = 3m_1$$

$$\vec{F} = (m_1 + m_2) a = m_1 a$$

الكتلة نفس $\vec{F}$ القوة ساوية

$$\text{but } m_2 = 3m_1$$

$$4m_1 a = m_1 \times 3$$

$$\therefore a = \frac{3}{4} = 0.75 \text{ m/s}^2$$

لأن كتلة كتلة الكتلة واحدة
الكتلة وبقيت القوة نفس : سرعة
تزداد مع كتلة الكتلة

5.1 :- A 3 kg object.

$\vec{a} = (2\hat{i} + 5\hat{j})$, Find the magnitude of the force.

Solu:-

$$\vec{F} = m \vec{a}$$

$$= 3(2\hat{i} + 5\hat{j})$$

$$= 6\hat{i} + 15\hat{j}$$

"components"

$$\Rightarrow |\vec{F}| = \sqrt{(6)^2 + (15)^2} = 16.2 \text{ N}$$

$$\Rightarrow \theta = \tan^{-1} \frac{15}{6} = 68^\circ 15'$$

5.2 :- Force $\vec{F}$ applied to $m_1 \rightarrow a_1 = 3 \text{ m/s}^2$

to $m_2 \rightarrow a_2 = 1 \text{ m/s}^2$

(a) Find the ratio $\frac{m_1}{m_2}$

(b) If m_1 and m_2 combined into one object what is its acceleration under the same $\vec{F}$.

القراءة لكل واحد من حيا على

١٠٠

جواب: 3 و 4 = 4 و 3 = 3 و 2 = 2 و 1 = 1

$$0. \text{ سوپر } 70 = 1.47 = 2 - 1.2$$

↓ ০০

© محمد الصفور شاع و م ا ر

صیغہ صیغہ سماع

$$\bar{C} \times C = \varphi \sim 3$$

في $v - 1 = l \times n \leq$ و منه الظاهري أقل منه

السؤال الثاني: $(-5) \times 7 = 7 \times \dots = 10$

$\frac{1}{2} \times 12 = 6$

(١٦) يوم السبت سابع عشر من شهر ربيع الثاني سنة ١٤٤٠ هـ، افتتح صاحب السمو الملكي الأمير

* صبحی صید شائع عام ہے، خیراتیہ حجر اللہ

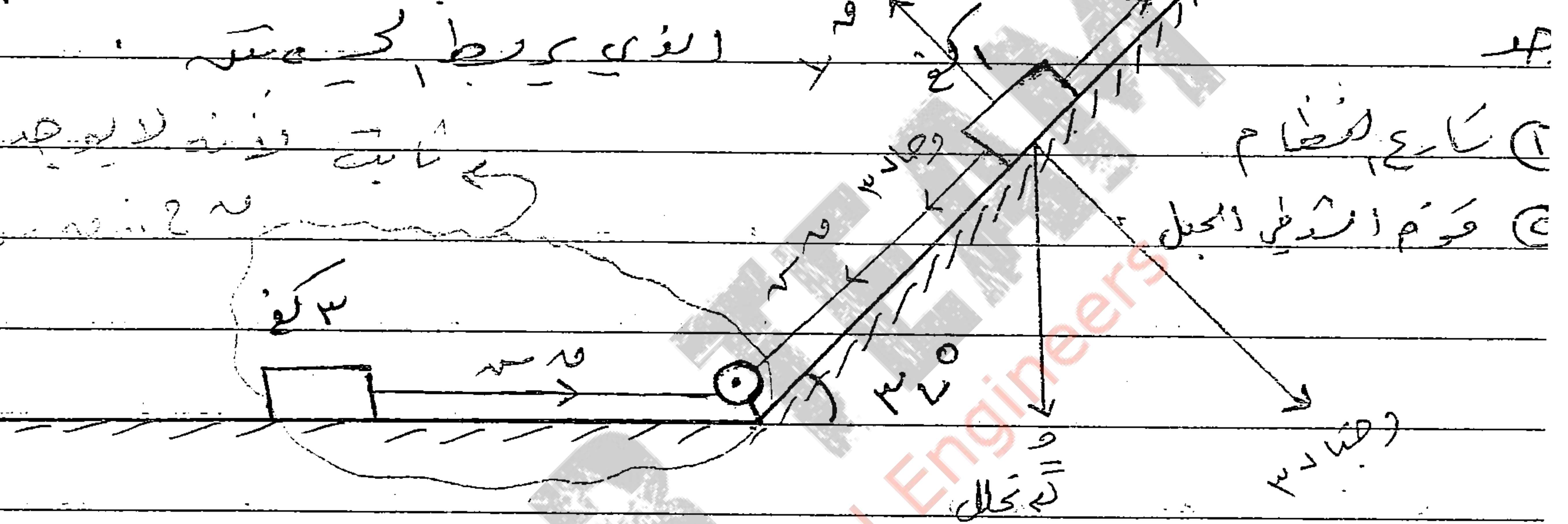
$$\sigma \times e = u \sim 3$$

5x0-3-12

$$\hookrightarrow C \times T' = T_{\text{un}} - 1 \text{ s} \hookrightarrow$$

$\frac{1}{2} \times 10 = 5$

Ex:- هم كتلة (ا كغ) موهنوع على مستوى أفقي يحل
تحت الضغط لأسفل كحامي الشكل، يتصل بحجم آخر
كتلة (3 كغ) موهنوع على الجزء السفلي منه
السوي، ويصل بينهما حبل يمر خلال بكر
عند الذبذبات، أثرت قوة (12) نيوتن بحجم (P)
سوقية لـ 12 كغ السوي الحامل، أم - السوي الحبل



1- حل

$$12 = (12 + 3) \times 1 = 15 \times 1 = 15$$

$$12 = (12 + 3) \times 1 = 15 \times 1 = 15$$

$$12 = 15 \times 1 = 15$$

$$12 = 15 \times 1 = 15$$

$$12 = 15 \times 1 = 15$$

$$12 = 15 \times 1 = 15$$

$$12 = 15 \times 1 = 15$$

$$12 = 15 \times 1 = 15$$

$$12 = 15 \times 1 = 15$$

التاريخ 26 / 10 / 08 Follow

الموضوع:

أي M_k لدينا الوسيط المربوطة في معادلة (1)

$$f_k = M_k n$$

from eq(2)

$$n = m_1 g - F \sin \theta$$

$$f_k = M_k n = M_k (m_1 g - F \sin \theta) \quad \text{--- (4)}$$

نفس

substitute eq(3) and (4) in eq(1)

$$F \cos \theta - f_k - T = m_1 a$$

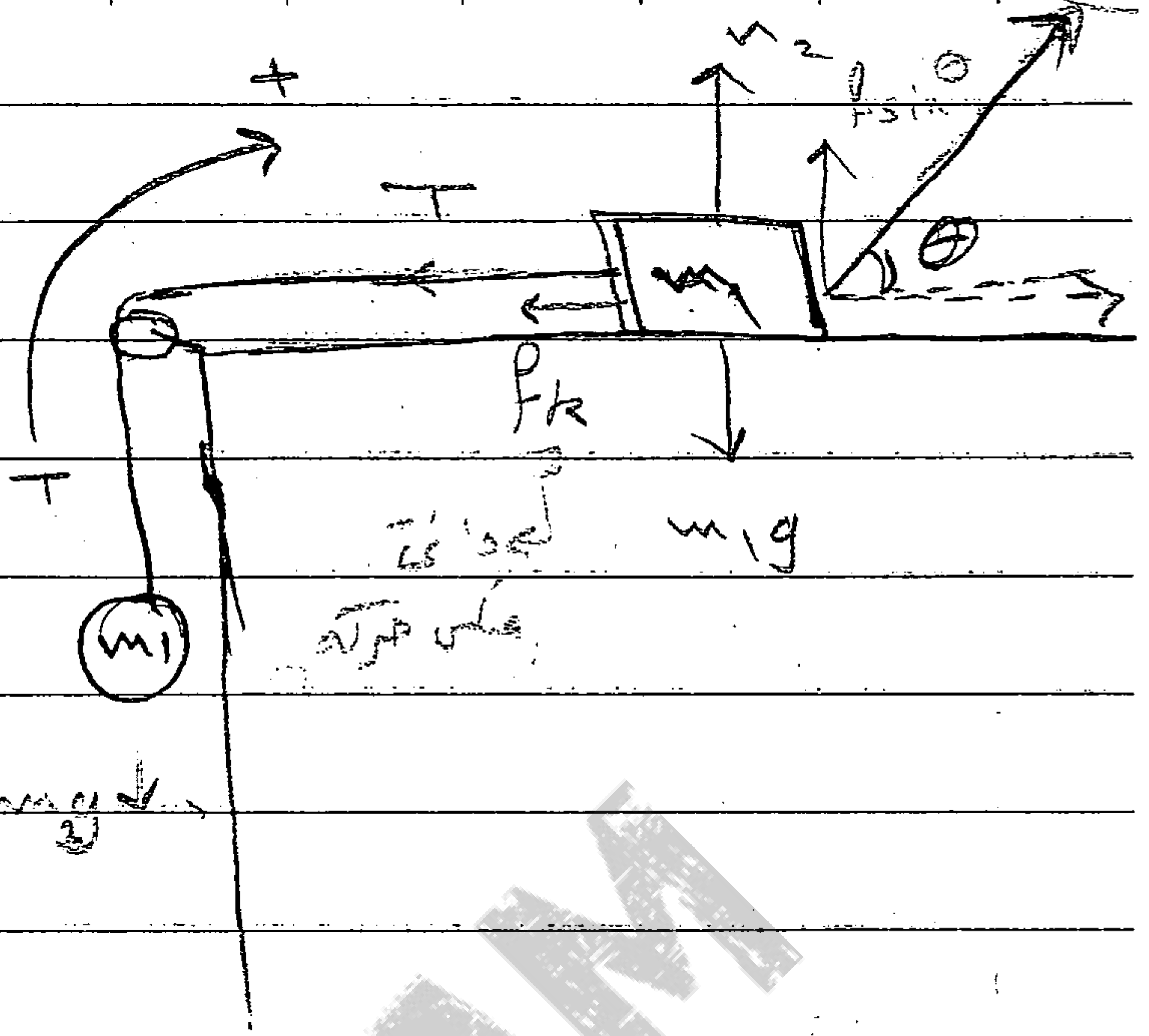
$$\Rightarrow F \cos \theta - M_k (m_1 g - F \sin \theta) - m_2 (a + g) = m_1 a$$

$$\Rightarrow a = \frac{F (\cos \theta + M_k \sin \theta) - (m_2 + M_k m_1) g}{m_1 + m_2}$$

26/10/09 Follow التاريخ

الموضوع:

Ex:-



- Find the magnitudes of acceleration of the two objects.
- there is friction.

Solution

« عاونا على القوى في بيان حل
أي سؤال »

* for m_1 :-

اعطاء m_1 مبرمج بنفس الكيفية لكانت

$$\sum F_x = m_1 a$$

السماح

$$F \cos \theta - T - F_k = m_1 a \quad \dots (1)$$

$$\sum F_y = 0$$

$$N + F \sin \theta - m_1 g = 0 \quad \dots (2)$$

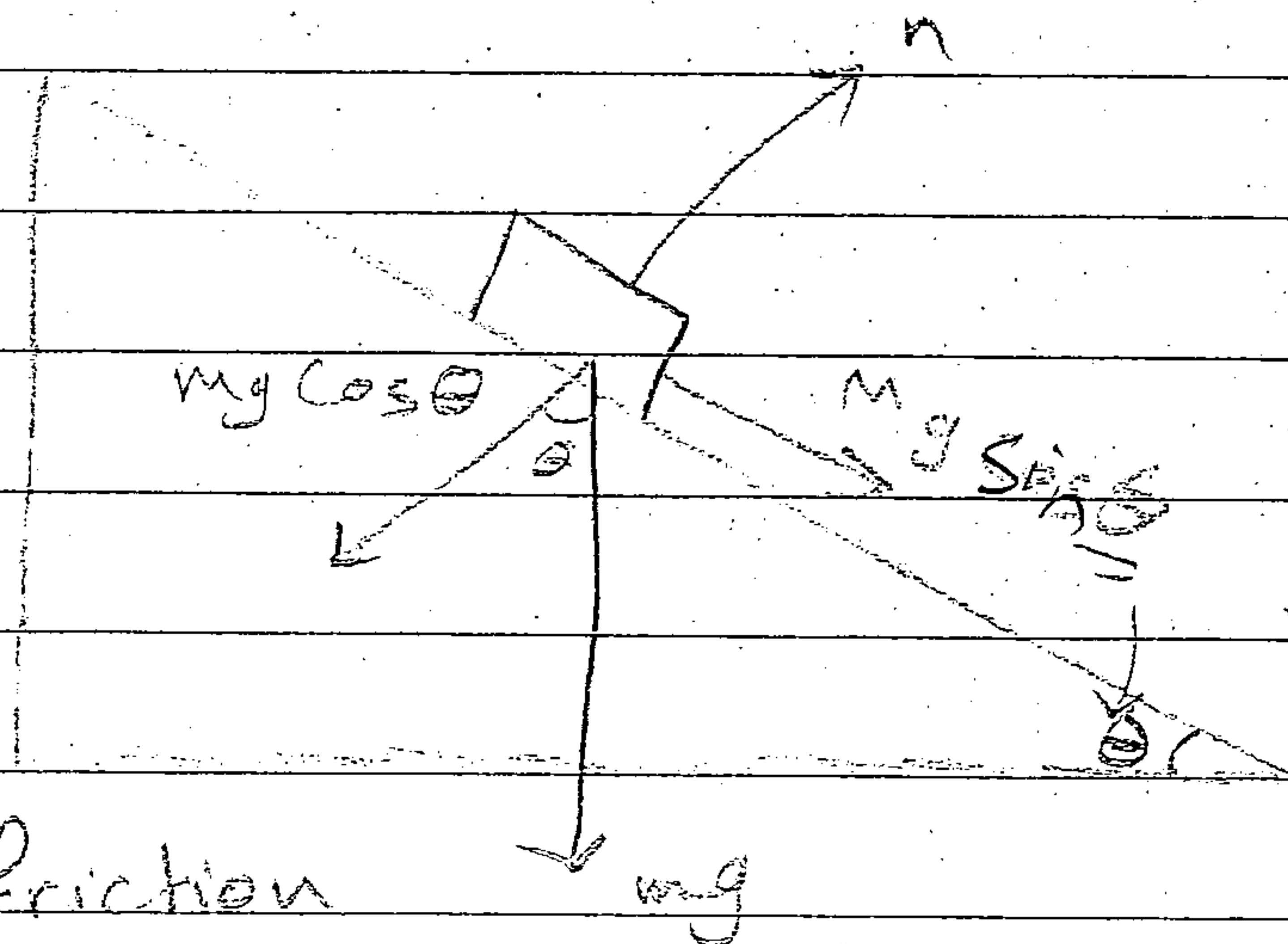
* for m_2 :-

$$\sum F_y = m_2 a$$

$$T - m_2 g = m_2 a \Rightarrow T = m_2 a + m_2 g \quad \dots (3)$$

Ex:-

ميكانيكا



Friction

Find μ_s as a function of θ .Is it possible to find μ_s without

Soln:-

$$\sum F_y = 0$$

$$n - mg \cos \theta = 0 \Rightarrow n = mg \cos \theta, \quad F_s = \mu_s n$$

$$F_s \Rightarrow \mu_s mg \cos \theta$$

$$\sum F_x = 0$$

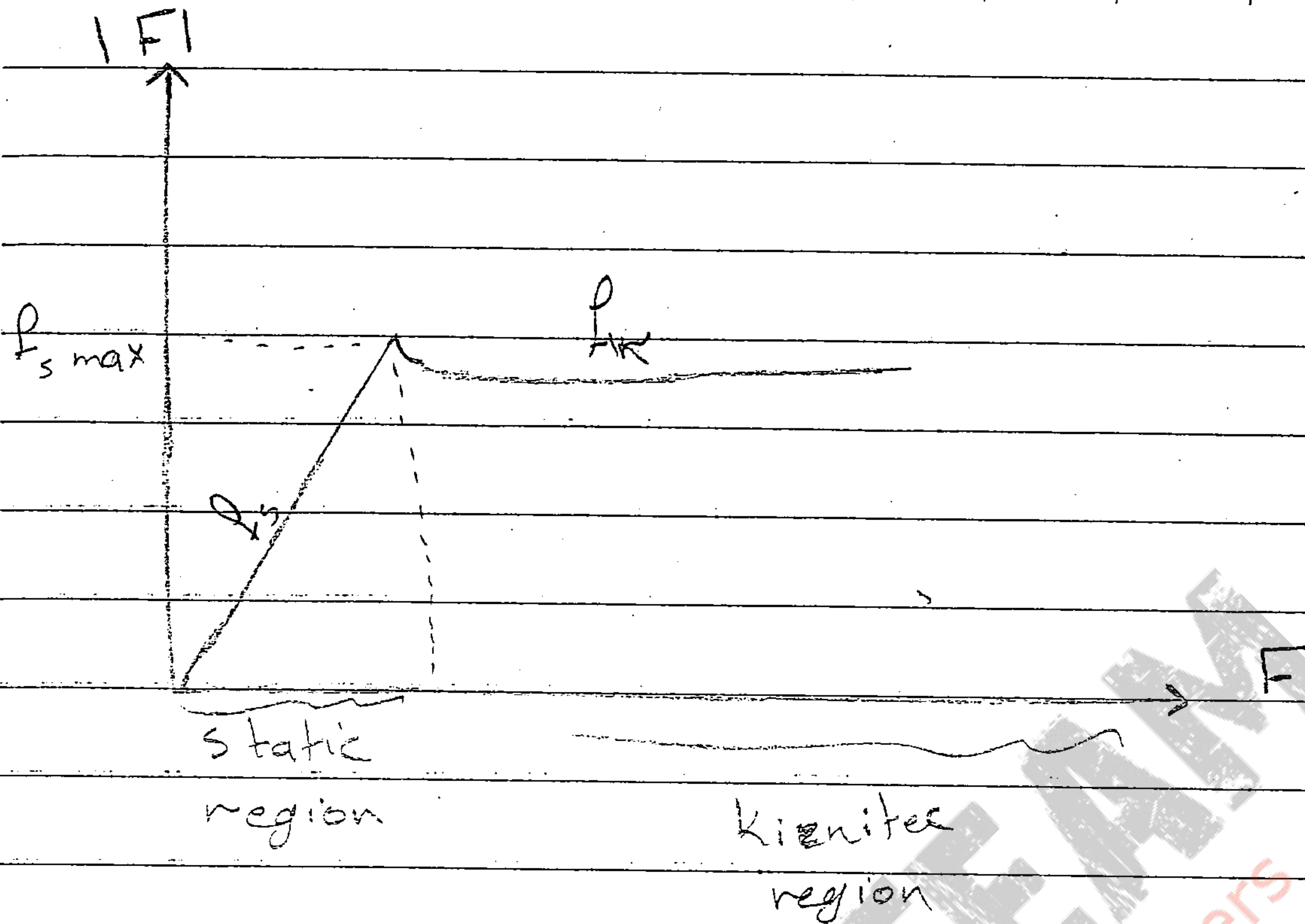
$$mg \sin \theta - F_s = 0$$

$$mg \sin \theta = F_s = \mu_s mg \cos \theta$$

$$\mu_s = \frac{\sin \theta}{\cos \theta} \Rightarrow \mu_s = \tan \theta$$

تاریخ 27/10/08 Follow

الموضوع:



* if $F - F_k \neq 0$, we have acceleration.

* if $F = f_k$, constant speed. $a = 0$

* $M_k < M_s$ generally.

* Typical values of M_k, M_s between

0.03 $\rightarrow$ 1

"القيمة الدنيا هي 0.03"

"القيمة العظمى هي 1"

$$* f_k = \mu_k n \rightarrow \text{magnitude of normal force}$$

↓
magnitude of force of kinetic friction

↓
coefficient of kinetic friction

معاظم

note:- عند التماس بين جسمين في البداية يتغير سائلنا. يأتي ذلك تولى الاحتكاك الكهربي بين الجسمين المطبقين

- بالعربي :-
- * $\mu_s = 0.4$ - معامل الاحتكاك الساكن
 - * $\mu_k = 0.3$ - معامل الاحتكاك الحركي
 - * $F = 10 \text{ N}$ - قوة الدفع
 - * $m = 10 \text{ kg}$ - كتلة الجسم
 - * $g = 10 \text{ m/s}^2$ - تسارع الجاذبية

* note:- عند زيادة القوة المؤثرة في جسم تزداد القوة الاحتكاكية حتى يتساوى مع القوة المؤثرة. بعد ذلك يتحرك الجسم ويتزايد الاحتكاك الحركي.

محاضرة 5.8

5.8 Forces of Friction

الموضوع: التاريخ 22/10/08

* When an object is in motion either on a surface or in a viscous medium such as air or water, there is resistance to the motion because the object interacts with its surroundings.

* we call such resistance a force of Friction.

* The direction of the friction force is opposite the direction of motion.

* The maximum force of static friction -

$F_{s, \max} \propto n$

static "ساكني" ← قوى الإحتكاك الساكني
 → normal

$F_s \leq \mu_s n \rightarrow$ magnitude of normal force.

↓ ↓
 magnitude coefficient "معامل الإحتكاك
 of static of static الساكني"
 friction friction

القانون الثاني لنيوتن

التاريخ 22 / 10 / 08

- add ① + ② :-

$$-m_1g + m_2g = m_1a + m_2a$$

$$a = \left(\frac{m_2 - m_1}{m_1 + m_2} \right) g \rightarrow \textcircled{3}$$

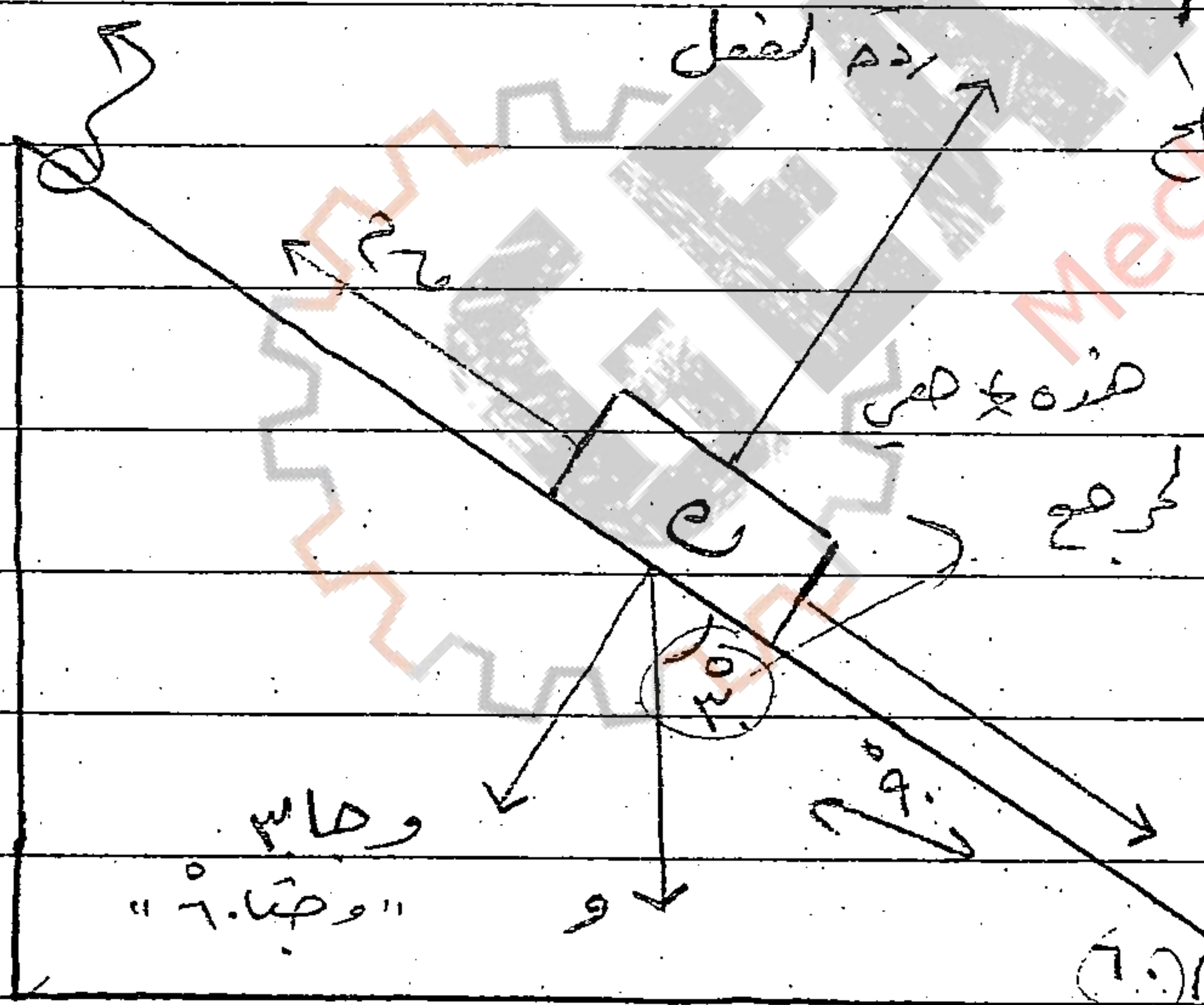
- substitute eq ③ in eq ① :-

$$T - m_1g = m_1a$$

$$T = m_1g + m_1a = \frac{2m_1m_2g}{m_1 + m_2}$$

طانه حسي
للا سفل

" Notes "



* دائما قوم هذه الفعل بكمه عمودية على السطح
نكس الحركة العمودية

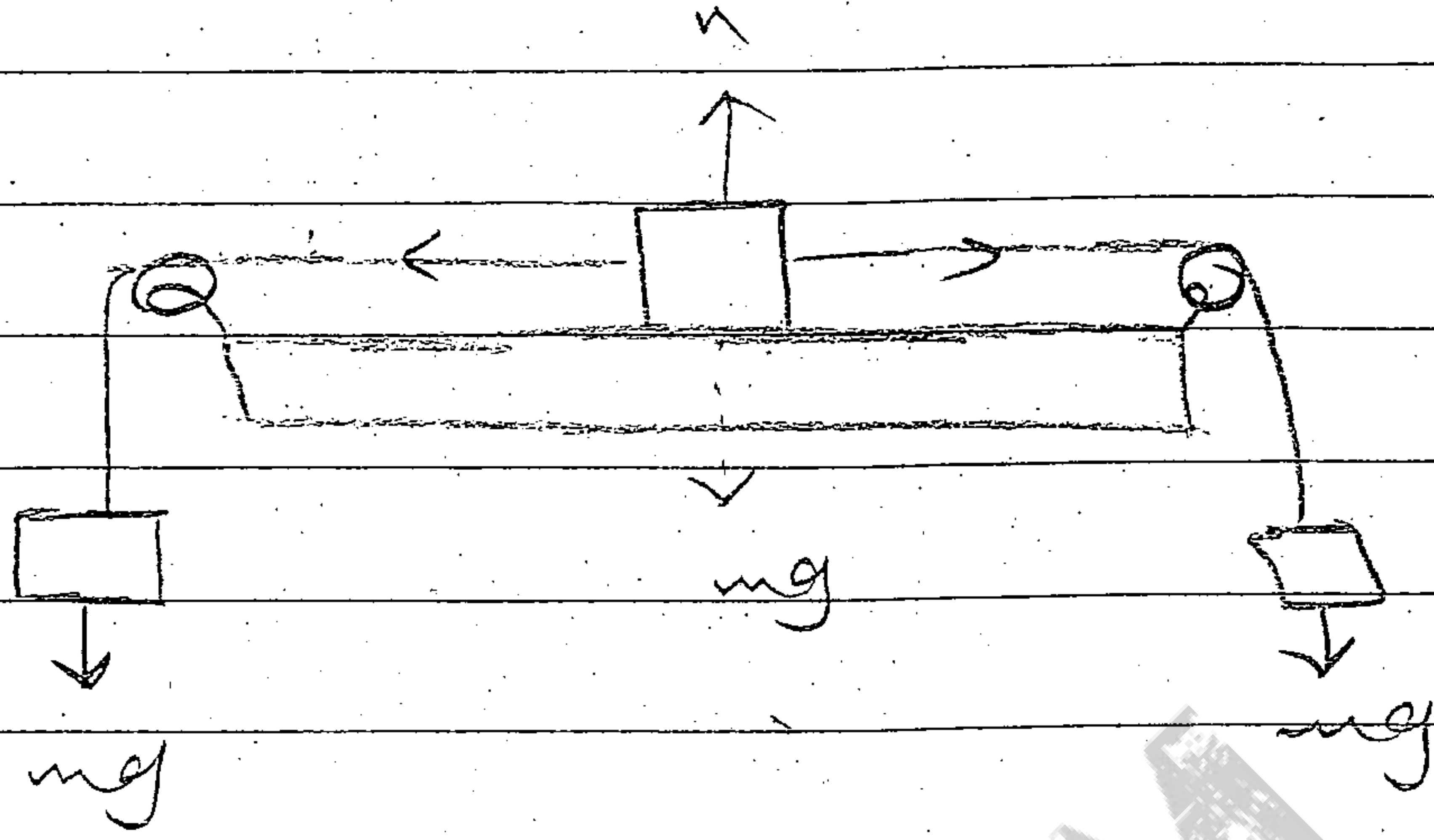
* رد الفعل تادي القوة المتطابقة
هذه هي المجموع
لذلك عندما نطلب هذه
افعال رد الفعل افعل الحركة السوية

* قوم الافعال بكاف
اتجاه الحركة

* دائما رده الجسم عموديه
على الاسطح السوية
كل زاوية تادي حثا محثا

التاريخ 26/ 10/08

5.38 :-



الموضوع: Newton's second law 6.1 التاريخ: 29/10/08

aparticle in Uniform circular motion

الحركة الدائرية المنتظمة

* Uniform circular Motion :-

→ constant speed v تكون السرعة ثابتة

مقداراً وتغير اتجاهها

* $a_c = \frac{v^2}{r}$ $\frac{\Sigma F}{m} = \frac{v^2}{r}$

where:-

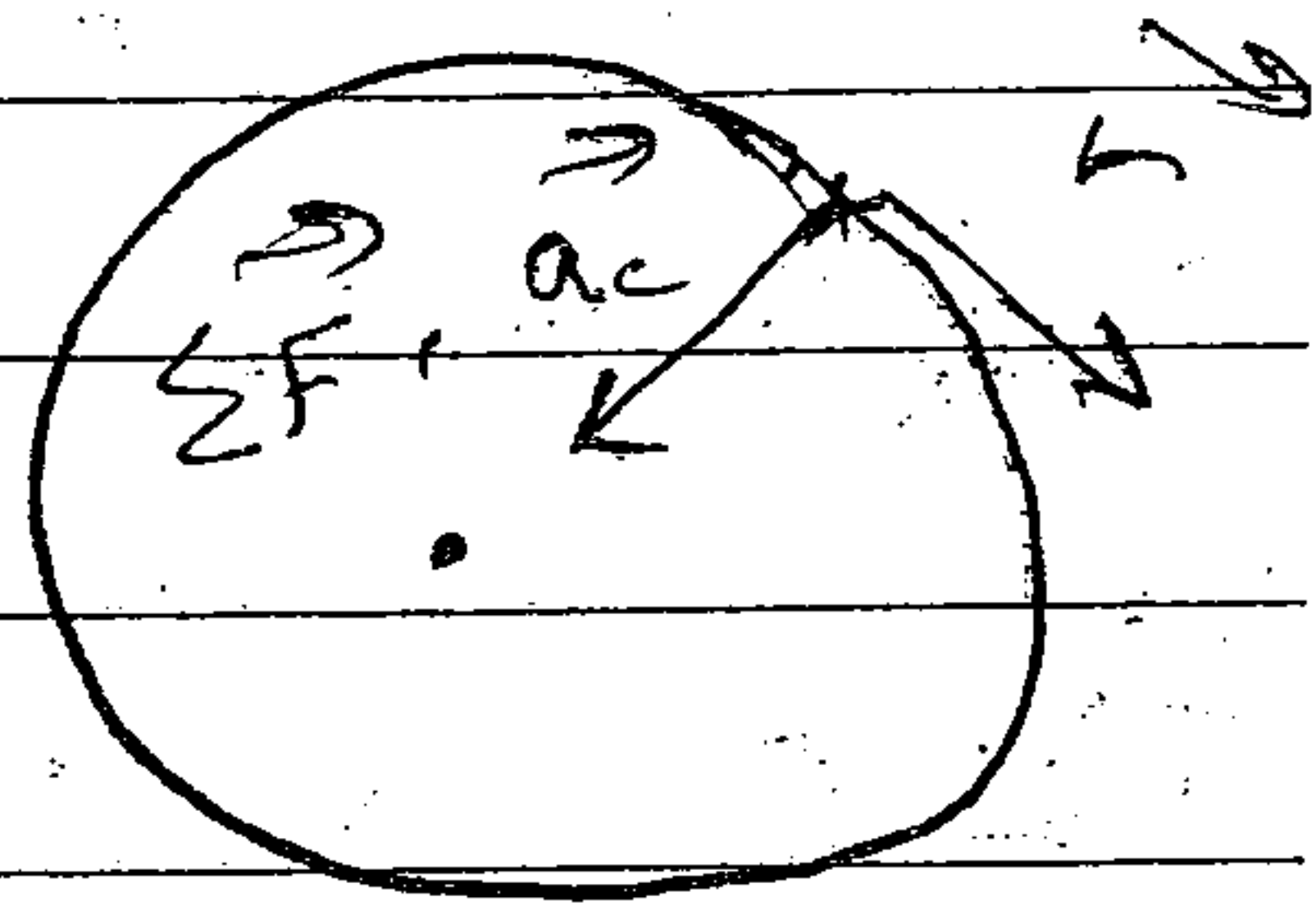
a_c centripetal acceleration

التسارع المركزي " يتجه باتجاه مركز الدائرة "

r :- radius of the circle

* $\Sigma F = m a_c = \frac{m v^2}{r}$

net force " القوة المركزية "



في الحركة الدائرية المنتظمة اتجاه التسارع الخطي يتغير باستمرار
السرعة، أما في الحركة الدائرية المنتظمة فباتجاه التسارع
غير متغير أي السرعة.

كلية عمان التقنية للدراسات والبحوث
مؤسسة جامعة عمان التقنية للدراسات والبحوث

حركة

$\Sigma F = m a_c = \frac{m v^2}{r}$ → follow

29/10/08 التاريخ Follow

الموضوع:

exp:- A small ball of mass "m" is
"suspended from a string of length
"L", The ball revolves with constant
speed v in a horizontal circle of
radius "r". Find an expression for
v.

Soln:-

$$\sum F_y = 0 \Rightarrow T \cos \theta - mg = 0$$

$$\Rightarrow T = \frac{mg}{\cos \theta}$$

$$\sum F_x = ma_c$$

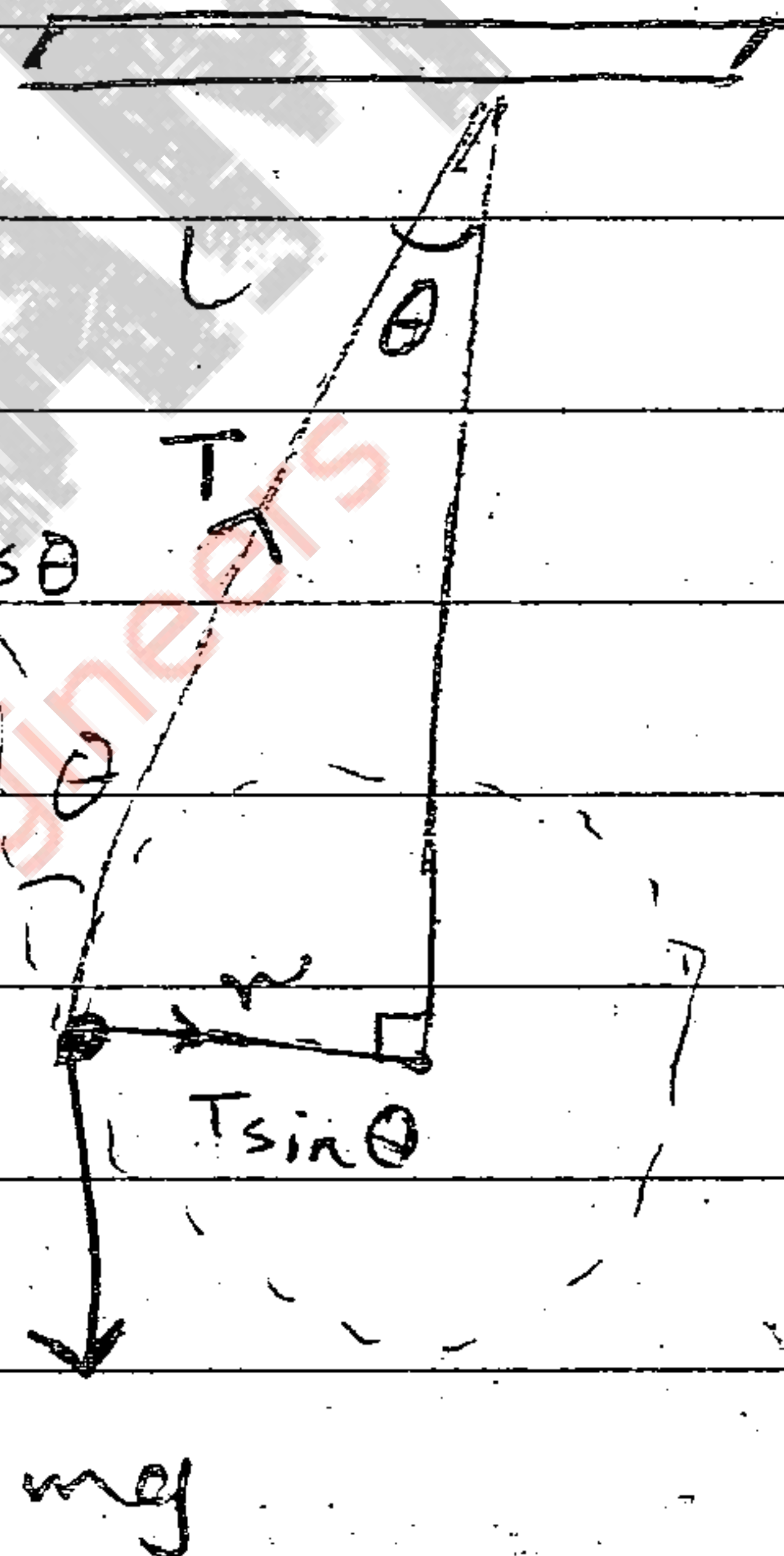
$$T \sin \theta = m \frac{v^2}{r}$$

$$\Rightarrow v = \sqrt{rg \tan \theta}$$

$$\frac{mg \sin \theta}{\cos \theta} = m \frac{v^2}{r}$$

$$v = \sqrt{Lg \sin \theta \tan \theta}$$

$$\Rightarrow \text{in } \Delta \quad \sin \theta = \frac{r}{L} \quad \therefore r = L \sin \theta$$



التاريخ 1/10/08 Follow

الموضوع:

exp 6.3:-

A car with 1500 kg radius of the curve of the road 35 m, $\mu_s = 0.523$

find the maximum speed the car can have and still make the turn successfully

Solu:-

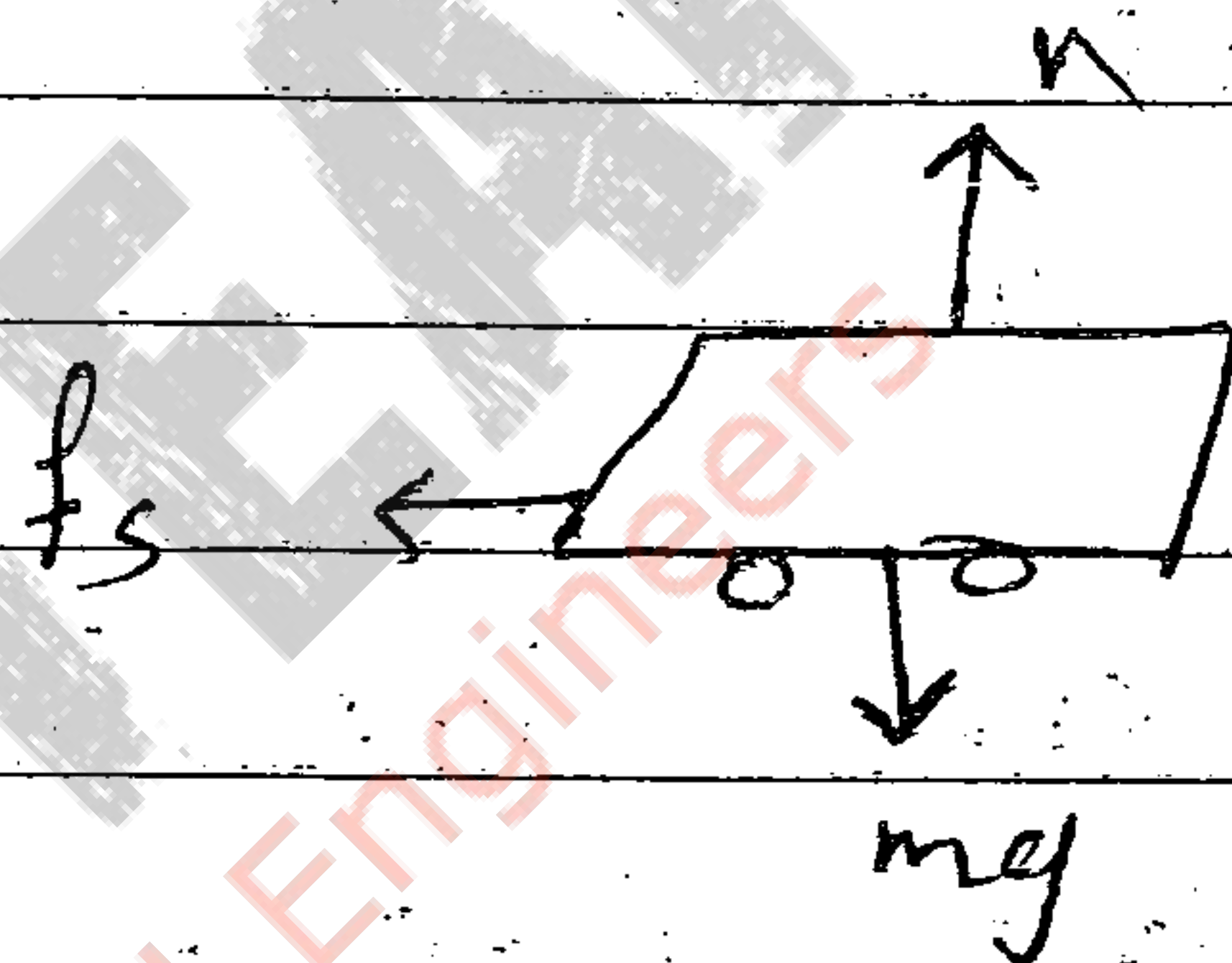
$$f_{s, \max} = \frac{m v^2}{r}$$

$$\mu_s mg = \frac{m v^2}{r}$$

$$v = \sqrt{\mu_s g r}$$

$$\Rightarrow \sqrt{(0.523) \times (9.8) (35)}$$

$$= 13.4 \text{ m/s}$$



29/10/08 التاريخ Follow

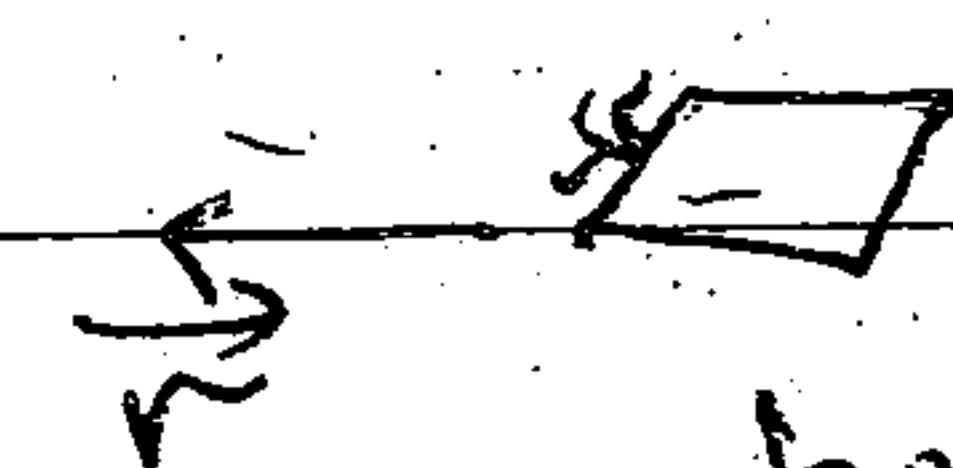
exp:- A pilot of mass M in a jet aircraft executes a loop-the-loop. The aircraft moves in a vertical circle of radius 2.7 km at a constant speed of 225 m/s .

(a) Determine the force exerted by the seat on the pilot at the bottom of the loop. "n bottom"

(b) Determine the Force exerted by the seat on the pilot at the top of the loop.

top  "n top"

2.7 km

 bottom

$r = 2.7 \text{ km}$

$v = 225 \text{ m/s}$

constant speed.

Follow

Solution:-



$$\Sigma F_{\text{bottom}} = n - mg = \frac{mv^2}{r} \Rightarrow \text{القوة المركزية}$$

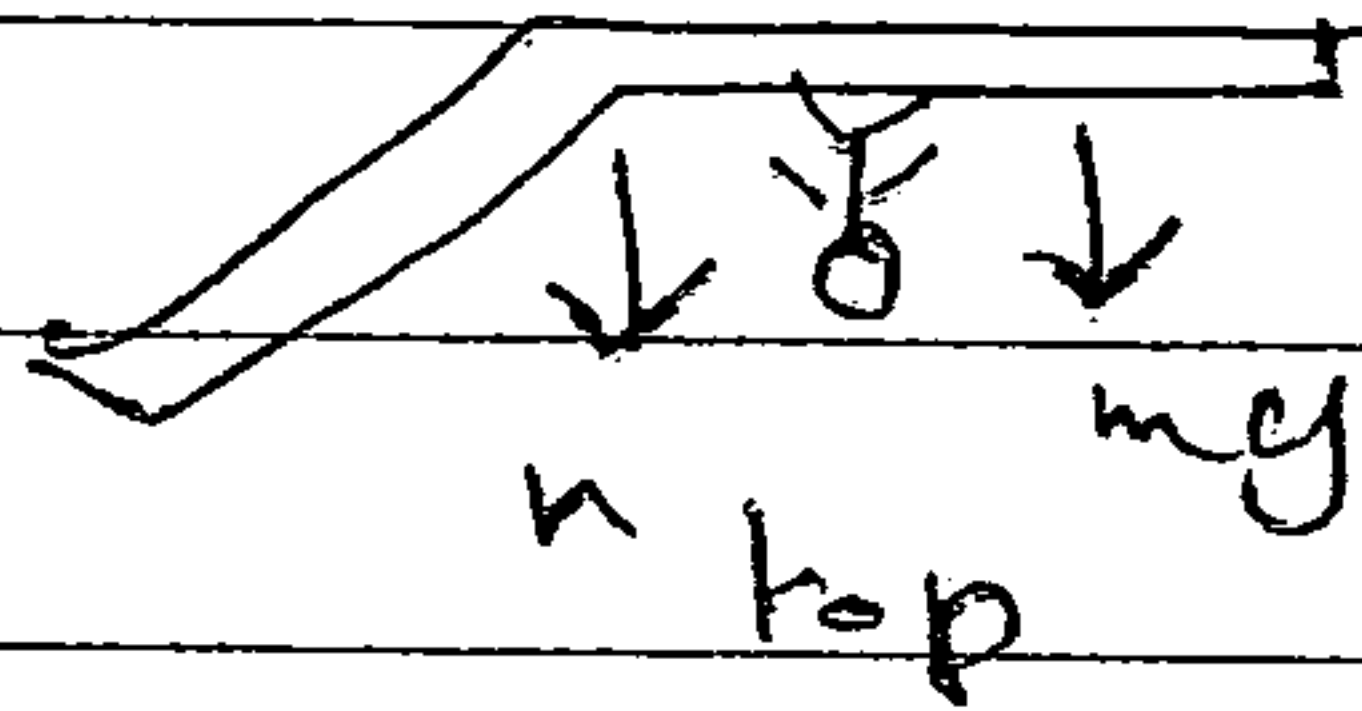
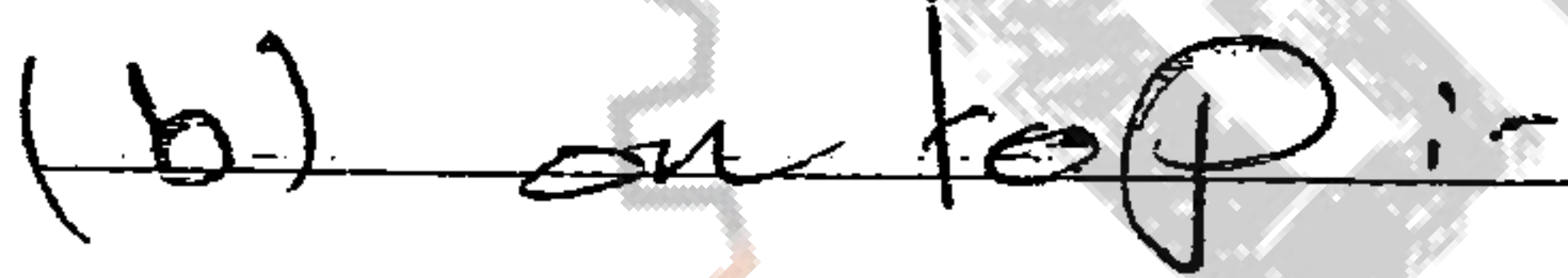
$$n_{\text{bottom}} = mgy + \frac{mv^2}{r}$$

الفريق السيد الشيخ هو

$n_1 mg \left(\frac{1 + v^2}{ng} \right)$ if at bottom
if at Top

$$= \frac{mg (1 + (225)^2)}{(2.73 \times 10^3) (9.8)}$$

$$w_{\text{bottom}} = 2.91 \text{ mg}$$



$$\sum F = n_{\text{top}} + mg = \frac{mv^2}{r}$$

$$N_{top} = mg \left(\frac{v^2}{rg} - 1 \right)$$

$$m_{\text{top}} = 0.413 \text{ mg}$$

→ follow

ex1:- حدد الساع المركزي والفرص لدوري لحجم سير
 حارس دائري رضى قطره (م) سرعة
 ثمانية ففداهما 11م 8
 ن م = 11 ، م = 11 ، لفة م

solu:-

ح - 8 م 11

$$\textcircled{1} \text{ ن م} = \frac{11 \times 8}{2} = 44 \text{ م 11}$$

$$\textcircled{2} \text{ م} = \frac{44 \times 2}{8} = 11 \text{ م 11}$$

ex2:- يور حجر فست لطف فقط حارس دائري سرعة المرات
 اذا كانت في الحجر (م) في اربعة
 صاحب 1 طول الخط 2 الساع المركزي الحجر

solution:-

$$\textcircled{1} \text{ ح} = \frac{\text{عدد الدوران} \times \text{ن م}}{\text{ر}} = \frac{4 \times 11}{2} = 22 \text{ م 11}$$

$$\textcircled{2} \text{ ح} = \frac{11 \times 2}{4} = 5.5 \text{ م 11}$$

$$\textcircled{3} \text{ ن م} = \frac{22 \times 2}{11} = 4 \text{ م 11}$$

الموضوع: Non uniform Circular motion
الحركة الدائرية الغير منتظمة

6.2 التاريخ 2 / 11 / 08

motion

تغير السرعة متغيرة في المقدار والاتجاه

* varying speed :- القوة المركزية :- هي القوة

التي تسبب التسارع المركزي

ويكون اتجاهها نحو المركز
 $\vec{a} = \vec{a}_n + \vec{a}_t$

where

$\vec{a}$:- total acceleration

$\vec{a}_n$:- radial acceleration

$\vec{a}_t$:- tangential acceleration

تسارع مماسي

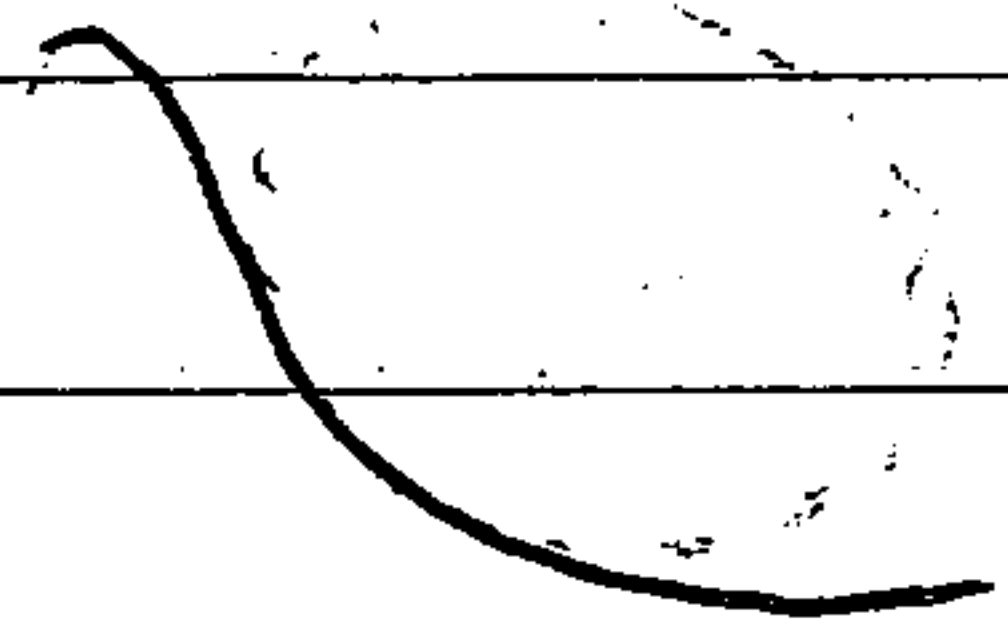
* The total Force exerted on a particle :-

$$\vec{F} = \vec{F}_n + \vec{F}_t$$

السبب في ظهور $\vec{a}_t$ هو وجود تغير في سرعة الجسيم

يكون اتجاه هذه القوايسين في دائرة خاطئة اولى

فيكون في اتجاه



6.1 :-

$$\underline{m = 25 \text{ kg}}, \quad r = 0.8 \text{ m}$$

$$\underline{m = 3 \text{ kg}}$$

Find the range of speeds can the object have before the string breaks?!!

Solution:-

$$T_{\max} = (25 \text{ kg}) \cdot g = 25 \times 9.8$$

$$= 245 \text{ N}$$

أي سرعة قبل أن تنقطع

$$T = \frac{mv^2}{r}$$

$$v^2 = \frac{rT}{m}$$

$$v_{\max} = \sqrt{\frac{r T_{\max}}{m}}$$

$$= \sqrt{\frac{(0.8)(245)}{3}} \Rightarrow 8.0808 \text{ m/s}$$

→ follow

V.I.:
6.5: ~ 1.2

$$\vec{v} = 50 \text{ cm/s} \Rightarrow 0.5 \text{ m/s}$$

سید شافق حسین

اکھلوں اُنکاو سچا

۱۔ جنگالہ سکوئی

normal force



mg

$$F_s = M_s a \Rightarrow F_s = \frac{M_s v^2}{r} \Rightarrow m M_s = \frac{m v^2}{r}$$

$$M_s \omega_g = \frac{\omega \sqrt{2}}{r}$$

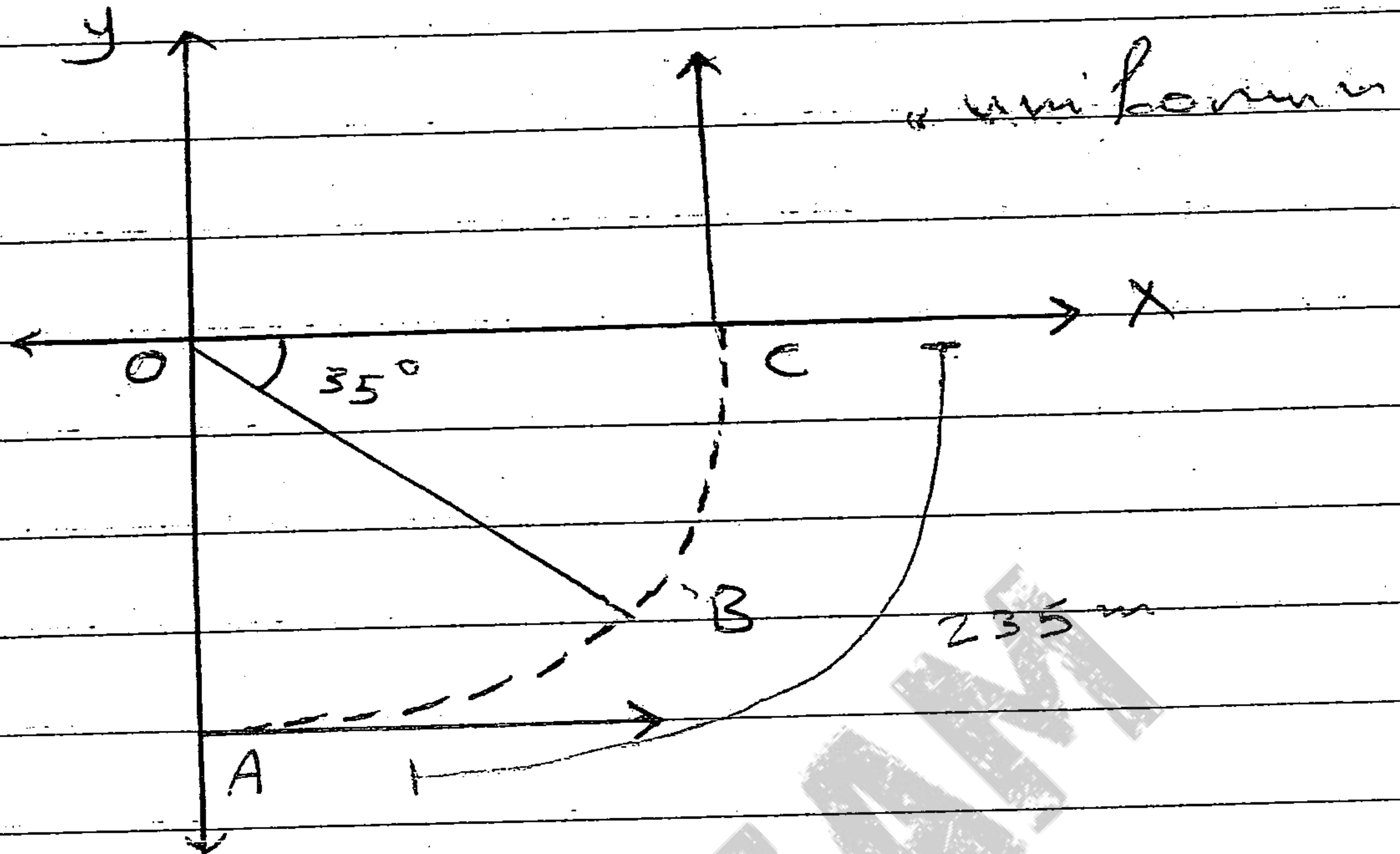
$$M_s = \frac{V^2}{hg}$$

$$s = \frac{(0.5)^2}{0.3 \times 9.8} \Rightarrow 0.085$$

لا يَكُنْ أَنْ تَتَفَكَّرَ M_2 ، M_3 ، M_4 ، M_5 ، M_6 ، M_7 ، M_8 ، M_9 ، M_{10} ، M_{11} ، M_{12} ، M_{13} ، M_{14} ، M_{15} ، M_{16} ، M_{17} ، M_{18} ، M_{19} ، M_{20} ، M_{21} ، M_{22} ، M_{23} ، M_{24} ، M_{25} ، M_{26} ، M_{27} ، M_{28} ، M_{29} ، M_{30} ، M_{31} ، M_{32} ، M_{33} ، M_{34} ، M_{35} ، M_{36} ، M_{37} ، M_{38} ، M_{39} ، M_{40} ، M_{41} ، M_{42} ، M_{43} ، M_{44} ، M_{45} ، M_{46} ، M_{47} ، M_{48} ، M_{49} ، M_{50} ، M_{51} ، M_{52} ، M_{53} ، M_{54} ، M_{55} ، M_{56} ، M_{57} ، M_{58} ، M_{59} ، M_{60} ، M_{61} ، M_{62} ، M_{63} ، M_{64} ، M_{65} ، M_{66} ، M_{67} ، M_{68} ، M_{69} ، M_{70} ، M_{71} ، M_{72} ، M_{73} ، M_{74} ، M_{75} ، M_{76} ، M_{77} ، M_{78} ، M_{79} ، M_{80} ، M_{81} ، M_{82} ، M_{83} ، M_{84} ، M_{85} ، M_{86} ، M_{87} ، M_{88} ، M_{89} ، M_{90} ، M_{91} ، M_{92} ، M_{93} ، M_{94} ، M_{95} ، M_{96} ، M_{97} ، M_{98} ، M_{99} ، M_{100} ، M_{101} ، M_{102} ، M_{103} ، M_{104} ، M_{105} ، M_{106} ، M_{107} ، M_{108} ، M_{109} ، M_{110} ، M_{111} ، M_{112} ، M_{113} ، M_{114} ، M_{115} ، M_{116} ، M_{117} ، M_{118} ، M_{119} ، M_{120} ، M_{121} ، M_{122} ، M_{123} ، M_{124} ، M_{125} ، M_{126} ، M_{127} ، M_{128} ، M_{129} ، M_{130} ، M_{131} ، M_{132} ، M_{133} ، M_{134} ، M_{135} ، M_{136} ، M_{137} ، M_{138} ، M_{139} ، M_{140} ، M_{141} ، M_{142} ، M_{143} ، M_{144} ، M_{145} ، M_{146} ، M_{147} ، M_{148} ، M_{149} ، M_{150} ، M_{151} ، M_{152} ، M_{153} ، M_{154} ، M_{155} ، M_{156} ، M_{157} ، M_{158} ، M_{159} ، M_{160} ، M_{161} ، M_{162} ، M_{163} ، M_{164} ، M_{165} ، M_{166} ، M_{167} ، M_{168} ، M_{169} ، M_{170} ، M_{171} ، M_{172} ، M_{173} ، M_{174} ، M_{175} ، M_{176} ، M_{177} ، M_{178} ، M_{179} ، M_{180} ، M_{181} ، M_{182} ، M_{183} ، M_{184} ، M_{185} ، M_{186} ، M_{187} ، M_{188} ، M_{189} ، M_{190} ، M_{191} ، M_{192} ، M_{193} ، M_{194} ، M_{195} ، M_{196} ، M_{197} ، M_{198} ، M_{199} ، M_{200} ، M_{201} ، M_{202} ، M_{203} ، M_{204} ، M_{205} ، M_{206} ، M_{207} ، M_{208} ، M_{209} ، M_{210} ، M_{211} ، M_{212} ، M_{213} ، M_{214} ، M_{215} ، M_{216} ، M_{217} ، M_{218} ، M_{219} ، M_{220} ، M_{221} ، M_{222} ، M_{223} ، M_{224} ، M_{225} ، M_{226} ، M_{227} ، M_{228} ، M_{229} ، M_{230} ، M_{231} ، M_{232} ، M_{233} ، M_{234} ، M_{235} ، M_{236} ، M_{237} ، M_{238} ، M_{239} ، M_{240} ، M_{241} ، M_{242} ، M_{243} ، M_{244} ، M_{245} ، M_{246} ، M_{247} ، M_{248} ، M_{249} ، M_{250} ، M_{251} ، M_{252} ، M_{253} ، M_{254} ، M_{255} ، M_{256} ، M_{257} ، M_{258} ، M_{259} ، M_{260} ، M_{261} ، M_{262} ، M_{263} ، M_{264} ، M_{265} ، M_{266} ، M_{267} ، M_{268} ، M_{269} ، M_{270} ، M_{271} ، M_{272} ، M_{273} ، M_{274} ، M_{275} ، M_{276} ، M_{277} ، M_{278} ، M_{279} ، M_{280} ، M_{281} ، M_{282} ، M_{283} ، M_{284} ، M_{285} ، M_{286} ، M_{287} ، M_{288} ، M_{289} ، M_{290} ، M_{291} ، M_{292} ، M_{293} ، M_{294} ، M_{295} ، M_{296} ، M_{297} ، M_{298} ، M_{299} ، M_{300} ، M_{301} ، M_{302} ، M_{303} ، M_{304} ، M_{305} ، M_{306} ، M_{307} ، M_{308} ، M_{309} ، M_{310} ، M_{311} ، M_{312} ، M_{313} ، M_{314} ، M_{315} ، M_{316} ، M_{317} ، M_{318} ، M_{319} ، M_{320} ، M_{321} ، M_{322} ، M_{323} ، M_{324} ، M_{325} ، M_{326} ، M_{327} ، M_{328} ، M_{329} ، M_{330} ، M_{331} ، M_{332} ، M_{333} ، M_{334} ، M_{335} ، M_{336} ، M_{337} ، M_{338} ، M_{339} ، M_{340} ، M_{341} ، M_{342} ، M_{343} ، M_{344} ، M_{345} ، M_{346} ، M_{347} ، M_{348} ، M_{349} ، M_{350} ، M_{351} ، M_{352} ، M_{353} ، M_{354} ، M_{355} ، M_{356} ، M_{357} ، M_{358} ، M_{359} ، M_{360} ، M_{361} ، M_{362} ، M_{363} ، M_{364} ، M_{365} ، M_{366} ، M_{367} ، M_{368} ، M_{369} ، M_{370} ، M_{371} ، M_{372} ، M_{373} ، M_{374} ، M_{375} ، M_{376} ، M_{377} ، M_{378} ، M_{379} ، M_{380} ، M_{381} ، $M_{382}</$

التاريخ 3 / 11 / 08 follow

6.10:-



arc $ABC = 235 \text{ m}$, the car

completes the turn in 36 s

(a) what is the $\vec{a}$ when the car is located B.

(b) average speed (c) average $\vec{a}$ during the 36 s .

solution:-

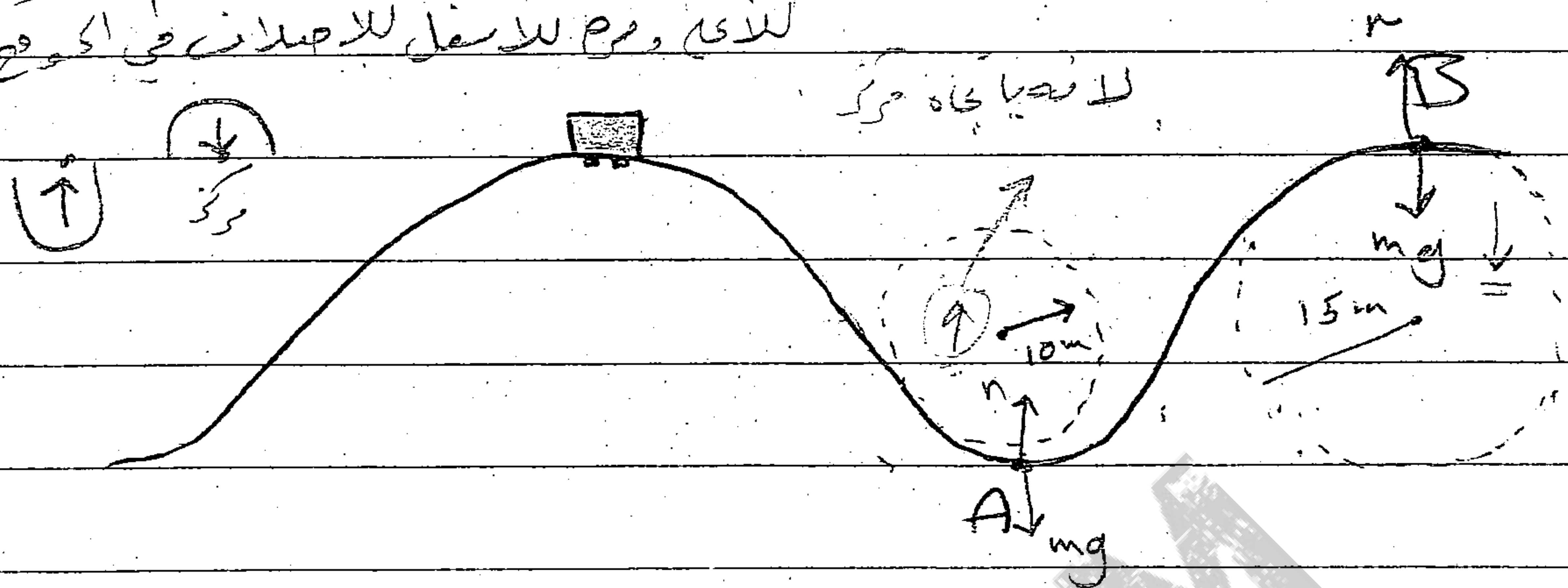
$$(a) \quad a_c = \frac{v^2}{r} \Rightarrow v = \frac{235}{36}$$

$$= 6.5 \text{ m/s}$$

$$\text{arc} = \frac{1}{\alpha} \hat{e}_r$$

$$s =$$

مكتم الزوال هو انه اياه دايما كذا كذا فخرم بيكونه - 14.6
للاي ورمم للاسفل للاصلان في الكوچ 1000



$$m = 500 \text{ kg}, \quad V_A = 20 \text{ m/s}$$

(a) Find the Force exerted by the track on the car

(b) $V_{\max}$ at (B) without Flying.

Solution:

$$(a) \sum F_r = \frac{mv^2}{r}$$

$$n - mg = \frac{mv^2}{r} \Rightarrow n = mg + \frac{mv^2}{r} = (500)(9.8) + \frac{(500)(20)^2}{10}$$

$$\Rightarrow n = 2.49 \times 10^4 \text{ N}$$

$$(b) n - mg = -\frac{mv^2}{r}$$

the maximum speed at B corresponds to 0

$$mg = \frac{mv^2}{r}$$

$$v_{\max} = \sqrt{rg}$$

$$\rightarrow \text{follow} = \sqrt{(15)(9.8)} \Rightarrow 12.1 \text{ m/s}$$

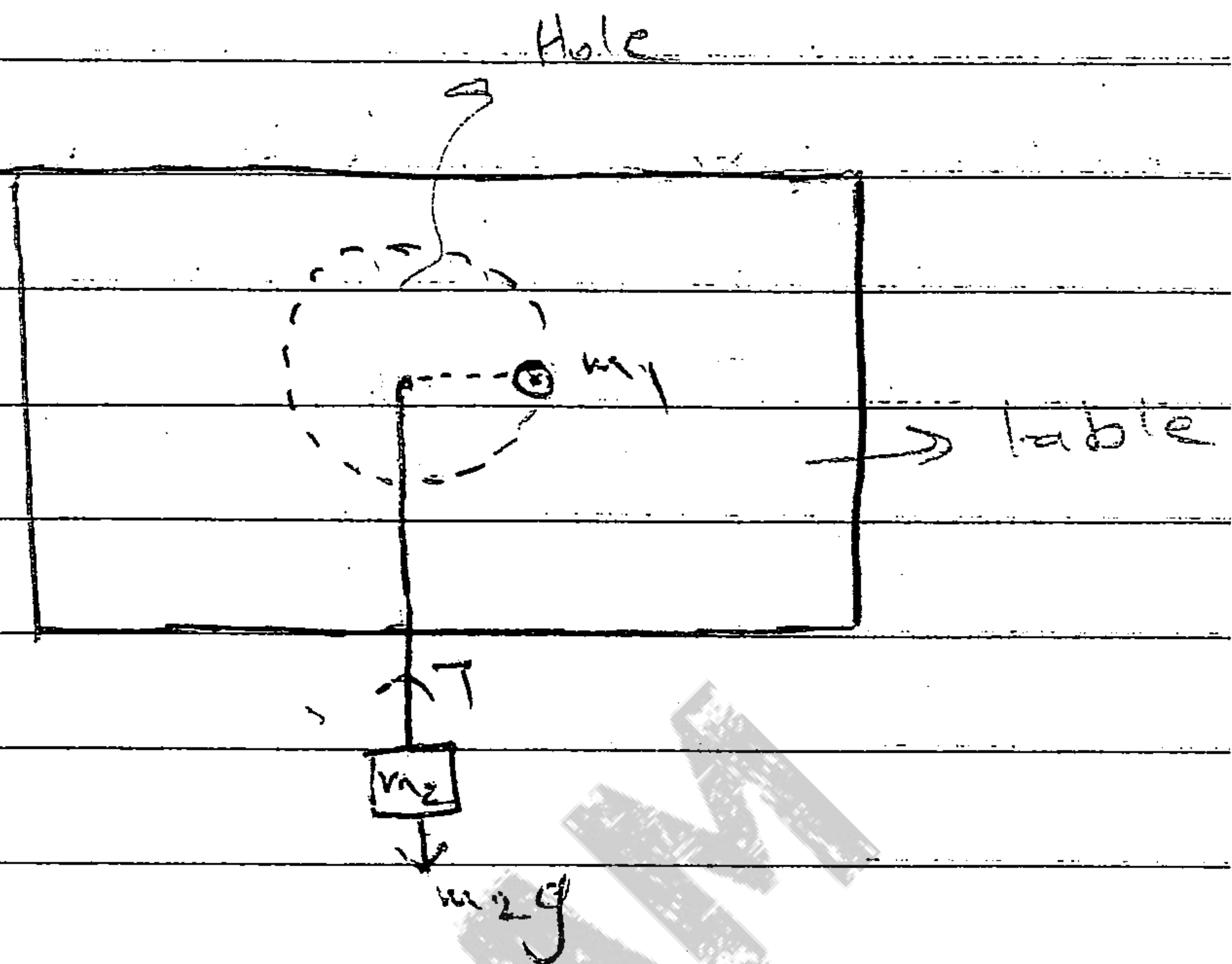
6.44:-

قزينة

 m_2 is equalberum m_1 is rounding

solution:- تلف

(a) $T = m_2 g$



$$\sum F_y = 0 \quad \text{because } m_2 \text{ is in equilibrium.}$$

$$T - m_2 g = 0$$

$$T = m_2 g$$

(b) $T = F_c = m_2 g$. "T" Tension. الـ سبب في قزينة الـ توتنج

(c) $F = \frac{m_1 v^2}{r} \Rightarrow T = \frac{m_1 v^2}{r}$

$$m_2 g = \frac{m_1 v^2}{r} \Rightarrow v = \sqrt{\frac{m_2}{m_1}} g r \quad v \propto m$$

(d) كلما زاد m_2 فزاد v فبقيت m_1 في Hole وتقل r .

(e) كلما قل m_2 قل v فبقيت m_1 في Hole وزاد r .

7.2 work done by a constant force

7.1

7/3

* A system:- is most often a single particle, collection of particles, or a region of space, and may vary in size and shape.

* A system boundary separates the system from the environment.

- الهدف من أخذ النظام "system" هو إكمال كامل القوى
حالا 2. هذا النظام ليس له بداية في بعض الأحيان

7.2 work done by a constant force:-

$$W = \vec{F} \cdot \Delta \vec{r} = |\vec{F}| |\Delta \vec{r}| \cos \theta$$

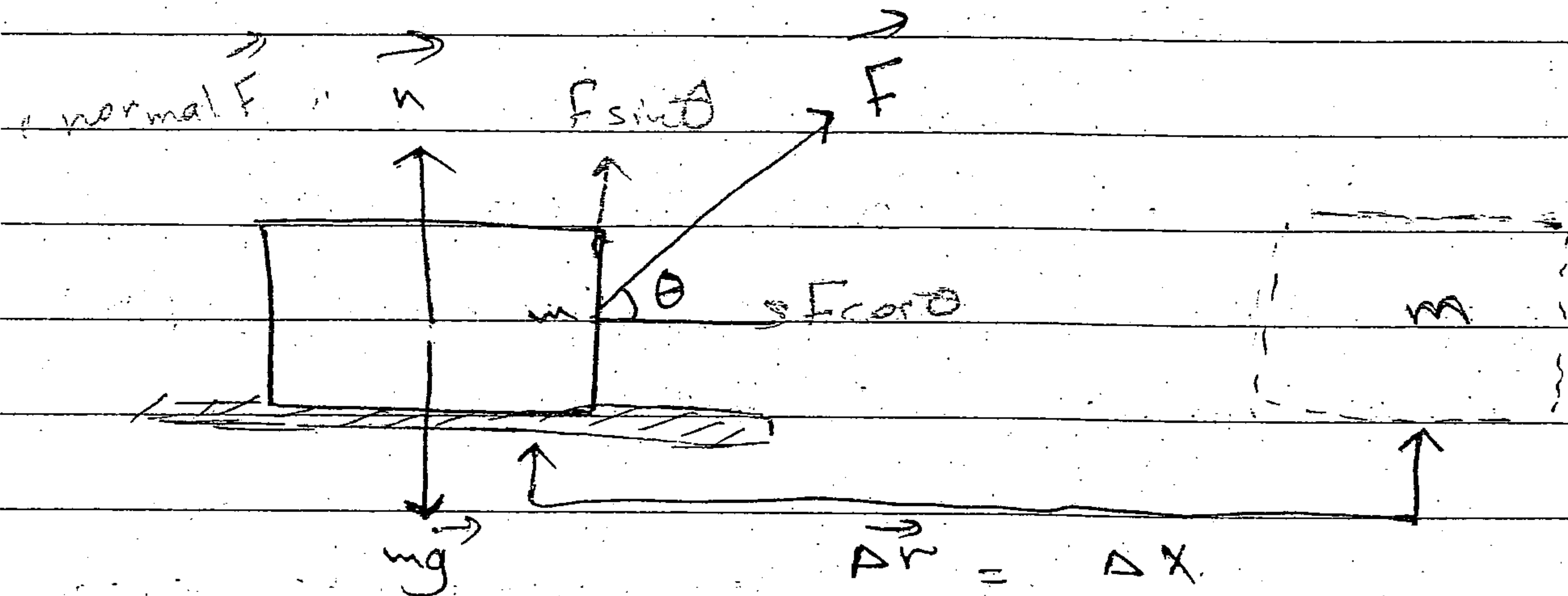
↓ ↓ ↓
work done by constant force on system constant force displacement of the point of application of the force

θ :- angle between $\vec{F}$ and $\Delta \vec{r}$

وهذا الزاوية هي الزاوية بين القوة والازاحة

6/11/2018 follow التاريخ

الموضوع:



No friction لا يوجد احتكاك

notes:-

- (1) $w = 0$, If $\Delta \vec{r} = 0$ القوة العمودية على الإزاحة لا تبذل
(2) $w = 0$, if $\theta = 90^\circ$ تستقل

$w_n = 0$ لأنه $\theta = 90^\circ$

$w_{mg} = 0$ لأنه $\theta = 90^\circ$

work of gravitational force.

- (3) if $\vec{F}$ and $\Delta \vec{r}$ in the same direction w is positive. لأنه $\cos 0 = 1$

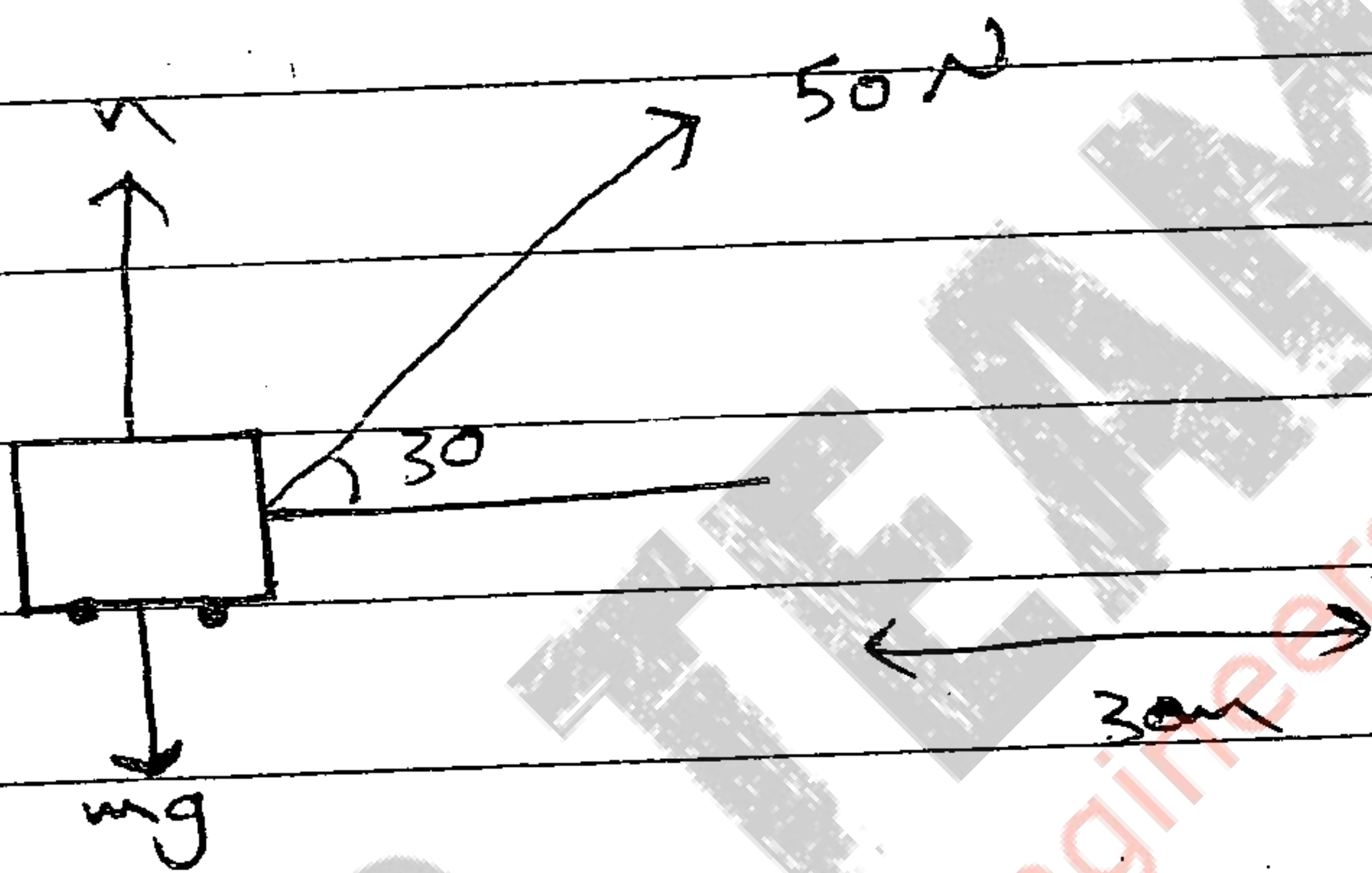
$[w] = n \cdot m = J = \text{Joule}$

هذه الطاقة التي تبذل سطرًا سطرًا هي القوة المركزية لأنها تعاكس الإزاحة دوماً

6 / 11 / 08 التاريخ follow

exp:- A man cleaning floor pulls a vacuum cleaner with a force 50 N at angle 30° with the horizontal. Find the work done by the force on the cleaner moves 3 m to the right.

solution:-



$$W_n = 0$$

لأنه عمودي على الاتجاه

$$\cos 90 = 0$$

$$W_{mg} = 0$$

لأنه عمودي على الاتجاه

$$\cos 90 = 0$$

$$\begin{aligned}
 W_F &= |F| |Aw| \cos \theta \\
 &= (50)(3) \cos 30 \\
 &= 130 \text{ Jole.}
 \end{aligned}$$

مثال ١- أوجد قوة (C) طوية (م) جسم باتجاه سطح زاوية (٦٠°)،
حاذية أفقية موقوفة (٢٢ م)، اكتب شغل هذه القوة.

$$\text{الحل ١- } \vec{C} = \vec{N} = 50 \text{ م} \rightarrow 50 \text{ م} \times 50 \text{ م} = 2500 \text{ م}^2$$

$$C = 50 \text{ م}$$

لاحظ أنه شغل مركبة القوة العمودية للاتجاه الحركة
(٦٠°) يادي صفرًا لأنها لم تحرك الجسم

ملاحظة: لذلك عند إيجاد الشغل لا نأخذ سوى المركبة الأفقية
لأنه العمودية لا تحركه. $\vec{C} = 50 \text{ م}$

مثال ٢- يسحب صندوق بسرعة ثابتة على سطح أفقي فسه بتأثير قوة
شد. فإذا أخرجت قوة الشد شغلًا مقداره (٥٤ جول)
حينما انزاح الصندوق (٢٩ م) باتجاه اليمين، ما مقدار
قوة الاحتكاك بين الصندوق والسطح.

الحل ١- بما أنه الصندوق يتحرك بسرعة ثابتة، فإنه القوة المحصلة فيه صفر
صفر وبالتالي فالشغل الكلي المحصول يادي صفرًا
شغل كلي = شغل قوة X الانزاحة = صفر

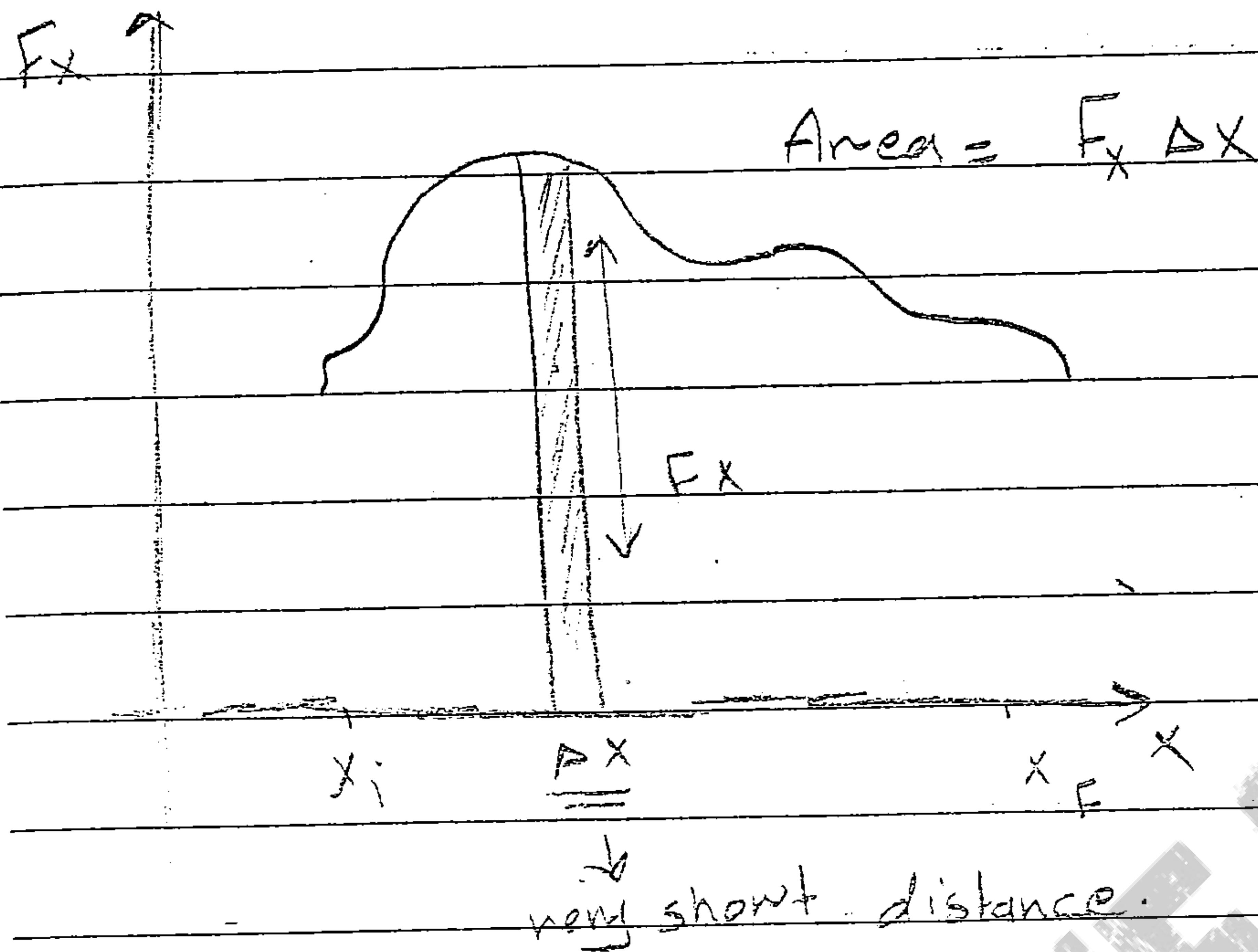
$$\text{شغل قوة الشد} + \text{شغل قوة الاحتكاك} = \text{صفر} \rightarrow 54 + \text{شغل قوة الاحتكاك} = 0$$

$$\text{شغل قوة الاحتكاك} = -54 \text{ م} \times 29 \text{ م} = -1566 \text{ م}^2$$

$$\text{شغل قوة الاحتكاك} = -1566 \text{ م}^2$$

لأنه شغل قوة الاحتكاك يساوي صفرًا

الموضوع: 7.4 work done by a varying Force. التاريخ 6/11/08



$W = \text{Area under the curve.}$

* IF the Varying Force does work on a particle as the particle moves along the x-axis from x_i to x_f , the work done by the force on the particle.

$$W = \int_{x_i}^{x_f} F_x \cdot dx$$

أي x_i إلى x_f هي محاذ
من إيجاد التكامل المحدود
للإزاحة المطلوبة.

الشغل الكلي (عملياً) = المسافة الكلية تحت منحنى (الإزاحة - القوة)

6 / 10 / 08 التاريخ Follow

موضوع:

Ex:- Find the work in the following equation:-

$$F_x = 3x^2 + 5x + 2$$

$$x = 0 \text{ to } x = 2 \text{ m}$$

- Find work of F_x on the particle.

Sol:-

$$W = \int_{x_i}^{x_f} F_x \cdot dx$$

$$W = \int_0^2 (3x^2 + 5x + 2) \cdot dx$$

$$W = \left(x^3 + \frac{5x^2}{2} + 2x \right) \Big|_0^2 = 22 \text{ Jole}$$

smooth surface:-

وحيث أن
Friction less

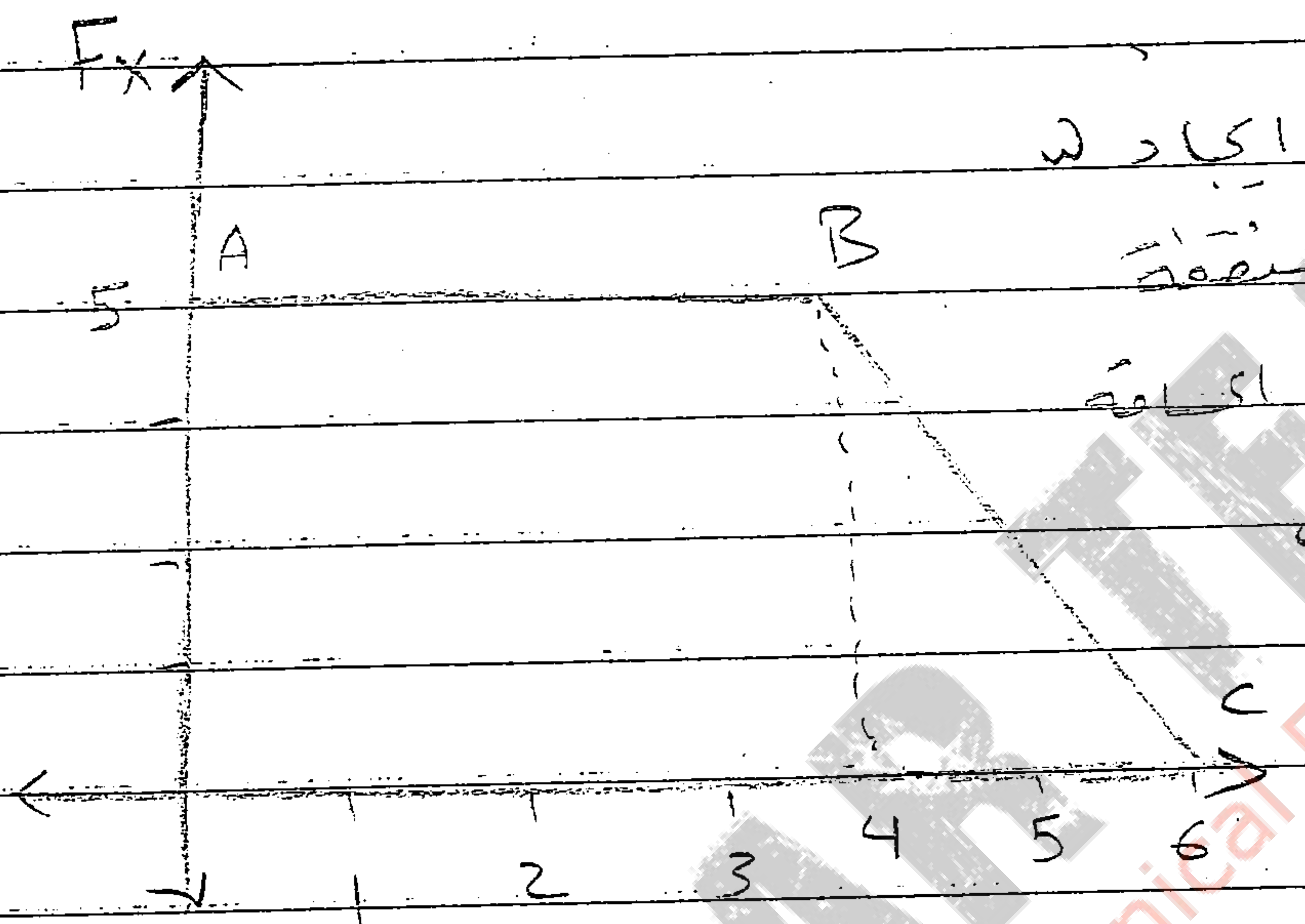
التاريخ 6 / 11 / 08 Follow:-

Exp:- A Force acting on a particle

varies with x as shown

calculate the work done by the Force

on the particle as it moves

from $x = 0$ to $x = 6 \text{ m}$ 

طريقة الحل :- عند معرفة أي دالة

من خلال تقسيم المسافة

إلى مسافات صغيرة أي دالة

بأنسب (ممكن) فكل دالة

العمل

solu:-

$$W_{AB} = (5)(4) = 20 \text{ J}$$

مساحة مستطيلة

$$W_{BC} = \frac{1}{2} \times (5)(2) = 5 \text{ J}$$

مساحة مثلثية

total work W_{AC}

$$W_{AC} = W_{AB} + W_{BC}$$

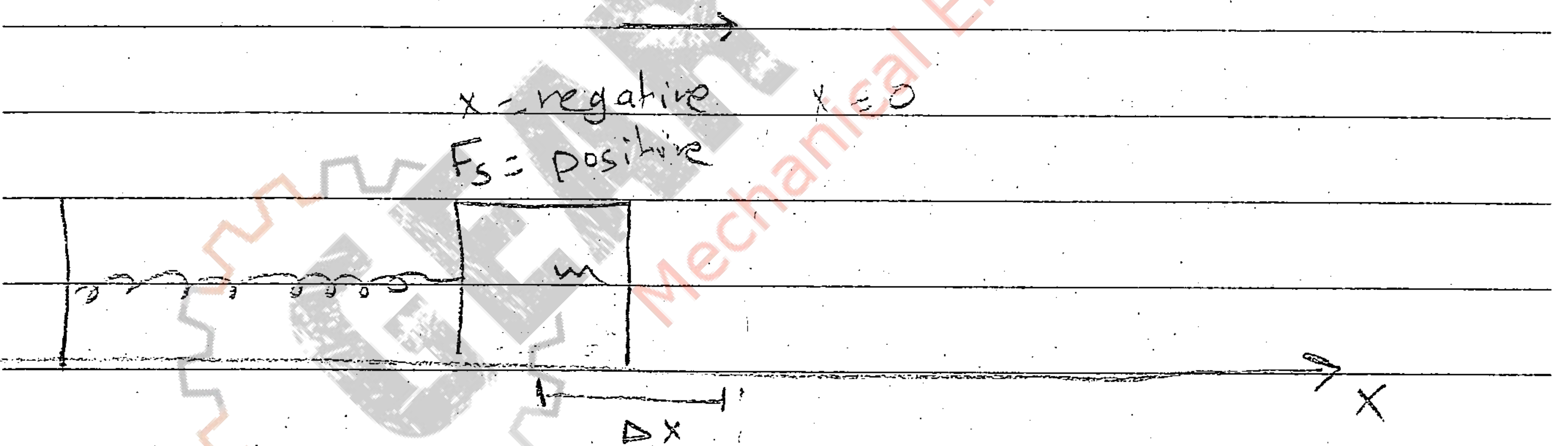
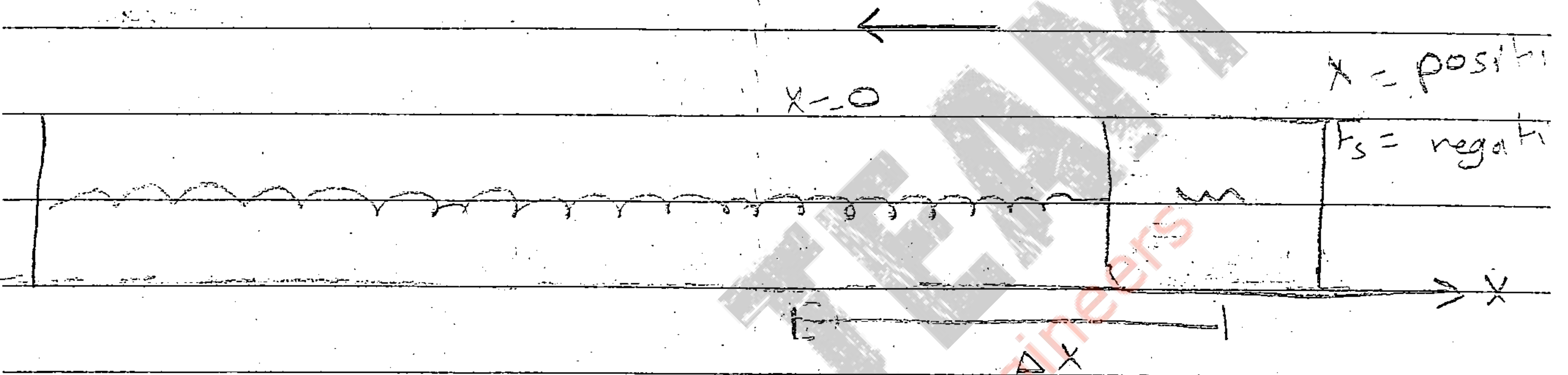
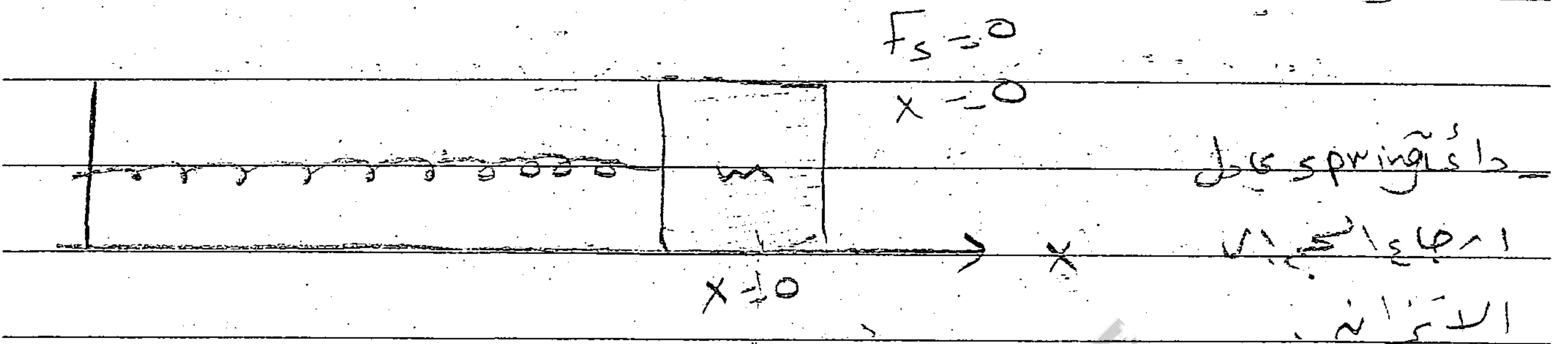
$$= 20 + 5 = 25 \text{ Joule}$$

تاريخ 6/11/08 Follow

الموضوع:

work done by spring:-

$F_s = -kx$
قوة مروية



* الشغل الذي تبذره القوة المروية "ش السطح"
ش السطح = $\int_{x_1}^{x_2} F_s dx$

أما الشغل الذي تبذره القوة الخارجية على السطح:
ش السطح = $\int_{x_1}^{x_2} F_{ext} dx$

* Hooke's law:-

يتناسب ما يولد هذا الجاد

النقطة الخصية مع « F_s »

$$\vec{F}_s = -k \times \vec{x}$$

where:-

لأنه F_s متجه F_s : spring force

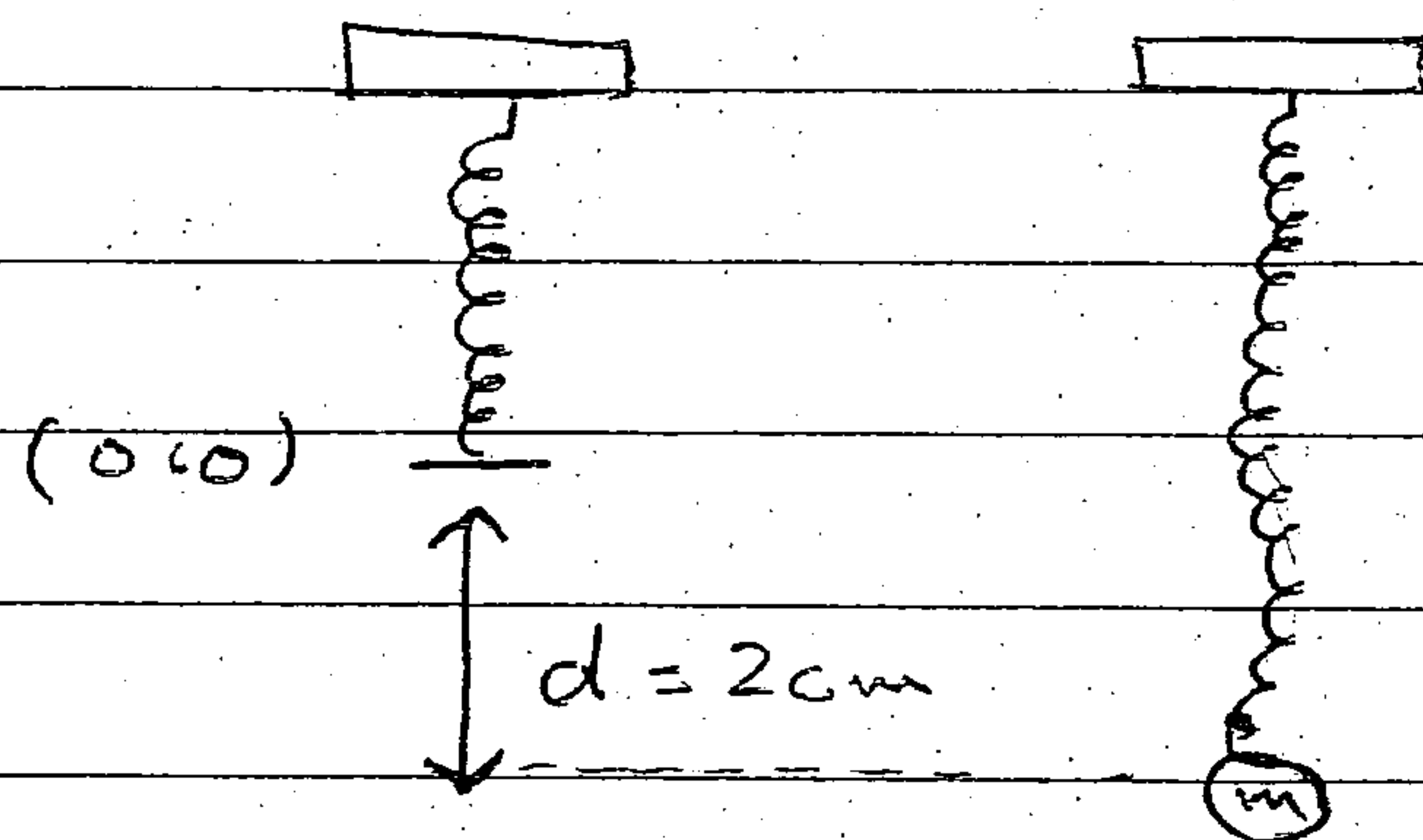
R:- Force constant or spring constant

«ثابت المرونة»

* The negative sign signifies that the force exerted by the spring is always directed opposite the displacement from equilibrium.

بالعربي «مع - ث س» $F_s = -k \times x$

exp:-



$$m = 0.55 \text{ kg}$$

- (a) find the force constant of the spring.
- (b) work done by the spring on the mass

Solu:-

(a)

$$T = F_s = kd$$

$$\sum F_y = 0$$

$$mg = kd$$

$$k = \frac{mg}{d}$$

$$= \frac{(0.55)(9.8)}{(0.02)} = 2.7 \times 10^2 \text{ N/m}$$

$$(b) W = \frac{1}{2} k y_i^2 - \frac{1}{2} k y_f^2$$

$$= 0 - \frac{1}{2} (2.7 \times 10^2) * (-0.02)^2$$

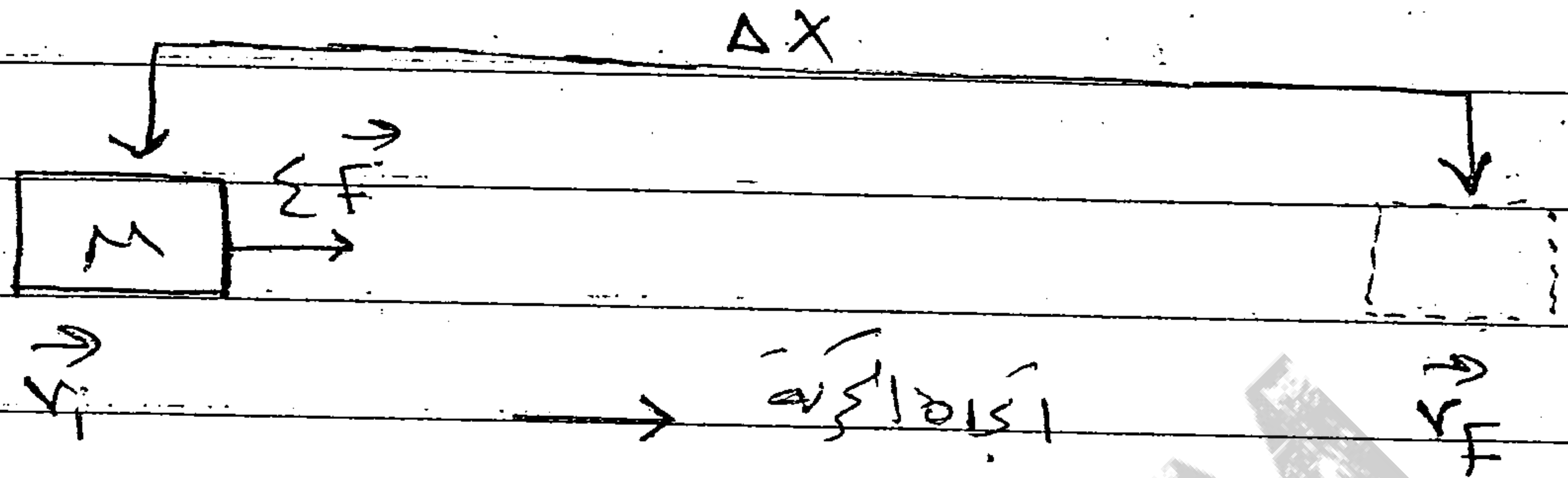
$$= -5.4 \times 10^{-2} \text{ J}$$

الموضوع: طاقة الحركة Kinetic energy and 7.5 التاريخ 10/1/08

the work - kinetic energy Theorem

مبرهنة الشغل - الطاقة

الطاقة الحركية: هي الطاقة التي تكتسب الجسم نتيجة حركته.



$$W_{\text{net}} = \int_{x_i}^{x_F} (\Sigma \vec{F}) dx = \int_{x_i}^{x_F} (ma) dx$$

net work done on the block.

مع قانون نيوتن II

$\Sigma F = ma$

$$W_{\text{net}} = \int m \frac{dv}{dt} dx$$

بالتعويض بـ $dx = v dt$

$$= \int m \frac{dv}{dt} \cdot v dt$$

$v = \frac{dx}{dt}$

$$= \int_{v_i}^{v_F} m \cdot v dv$$

$$W_{\text{net}} = \frac{1}{2} m v_F^2 - \frac{1}{2} m v_i^2$$

work → Kinetic energy

$$W_{\text{net}} = K_F - K_i = \Delta K$$

$$* K = \frac{1}{2} \times m \times v^2$$

$\downarrow$ kinetic energy $\downarrow$ mass $\rightarrow$ velocity.

kinetic energy

طاقة حركية

$$W_{net} = \Delta K$$

$$W_{net} = 2 J$$

$$\frac{1}{2} \times 2 \times v^2 = 2 J$$

* when work is done on a system and the only change in the system is in its speed, the net work done on the system equals the change in kinetic energy of the system.

« مقدار العمل هو التغير في الطاقة الحركية »

exp:- A 6 kg block initially at rest is pulled to the right along a horizontal, frictionless surface by constant horizontal force 12 N.

Find the block's speed after it has moved 3 m.

طاقة الوضع

الموضوع: potential energy of 7.6 التاريخ 10/11/08

a system :

طاقة الوضع :- هي الطاقة الكامنة في الجسم بسبب ارتفاعه عن سطح الأرض.

* If a particle of mass "m" is at a distance y above the earth's surface, the gravitational potential energy of the particle-earth system is :-

$$U_g = m \times g \times y$$

حيث :-
 U :- طاقة الوضع
 m :- كتلة الجسم
 g :- تسارع الجاذبية الأرضية
 y :- الارتفاع عن سطح الأرض

* The elastic potential energy stored in a spring of force constant k is :-

$$U_s = \frac{1}{2} \times k \times x^2$$

Spring constant "مطردنة الزنبرك"

أي إذا طالت طاقة الوضع المخزنة في الزنبرك
 فإنه ستزيد مقدار السعة x

$$w_{\text{net}} = -\Delta U_s \quad V_i = V_f$$

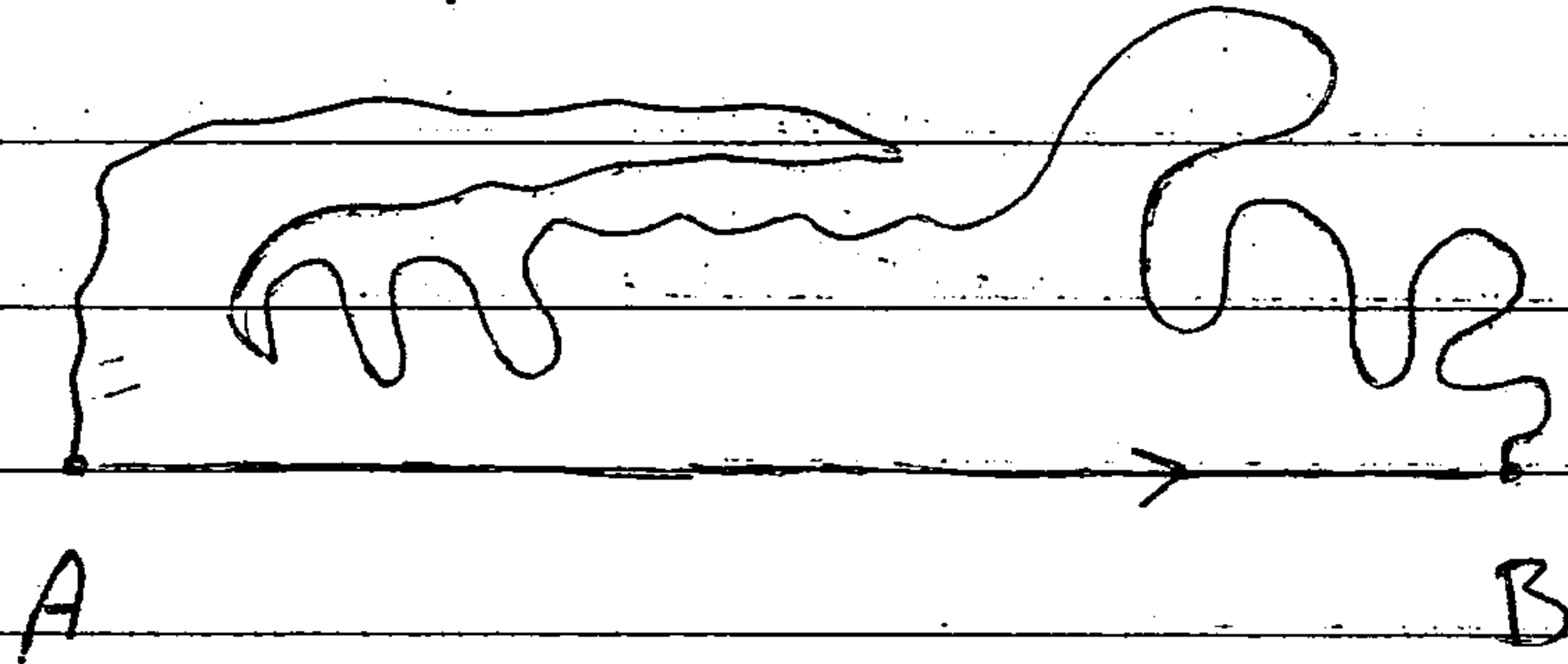
٥) مرحلة النقل - الطامة

لے کر تھیں انہیں ۲۰ جول سے $\frac{1}{C} \times ۲۰ \times ۱۰^۹$ جول
 الطاقہ الکافیہ فی الساعت
 قدر کے بغیر نہیں

HAJEER Notebook

الموضوع: 7.7 conservative and non conservative Forces:-
7.8 power.

7.8 power.



* A Force is conservative if the work it does on a particle that moving between any two points is independent of the path taken by the particle.

- the force is conservative :-

The work done by the force on a particle moving through any closed path is zero.

Notes:-
الحفاظة لا تفقد في المسار
تظل القوة المحافظة لا تفقد في المسار سواء في اتجاه أو عكس اتجاه
على المسار لا يتبدل، والسبب أي الإزالة.

* A Force that does not meet these criteria is said to be non conservative Force.

Example of conservative Forces

① is the Force of gravity

$$F_g = mg$$

$$W_g = \vec{F} \cdot \Delta \vec{r}$$

$$= (mg \hat{j}) \cdot (y_f \hat{j} - y_i \hat{j})$$

$$= -mg(y_f - y_i)$$

$$W_g = mgy_i - mgy_f = -\Delta U_g$$

$$= U_i - U_f = -\Delta U_g$$

② القوة الكهربية هي قوة محافظة

التاريخ 108 / 11 / 201

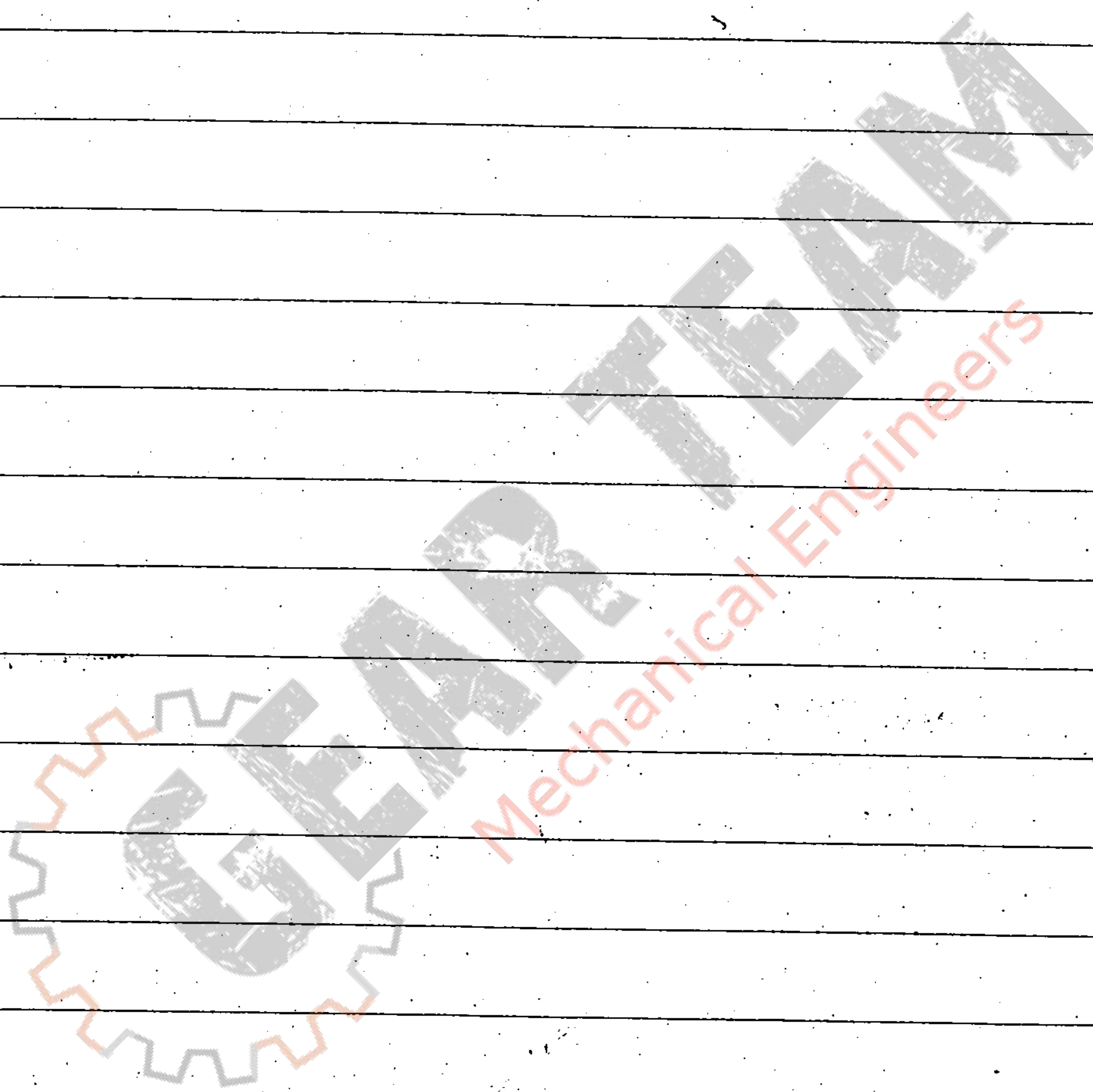
exp 9.7

مجموع

240

حالة

chapter 9



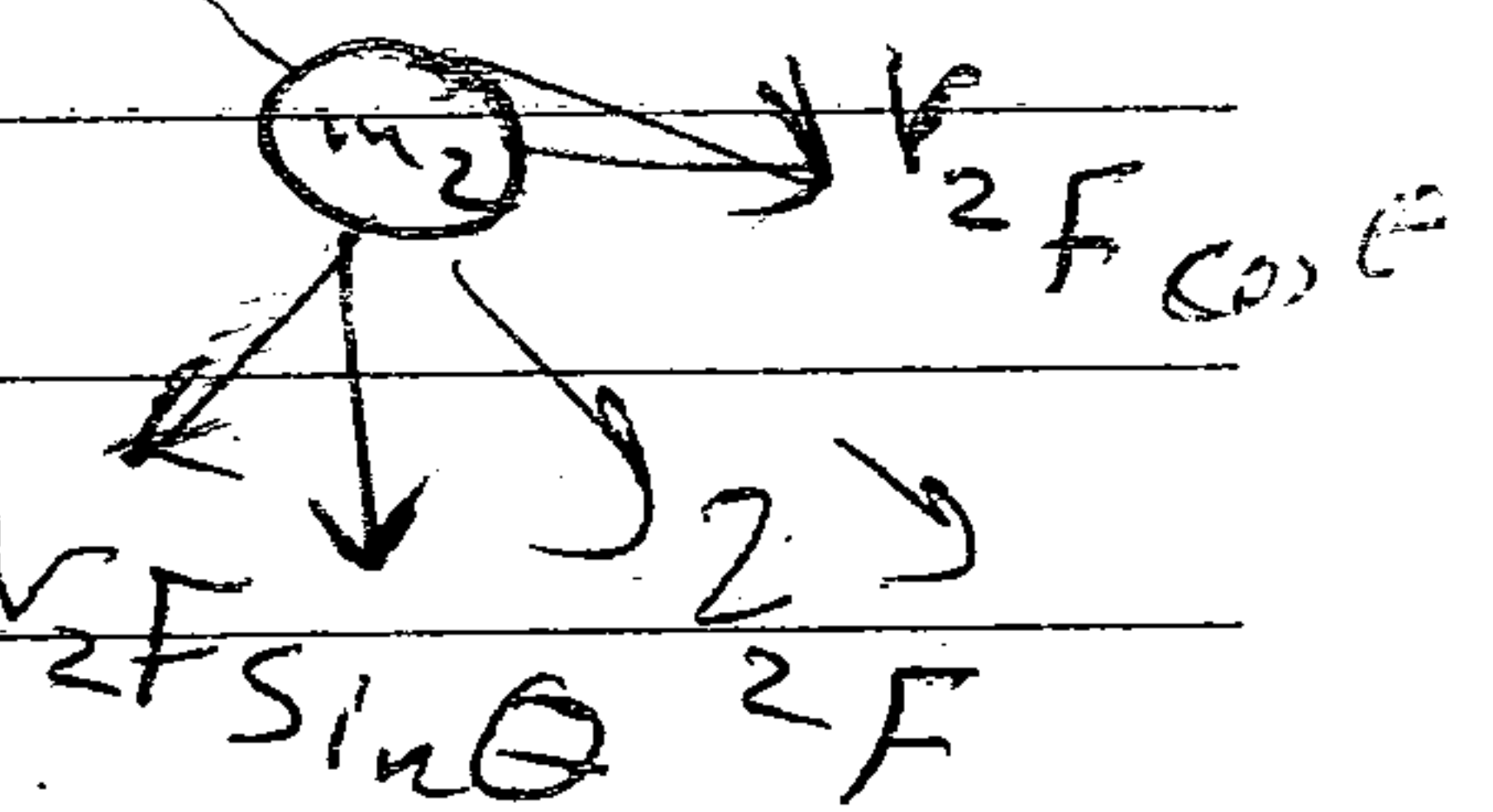
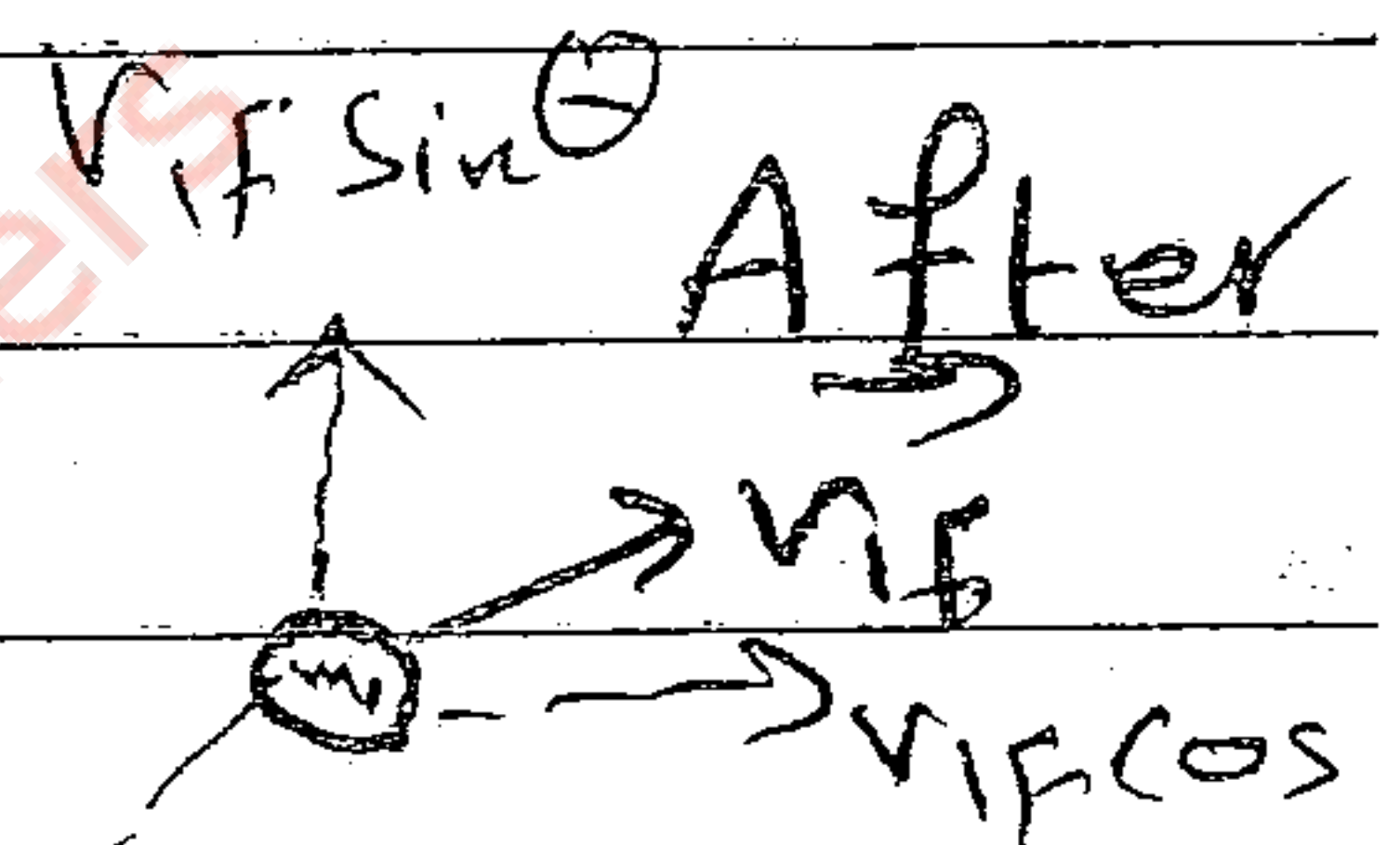
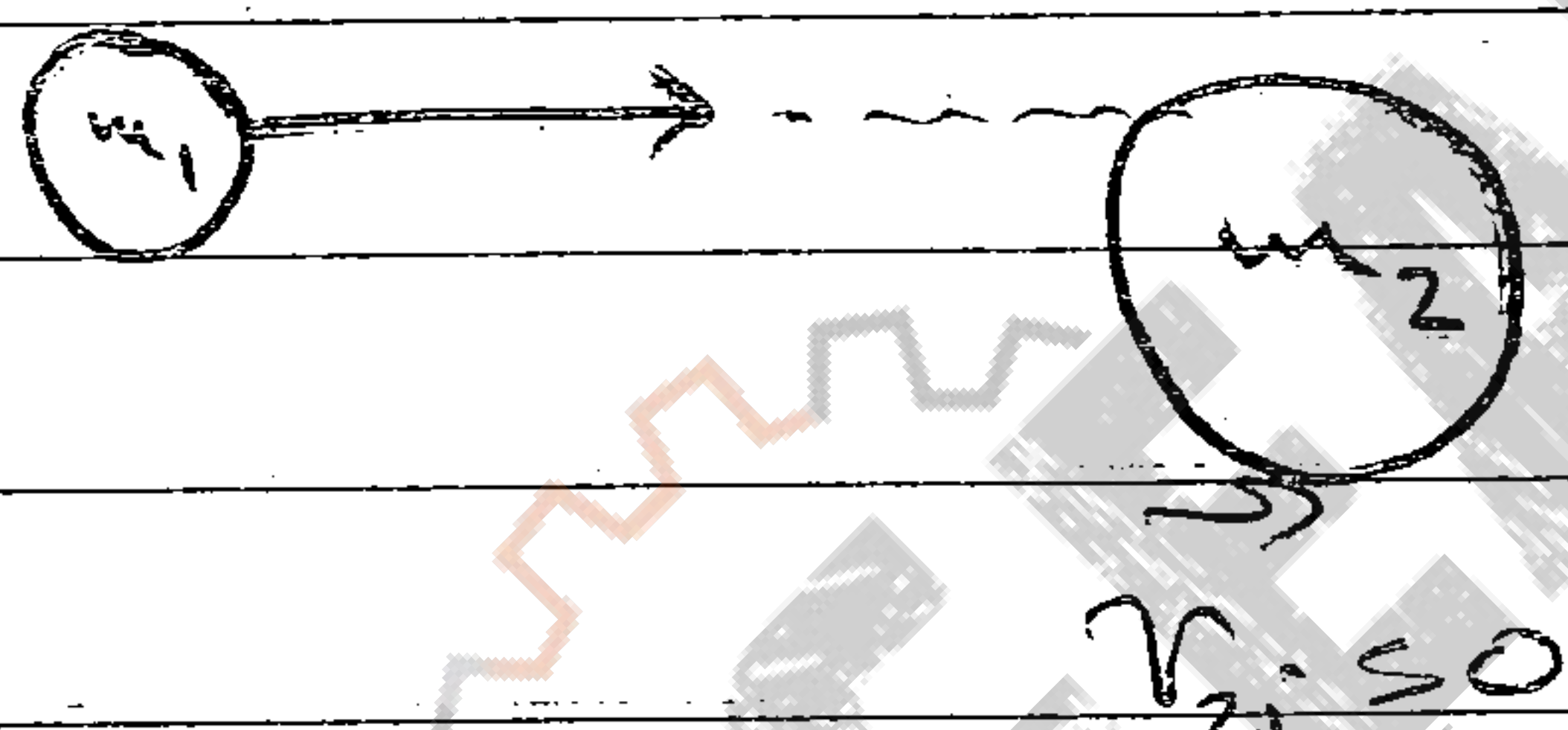
الموضوع: التحولات في بعدين
9.4 Collision in two dimensions.
التاريخ 30/11/08

* general Conservation of momentum:

$$m_1 v_{1ix} + m_2 v_{2ix} = m_1 v_{1fx} + m_2 v_{2fx}$$

$$m_1 v_{1iy} + m_2 v_{2iy} = m_1 v_{1fy} + m_2 v_{2fy}$$

before collision



* Conservation of momentum:-

$$m_1 v_{1i} + 0 = m_1 v_{1f} \cos \theta + m_2 v_{2f} \cos \theta$$

$$0 + 0 = m_1 v_{1f} \sin \theta - m_2 v_{2f} \sin \theta$$

هذا لنا في الحالتين في مقدار السرعة فقط

التاريخ 30/11/08 Follow

الموضوع:

* IF the collision is elastic:-

⊕

$$\frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2$$

السرعة قبل

السرعة بعد

سرعة السارية قبل أو بعد

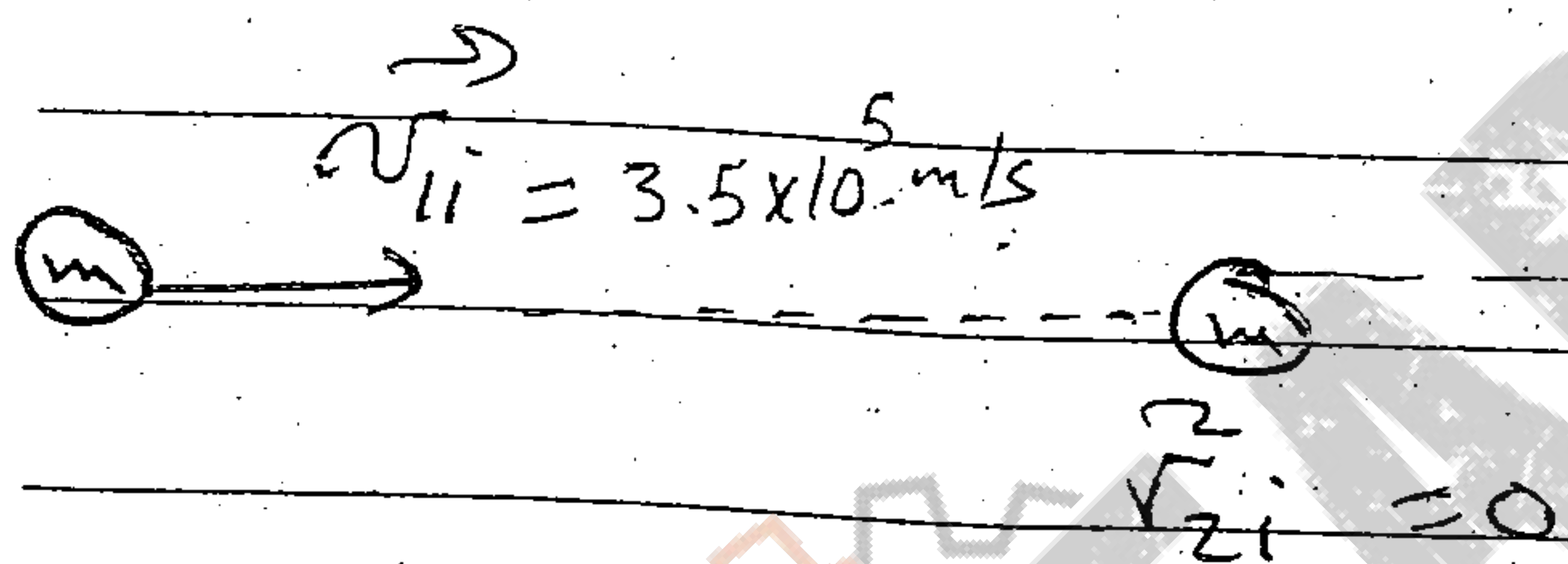
سرعة

exp 9.9:-

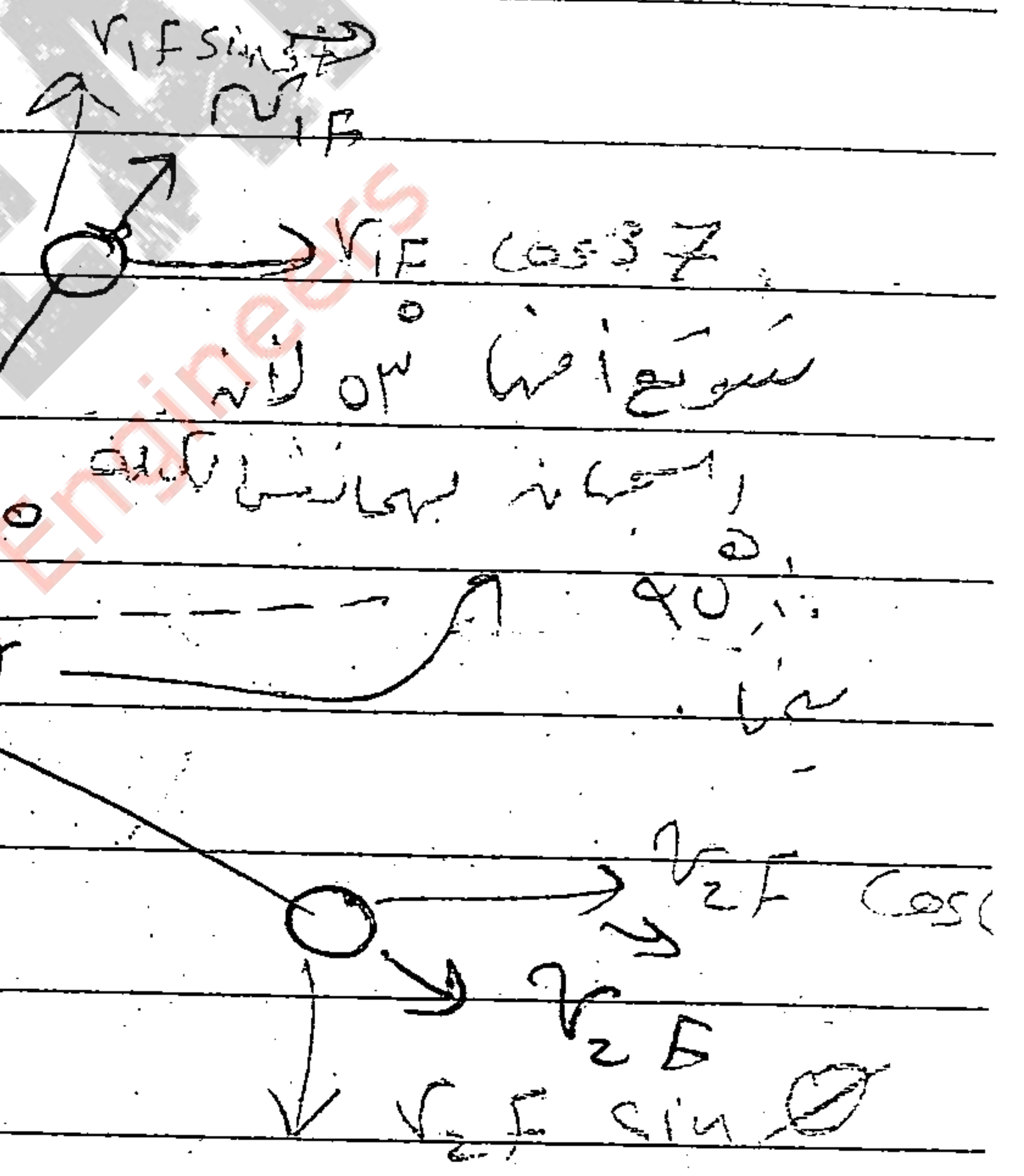
elastic collision

Two protons collide with each other

* before collision



$m_1 = m_2 = m$
 find v_{1f} , v_{2f} , θ



Solution:- X-direction

$$m_1 v_{1ix} + m_2 v_{2ix} = m_1 v_{1fx} + m_2 v_{2fx}$$

$$\therefore 3.5 \times 10^5 = v_{1f} \cos 37^\circ + v_{2f} \cos \theta \quad \text{--- (1)}$$

Follow

30/11/08 التاريخ follow

y-direction:-

$$0 = m v_{1f} \sin 37 - m v_2 \sin \theta \quad \text{--- (2)}$$

* conservation of kinetic energy:-

$$\frac{1}{2} m v_{1i}^2 + 0 = \frac{1}{2} m v_{1f}^2 + \frac{1}{2} m v_2^2$$

$$1.23 \times 10^{11} = v_{1f}^2 + v_2^2 \quad \text{--- (3)}$$

Rearrange eq (1) and (2):-

$$v_2 \cos \theta = 3.5 \times 10^5 - v_{1f} \cos 37$$

$$v_2 \sin \theta = v_{1f} \sin 37$$

Square them and add them:-

$$\Rightarrow v_2^2 = 1.23 \times 10^{11} - 5.59 \times 10^5 v_{1f} + v_{1f}^2 \quad \text{--- (4)}$$

التاريخ 30/ 11/ 08

Substitute eq (4) into eq (3)

$$(2 v_{1F} - 5.59 \times 10^5) v_{1F} = 0$$

$$\Rightarrow v_{1F} = 0$$

$$\Rightarrow 2v_{1F} - 5.59 \times 10^5 = 0$$

$$\therefore v_{1F} = 2.8 \times 10^5 \text{ m/s}$$

use eq (3) :-

$$v_{2F} = 2.11 \times 10^5 \text{ m/s}$$

use eq (2) :-

$$\phi = \sin^{-1} \left(\frac{v_{1F} \sin 37}{v_{2F}} \right)$$

$$= 53^\circ$$

note:- عند استخدام جيب التمام في معادلة كينما فنحن

النتيجة بأنه مخرج الزاوية بيننا وبين المصاعم هو

ط ١٠

الموضوع: The center of Mass: - التاريخ 21/12/08

موقع مركز الكتلة

* The position vector of center of mass of a system of particles is:-

$$\vec{r}_{cm} = \frac{1}{M} \sum_i m_i \vec{r}_i, \text{ where:-}$$

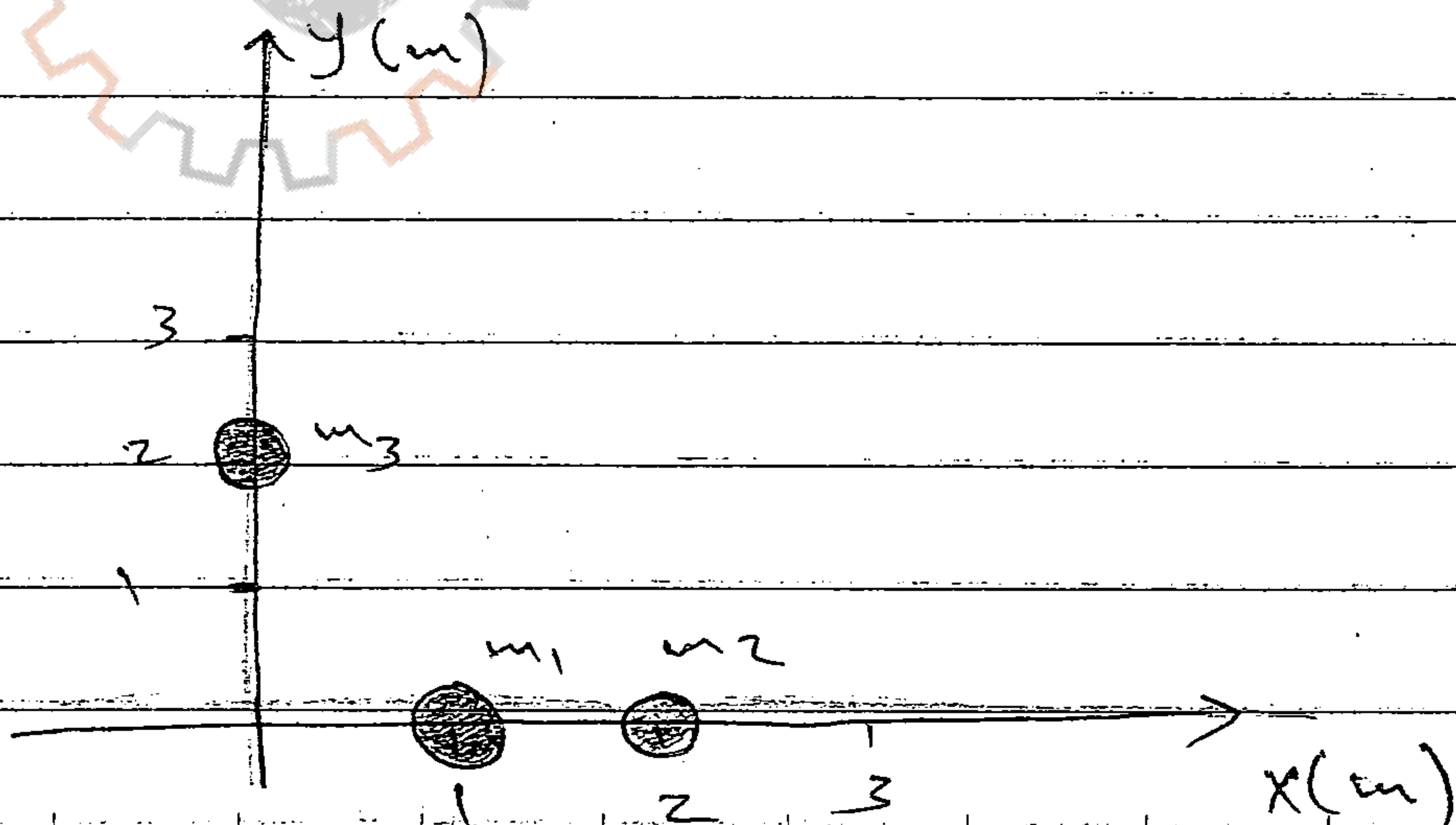
$\vec{r}_{cm}$ Central of mass

$M = \sum_i m_i$ is the total mass of the system.

$\vec{r}_i$ is the position vector of the i th particle.

$\vec{r}_i$ موقع الجسيم i

exp: A system consists of 3 particles
Find the center of mass of the system:-



3 / 12 / 2018 التاريخ

الموضوع:

Soln -

$$x_{cm} = \frac{1}{M} \sum m_i x_i = \frac{1}{4} [(1)(1) + (1)(2) + (2)(0)]$$

$$= \frac{3}{4} = 0.75 \text{ m}$$

$$y_{cm} = \frac{1}{M} \sum m_i y_i = \frac{1}{4} [(1)(0) + (1)(0) + (2)(2)]$$

$$= 1 \text{ m}$$

$$\vec{r}_{cm} = x_{cm} \hat{i} + y_{cm} \hat{j}$$

$$= [0.75 \hat{i} + 1 \hat{j}] \text{ m}$$

الموضوع: 9.6 Motion of a system of particles
التاريخ 21/12/08

يسرى النظام ككل

$$* \vec{v}_{cm} = \frac{1}{M} \sum_i m_i \vec{v}_i$$

where:-

$\vec{v}_{cm}$:- velocity of the center of mass for a system of particles.

$\vec{v}_i$:- velocity of one of them or for each other

$$* \vec{a}_{cm} = \frac{1}{M} \sum_i m_i \vec{a}_i$$

acceleration of the center of mass

* Newton's second law applied to a system of particles:-

$$(iii) \sum \vec{F}_{ext} = M \vec{a}_{cm}$$

$$(iii) \vec{P}_{total} = \sum \vec{p}_i = M \vec{v}_{cm}$$

أي بدل ما تجمع الزخم لكل الجسيمات في قمتها يسوية $\vec{v}_{cm}$ في قمتها يسوية
total linear momentum

→ follow

سوال 2 / 12 / 08 تاریخ

موضوع:

$$\textcircled{III} \quad I = \Delta p$$

$\hookrightarrow$ Impulse total

$$* \text{ If } \sum \vec{F}_{\text{ext}} = 0$$

$$0 = M \vec{a}_{\text{cm}}$$

$$0 = M \frac{d \vec{v}_{\text{cm}}}{dt}$$

$$0 = \frac{d}{dt} (M \vec{v}_{\text{cm}})$$

$$\Rightarrow M \vec{v}_{\text{cm}} = \vec{p}_{\text{total}} = \text{constant}$$

note: - The center mass of the system is always at the same position and moves with a constant velocity. It is not necessary that the center mass is at the geometric center of the system.

* note: - 252 UP 9.14 حل سوال

الموضوع: "chapter IX" problems التاريخ 21/12/08

Q.1 :-

$$M = 3 \text{ kg}$$

$$\vec{v} = (3\hat{i} - 4\hat{j}) \text{ m/s}$$

(a) Find the x and y components of the linear momentum.

(b) Find the magnitude and direction of the linear momentum.

Solution:-

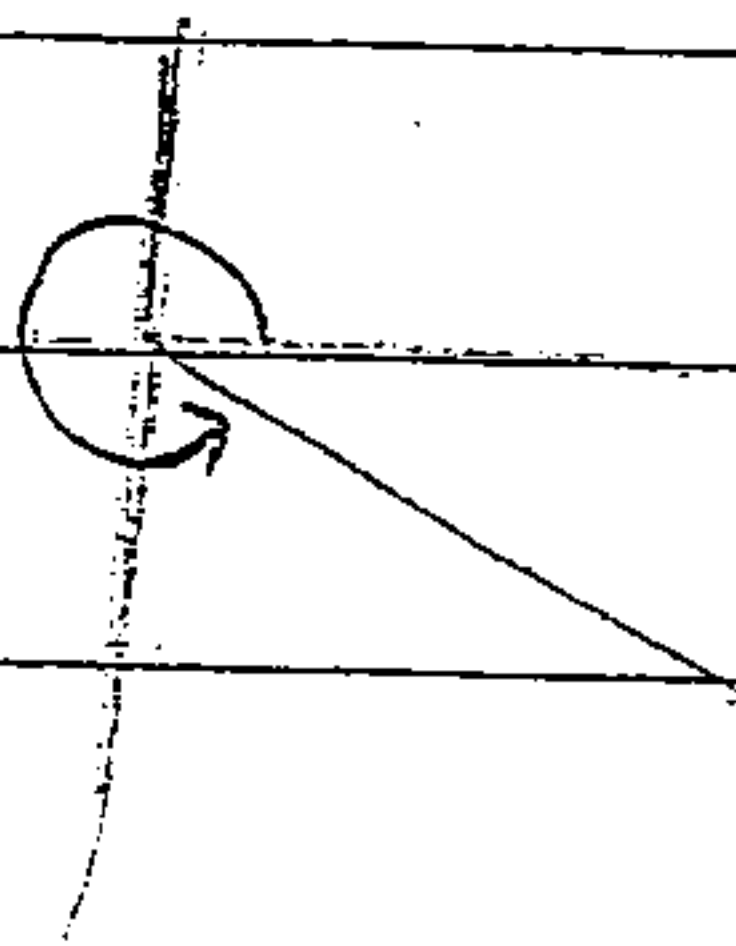
$$\begin{aligned} (a) \vec{p} &= M \vec{v} = 3(3\hat{i} - 4\hat{j}) \\ &= 9\hat{i} - 12\hat{j} \end{aligned}$$

$$P_x = 9 \text{ kg} \cdot \frac{\text{m}}{\text{s}}$$

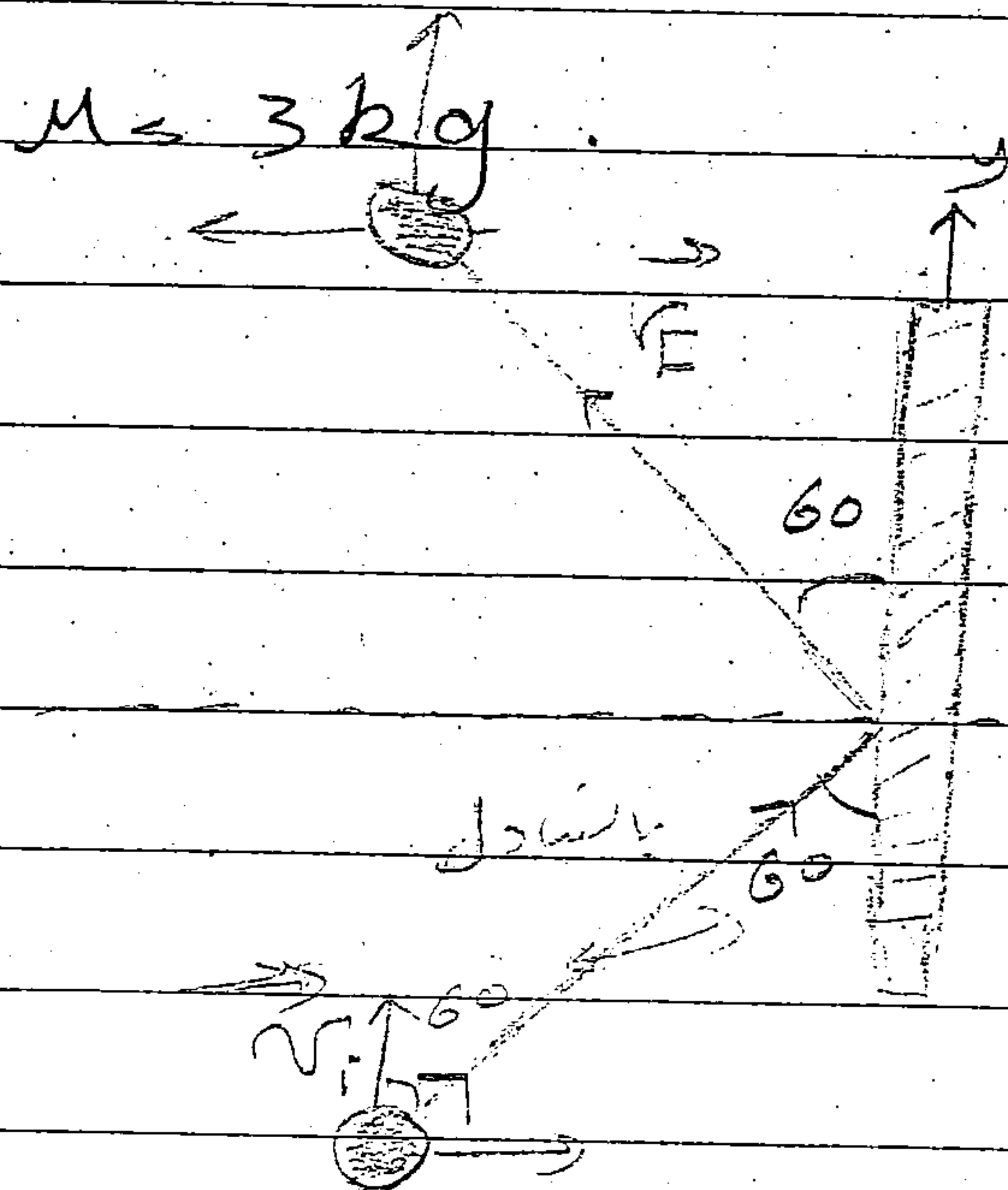
$$P_y = 12 \text{ kg} \cdot \frac{\text{m}}{\text{s}}$$

$$\begin{aligned} (b) |\vec{p}| &= \sqrt{(9)^2 + (12)^2} \\ &= 15 \text{ kg} \cdot \frac{\text{m}}{\text{s}} \end{aligned}$$

$$\theta = \tan^{-1} \frac{12}{9} = 30.7^\circ$$



q-7 1-



$$v_i = 10 \text{ m/s}, \quad v_f = 10 \text{ m/s}$$

$$\Delta t = 0.2 \text{ sec.}$$

what is the average Force exerted by the wall on the ball?

ما هي القوة المتوسطة التي يمارسها الجدار على الكرة؟

Soln:-

$$\vec{I} = \Delta \vec{p} = \vec{F}_{\text{avg}} \Delta t$$

$$F_{\text{avg}} = \frac{I}{\Delta t}$$

$$\begin{aligned} \Delta y = p_{fy} - p_{iy} &= m(v_{fy} - v_{iy}) \\ &= m(v \cos 60 - v \cos 60) \\ &= 0 \end{aligned}$$

$$\Delta P_x = m(v_{fx} - v_{ix})$$

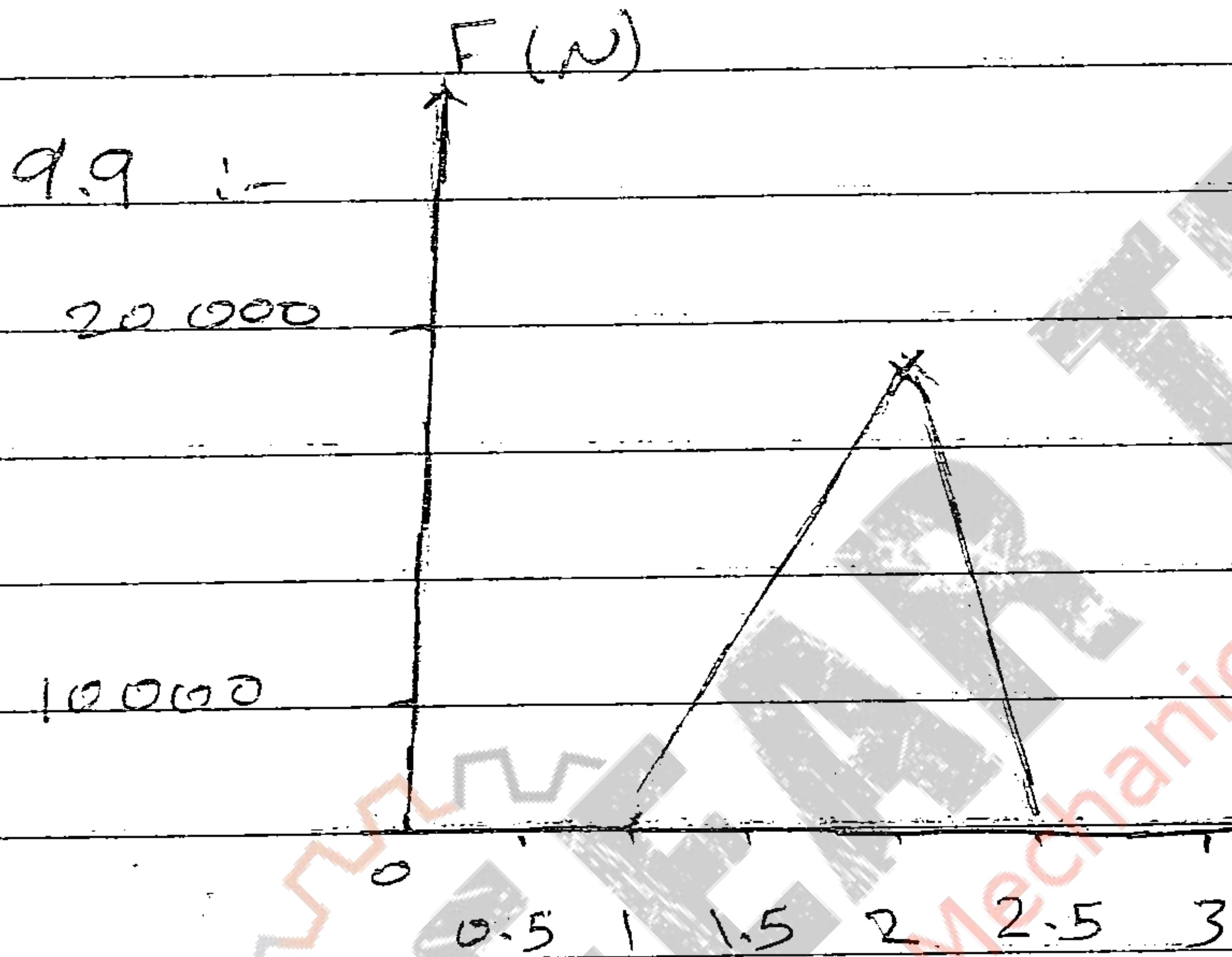
$$= m[-v \sin 60 - v \sin 60]$$

$$= -2m v \sin 60$$

$$= -2(3)(10)(\sin 60) = -52 \text{ kg} \cdot \frac{\text{m}}{\text{s}}$$

$$\therefore F_{\text{avg}} = \frac{\Delta P_x}{\Delta t} = \frac{-52}{0.2} = -260 \text{ N}$$

تجاه طاف في اتجاه



إذا كانت F مع t

طاف في اتجاه

تجاه طاف في اتجاه

إذا كانت F مع t

طاف في اتجاه

تجاه طاف في اتجاه

Find

- the Impulse.
- average force.
- peak force.

نوبه - انتبهوا اي الحمار ووجهه
الحمار

solu:-

مسألة

Ms

$$(a) I = \text{area} = \frac{1}{2} (1.5 \times 10^{-3}) (1800) = 13.5 \text{ N} \cdot \text{sec.}$$

$$(b) F = \frac{I}{\Delta t} = \frac{13.5}{1.5 \times 10^{-3}} = 9 \times 10^3 \text{ N} = 9 \text{ kN}$$

سعة كيلونيوتن

$$(c) F_{\text{max}} = 18000 \text{ N} = 18 \text{ kN}$$

q.14 :- $m_1 = 2.5 \times 10^4 \text{ kg}$, $v_{1i} = 4 \text{ m/s}$

$v_{2i} = 2 \text{ m/s}$

find:-

a) v_F

b) How much energy transferred into internal energy

solution:- perfectly inelastic ، تصادم
لأنه لم يبقا بعد

$$(a) m_1 v_{1i} + 3 m_2 v_{2i} = 4 m_F v_F$$

$$v_F = \frac{v_{1i} + 3 v_{2i}}{4} \Rightarrow \frac{(4) + (3)(2)}{4}$$

$$= 2.5 \text{ m/s}$$

4 / 12 / 08 التاريخ

$$(b) \Delta K = K_f - K_i$$

$$= \frac{1}{2} (4m) v_f^2 - \left[\frac{1}{2} m v_{1i}^2 + \frac{1}{2} (3m) v_{2i}^2 \right]$$

$$= \frac{1}{2} m [4 v_f^2 - v_{1i}^2 - 3 v_{2i}^2]$$

$$= -3.75 \times 10^4 \text{ J}$$

$$= 37.5 \text{ kJ}$$

التاريخ 4 / 12 / 08 Follow

الموضوع:

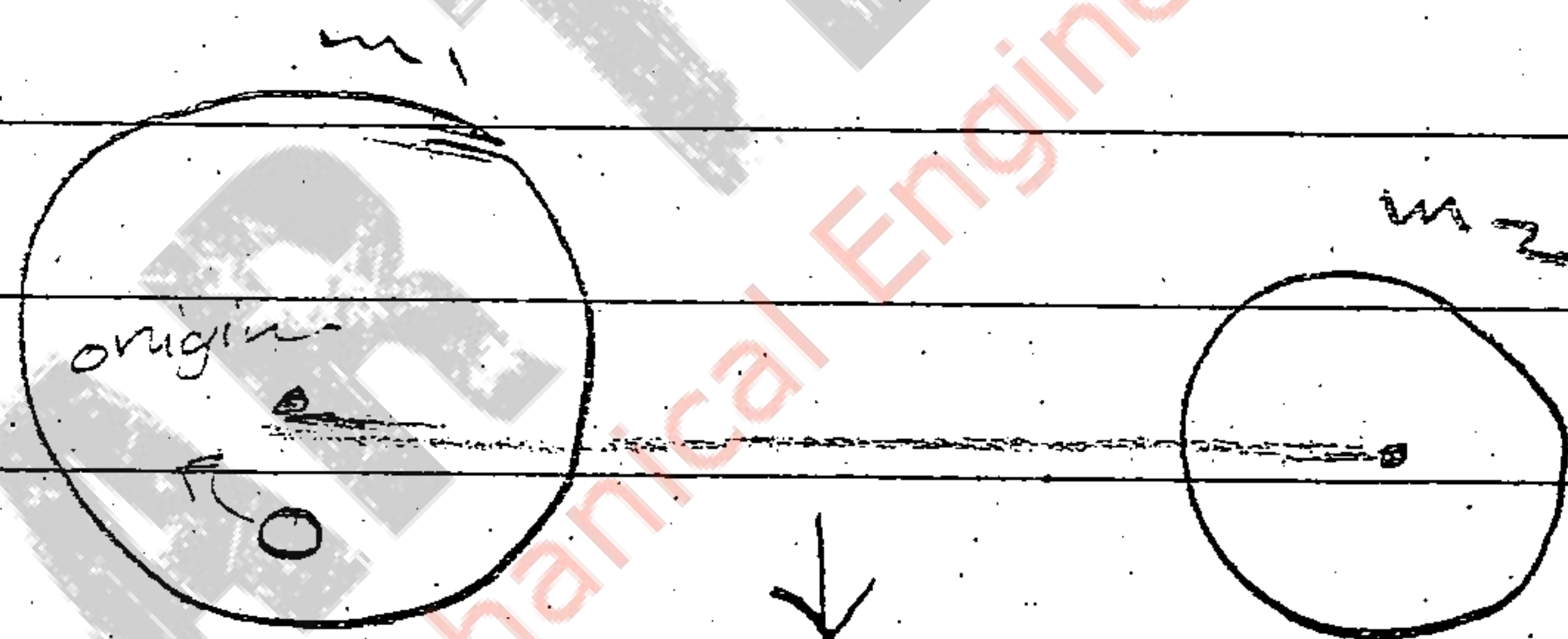
9341- $m_1 = 5.98 \times 10^{24} \text{ kg}$ (earth)

$m_2 = 7.36 \times 10^{22} \text{ kg}$ (moon)

distance between earth and moon center
 $= 3.84 \times 10^8 \text{ m}$.

Find center of mass of the earth-moon system measured from the center of earth

solution:-



$3.84 \times 10^8 \text{ m}$

$$X_{cm} = \frac{1}{M} \sum m_i x_i$$

الأرض ← الأرض ← الأرض

$$= \frac{(5.98 \times 10^{24})(0) + (7.36 \times 10^{22})(3.84 \times 10^8)}{6.05 \times 10^{24}}$$

$X_{cm} = 4.67 \times 10^6 \text{ m}$ from the earth's center.

14/12/08 follow التاريخ

الموضوع:

* Angular Acceleration:-

$$\alpha_{\text{avg}} = \frac{\Delta \omega}{\Delta t}$$

average angular acceleration. (ألفا)

$$\alpha = \lim_{\Delta t \rightarrow 0} \frac{\Delta \omega}{\Delta t} = \frac{d\omega}{dt}$$

instantaneous

angular

acceleration.

$$[\theta] = \underline{\text{rad}} \text{ or } \underline{\text{deg}}$$

$$\theta(\text{rad}) = \frac{\pi}{180} (\text{deg})$$

$$1 \text{ rad} = \frac{360}{2\pi} = 57.3^\circ$$

$$1 \text{ rad} = 57.3^\circ$$

* Average and Ins. velocity :-

$$\omega_{\text{avg}} = \frac{\Delta\theta}{\Delta t} = \frac{\theta_F - \theta_i}{t_F - t_i}$$

average angular
speed (السرعة الزاوية)

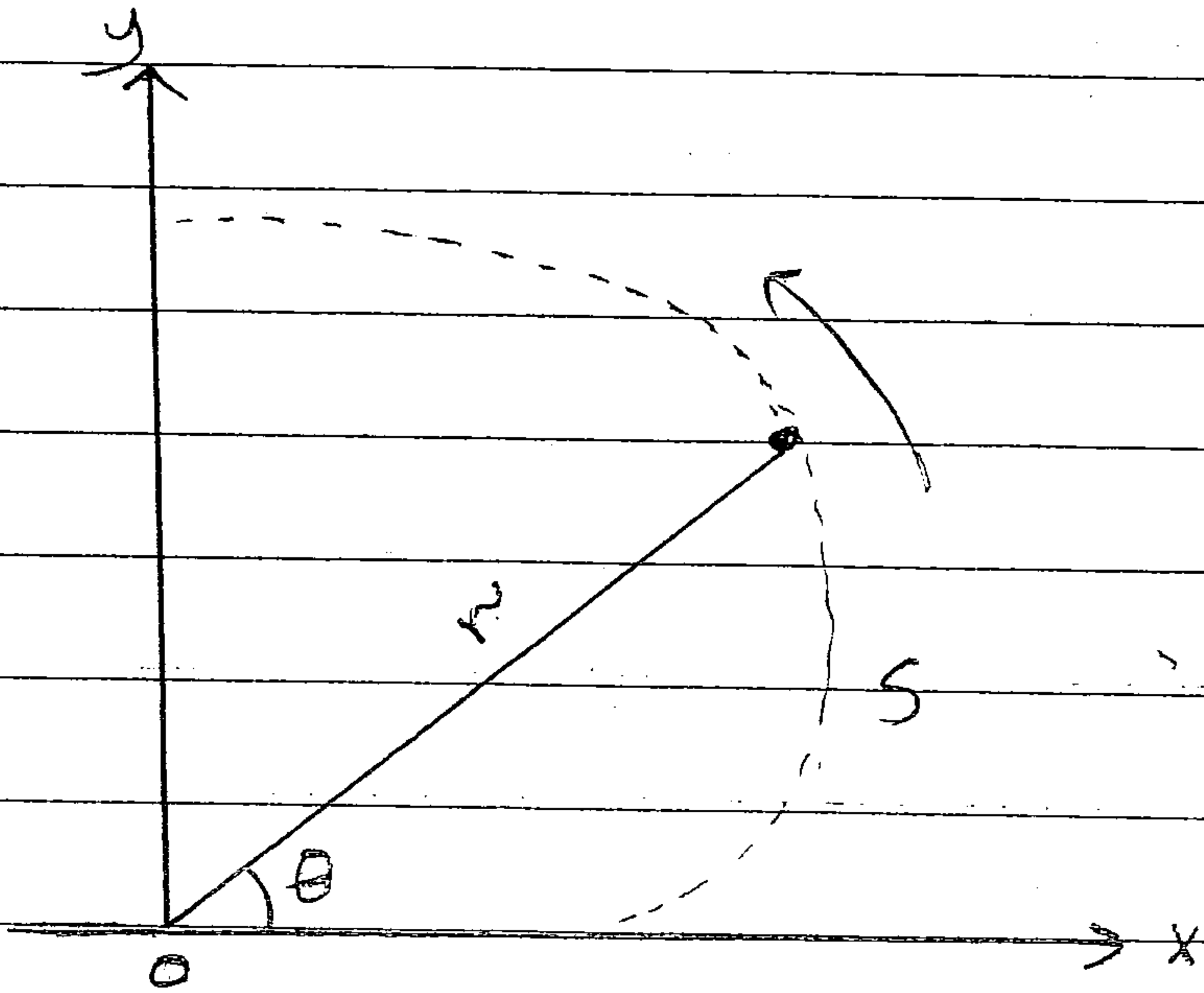
$$\omega_{\text{inst.}} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$$

instantaneous

angular speed

$$[\omega] = \text{rad/s}$$

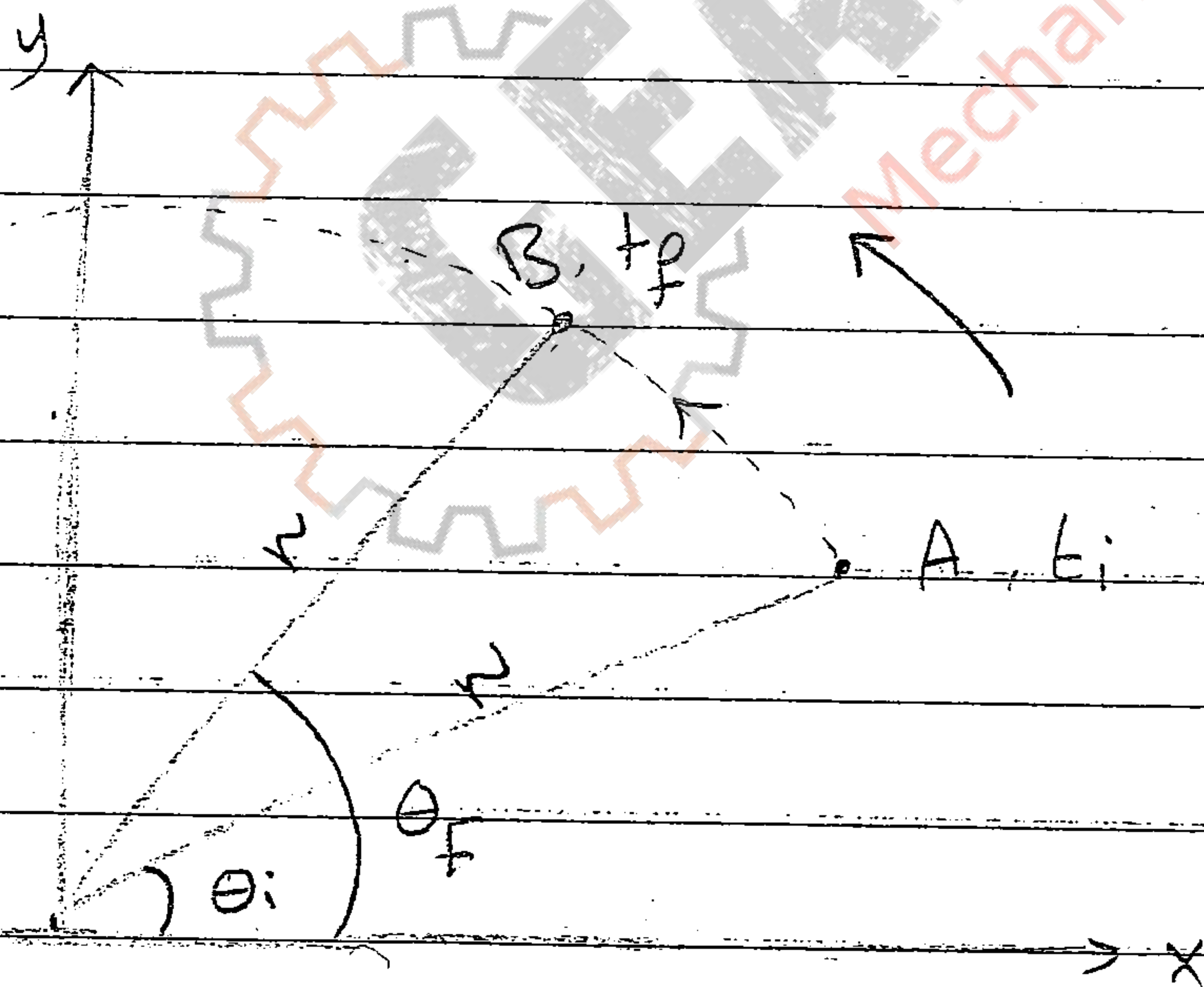
الموضوع: Angular position, velocity, Acceleration
 10.1 التاريخ 14/ 12/ 08



θ :- angular position.

$$s = r\theta$$

↓
translational position



$$\Delta\theta = \theta_f - \theta_i$$

Angular displacement → follow

3/12/08 التاريخ

الموضوع:

Q.35 :- located at y-axis

2 kg located at 3 m.

3 kg " " 2.5 m

2.5 kg " " the origin.

4 kg " " -0.5 m

Find the center of mass :-

Solution :-

$$y_{cm} = \frac{(2)(3) + (3)(2.5) + (2.5)(0) + (4)(-0.5)}{2 + 3 + 2.5 + 4}$$

$$= 1 \text{ m}$$

$$\therefore x_{cm} = 0$$

الموضوع: Rotational kinematics 10.2 التاريخ 12/08/19

Rotational motion with
constant angular acceleration

- For Applied Rotational Kinematics
must have constant α .

من جهة واحدة الحركة الدورانية يجب أن تكون

constant α

Transitional (r) motion Rotational
حركة التناوب motion about a fixed axis.

$$* v_F = v_i + a t$$

$$* \omega_F = \omega_i + \alpha t$$

$$* x_F = x_i + v_i t + \frac{1}{2} a t^2$$

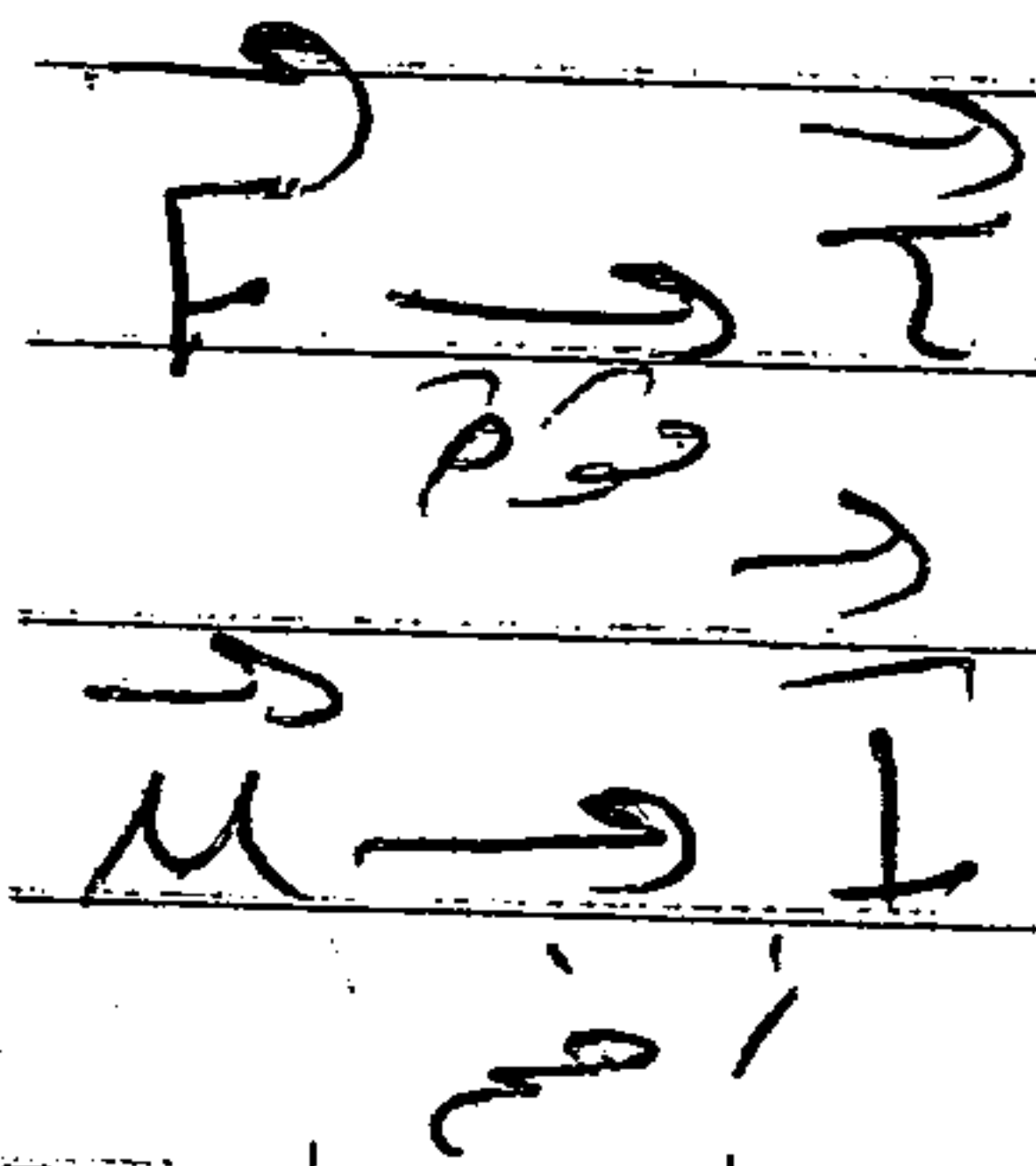
$$* \theta_F = \theta_i + \omega_i t + \frac{1}{2} \alpha t^2$$

$$** v_F^2 = v_i^2 + 2a(x_F - x_i)$$

$$** \omega_F^2 = \omega_i^2 + 2\alpha(\theta_F - \theta_i)$$

$$*** x_F = x_i + \frac{1}{2} (v_i + v_F) t$$

$$*** \theta_F = \theta_i + \frac{1}{2} (\omega_i + \omega_F) t$$



T R
 $x \rightarrow \theta$
 $v \rightarrow \omega$
 $a \rightarrow \alpha$

من جهة واحدة الحركة الدورانية
من جهة واحدة الحركة الدورانية
من جهة واحدة الحركة الدورانية

exp:- A wheel rotates with a constant angular acceleration of $3.5 \frac{\text{rad}}{\text{s}^2}$

(a) Find if $\omega_i = 2 \frac{\text{rad}}{\text{s}}$ at $t_i = 0$
 $\Delta\theta$ at $t = 2 \text{ sec}$

(b) How many revolutions has the wheel turned during this time.

(c) Find the angular speed at $t = 2 \text{ sec}$

Solution:- $\alpha = 3.5 \frac{\text{rad}}{\text{s}^2}$, $\omega_i = 2 \frac{\text{rad}}{\text{s}}$

$$\begin{aligned} (a) \Delta\theta &= \theta_f - \theta_i = \omega_i t + \frac{1}{2} \alpha t^2 \\ &= (2)(2) + \frac{1}{2} (3.5)(2)^2 \\ \Delta\theta &= 11 \text{ rad} = 11 \times 57.3^\circ \\ &= 630^\circ \end{aligned}$$

$$(b) \Delta\theta = 630^\circ \times \left(\frac{\text{revolution}}{360^\circ} \right) = 1.75 \text{ revolutions}$$

$$\begin{aligned} (c) \omega_f &= \omega_i + \alpha t \\ &= 2 + (3.5)2 \\ &= 9 \text{ rad/s} \end{aligned}$$

الموضوع: "linear" Angular and transitional 10-3 التاريخ 14/12/08

Quantities

* when a rigid object rotates about a fixed axis.

$$s = r\theta$$

Transitional position radius of circle angular position

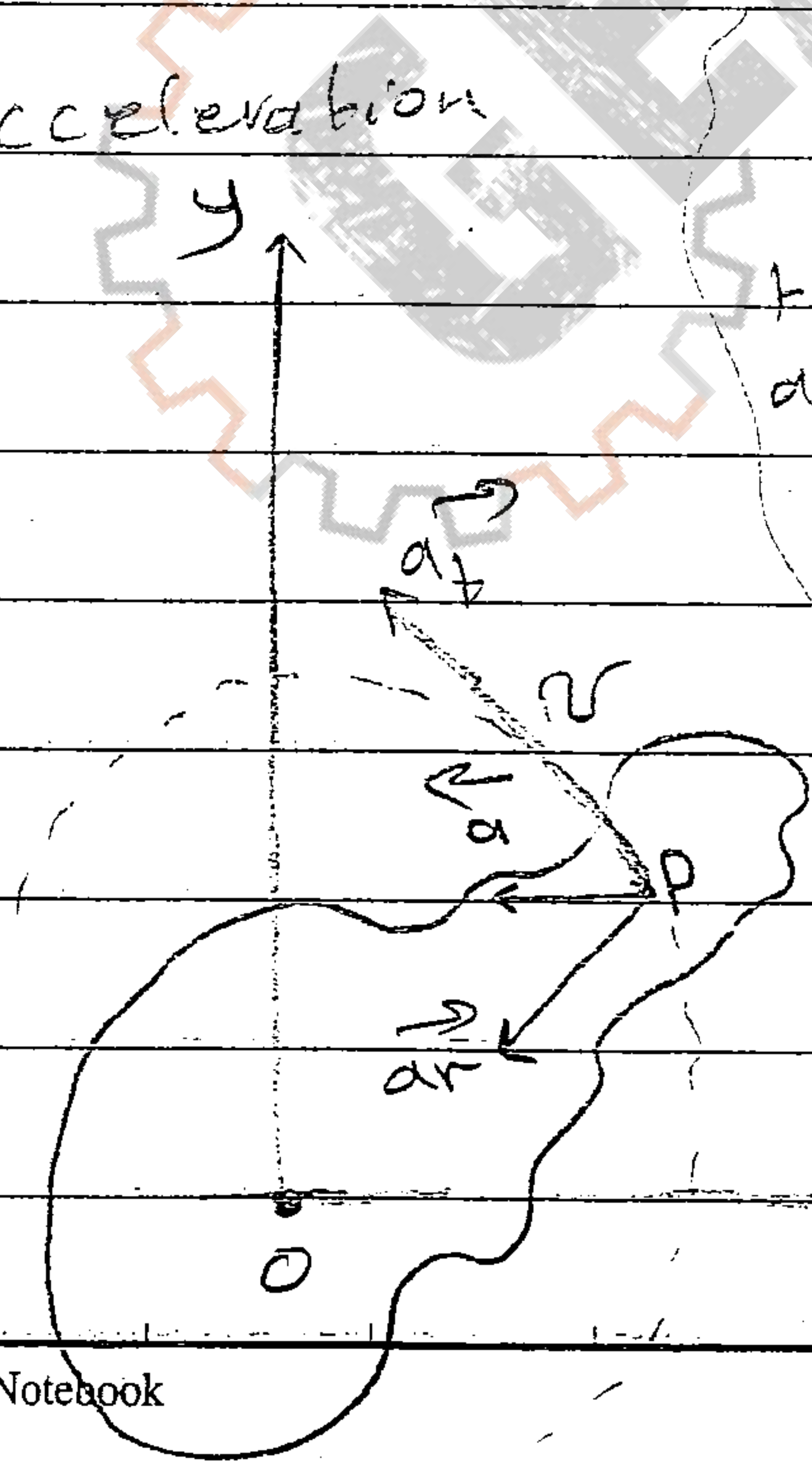
$$v = r\omega$$

Transitional speed angular speed

$$a = r\alpha$$

Transitional acceleration angular acceleration

Transitional acceleration



$$\vec{a} = \vec{a}_t + \vec{a}_r$$

total acceleration tangential acceleration radial acceleration

$$a_c = a_r = \frac{v^2}{r} = \frac{(r\omega)^2}{r} = r\omega^2$$

$$a_t = r\alpha$$

$$|\vec{a}| = \sqrt{r^2\alpha^2 + r^2\omega^4}$$

x

→ follow

Follow التاريخ 12/08/14

الموضوع:

$$\therefore |\bar{a}| = \omega \sqrt{a^2 + \omega^4}$$

↓
magnitude
of total
acceleration

الموضوع: Rotational Kinetic Energy 10.4 التاريخ 14/12/08

energy

$$I = \sum_i m_i r_i^2$$

moment

of inertia

«ممانعة الجسم للحركة»
العزم الذاتي

المسافة

من المحور

$$[I] = \text{kg} \cdot \text{m}^2$$

$$K_R = \frac{1}{2} I \omega^2$$

Rotational

moment

velocity.

Kinetic energy of inertia

exp: - Four spheres

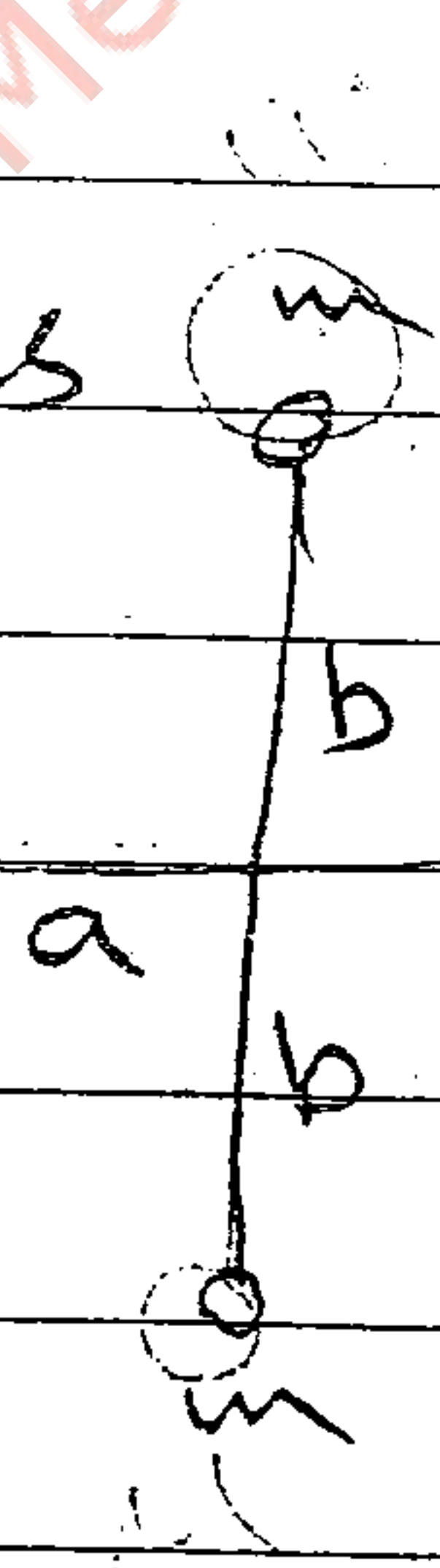
to the end

of the

rods

of

the negligible



are fastened

of the two

of

mass.

(a) if the system rotates about the y-axis with ω , find the I , K_R about y-axis.

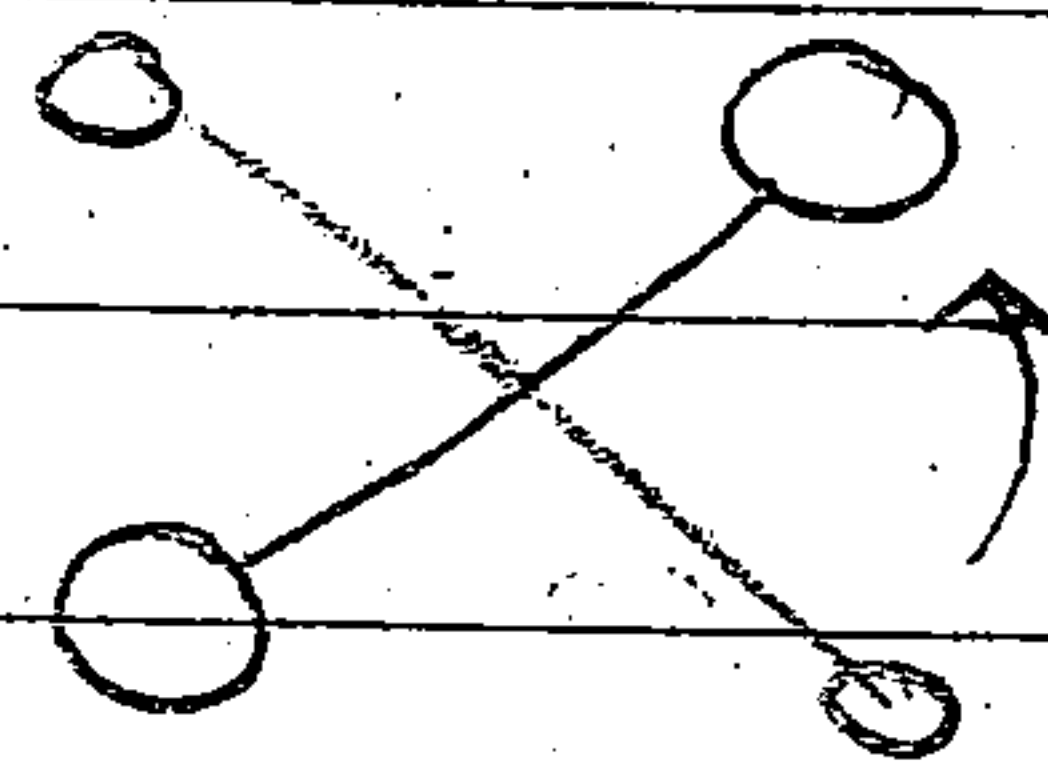
(b) if the system rotates in x-y plane about z-axis, find I , K_R about z-axis.

solution:-

$$\begin{aligned} (a) I_y &= \sum m_i r_i^2 \\ &= Ma^2 + Ma^2 + m(0)^2 + m(0)^2 \\ &= 2Ma^2 \end{aligned}$$

$$\therefore K_R = \frac{1}{2} I_y \omega^2 = Ma^2 \omega^2$$

(b)



$$\begin{aligned} I_z &= \sum m_i r_i^2 \\ &= Ma^2 + Ma^2 + mb^2 + mb^2 \\ I_z &= 2Ma^2 + 2mb^2 \end{aligned}$$

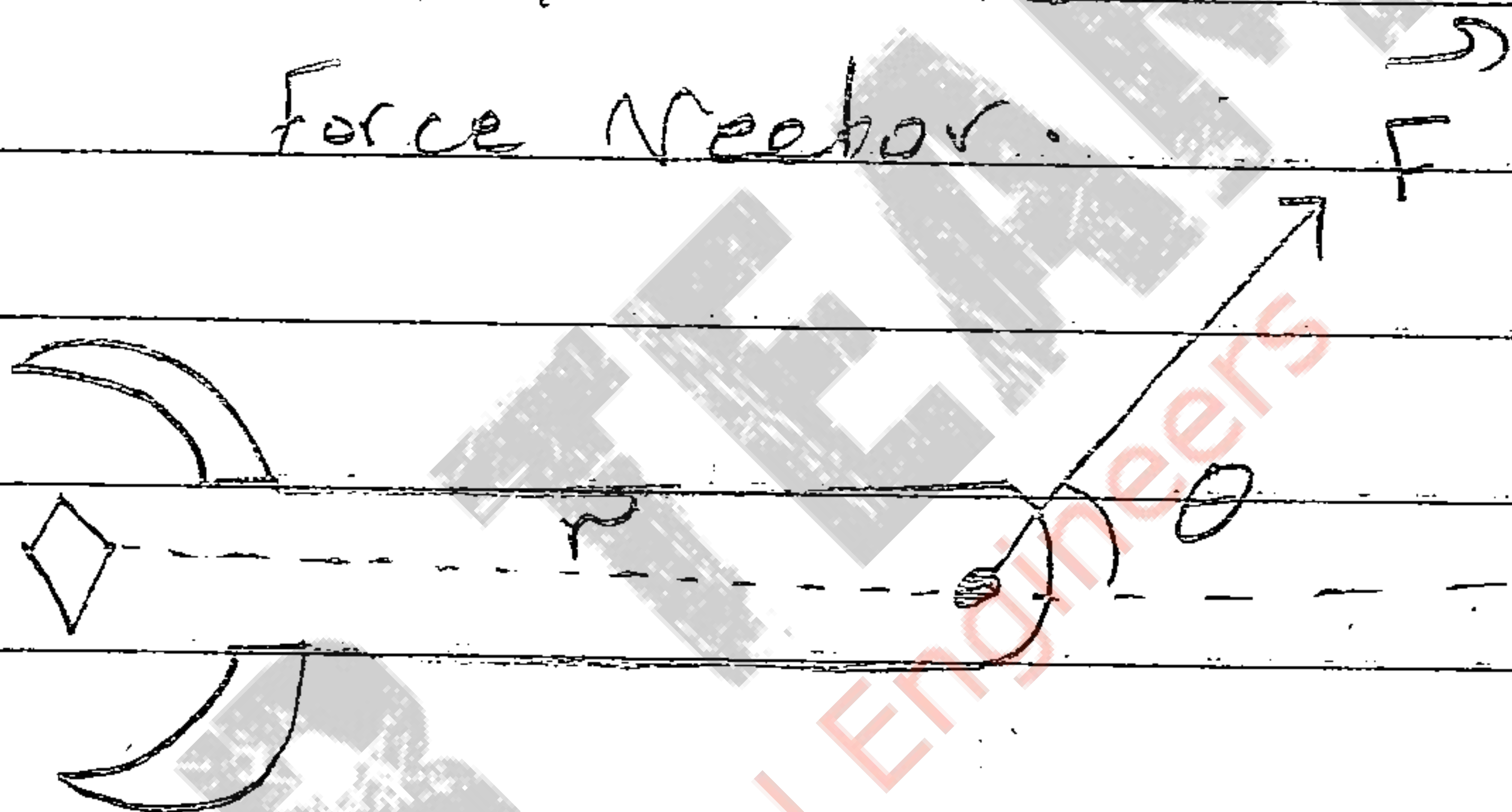
$$\therefore K_R = \frac{1}{2} I_z \omega^2 = (Ma^2 + mb^2) \omega^2$$

الموضوع: عزم الدوران Torque 10.6 التاريخ 14/12/08

$$\vec{\tau} = \vec{r} \times \vec{F} = |\vec{r}| |\vec{F}| \sin \theta$$

Torque associated with the force

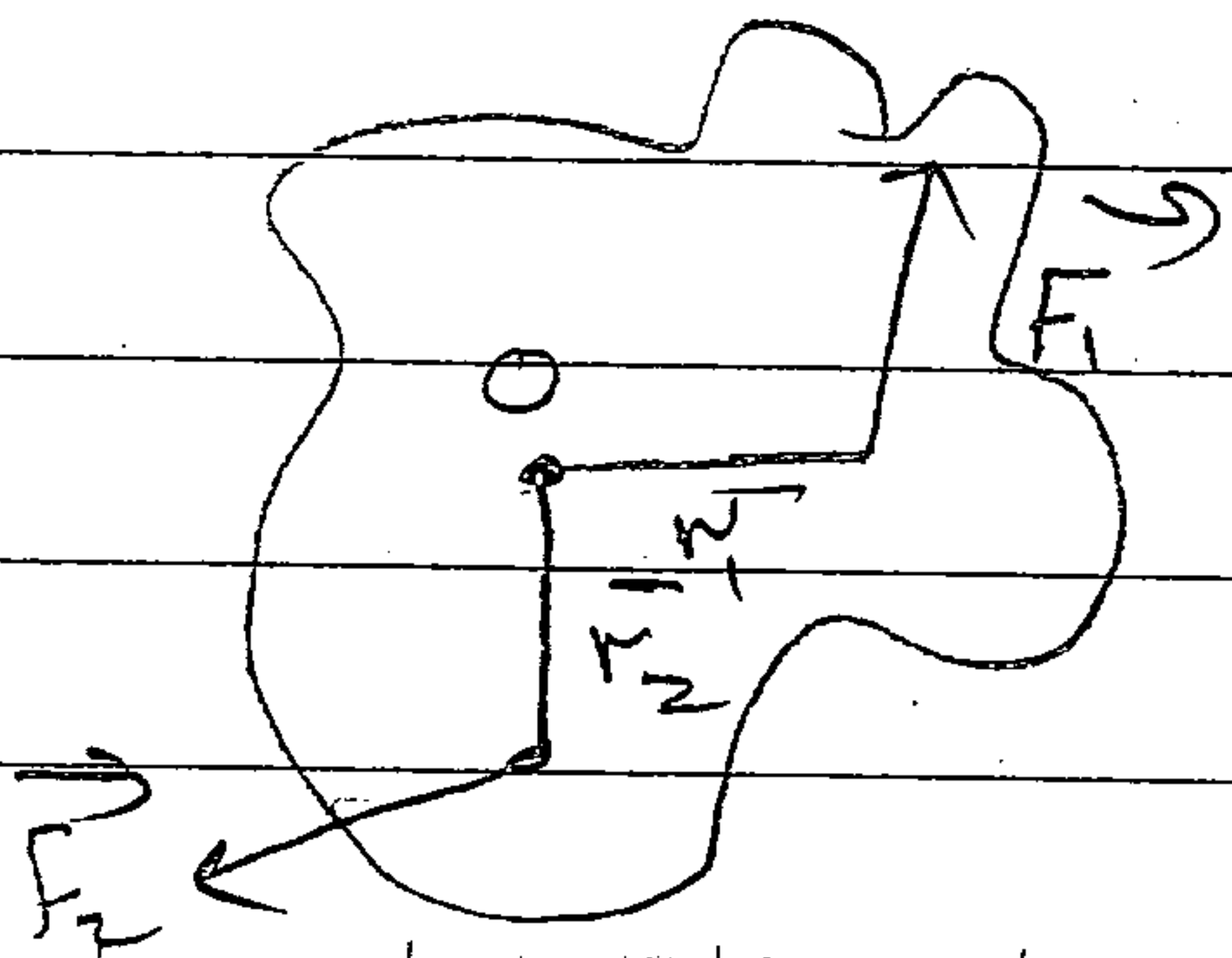
angle between the position vector of the point of application of force and the force vector.



* $\vec{\tau}$ is negative if turning tendency is clockwise
مع اتجاه عقارب الساعة

$\vec{\tau}$ is positive if turning tendency is counter clockwise.

عكس اتجاه عقارب الساعة



$$\sum \vec{T} = \vec{T}_1 + \vec{T}_2$$

$$\sum T = T_1 - T_2$$

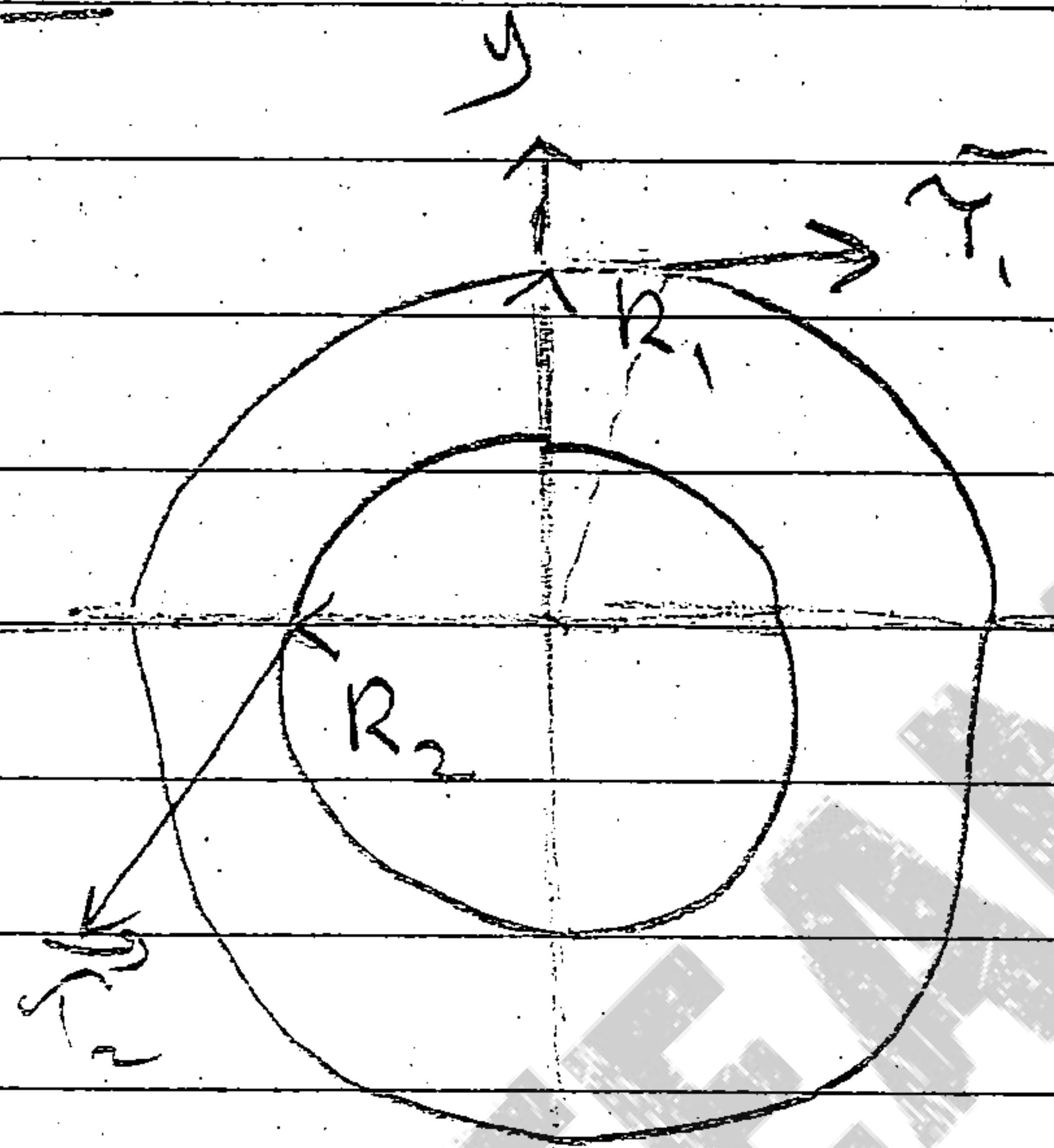
لتحقيق

→ follow the sign convention

exp 10-7 :- ^{المهم}

Cylinder.

- Find the net
acting on
cylinder
the rotation



torque

the
about

(z-axis)

Solution :-

$$R_1 = 1m, R_2 = 0.5m.$$

$$T_1 = 5N \text{ Force, } T_2 = 15N.$$

$$\sum \bar{\tau} = \bar{\tau}_1 + \bar{\tau}_2$$

$$= R_1 T_1 + R_2 T_2$$

$$= R_2 T_2 - R_1 T_1 \rightarrow \text{لا تضع علامة}$$

$$= (0.5)(15) - (1)(5)$$

$$= 2.5 Nm$$

∴ counter clockwise

الموضوع: 14/12/2018 التاريخ 10-7 the Rigid object

under a Net Torque

$$\sum F_t = \sum F_r$$

$$\tau = (\sum F_r) r$$

$$= (m a) r$$

للإسقاط دالة

$$= m(r \alpha) r$$

$$= m r^2 \alpha$$

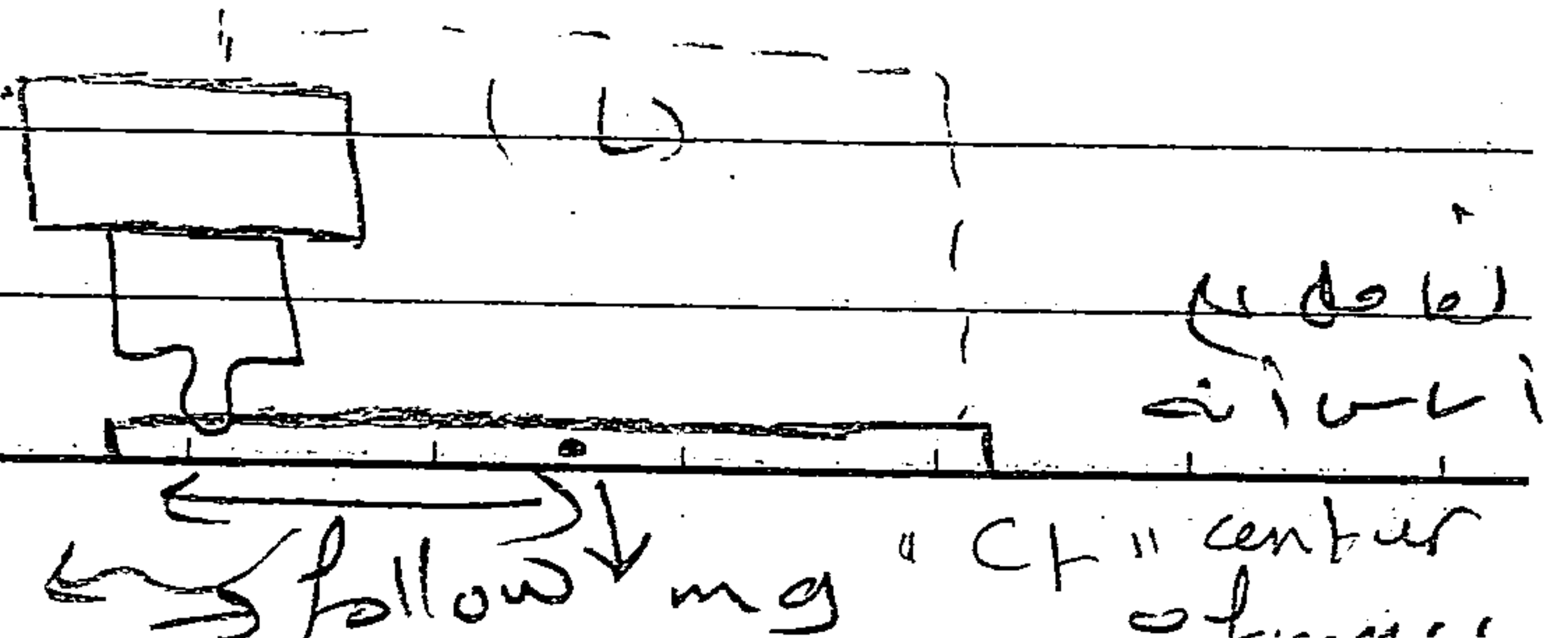
$$= I \alpha$$

$$\boxed{\sum \vec{\tau} = I \vec{\alpha}}$$

Newton second law.

radial Force $\sum F_r = 0$

exp 10.8 :- rod of length (L) and mass M, free rotate in the vertical plane. The rod is released from rest in the horizontal position. what is the initial angular acceleration of the rod and the initial transitional acceleration of its right end.



$$r = \frac{L}{2}$$

التاريخ 16/12/08 Follow

الموضوع:

Solution:-

$$\tau = Fr$$
$$(mg) \left(\frac{L}{2} \right)$$

$$\tau = I \alpha \rightarrow \alpha \text{ مطلوب}$$

$$\alpha = \frac{\tau}{I} = \frac{mg \frac{L}{2}}{\frac{1}{3} ML^2} = \frac{3g}{2L}$$

$$a_t = L \alpha = L \frac{3g}{2L} = \frac{3g}{2}$$

حل 10.10 exp 10.10

الموضوع: energy considration 10.8 التاريخ 21/12/08

in Rotational motion.

* The rate at which work is done by an external force in rotating rigid object about a fixed axis, or the power delivered is :-

$$P = \tau \omega$$

torque angular speed

$$dW = (\sum \tau) d\theta$$

work

$$\sum W = K_{RF} = K_{Ri}$$

$$= \frac{1}{2} I \omega_f^2 - \frac{1}{2} I \omega_i^2$$

work

angular speed.

work - kinetic energy theorem for rotational motion.

تتبع التاريخ 21/12/08

$$\vec{p} = m \vec{v}$$

linear momentum

$$\Rightarrow L = I \omega$$

angular momentum

notes:- Linear $\rightarrow$ angular

$$x \rightarrow \theta$$

$$m \rightarrow I$$

$$v \rightarrow \omega$$

$$F \rightarrow \tau$$

$$a \rightarrow \alpha$$

$$p \rightarrow L$$

قوانين حفظ الطاقة نفسها
قوانين الحركة نفسها

الكم

الكم

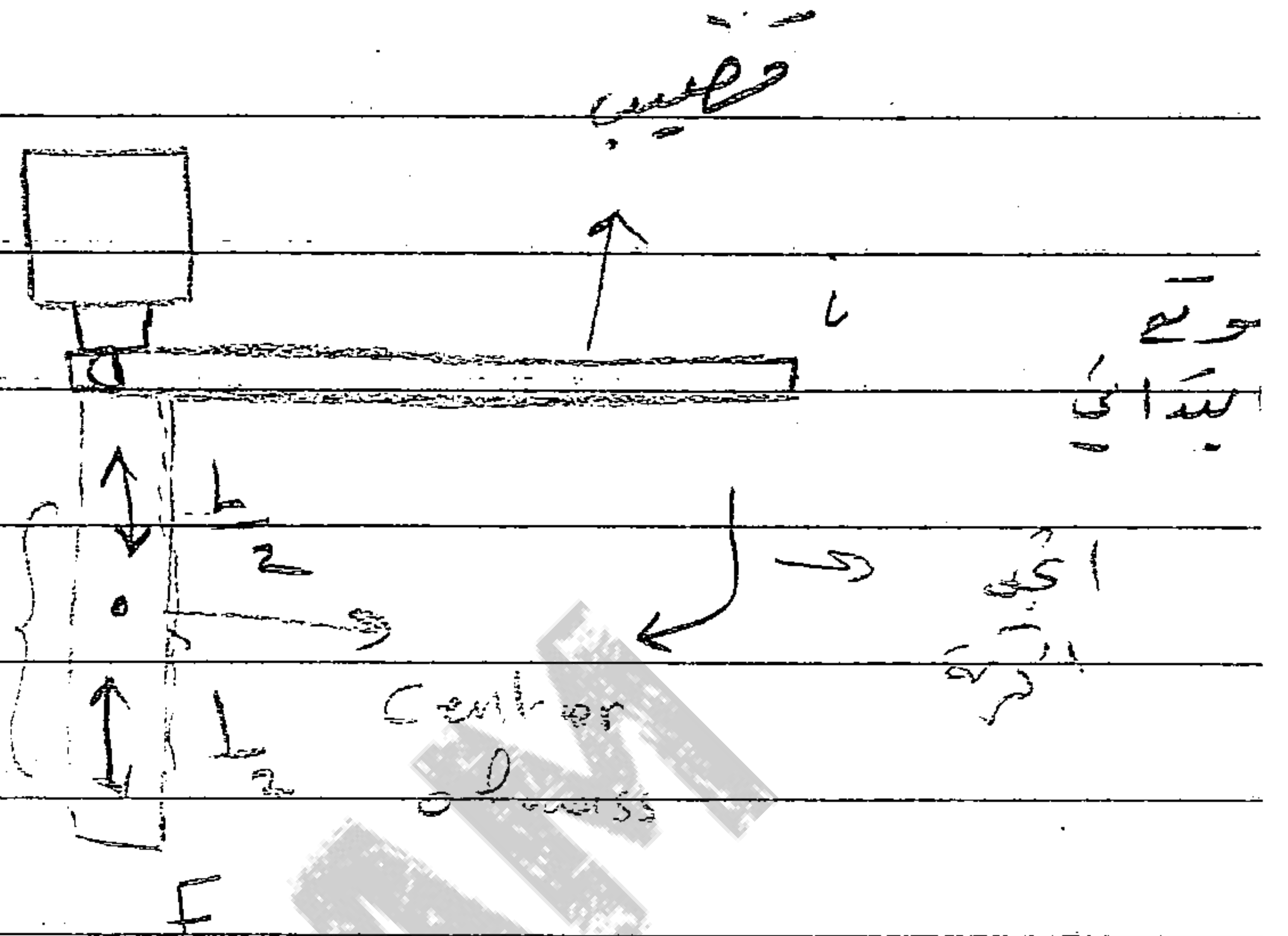
الكم

تتبع التاريخ 21/12/08

الموضوع:

exp 10.11 :-

rod قضيب
 $L, m \rightarrow$ وزنه
 $L \rightarrow$ طوله



(a) angular speed when
 the rod reaches its
 lowest position

(b) Tangential speed at center of mass
 and lowest point

solution:-

$U = 0$ in the vertical position.

نقطة انعطاف

(a)

الوضع العمودي

$$E_i = U_i + K_i$$

$$= mg \frac{L}{2} + 0$$

$$E_f = 0 + \frac{1}{2} I \omega^2$$

$$E_f = E_i \Rightarrow \frac{1}{2} I \omega^2 = mg \frac{L}{2}$$

$$\omega = \sqrt{\frac{mgL}{I}}$$

$$\omega = \sqrt{\frac{3g}{L}}$$

21/ 12/08 التاريخ

الموضوع:

(b) $v = r\omega$

$$v_{cm} = \frac{L}{2} \sqrt{\frac{3g}{L}}$$

$$= \frac{1}{2} \sqrt{3gL}$$

$$v = r\omega$$

$$= L\omega$$

$$= \sqrt{3gL}$$

10.1 :-

$$\theta = 5 + 10t + 2t^2$$

$\Rightarrow$ Angular position "rad"

find θ, ω, α at $t = 3 \text{ sec}$.

solution:-

$$\theta \Rightarrow \theta(t=3) = 5 + 30 + 18 = 53 \text{ rad.}$$

$$\omega = \frac{d\theta}{dt} = 10 + 4t$$

$$\Rightarrow \omega(t=3) = 10 + 12 = 22 \text{ rad/s.}$$

$$\alpha = \frac{d\omega}{dt} = 4 \Rightarrow \alpha(t=3) = 4 \text{ rad/s}^2.$$

23/12/108 follow التاريخ

الموضوع:

10.13 :-

$$r = 1 \text{ m}, \quad \alpha = 4 \text{ rad/s}^2 \text{ (constant)}$$

$$\omega_i = 0, \quad \theta_i = 57.3^\circ = \underline{\underline{1 \text{ rad}}}$$

at $t = 2 \text{ sec}$

find (a) ω (b) $v, |\vec{a}|$ total
(c) θ_F $\rightarrow$ tangential speed.

solution :-

$$(a) \omega_F = \omega_i + \alpha t$$

$$= 0 + 4 \times 2$$

$$= 8 \text{ rad/sec}$$

$$(c) \theta_F = \theta_i + \omega_i t + \frac{1}{2} \alpha t^2$$

$$= 1 + 0 + \frac{1}{2} (4) (2)$$

$$= \underline{\underline{9 \text{ rad}}}$$

$$(b) v = r \omega$$

$$= 1 \times 8 = 8 \text{ m/s}$$

$$|\vec{a}| = r \sqrt{\alpha^2 + \omega^2}$$

$\rightarrow$

مقدار التسارع

$$= 1 \sqrt{(4)^2 + (8)^2} = 64.1 \text{ m/s}^2$$

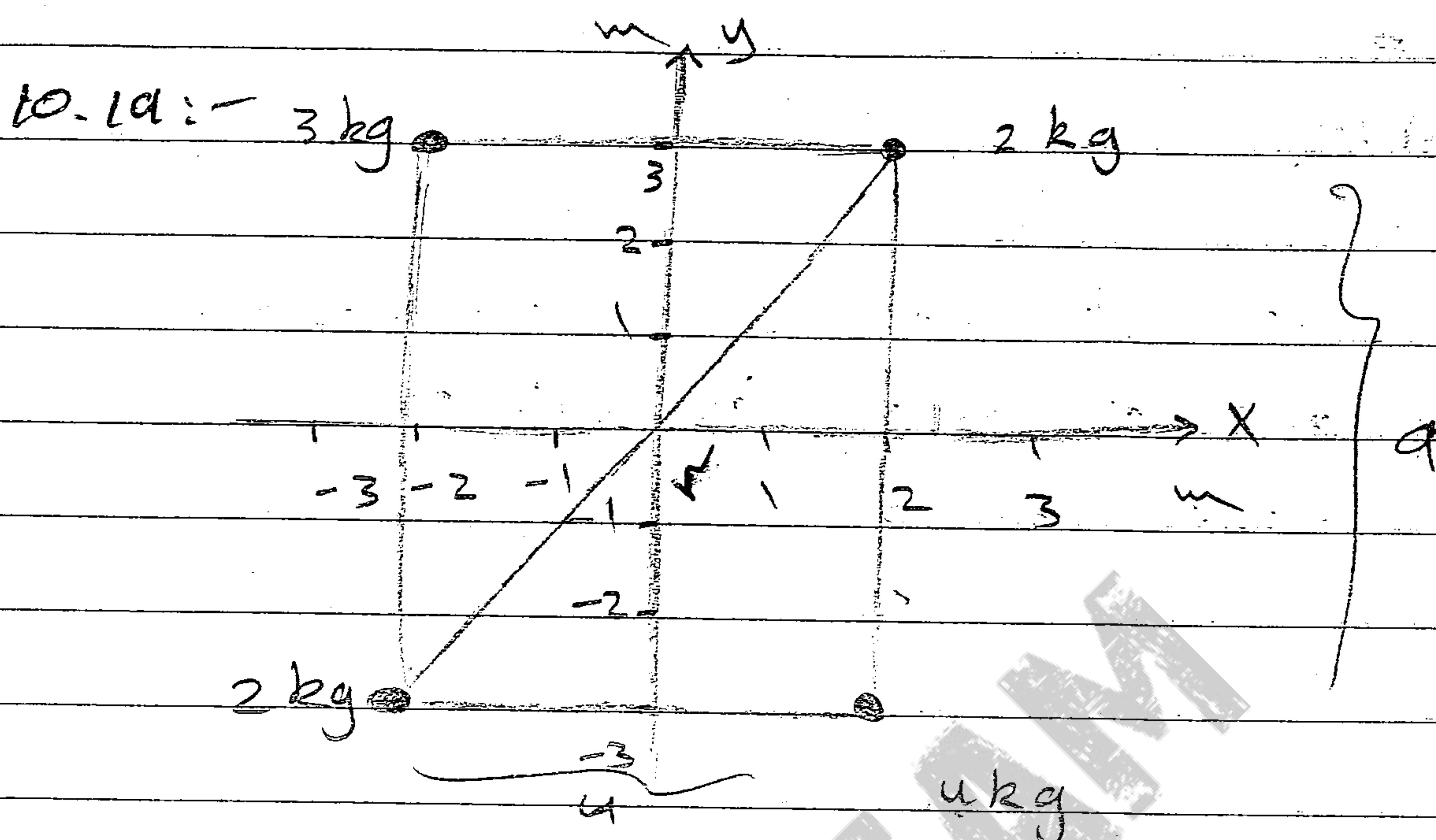
$$\phi = \tan^{-1} \frac{at}{ar}$$

$\rightarrow$ follow

(20)

23/12/08 التاريخ follow

الموضوع:



rotation about Z-axis.

$$\omega = 6 \text{ rad/s}$$

Find I_z , K_R .

Solution:-

$$r = \sqrt{4+4} = \sqrt{8}$$

$$\begin{aligned} I_z &= \sum m_i r_i^2 = 2r^2 + 4r^2 + 3r^2 + 2r^2 \\ &= 11r^2 \\ &= 11 \times 8 = 88 \text{ kg.m}^2 \end{aligned}$$

$$K_R = \frac{1}{2} I_z \omega^2$$

$$= \frac{1}{2} (88) (6)^2 = 1.584 \times 10^3 \text{ J}$$

23/ 12/08 Follow التاريخ

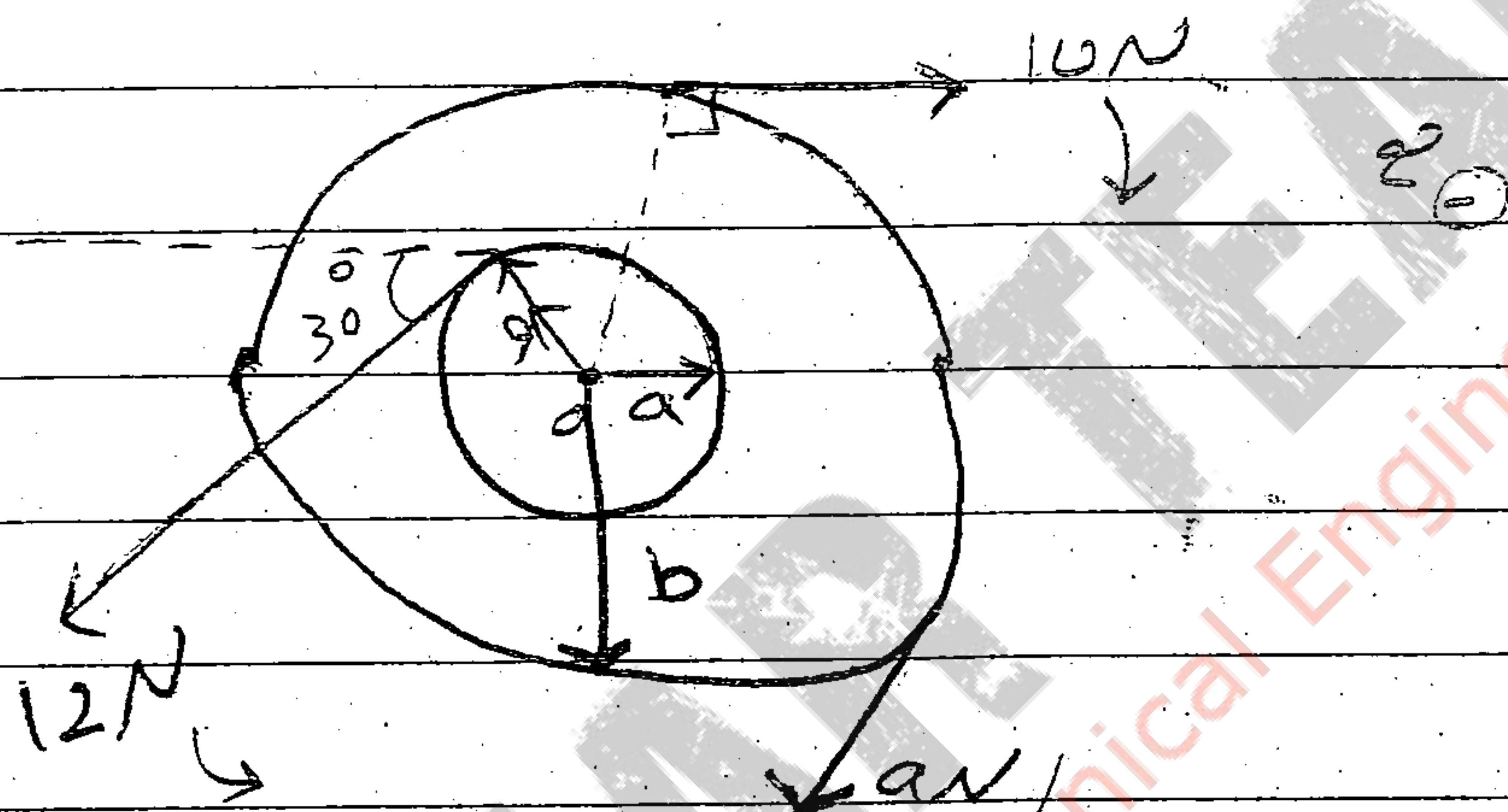
الموضوع:

10.2a1-

Find the net torque on the wheel about the axle through O.

taking $a = 10 \text{ cm}$

$b = 25 \text{ cm}$.



solution:-

$$\sum \tau = -(10)(b) - (a)(b) + (12)(a) \sin 90^\circ$$

$$= -10(0.25) - (9)(0.25) + (12)(0.1) \neq 1$$

$$= -3.55 \text{ N.m}$$

لے ای ۱۰۰۰ م
کتاب

الباقی

23/12/08 التاريخ Follow

الموضوع:

10-30 1-

$$r = 7 \text{ cm}$$

$$T = 0.6 \text{ m.m}$$

$$m = 2 \text{ kg}$$

$$\omega_F = 1200 \text{ rev/min.}$$

$$\omega_i = 0$$

Find the following

(a) t (b) θ "how many revolutions does it turn while accelerating?"
 كم زففة زففة
 زففة زففة

Solution:-

$$r = 0.07 \text{ m}$$

(a) $\sum \tau = I \alpha$ moment of inertia.

$$\therefore \alpha = \frac{\sum \tau}{I}$$

$$\therefore I = \frac{1}{2} m R^2 = \frac{1}{2} (2) (0.07)^2 = 4.9 \times 10^{-3} \text{ kg.m}^2$$

$$\therefore \alpha = \frac{0.6}{4.9 \times 10^{-3}} = 122 \text{ rad/s}^2$$

$$\therefore \alpha = \frac{\Delta \omega}{\Delta t} \Rightarrow \Delta t = \frac{\Delta \omega}{\alpha} = \frac{126}{122} = 1.03 \text{ sec.}$$

$$(b) \Delta \theta = \omega_i t + \frac{1}{2} \alpha t^2$$

$$= \frac{1}{2} (122) (1.03)^2 = 64.7 \text{ rad}$$

$$= 64.7 \left(\frac{57.3^\circ}{360} \right)$$

$$= 10.3 \text{ rev.}$$

CHAPTER 11

Angular momentum

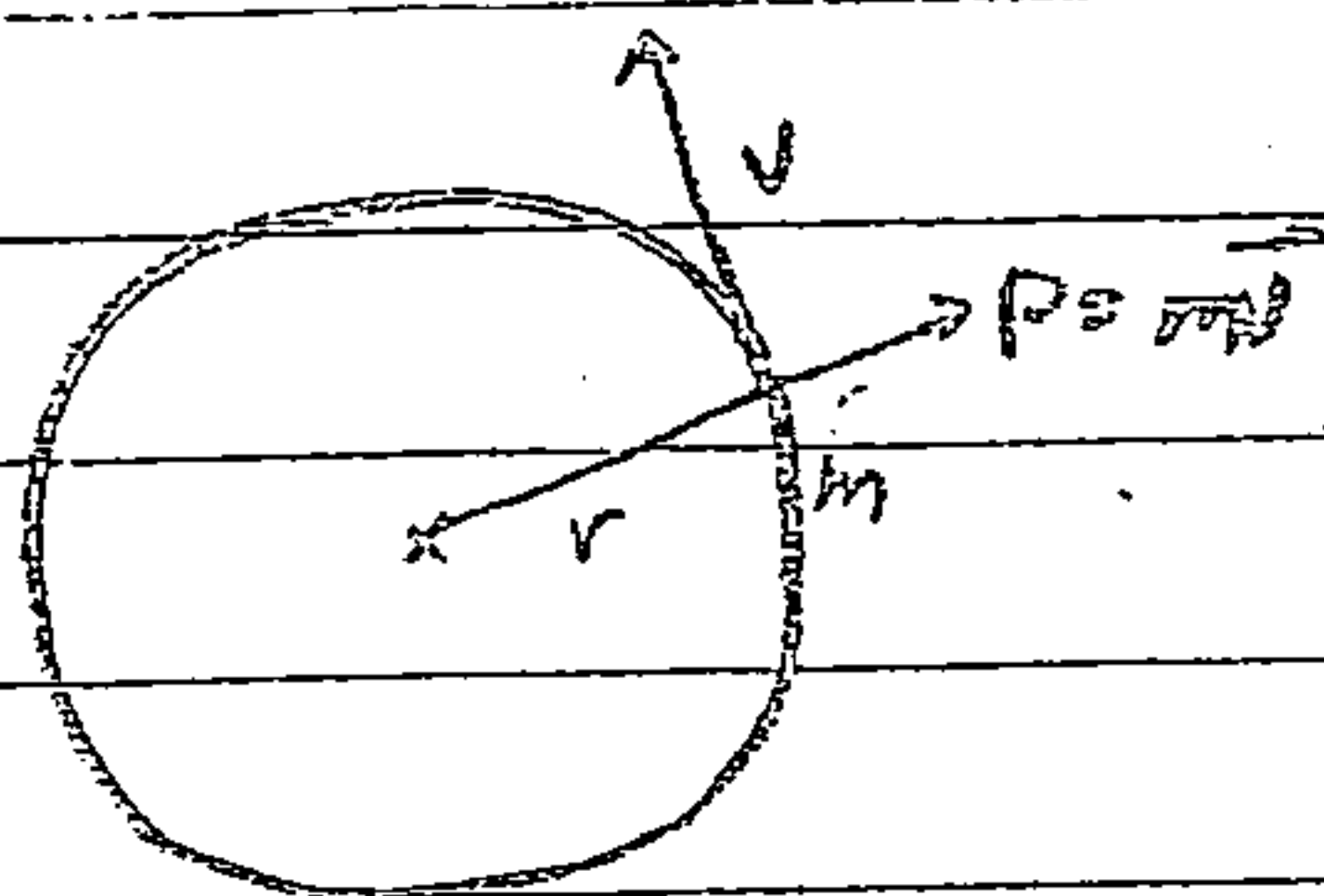
$$\vec{L} = \vec{r} \times \vec{p}$$

$$= r m v \sin \theta$$

$$L = m v r \sin \theta$$

θ (r, p)

unit $\text{Kg} \cdot \text{m}^2/\text{s}$

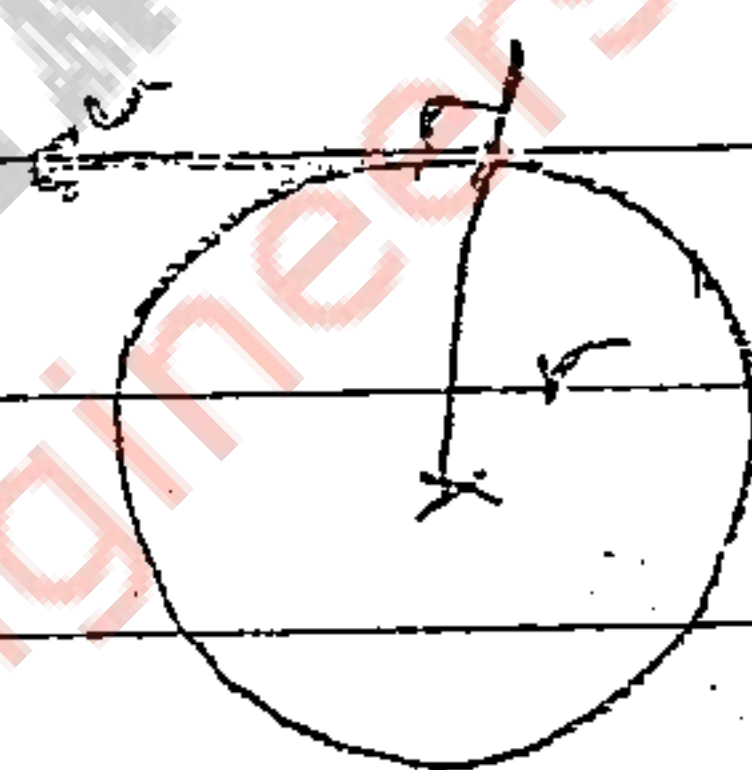


Ex: uniform circular motion of a particle with mass = 4 kg
calculate L relative to the center if $v = 4 \text{ m/s}$, $r = 2 \text{ m}$

$$L = m v r \sin \theta$$

$$= 4(4)(2) \sin 90$$

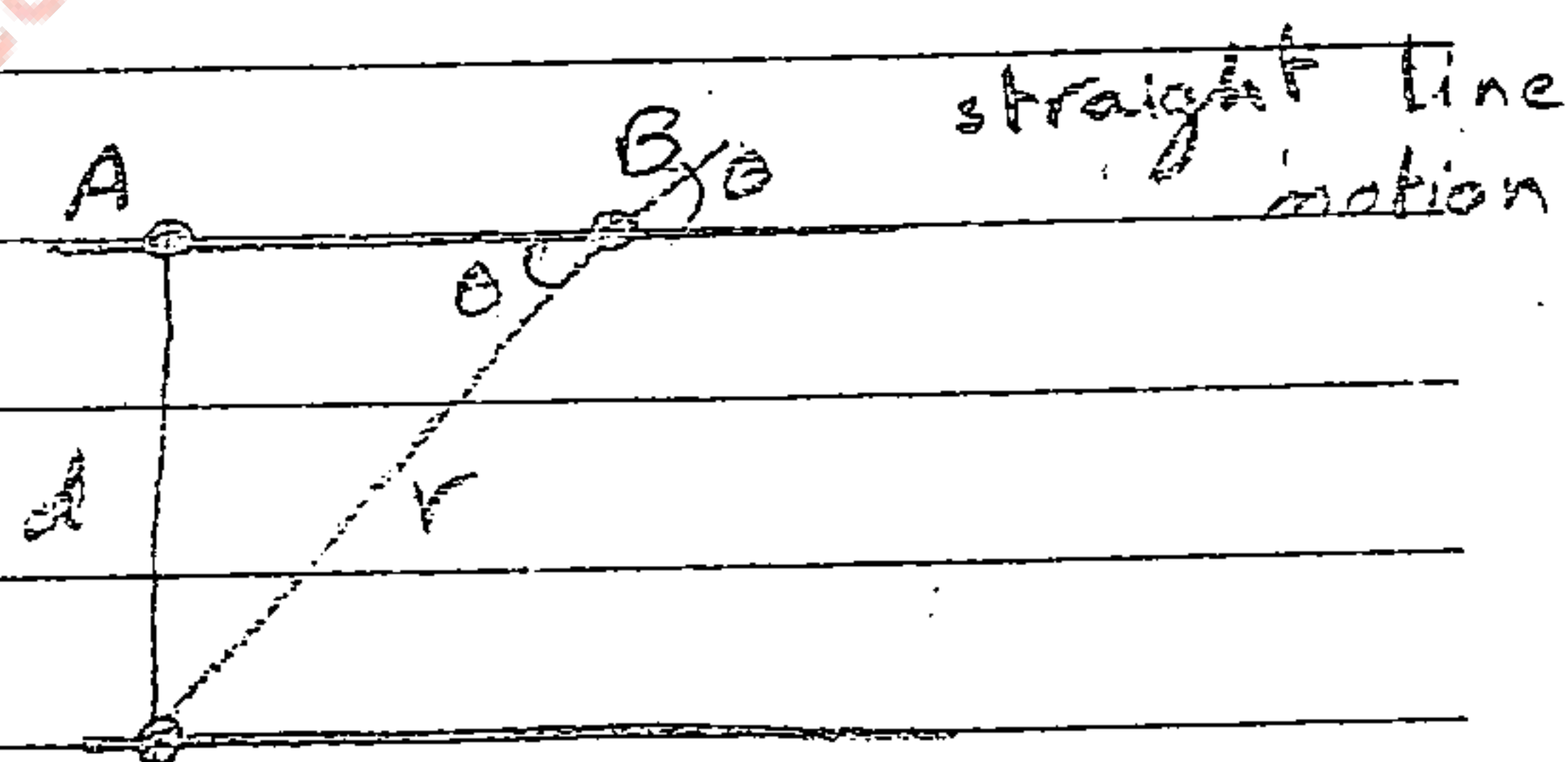
$$= 32 \text{ Kg} \cdot \text{m}^2/\text{s}$$



ex: $L_A = m v r \sin \theta$

$$= m v d \sin 90$$

$$= m v d$$

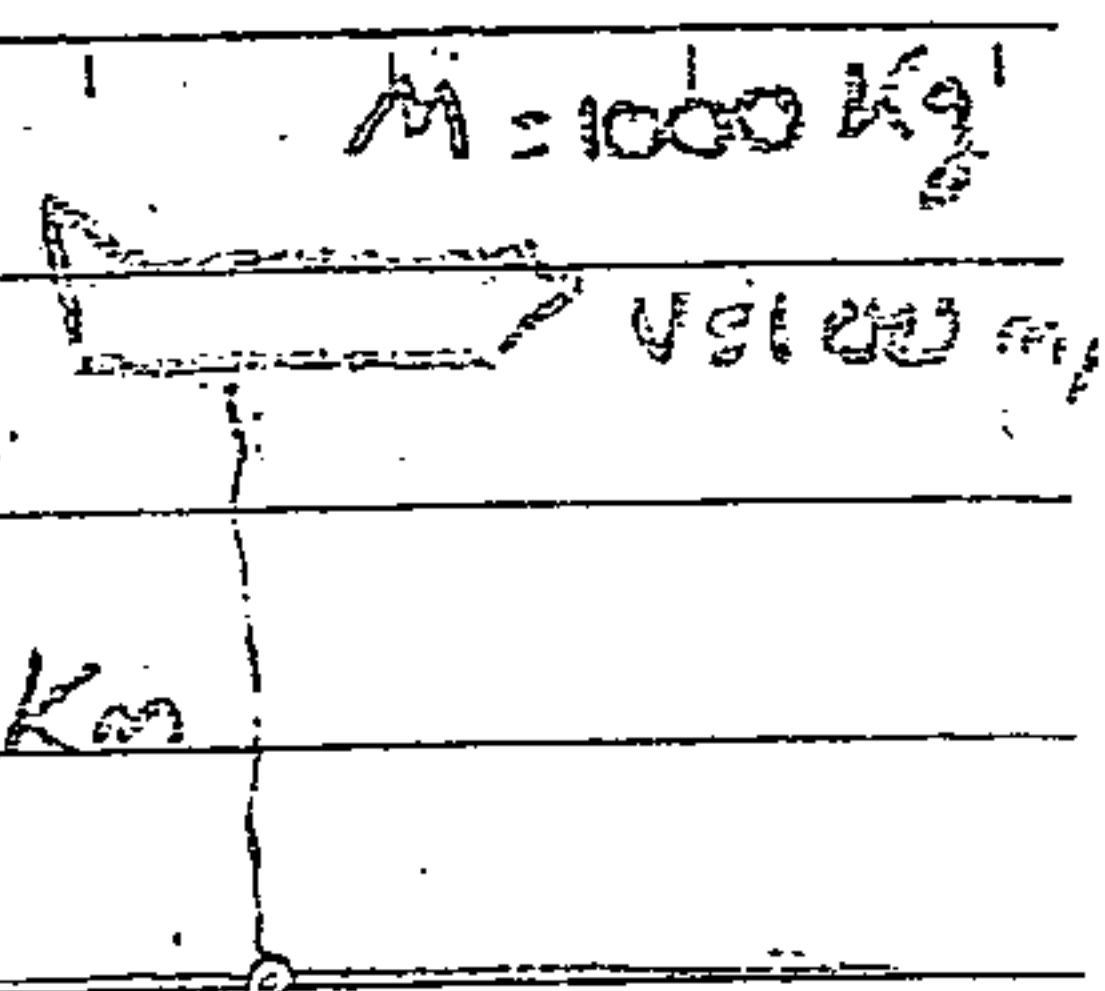


$$L_B = m v r \sin \theta$$

$$= m v r \frac{d}{r}$$

$$= m v d$$

ex: Calculate the angular momentum when the plane is directly above the reference point.



$$L = mvd$$

$$= 1000(100)(1000)$$

$$= 10^8 \text{ Kg} \cdot \text{m}^2/\text{s}$$

(b) Calculate L relative to the same point after 10 s .

$$L = mvd$$

$$= 10^8 \text{ Kg} \cdot \text{m}^2/\text{s}$$

$$\vec{L} = \vec{r} \times \vec{p}$$

$$L = rp \sin \theta$$

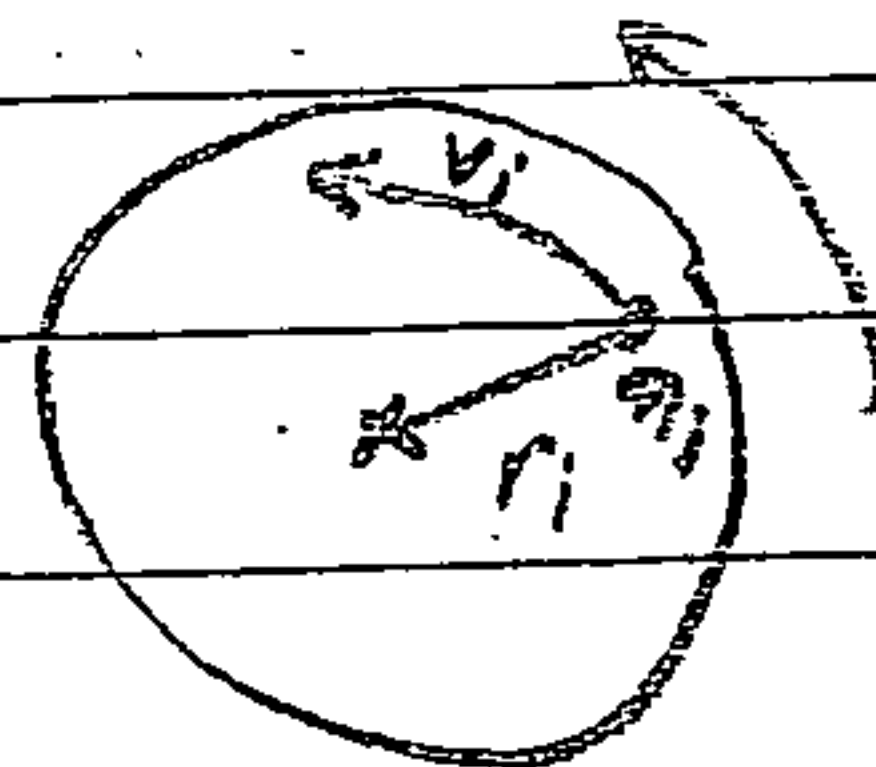
Direction

L plane of rotation

$\rightarrow$ of rotation is counterclockwise

$\rightarrow$ " " " clockwise

rotation of rigid body



$L??$

ex: $v_i = r_i \omega$

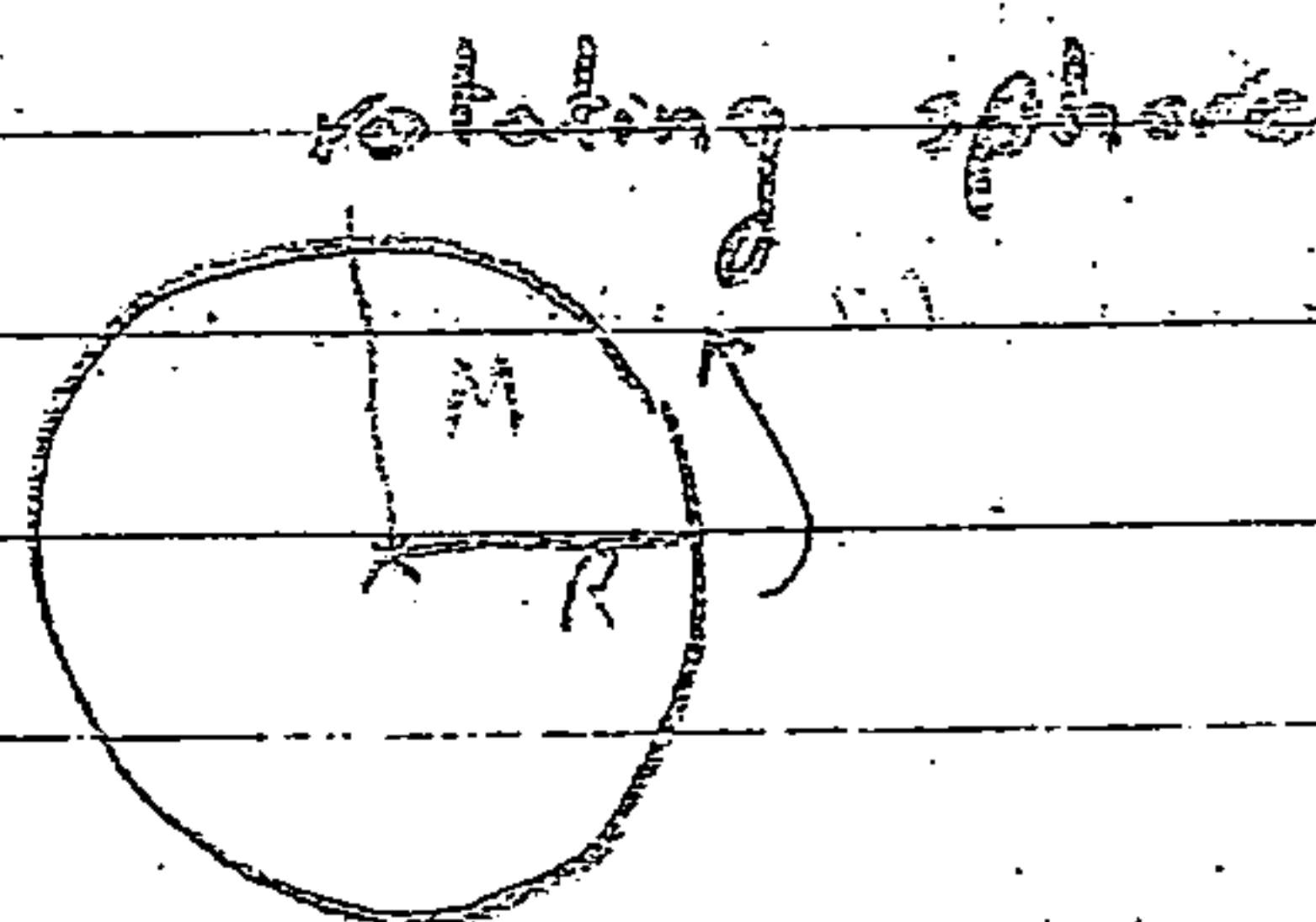
$$L_i = m_i v_i r_i$$

$$L = \sum m_i v_i r_i$$

$$\Rightarrow L = \underbrace{\left[\sum (m_i r_i^2) \right]}_I \omega$$

$$L = I\omega$$

ex: $L = I\omega$
 $= \frac{2}{5}MR^2\omega$

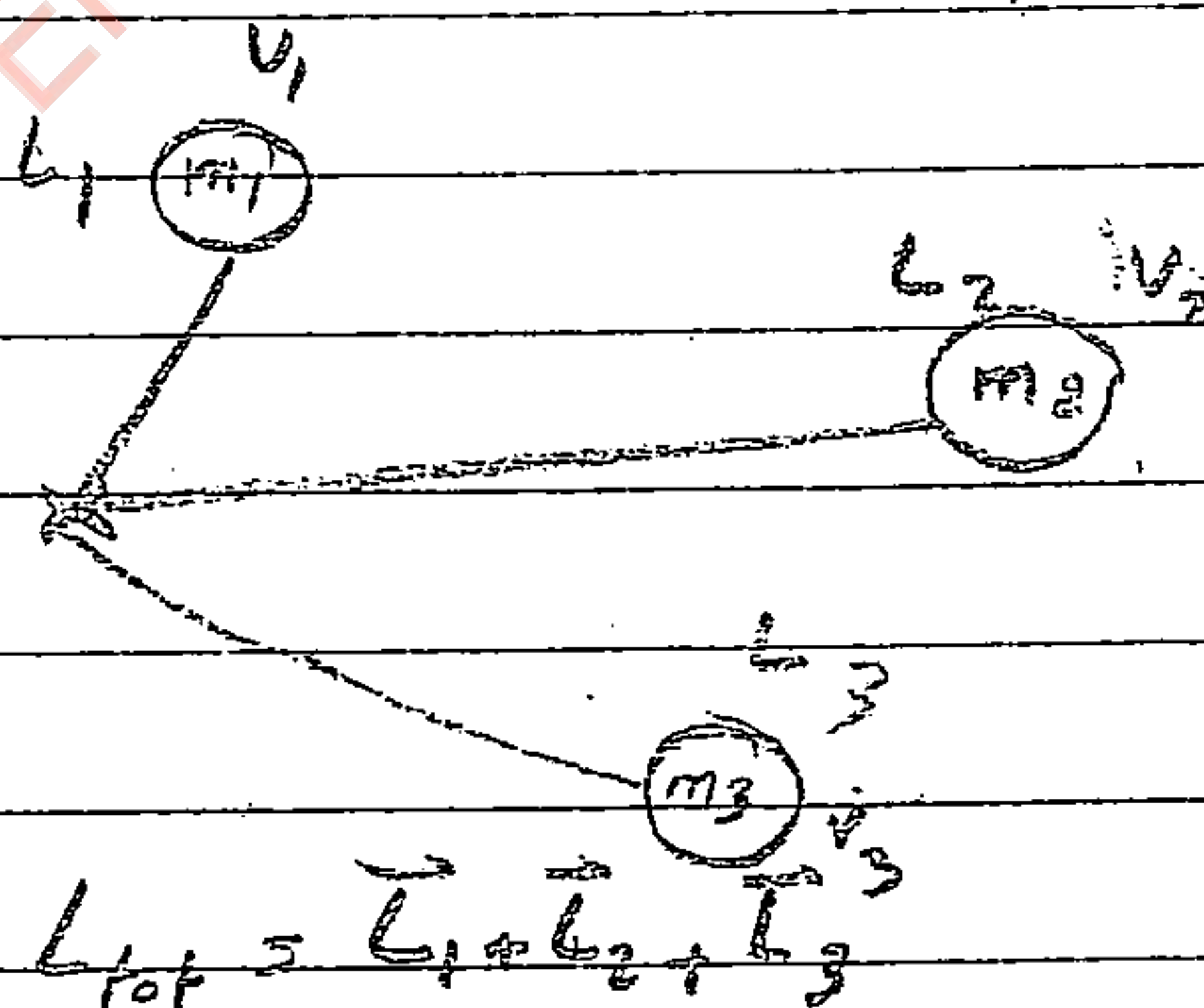


* angular momentum of a system of particles

$$\sum \vec{F}_{ext} = \frac{d\vec{P}_{tot}}{dt}$$

$$\sum \vec{\tau}_{ext} = \frac{d\vec{L}_{tot}}{dt}$$

if $\frac{d\vec{L}_{tot}}{dt} = 0 \Rightarrow \vec{L}_{tot} = \text{constant}$



ex: $L = L_1 + L_2 + L_3$
 $L_1 = I\omega$
 $= m_1 \frac{L^2}{4} \omega$

$$L_2 = I\omega$$

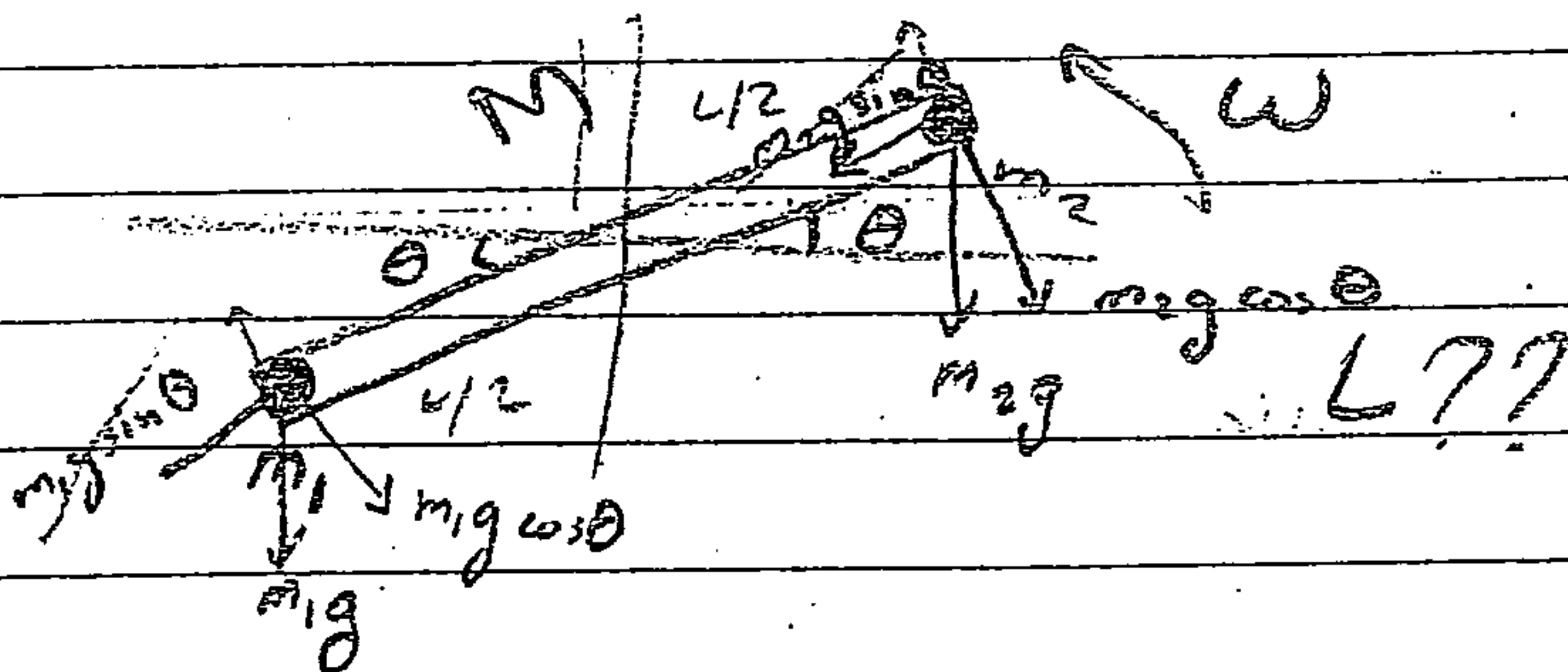
$$= m_2 \frac{L^2}{4} \omega$$

$$L_3 = I\omega$$

$$= \frac{1}{12} ML^2 \omega$$

or $I_{tot} = I_1 + I_2 + I_3$

$$L_{tot} = \frac{L^2}{4} (m_1 + m_2 + \frac{M}{3}) \omega$$



Calculate α when the system makes an angle θ with Horiz.

$$T_1 = m_1 g \frac{l}{2} \cos \theta$$

$$T_2 = m_2 g \frac{l}{2} \cos \theta$$

$$\Sigma T = I \alpha$$

$$m_1 g \frac{l}{2} \cos \theta - m_2 g \frac{l}{2} \cos \theta =$$

$$= \frac{l}{4} (m_1 + m_2 + \frac{M}{3}) \alpha$$

$$\Rightarrow \alpha = \frac{2(m_1 - m_2) g \cos \theta}{l (\frac{M}{3} + m_1 + m_2)}$$

* angular momentum of isolated system.

$$\frac{dL_{tot}}{dt} = 0$$

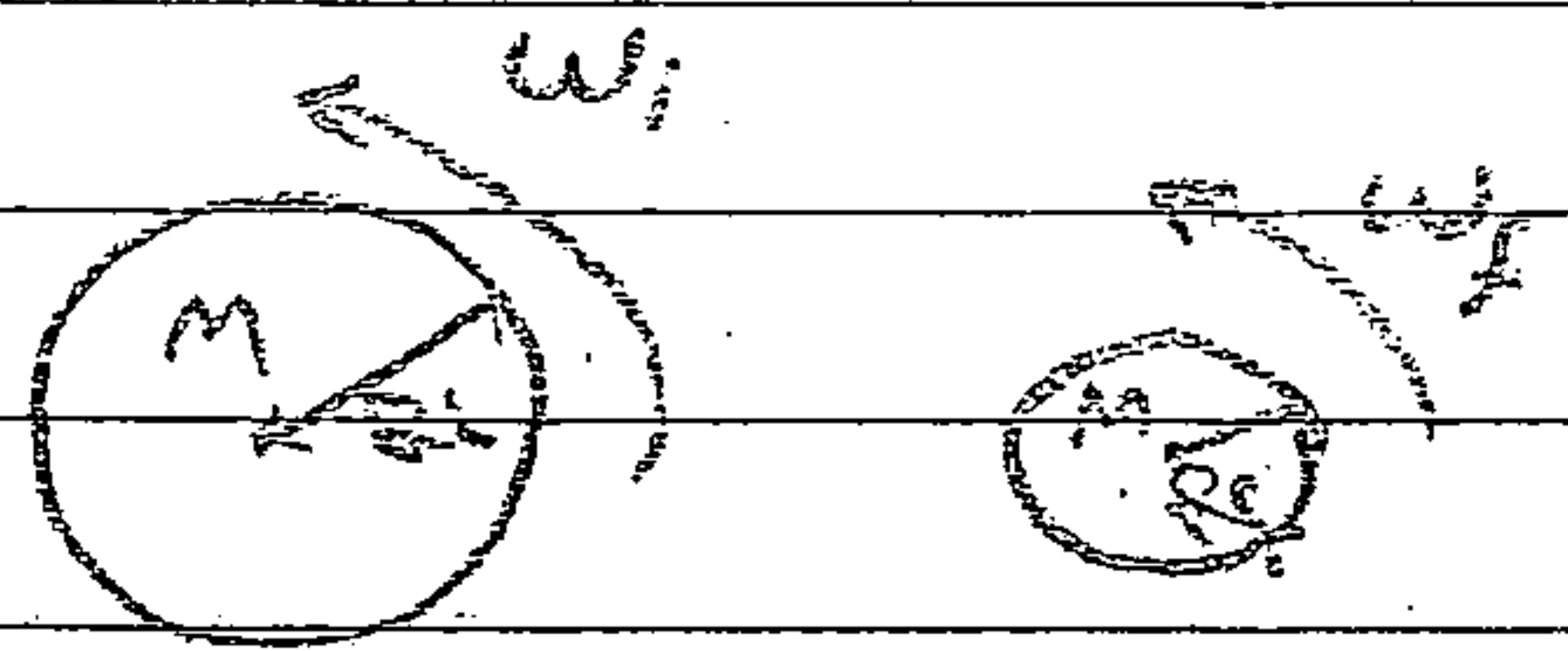
$$L_{tot} = \text{constant}$$

$$I_i \omega_i = I_f \omega_f$$

ex: $I_i \omega_i = I_f \omega_f$

$$\frac{2}{5} MR_i^2 \omega_i = \frac{2}{5} MR_f^2 \omega_f$$

$$\omega_f = \left(\frac{R_i}{R_f} \right)^2 \omega_i$$



16) $L = I\omega + mVR$
 $I = \frac{1}{2}MR^2$

$\Rightarrow L = \frac{1}{2}MR\omega^2 + mVR$
 $= \frac{1}{2}MR \frac{v}{R} + mVR$
 $= \frac{1}{2}MRv + mVR$

$L = \frac{1}{2}(2)(8 \times 10^{-2})v + 4v(8 \times 10^{-2}) = 0.4v$

* calculate the net external torque as a function of a ?

$T_{ext} = \frac{dL}{dt} = 0.4a \Rightarrow T = 0.4a$

* or?? $r \times F$
 $T_{ext} = mgr$

or $mg - T = ma$
 $TR = I\alpha$

$\Rightarrow mgr = 0.4a$

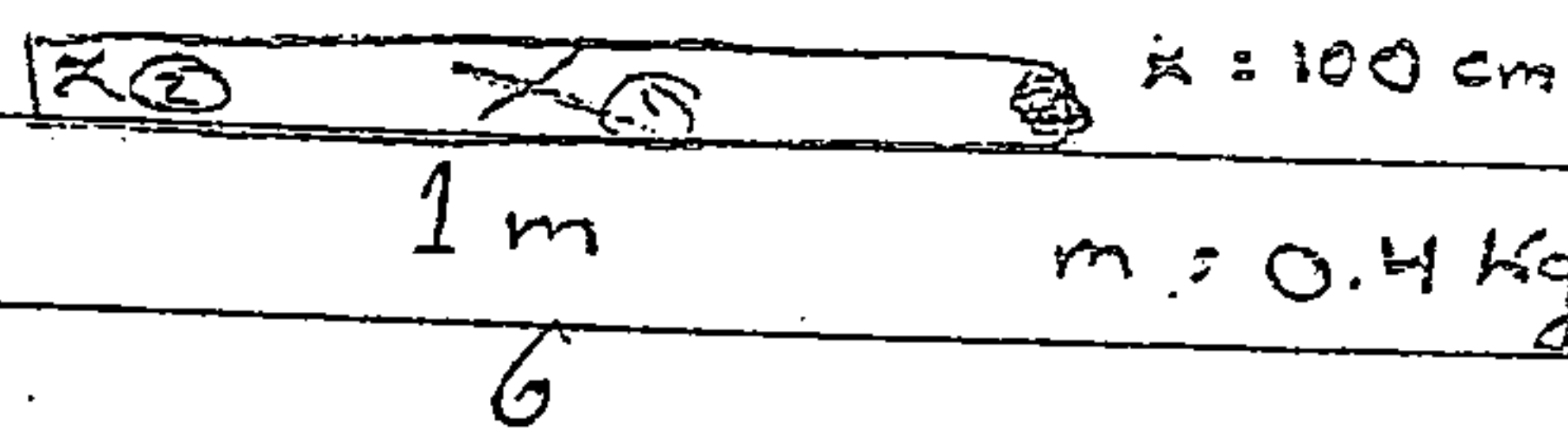
$4(9.8)(8 \times 10^{-2}) = 0.4a$

$\Rightarrow a = 7.85 \text{ m/s}^2$

$= \frac{1}{2}MR^2 \frac{a}{R}$
 $mg = \frac{1}{2}MR^2 \frac{a}{R} = ma \cdot R$

$\omega = 4 \text{ rad/s}$

$M = 0.1 \text{ Kg}$



27) ① L??

$L = I\omega + I_{par}\omega$

$I_{rod} = \frac{1}{12}MB^2$

$I_{par} = \sum m_i r_i^2$

$= m_1 r_1^2$

$= m \frac{B^2}{4}$

$L = \frac{1}{12}MB^2\omega + m \frac{B^2}{4}\omega = 0.433 \text{ Kg m}^2/\text{s}$

$$\Rightarrow$$

$$\textcircled{2} L = I_{\text{cm}} \omega + I_{\text{cm}} \omega$$

$$= \frac{1}{3} M b^2 \omega + m b^2 \omega$$

$$= 1.73 \text{ kg} \cdot \text{m}^2/\text{s}$$

$$\textcircled{52} L_i = L_f$$

$$I_i \omega_i = I_f \omega_f$$

$$\left(\omega_f = \frac{I_i \omega_i}{I_f} \right)$$

$$I_i = m r_i^2$$

$$I_f = m r_f^2$$

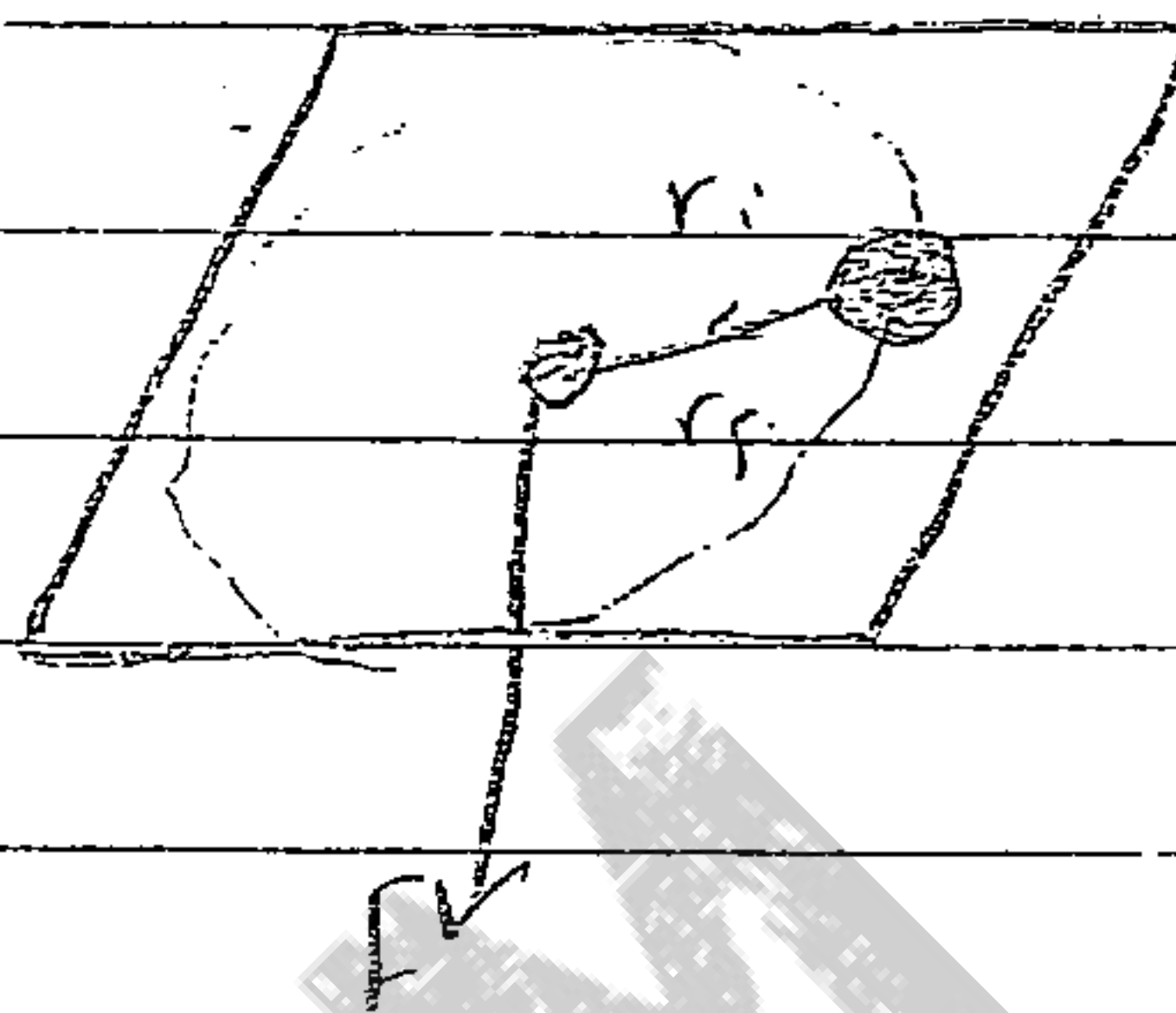
$$\omega_f = \frac{m r_i^2 \omega_i}{m r_f^2} = \frac{r_i^2}{r_f^2} \omega_i$$

$$\Rightarrow \frac{v_f}{r_f} = \frac{r_i^2}{r_f^2} \frac{v_i}{r_i} \Rightarrow \left(v_f = \frac{r_i v_i}{r_f} \right)$$

T?

$$T = \frac{m v_f^2}{r_f}$$

$$\Rightarrow T = \frac{m r_i^3 v_i^2}{r_f^3}$$



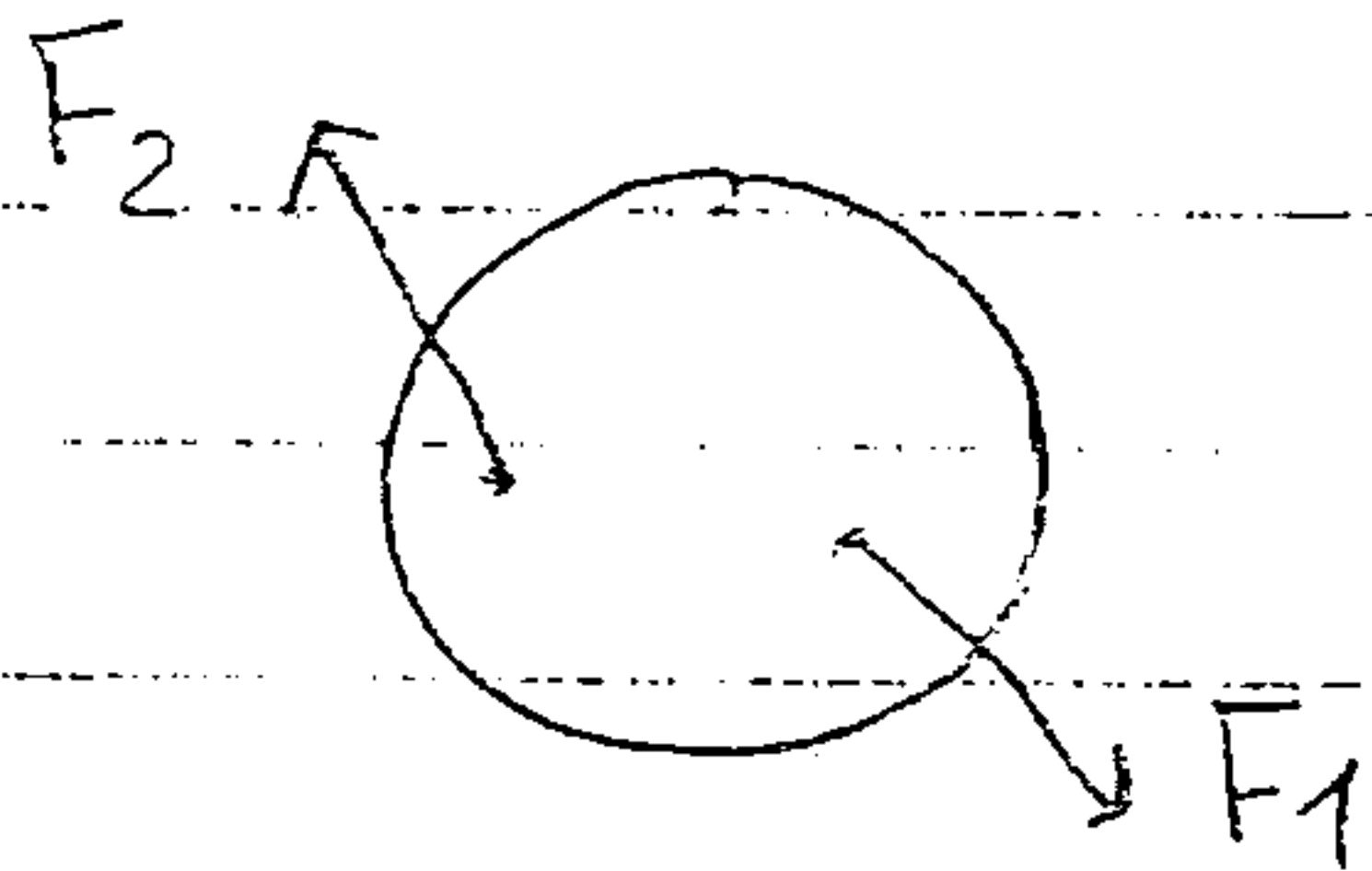
Static Equilibrium

1/9/35

* Two conditions for static equilibrium

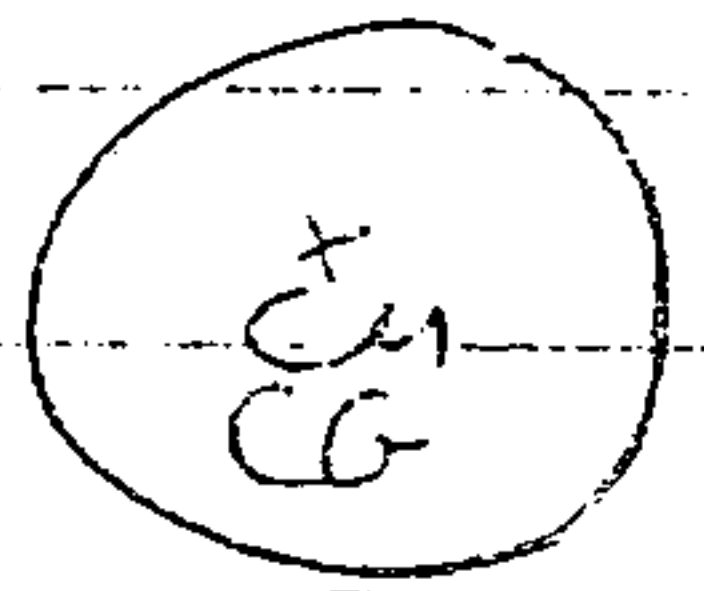
1) $\sum \vec{F} = 0$

2) $\sum \vec{\tau} = 0$ (about any axis)

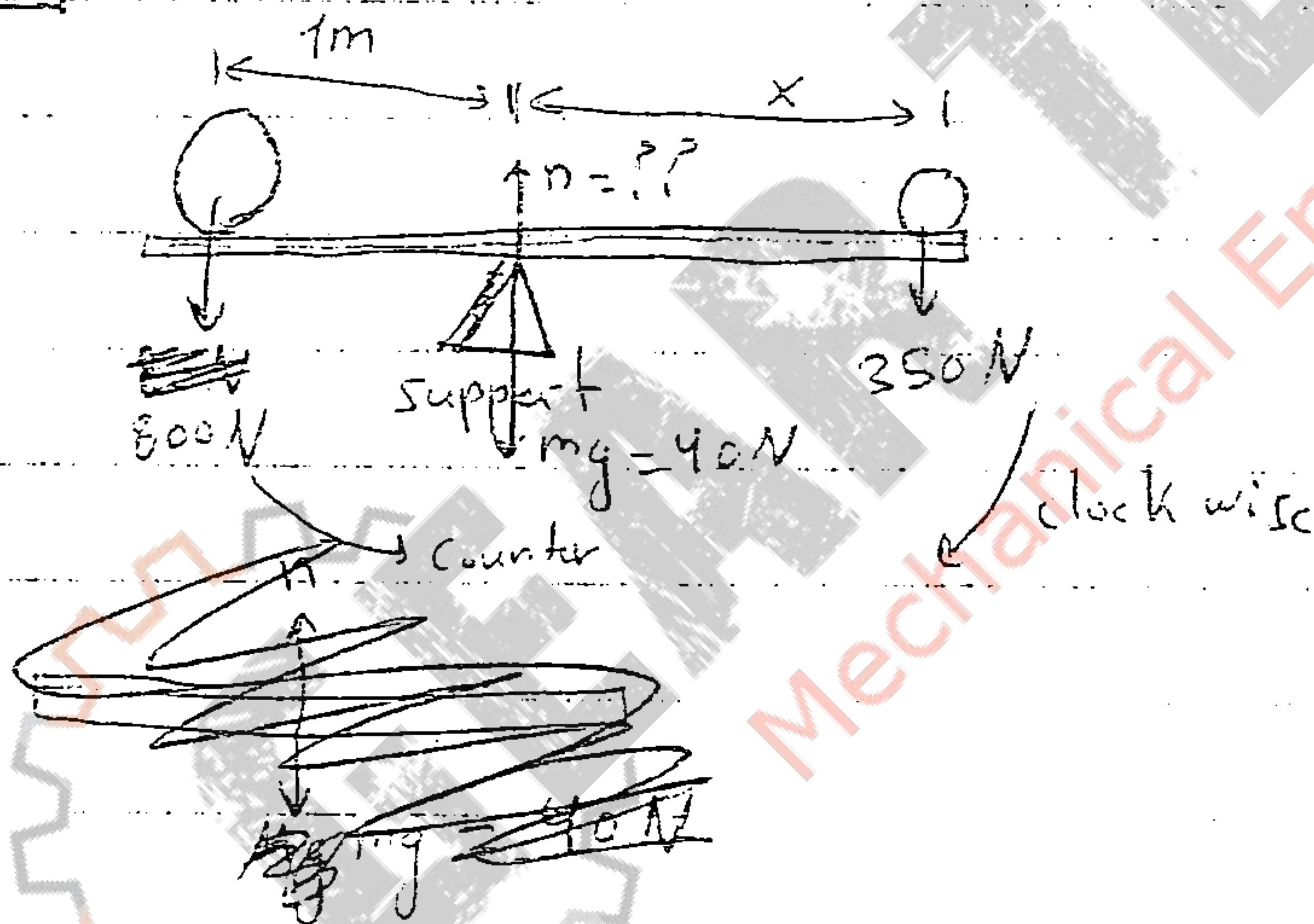


$F_1 = F_2$

Center of Gravity
Center of mass



Ex: 12.1



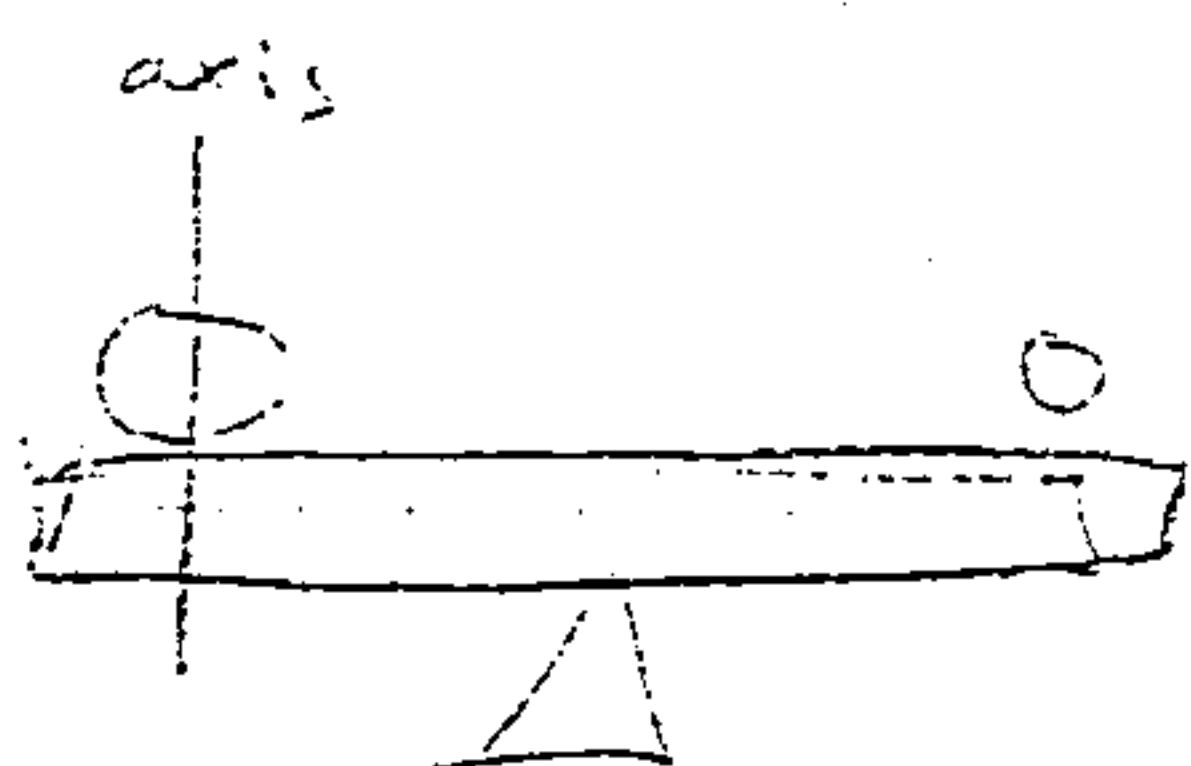
$n - 800 - 350 - 40 = 0$

$n = 40 + 350 + 800 = 1190 \text{ N}$ (normal force)

$(800)(1) = (350)(x)$

$x = \frac{800}{350} \approx 2.3 \text{ m}$

(البعد حتى يكون النظام متزن)

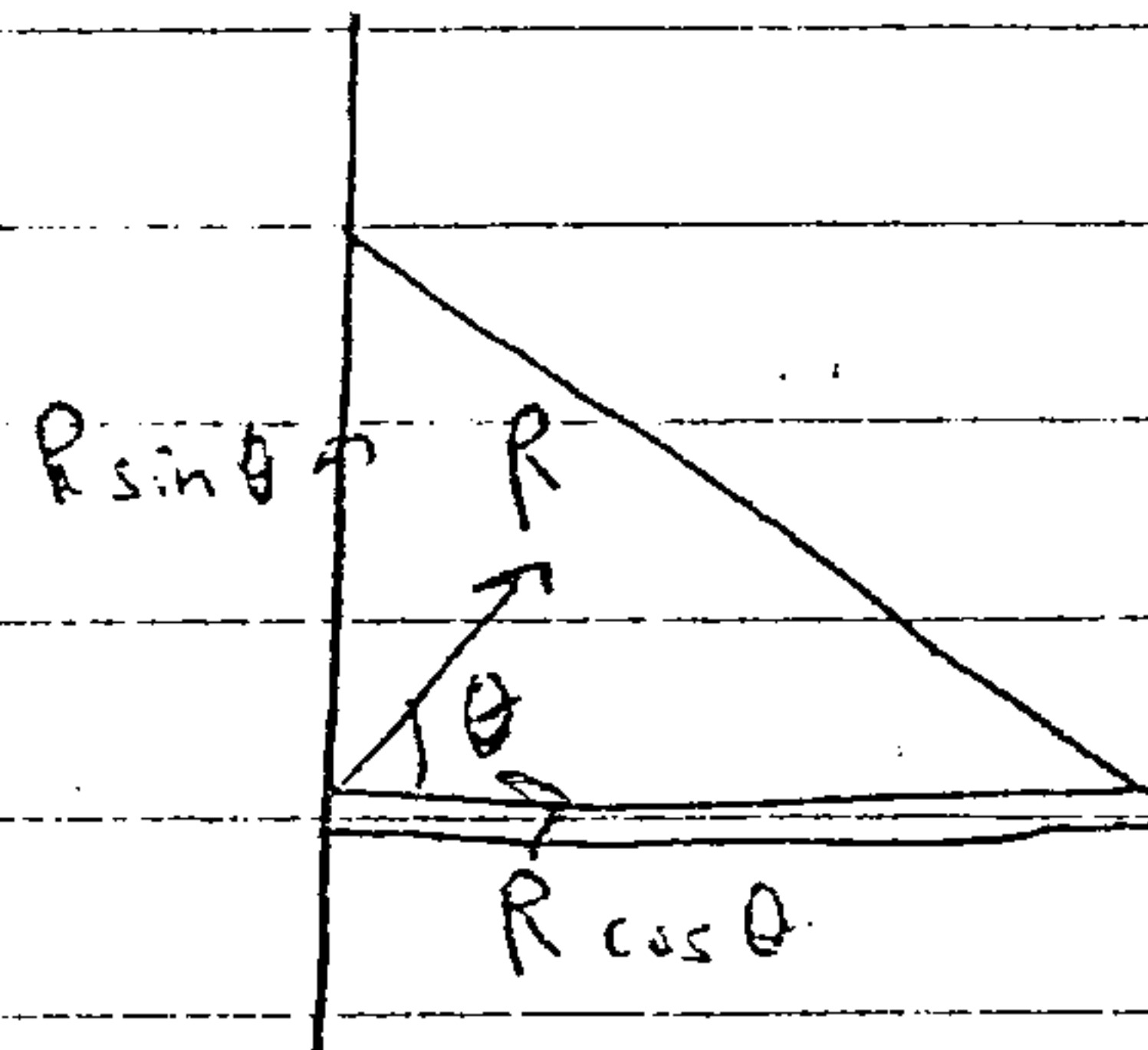
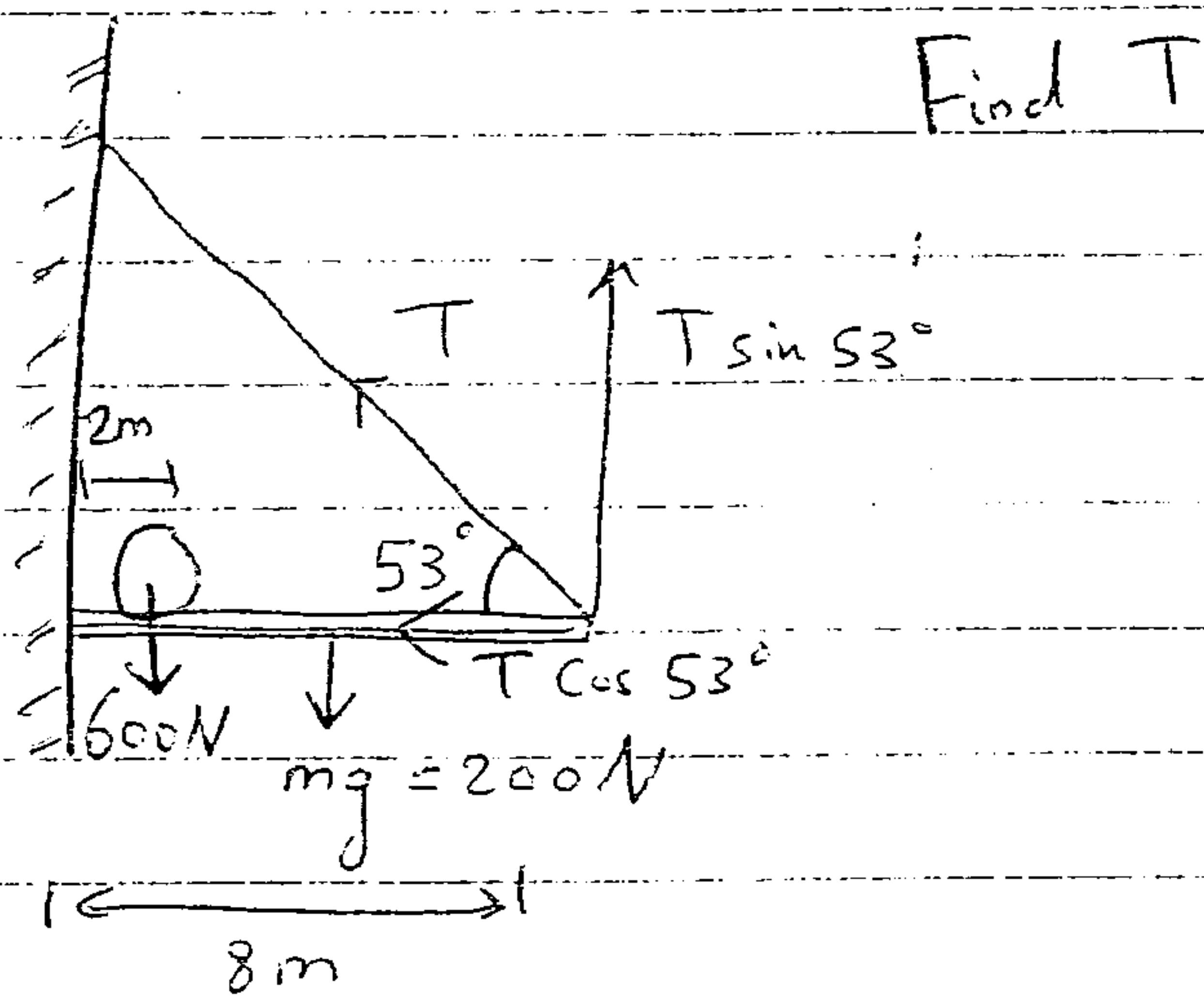


1190

$n(1) - (350)(1+x) - (40)(1) = 0$

(25)

12.3 Ex:-



~~$$T \sin 53^\circ + R \sin \theta - 600 - 200 = 0$$~~

$$T \sin 53^\circ + R \sin \theta - 600 - 200 = 0$$

$$R \cos \theta - T \cos 53^\circ = 0$$

$$\sum \vec{T} = 0$$

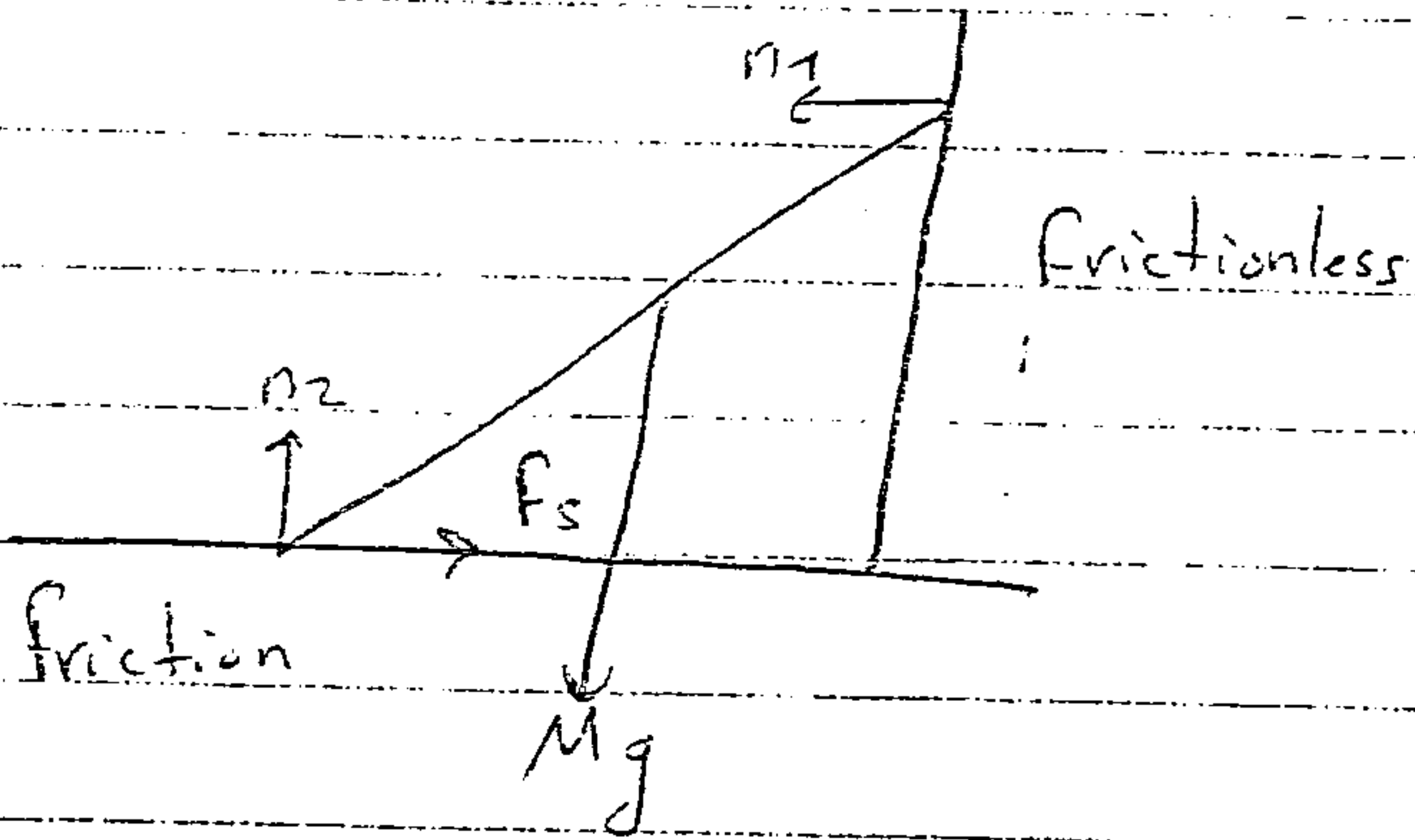
$$(T \sin 53^\circ)(8) - (600)(2) - (200)(4) = 0$$

$$T = 313 \text{ N}$$

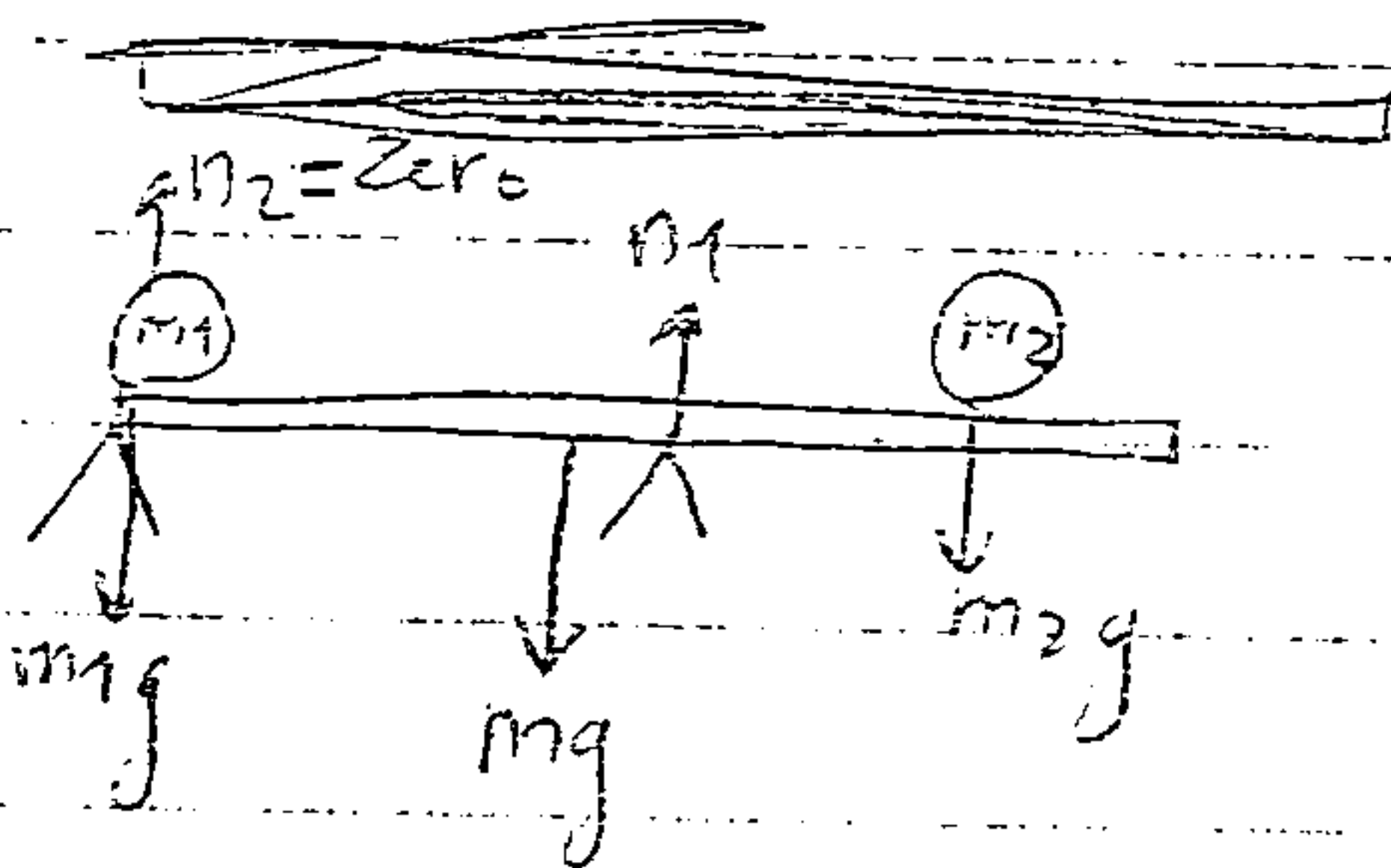
$$R \cos \theta = (313)(0.6) =$$

$$R \sin \theta =$$

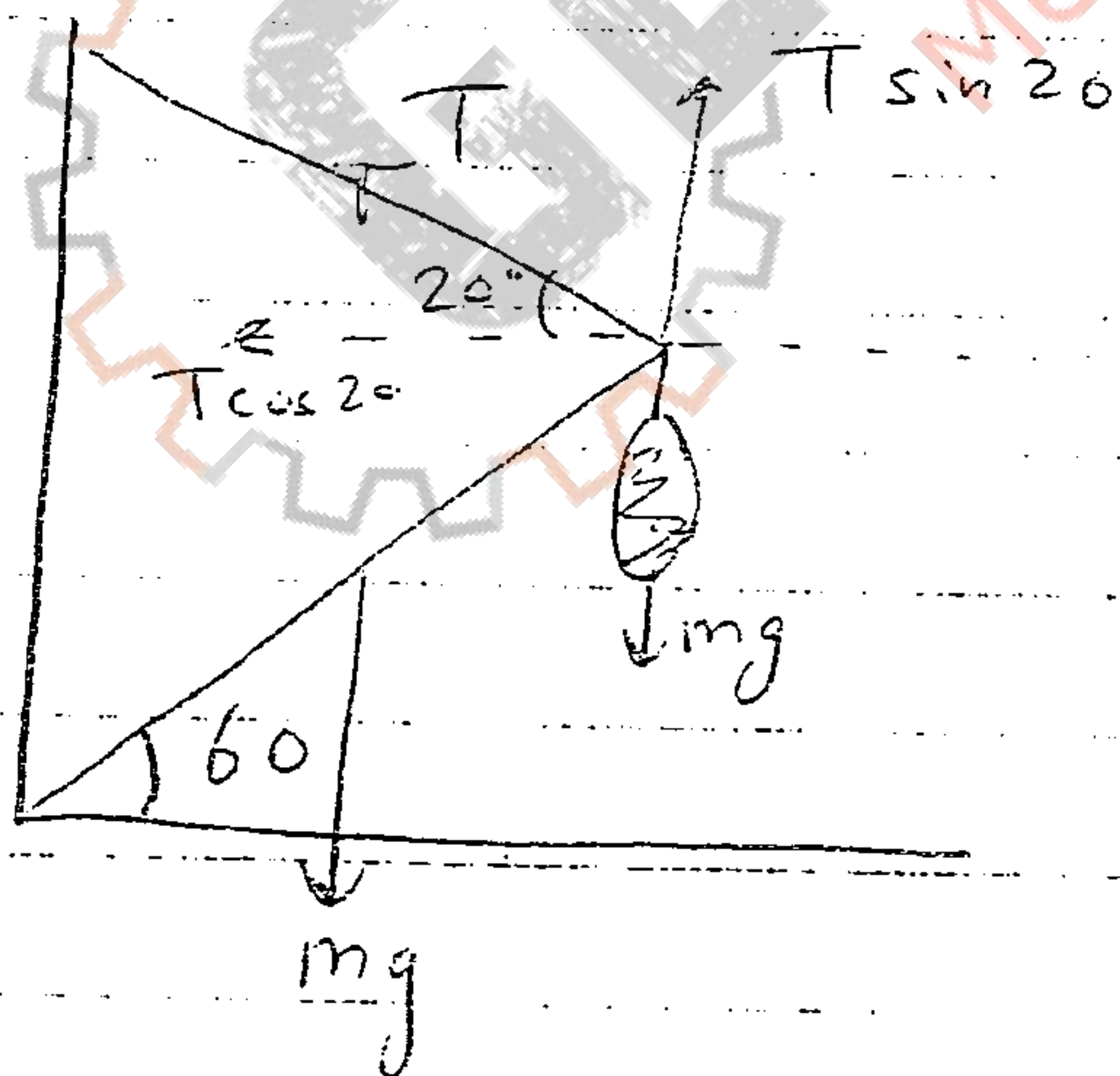
Ex:-

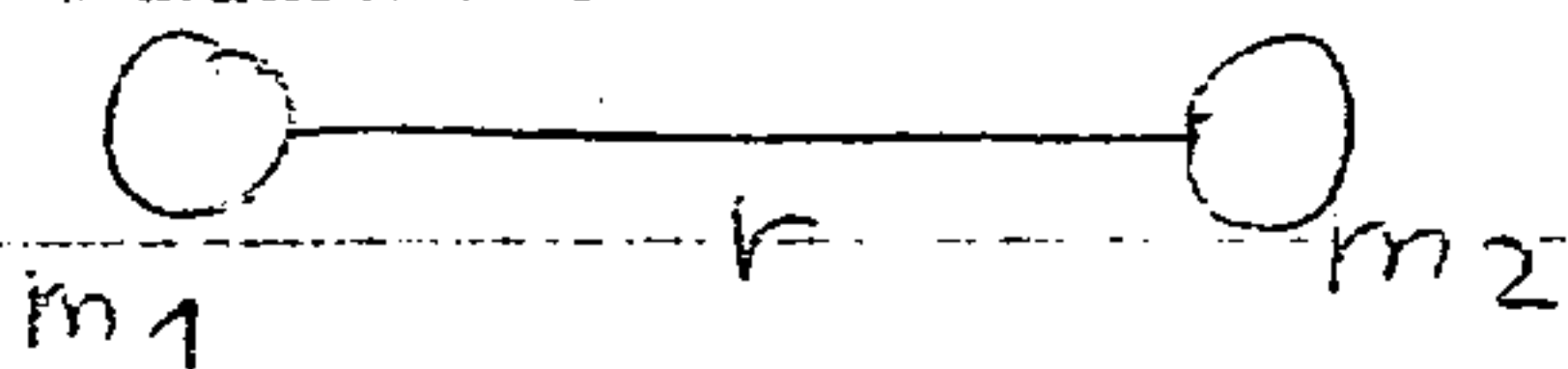


Problem 12.3

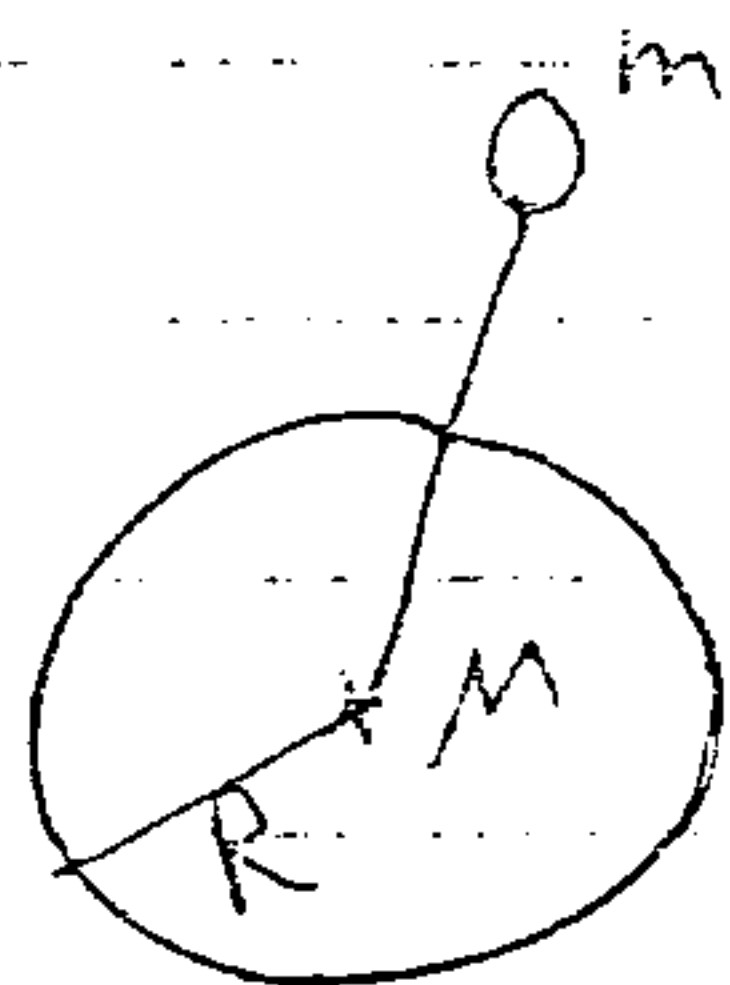


~~Problem 12.3~~





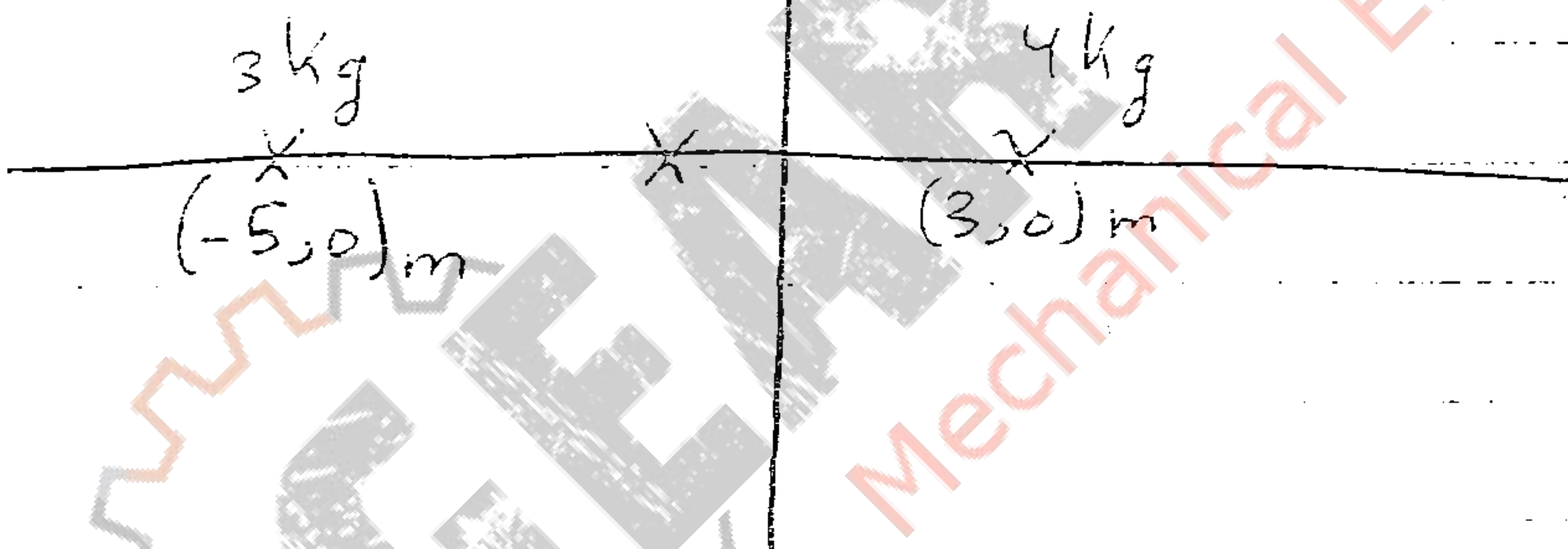
$$F_{\text{gravitational}} = G \frac{m_1 m_2}{r^2}$$



$$\frac{GmM}{r^2} = mg$$

$$g = \frac{G \cdot M}{r^2}$$

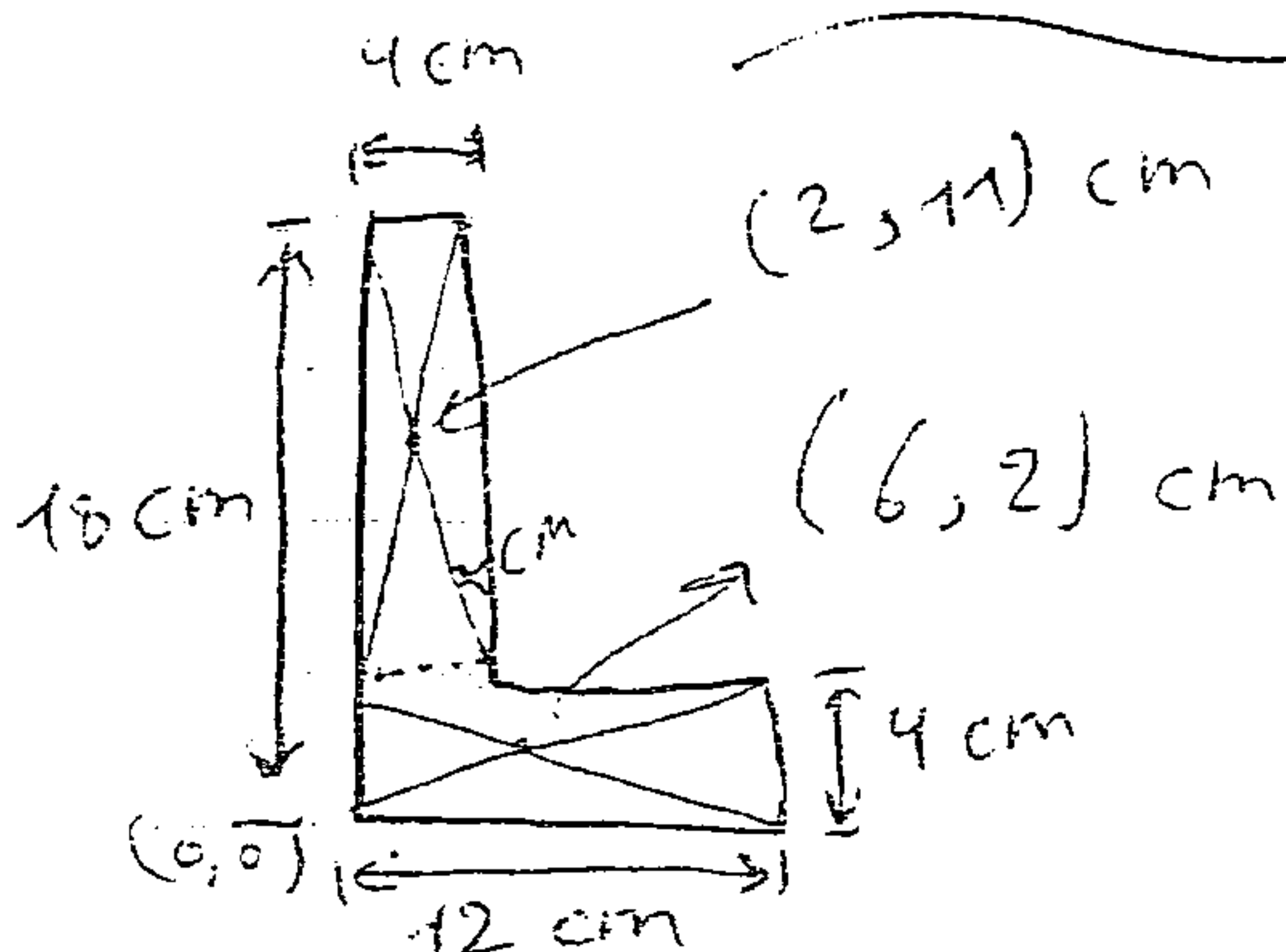
12.5



$$CM = \frac{(3)(-5) + (4)(3)}{7}$$

$$= \frac{-3}{7} = -0.43 \text{ m}$$

12.7



$$CM = \frac{(12)(4)(6) + (4)(4)(2)}{(12)(4) + (4)(4)}$$

$$= 3.85 \text{ cm}$$

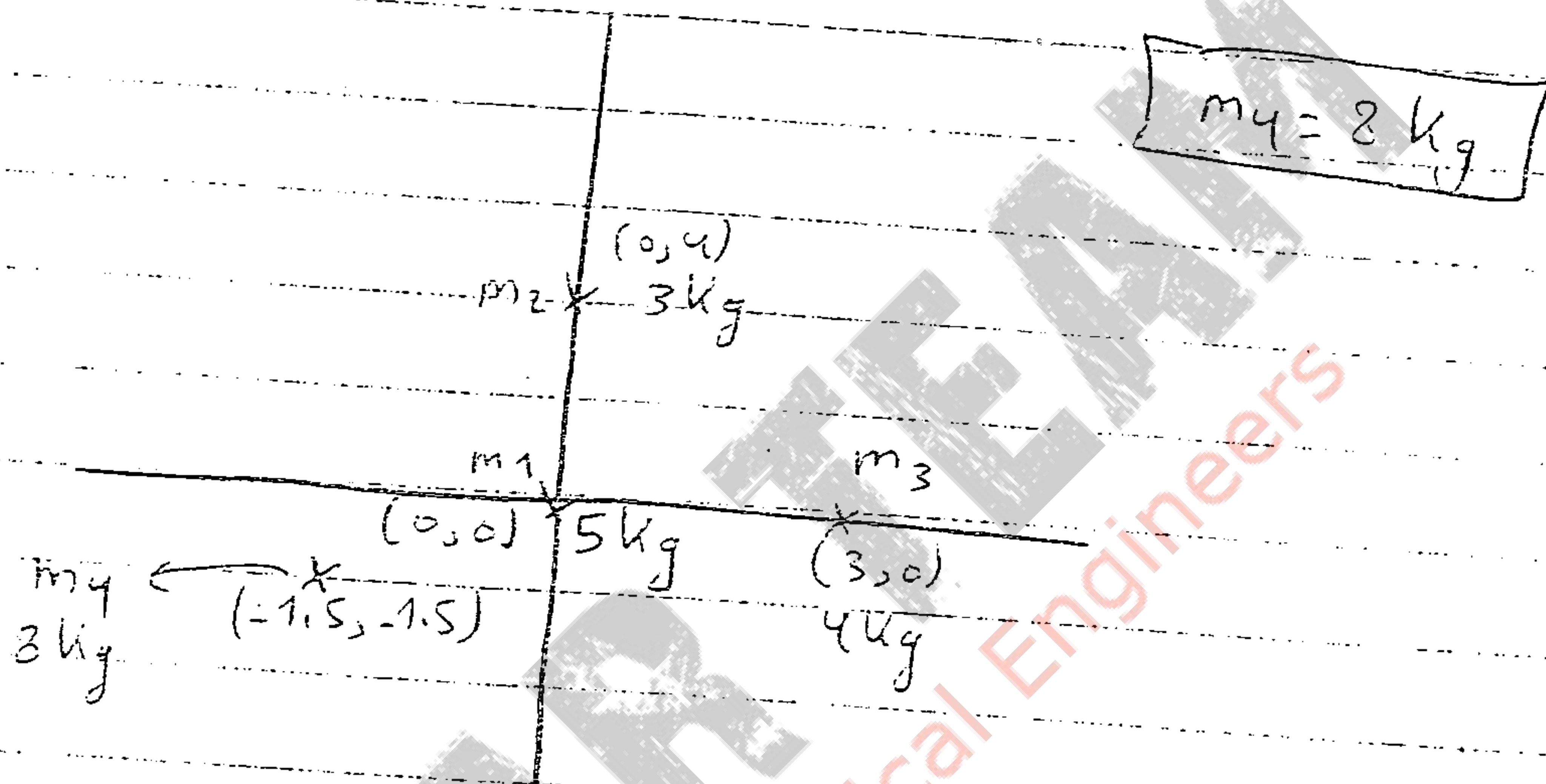
234

$$y(M) = \frac{(12)(4)(2) + (14)(4)(11)}{(12)(4) + (14)(4)} = \frac{24 + 154}{26} = \frac{178}{26} = 6.8$$

$$\vec{r}_{CM} = (3.85)\hat{i} + (6.85)\hat{j}$$

$$|\vec{r}_{CM}| = \sqrt{\quad}$$

12.9



$$x(M) = 0 = \frac{(5)(0) + (3)(0) + (3)(4) + (8)(x)}{5 + 3 + 4 + 8}$$

$$0 = \frac{12 + 8x}{20}$$

$$-12 + 8x = 0$$

$$8x = -12$$

$$x = \frac{-12}{8} = \frac{-3}{2} = -1.5$$

$$y(M) = 0 = \frac{(5)(0) + (3)(4) + (4)(0) + (8)(y)}{20} = \frac{12 + 8y}{20}$$

$$0 = 12 + 8y$$

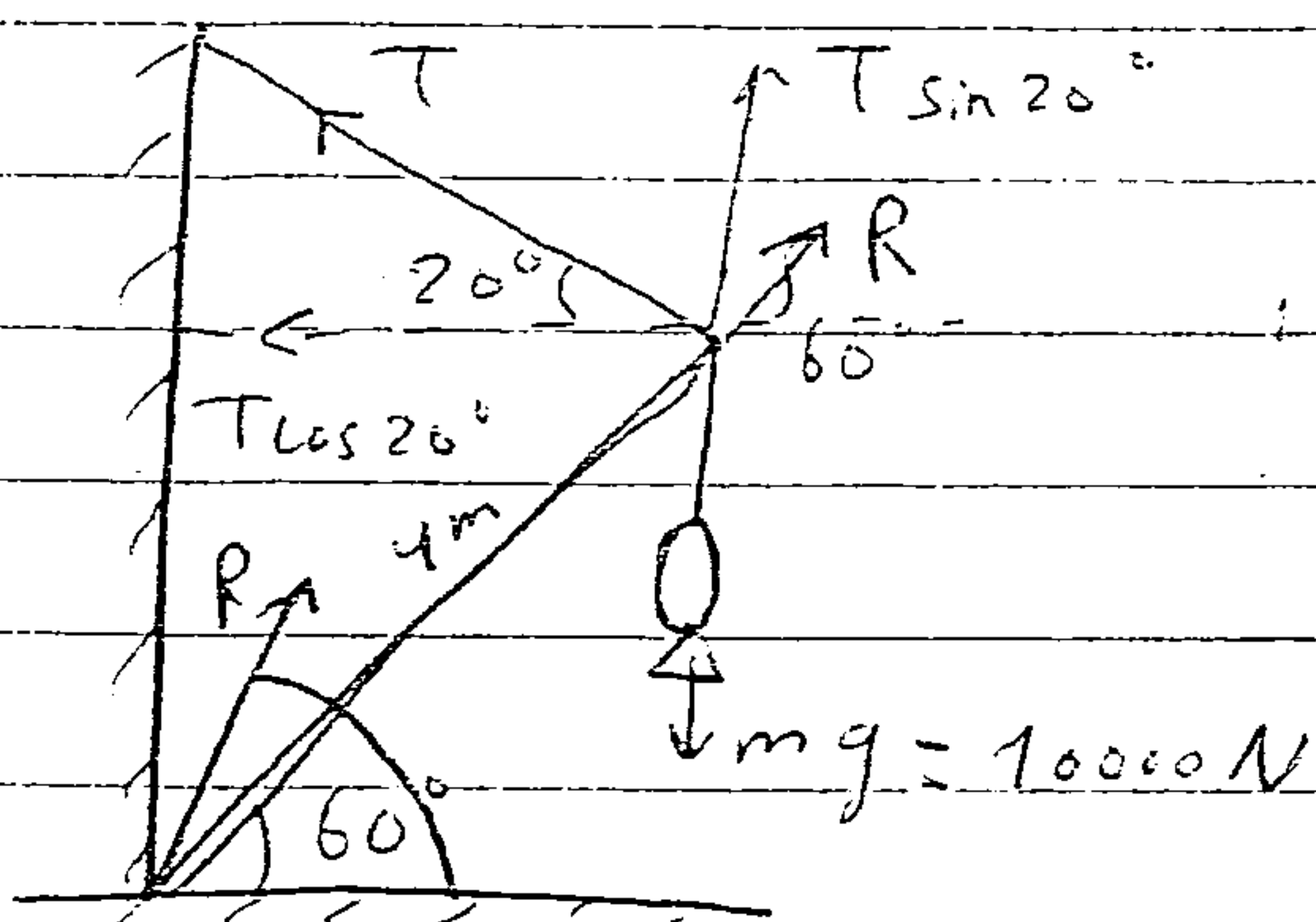
$$y = -1.5 \text{ cm}$$

(755)

$$\sum F_x = 0$$

$$\sum F_y = 0$$

12.47



$$R \cos 60^\circ = T \cos 20^\circ$$

$$T \sin 20^\circ + R \sin 60^\circ = 10000$$

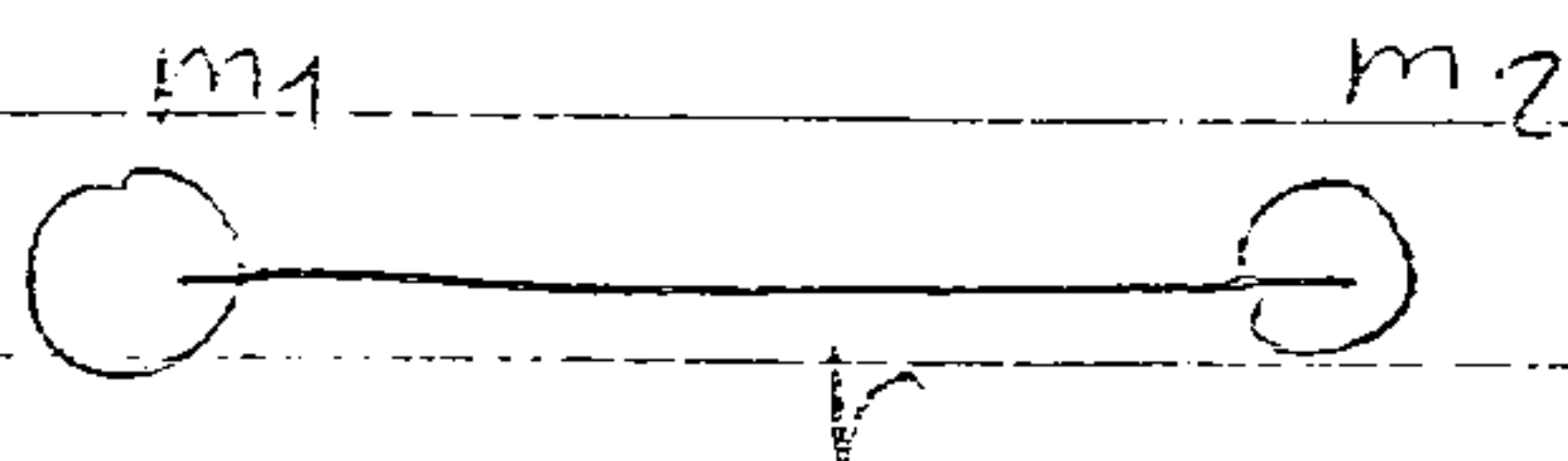
$$R \sin 60^\circ = 10000 - T \sin 20^\circ$$

$$\sqrt{3} = \frac{10000}{T \cos 20^\circ} - \tan 20^\circ$$

$$\frac{10000}{T \cos 20^\circ} = \sqrt{3} + \tan 20^\circ$$

Chapter 13

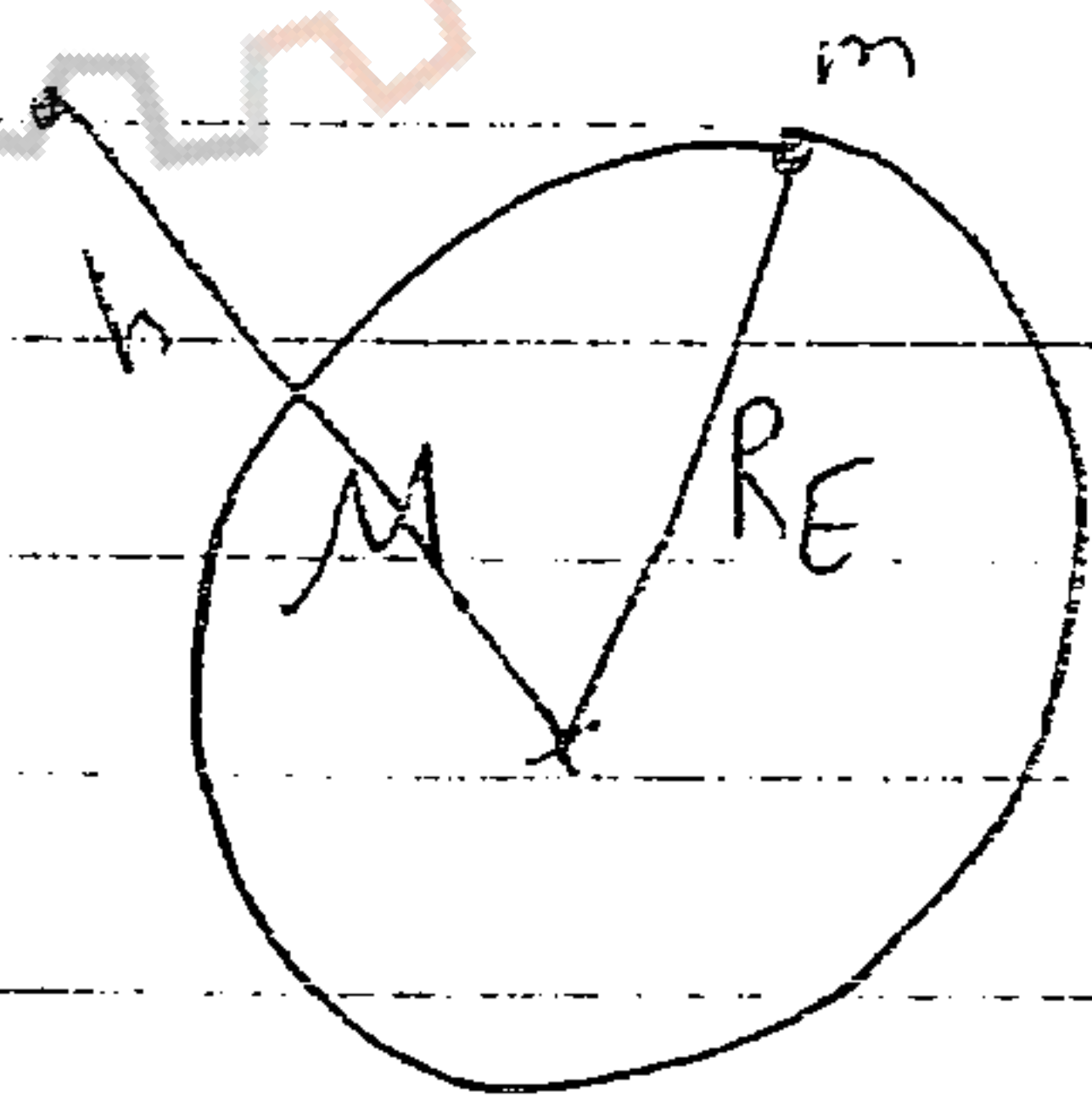
Universal Gravitation



$$F_{12} = G \frac{m_1 m_2}{r^2}$$

$$G = 6.67 \times 10^{-11}$$

$$F = \frac{6.67 \times 10^{-11} (100)(100)}{1^2} = 6.67 \times 10^{-7} \text{ N}$$



$$F_g = G \frac{mM}{R_E^2} = mg$$

$$g = GM$$

$$g' = \frac{GM}{(R_E + h)^2}$$

$$h = 350 \text{ km}$$

Space station

$$W = 4.22 \times 10^6 \text{ N} = mg$$

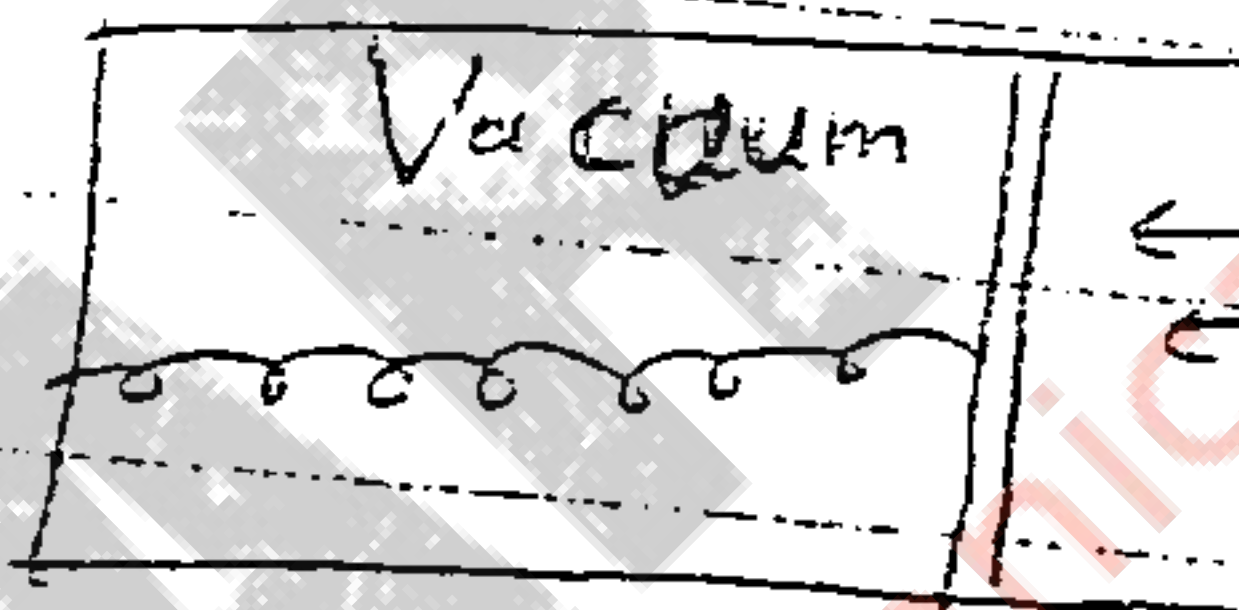
Chapter 14

Fluid Mechanics

$$\text{Pressure (P)} = \frac{dF}{dA}$$

$$dF = \int P dA$$

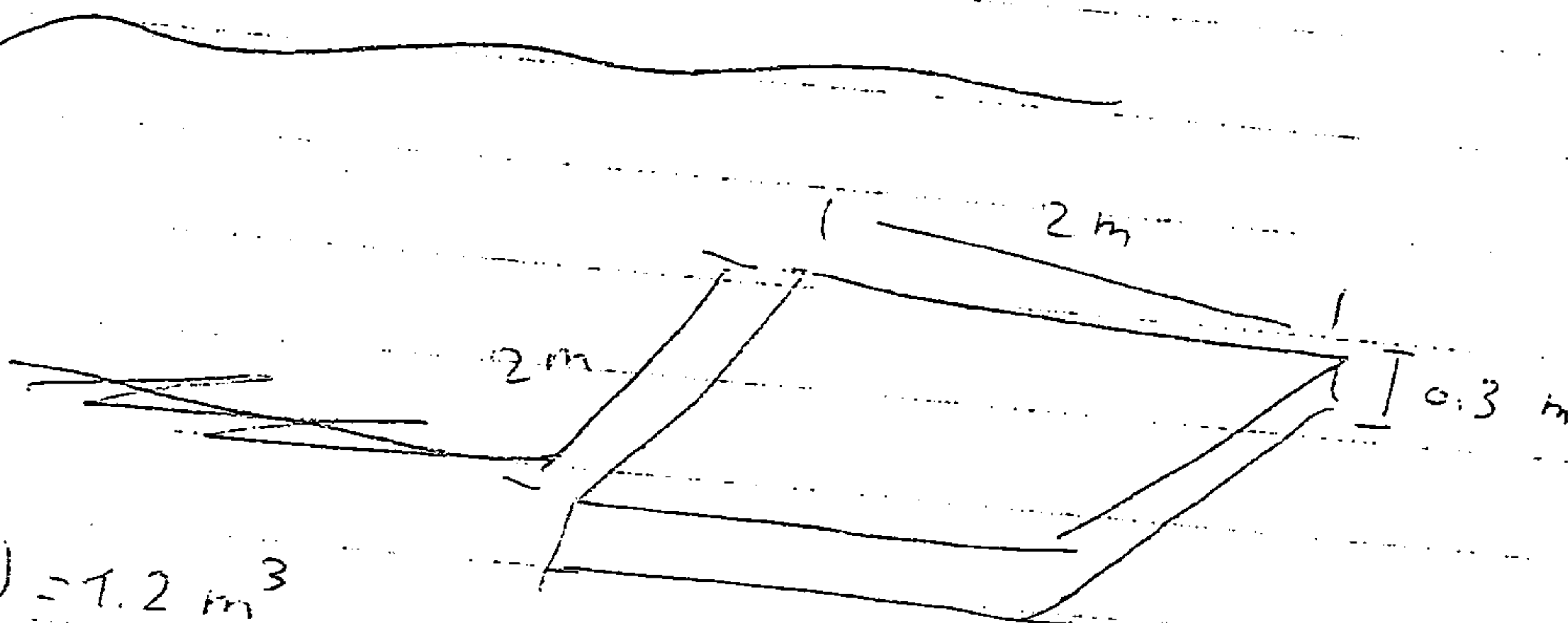
$$= \frac{F}{A} \leftarrow \text{Vector}$$



atmospheric pressure

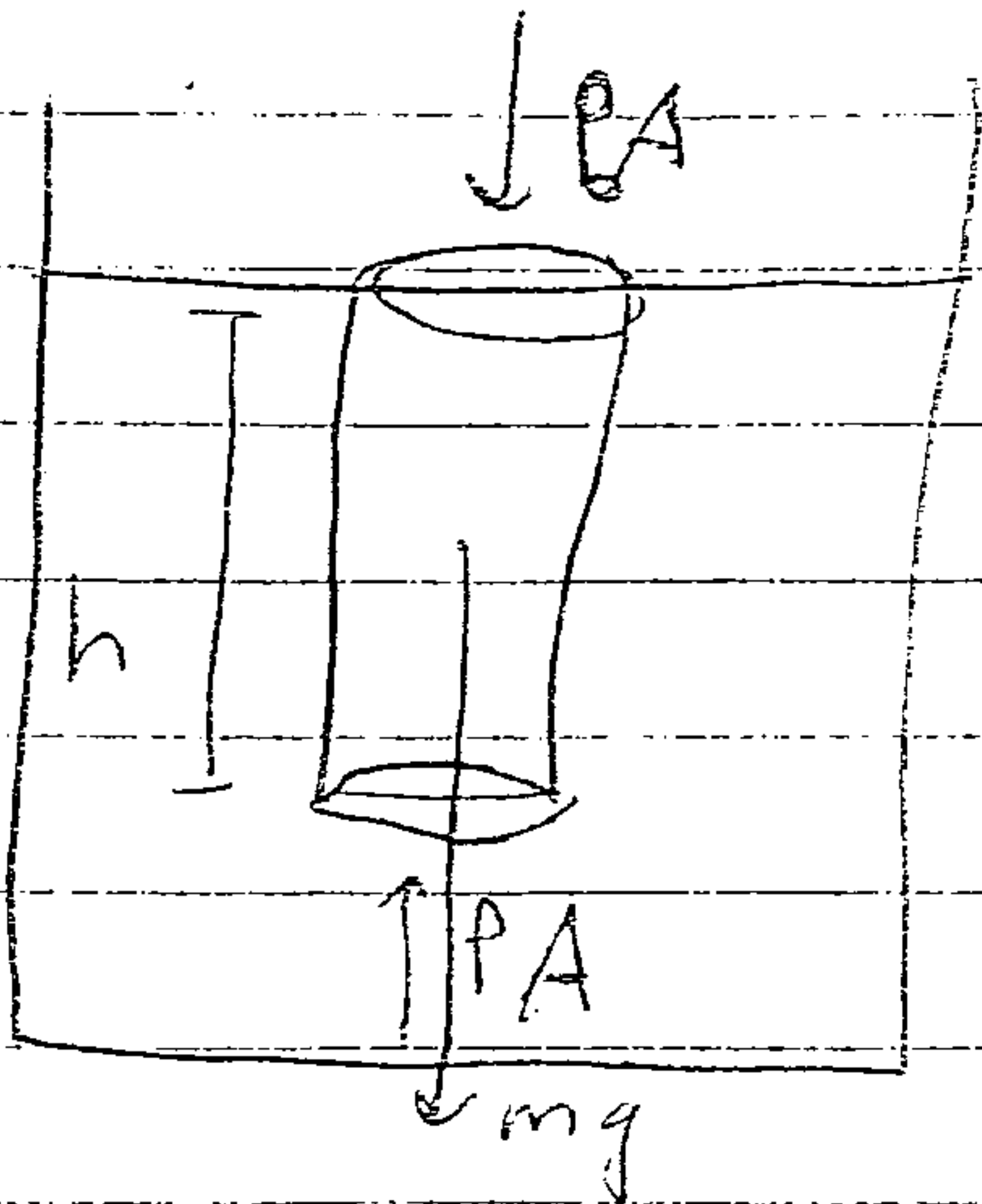
$$a = \text{N/m}^2$$

al



$$V = (2)(2)(0.3) = 1.2 \text{ m}^3$$

$$7000 \text{ kg/m}^3 \text{ (density)}$$

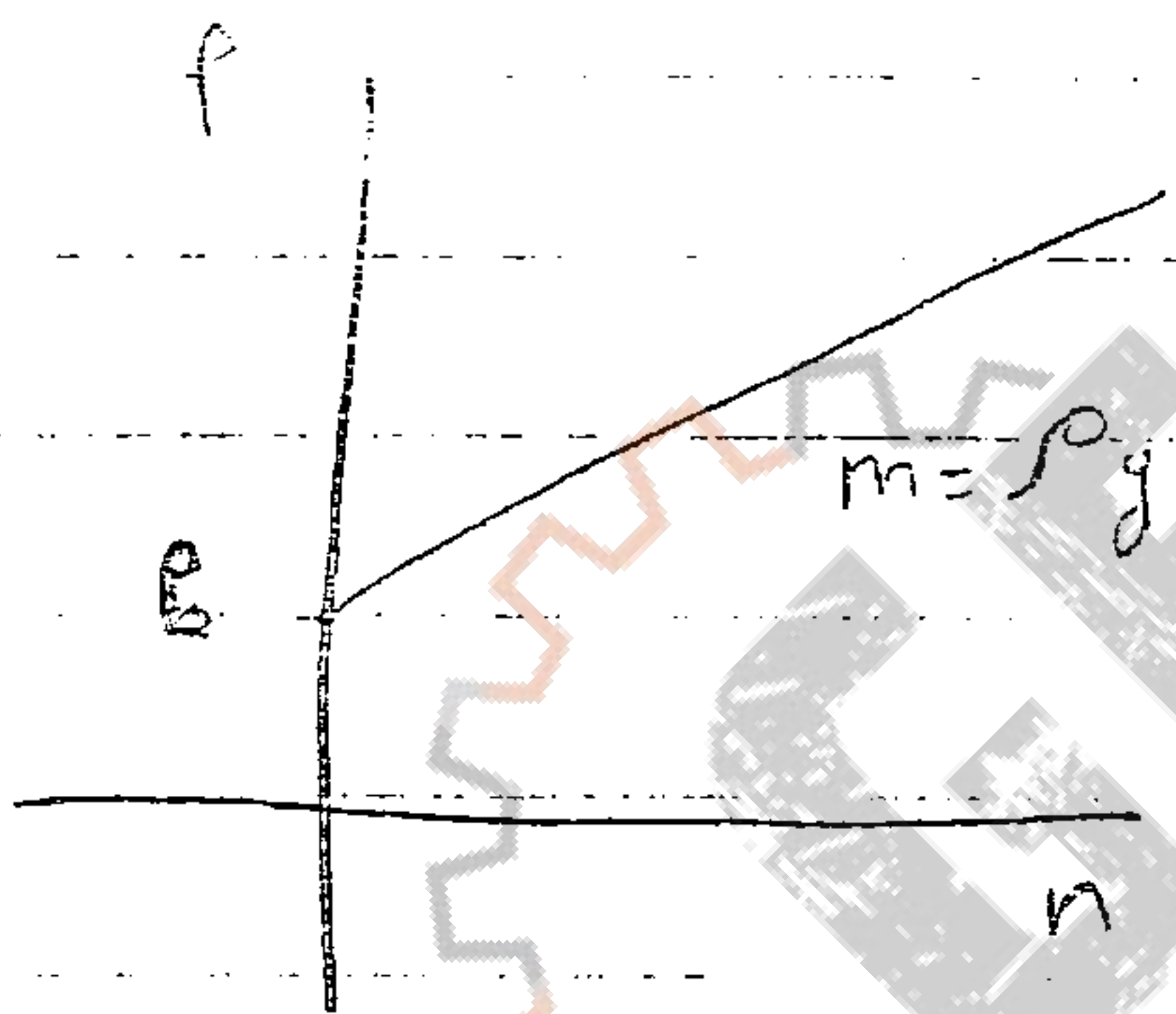


$$PA - Mg - P_0 A = 0$$

$$M = h A \rho$$

$$PA - \rho h A g - P_0 A = 0$$

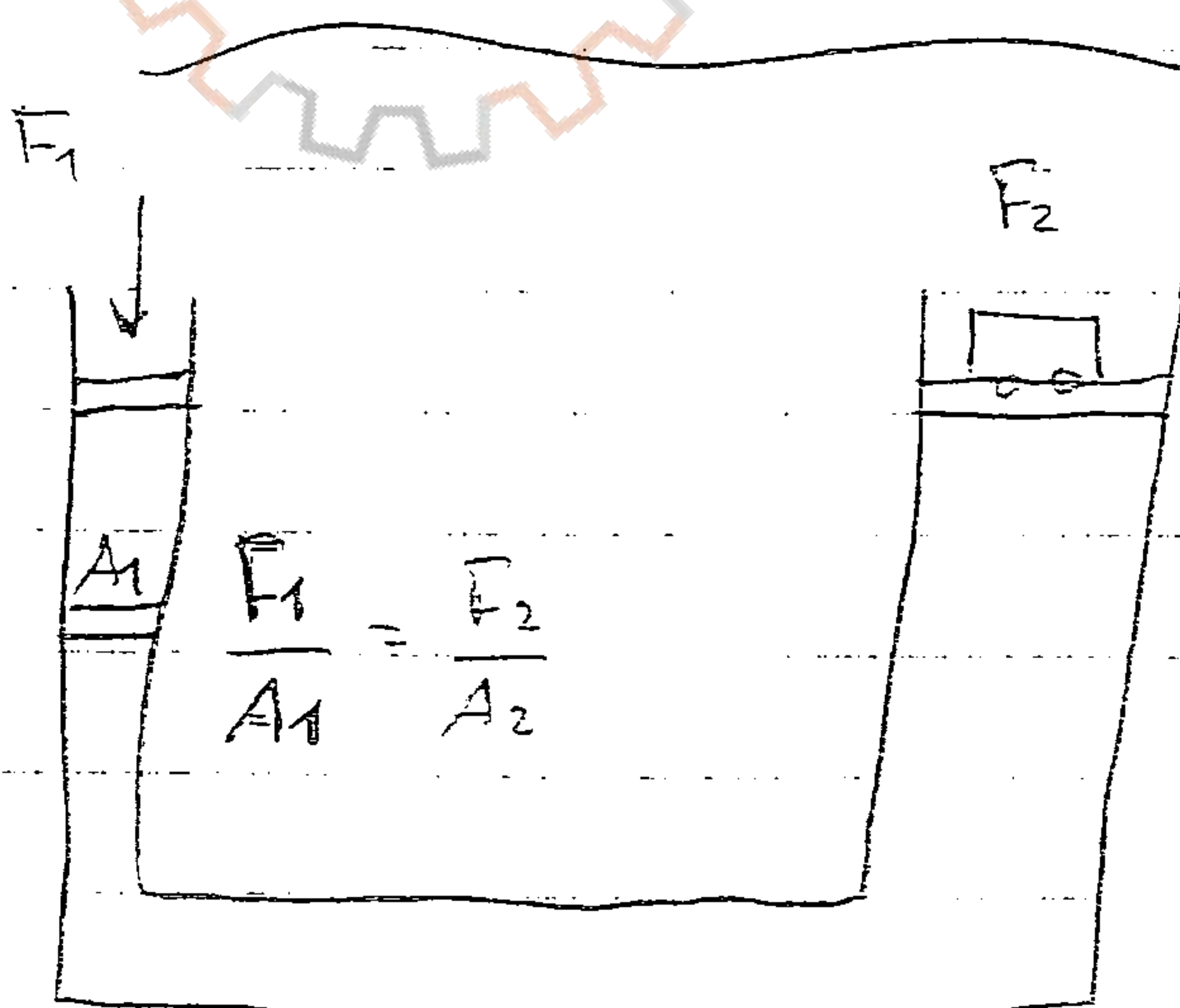
$$P = P_0 + \rho g h$$



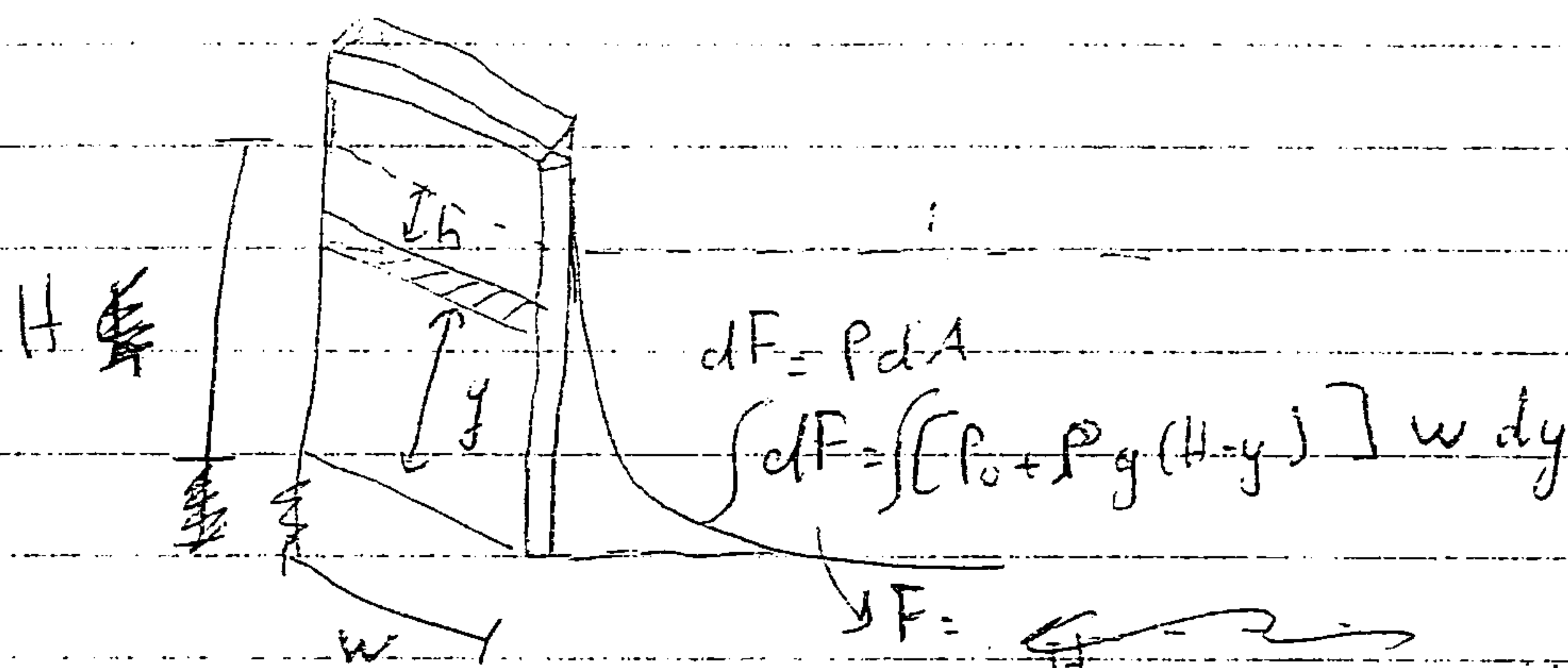
$P - P_0 =$ ~~gauge~~ ^{gauge} pressure

$P =$ Absolute pressure

$P_0 =$ atmospheric pressure



The Force on a Dam



$$p_y = p_0 + \rho_g (H - y)$$

$$\int_0^H \rho_g H w dy - \int_0^H \rho_g w y dy$$

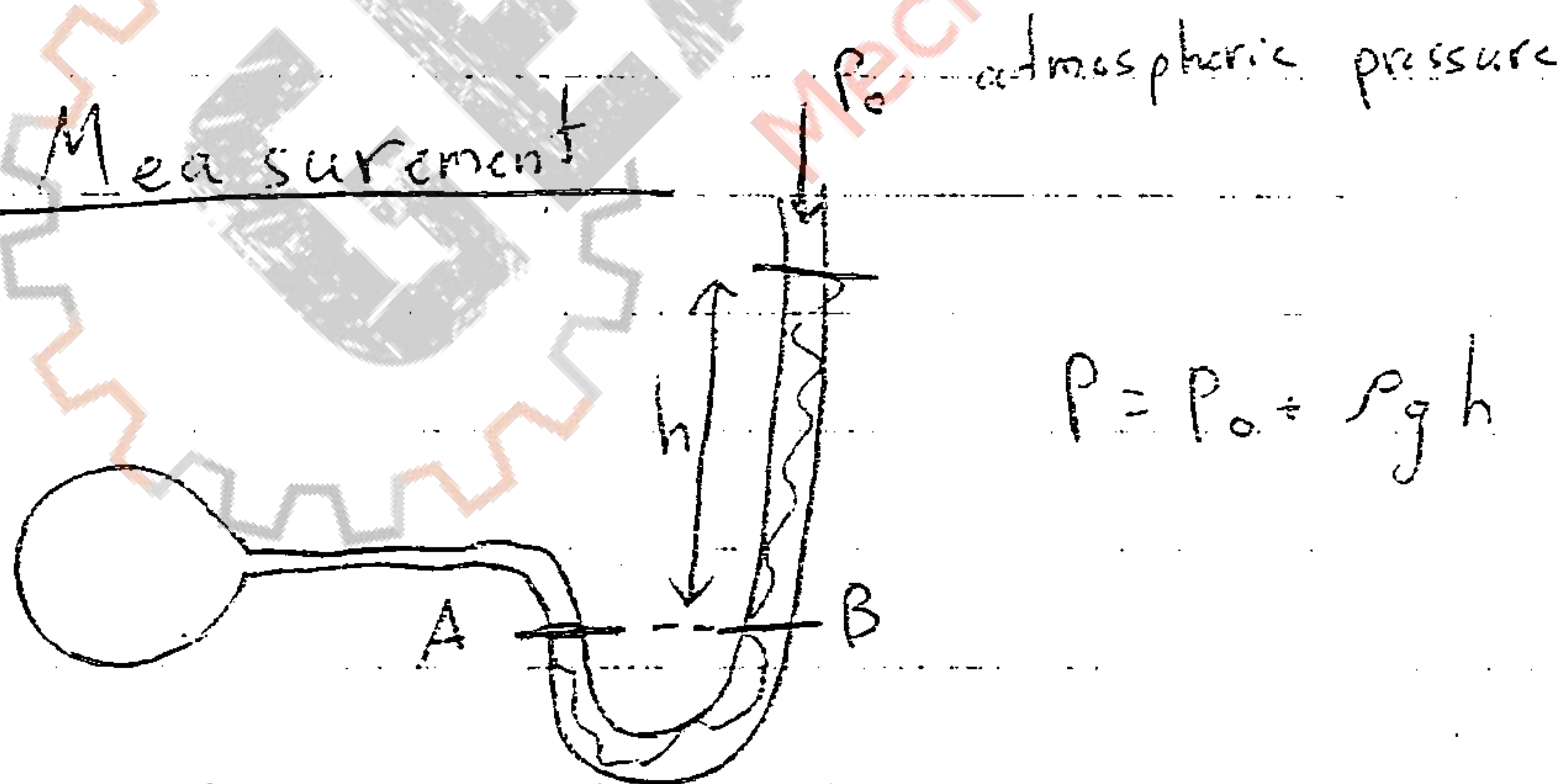
$$= \rho_g H w - \rho_g w \frac{H^2}{2}$$

$$= \rho_g w \frac{H^2}{2}$$

$$\frac{\text{Force}}{\text{Area}} = \rho_g (wH) \frac{H}{2}$$

$$\text{Average pressure} = \rho_g \left(\frac{H}{2} \right)$$

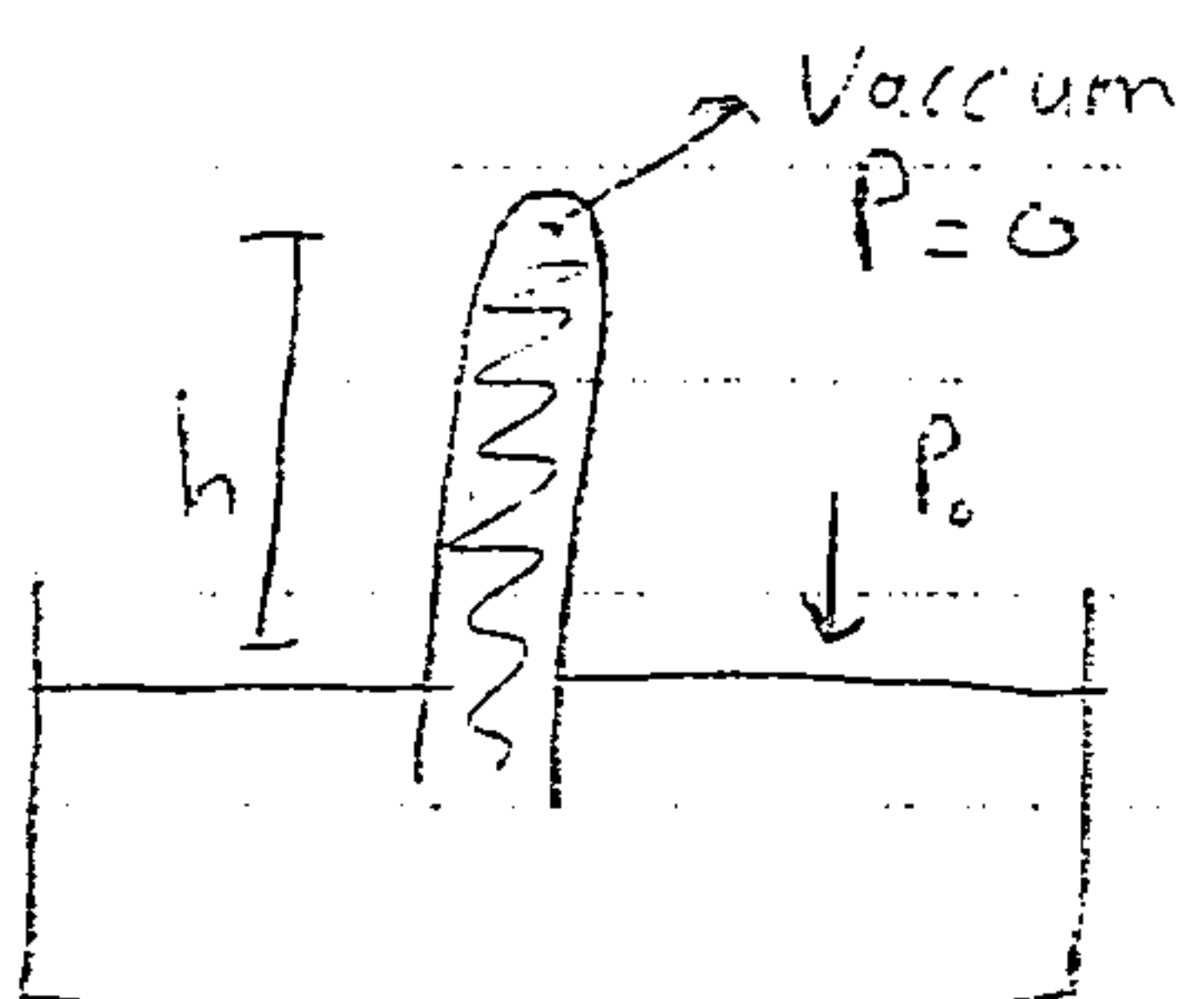
* Pressure Measurement



$$p = p_0 + \rho_g h$$

لقياس
ضغط الغاز

* Barometer



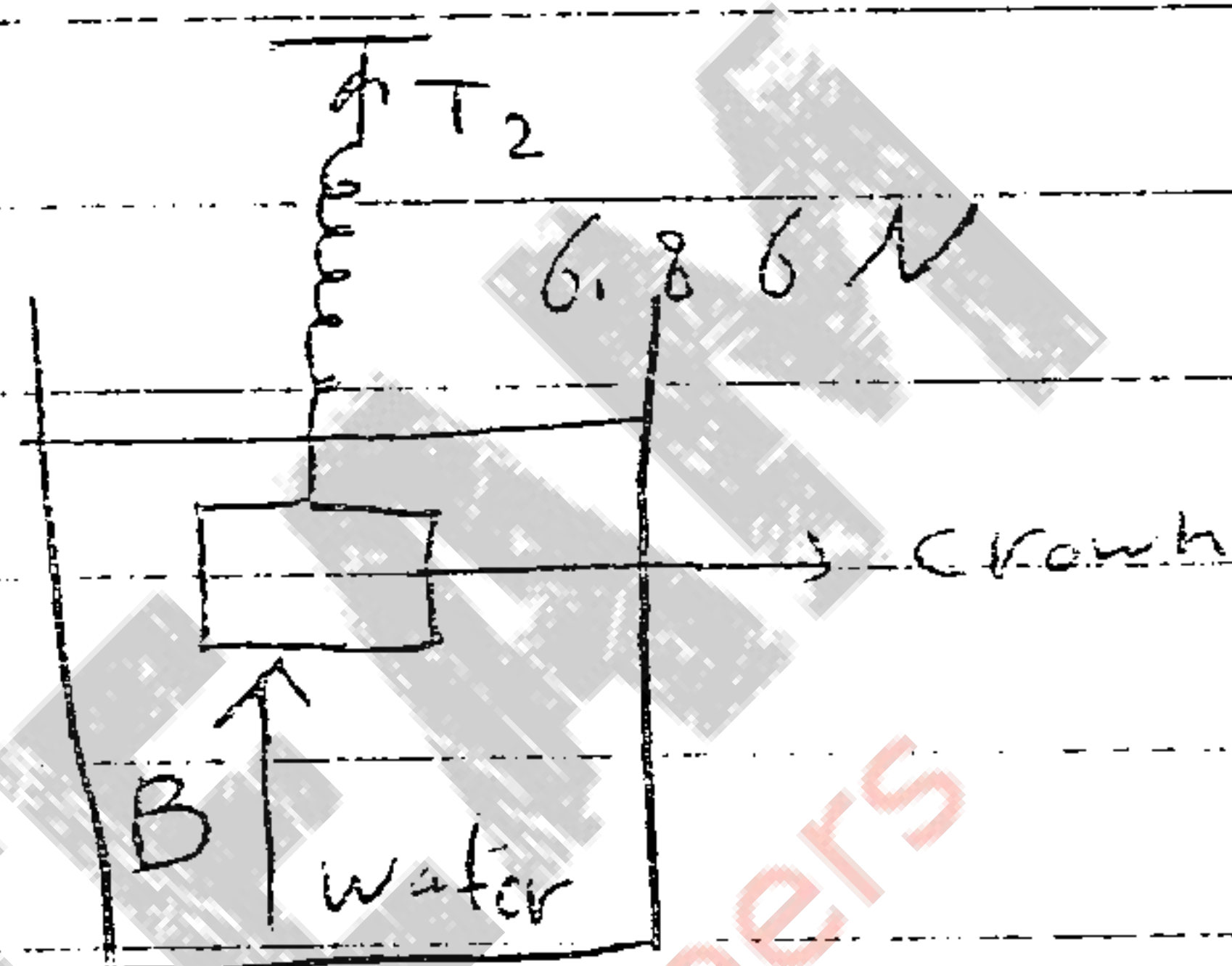
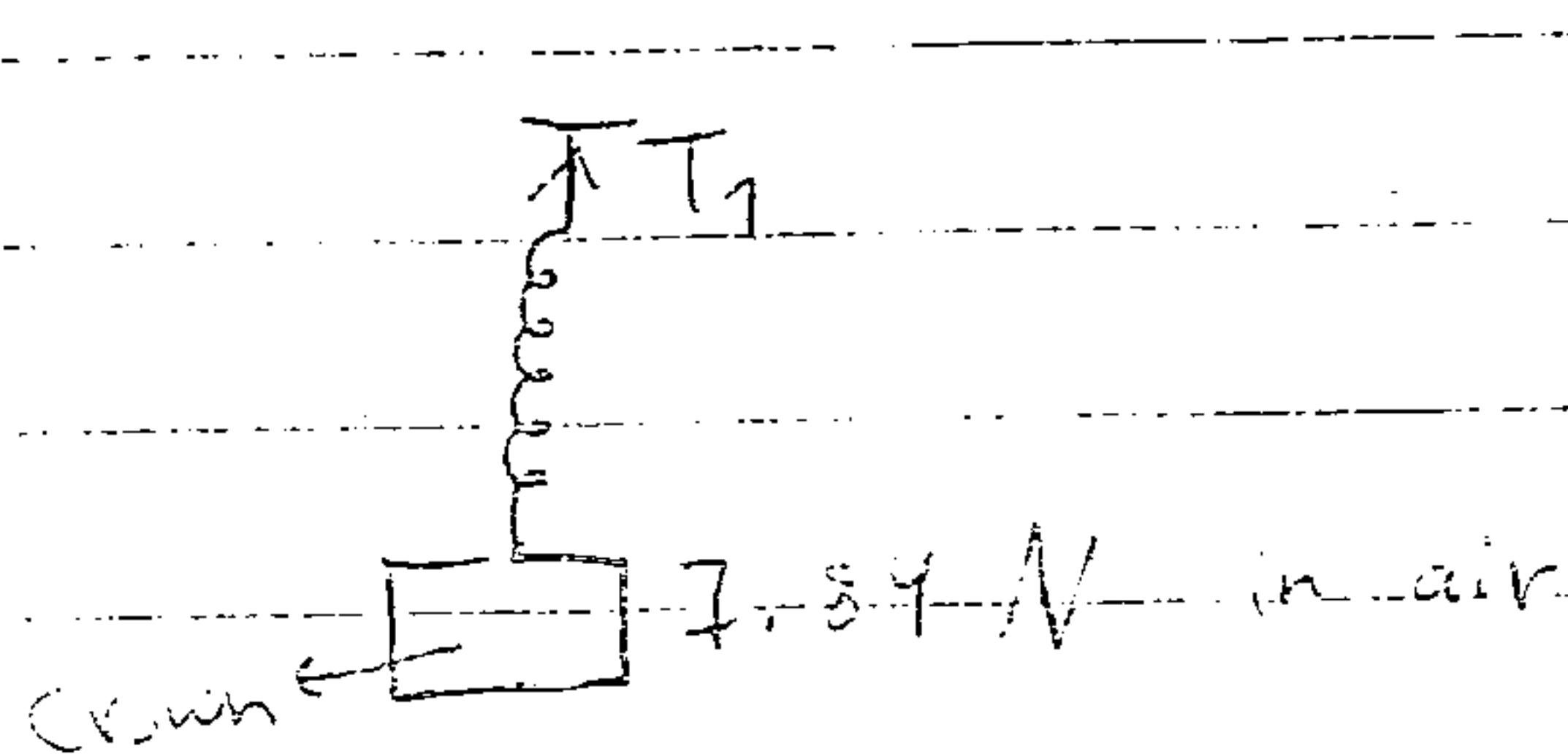
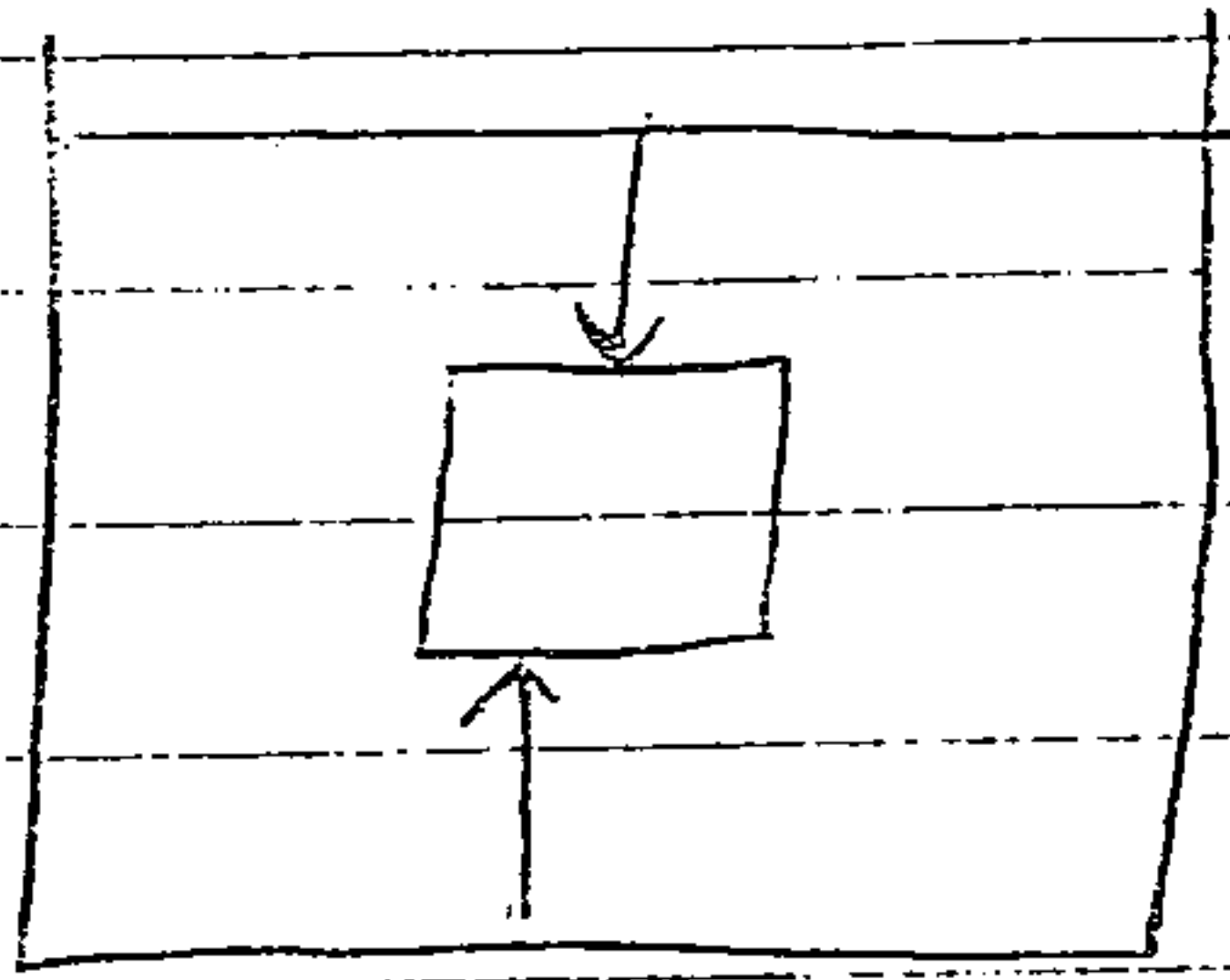
atmospheric
pressure

لقياس الضغط الجوي

P = p.h

Ex 1

* Archimedes's Principle



$$B = 7.84 - 6.86 = 0.98 \text{ N}$$

$$\rho_{\text{water}} V_{\text{crown}} g = B$$

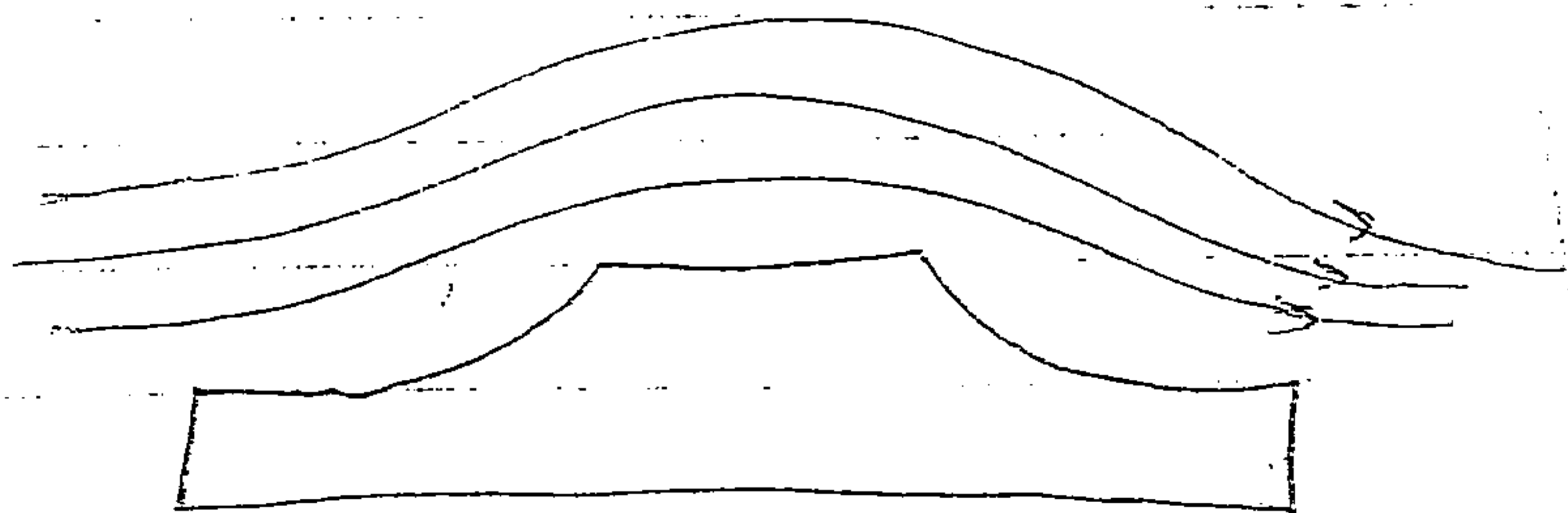
$$\rho_{\text{crown}} = \frac{M_{\text{crown}}}{V_{\text{crown}}}$$

* Ideal Fluid

- Non viscous (Internal friction is neglected)

- Steady flow (laminar)

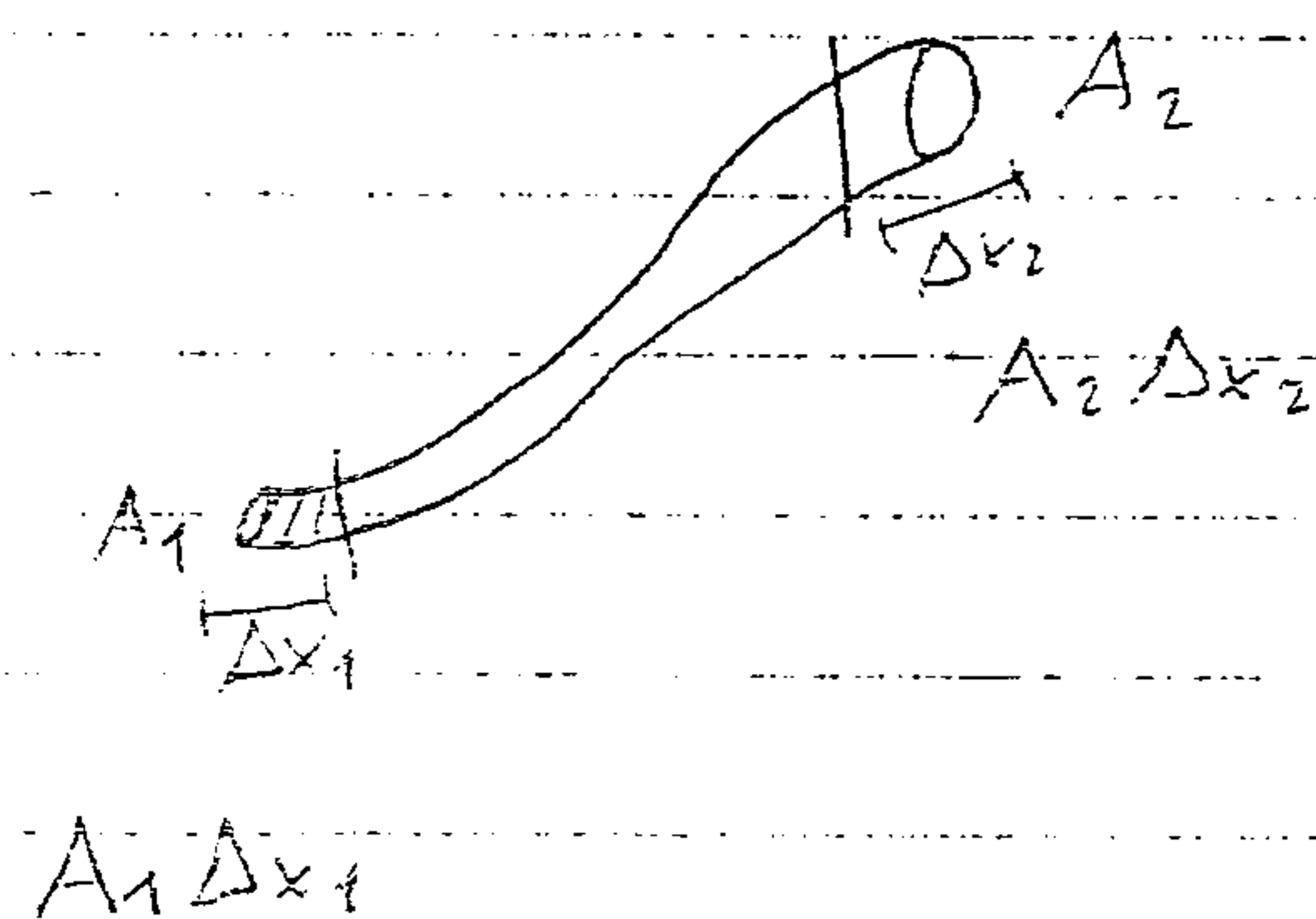
The velocity of the fluid at each point is constant



(160)

- The fluid is incompressible
- The flow is irrotational

+ Stream lines



$$\Delta x_1 = V_1 t$$

$$\Delta x_2 = V_2 t$$

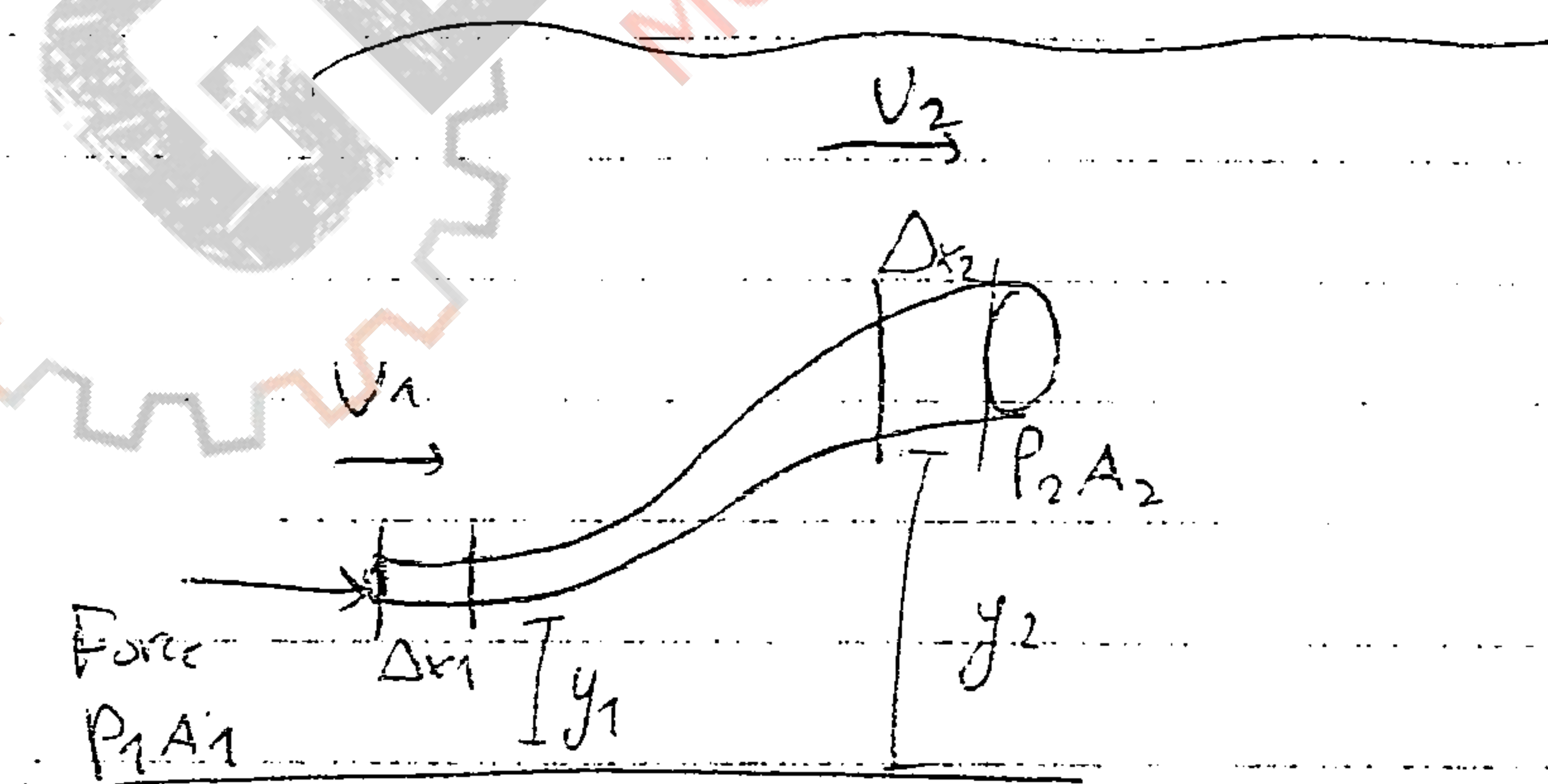
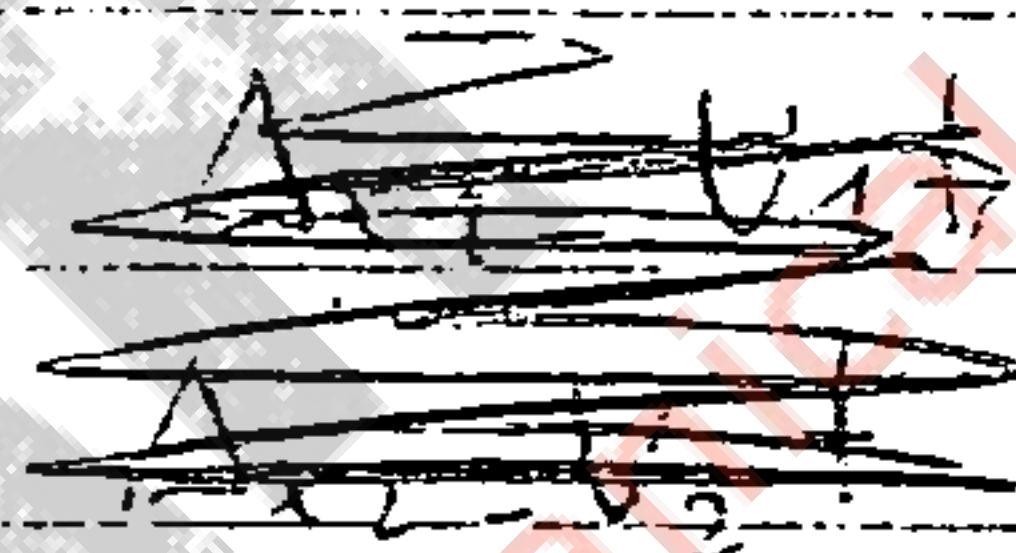
$$A_1 V_1 = A_2 V_2$$

Continuity equation

$$m_1 = (A_1 \Delta x_1) \rho$$

$$m_2 = (A_2 \Delta x_2) \rho$$

Volume



↓ (Area) (Δx)

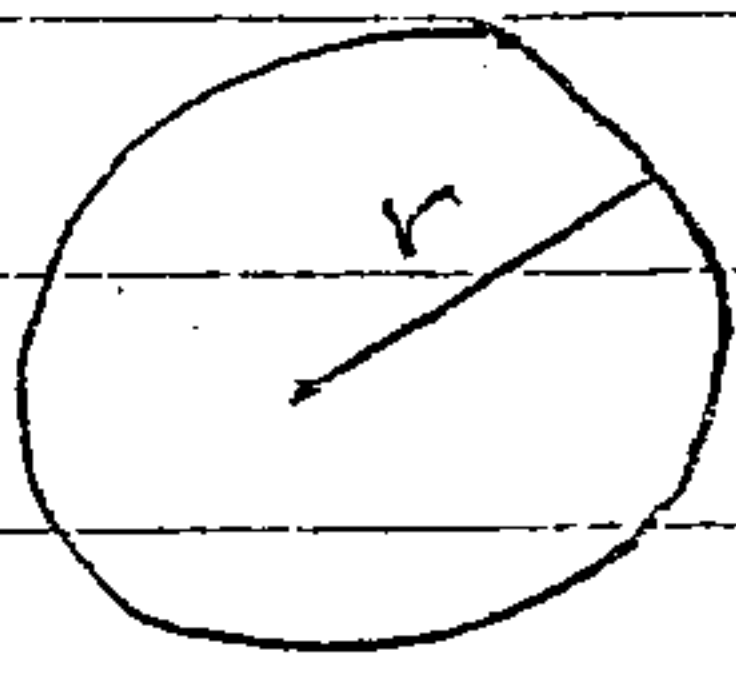
$$W = (P_1 - P_2) V = \frac{1}{2} m V_1^2 - \frac{1}{2} m V_2^2 + m g y_2 - m g y_1$$

~~$$W_{ext} = \Delta E = (P_1 - P_2) V = \frac{1}{2} m V_1^2 - \frac{1}{2} m V_2^2 + m g y_2 - m g y_1$$~~

$$P_1 + \frac{1}{2} \rho V_1^2 + \rho g y_1 = P_2 + \frac{1}{2} \rho V_2^2 + \rho g y_2$$

مراجعة
~~مراجعة~~

$$2r = 3 \text{ cm} = 3 \times 10^{-2} \text{ m}$$



$$r = 1.5 \times 10^{-2} \text{ m}$$

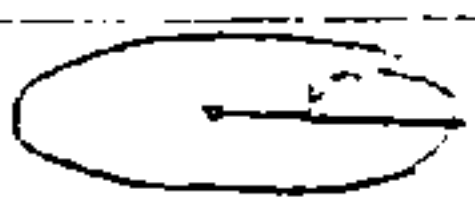
$$\text{Volume} = \frac{4}{3} \pi r^3$$

$$\text{Mass} = (\text{density}) (\text{Volume})$$

$$= (2.7 \times 10^3) \left(\frac{4}{3} \pi (1.5 \times 10^{-2})^3 \right) \text{ kg}$$

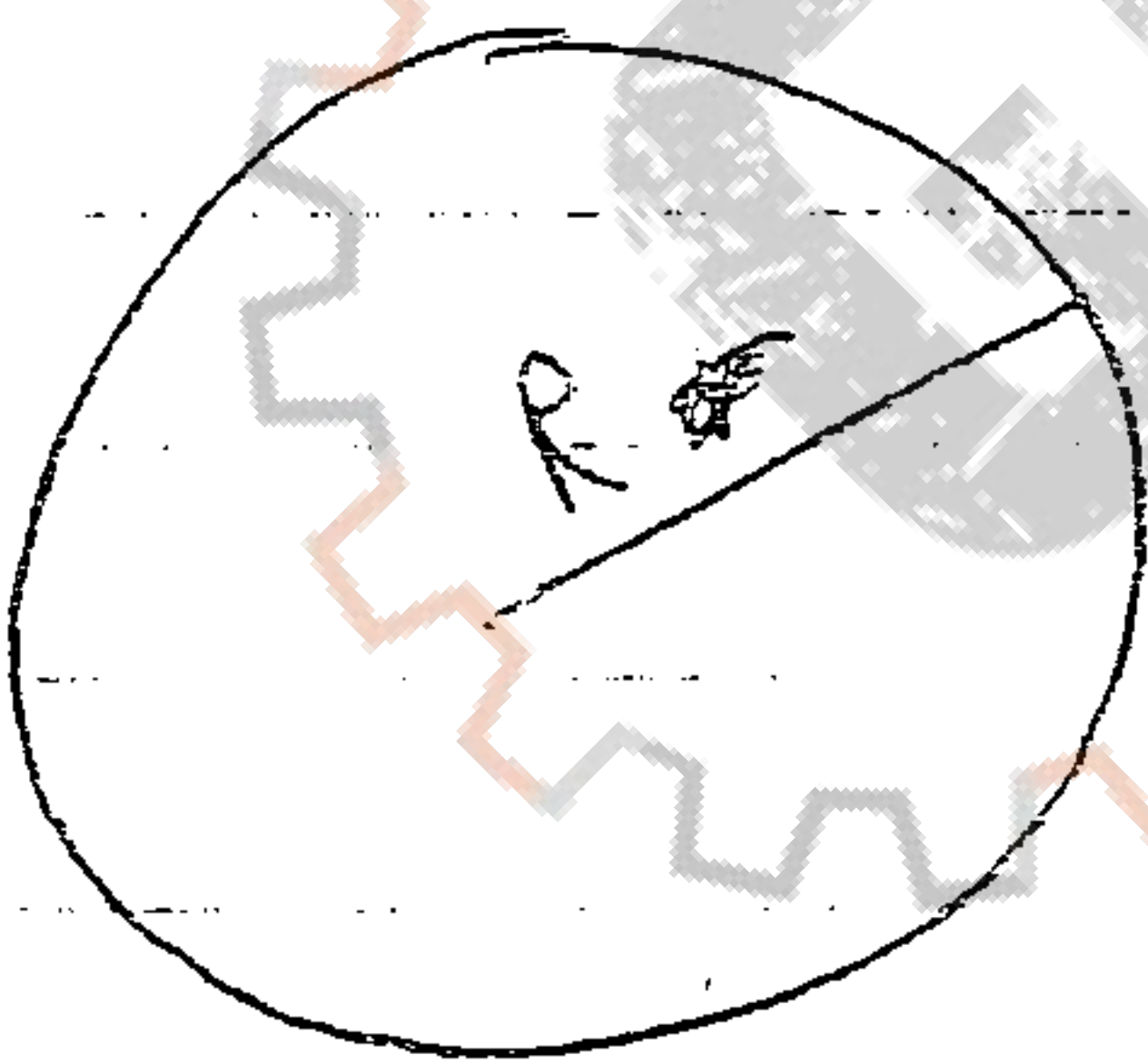
$$M = 50 \text{ kg}$$

$$r = 0.5 \text{ cm}$$



$$P = \frac{\text{Force}}{\text{Area}} = \frac{mg}{(0.5 \times 10^{-2})^2 \pi} = \frac{50 \times 10}{0.25 \times 10^{-4} \pi} \approx 6.2 \times 10^7 \text{ Pa}$$

$$m = ?$$



$$R = 6.37 \times 10^6 \text{ m}$$

$$P = 1.013 \times 10^5 \text{ N/m}^2$$

$$\text{Area} = 4\pi R^2$$

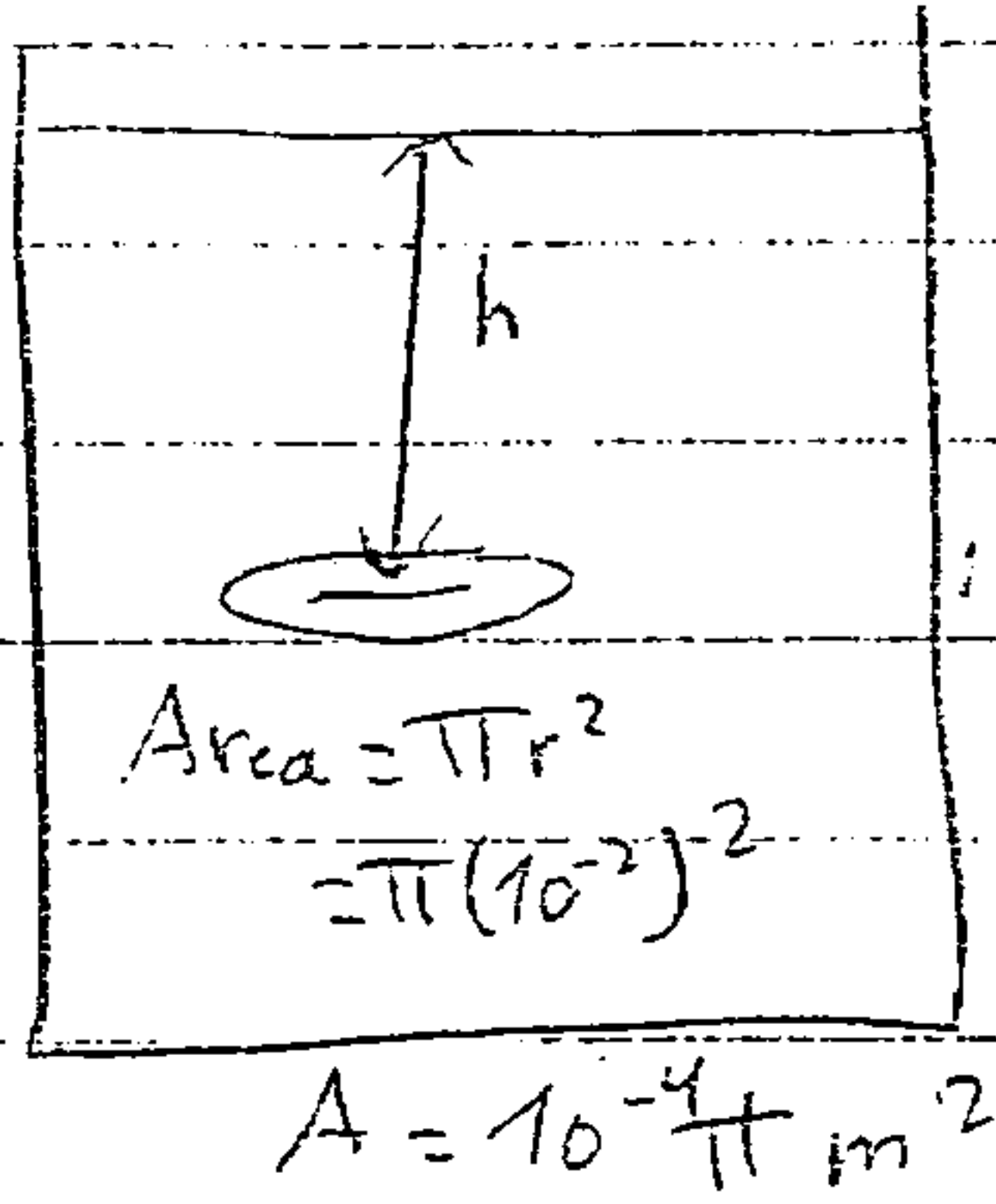
$$F = P \times A$$

$$= 1.013 \times 10^5 \times 4 \times (6.37 \times 10^6)^2 \times \pi$$

$$m = \frac{F}{g}$$

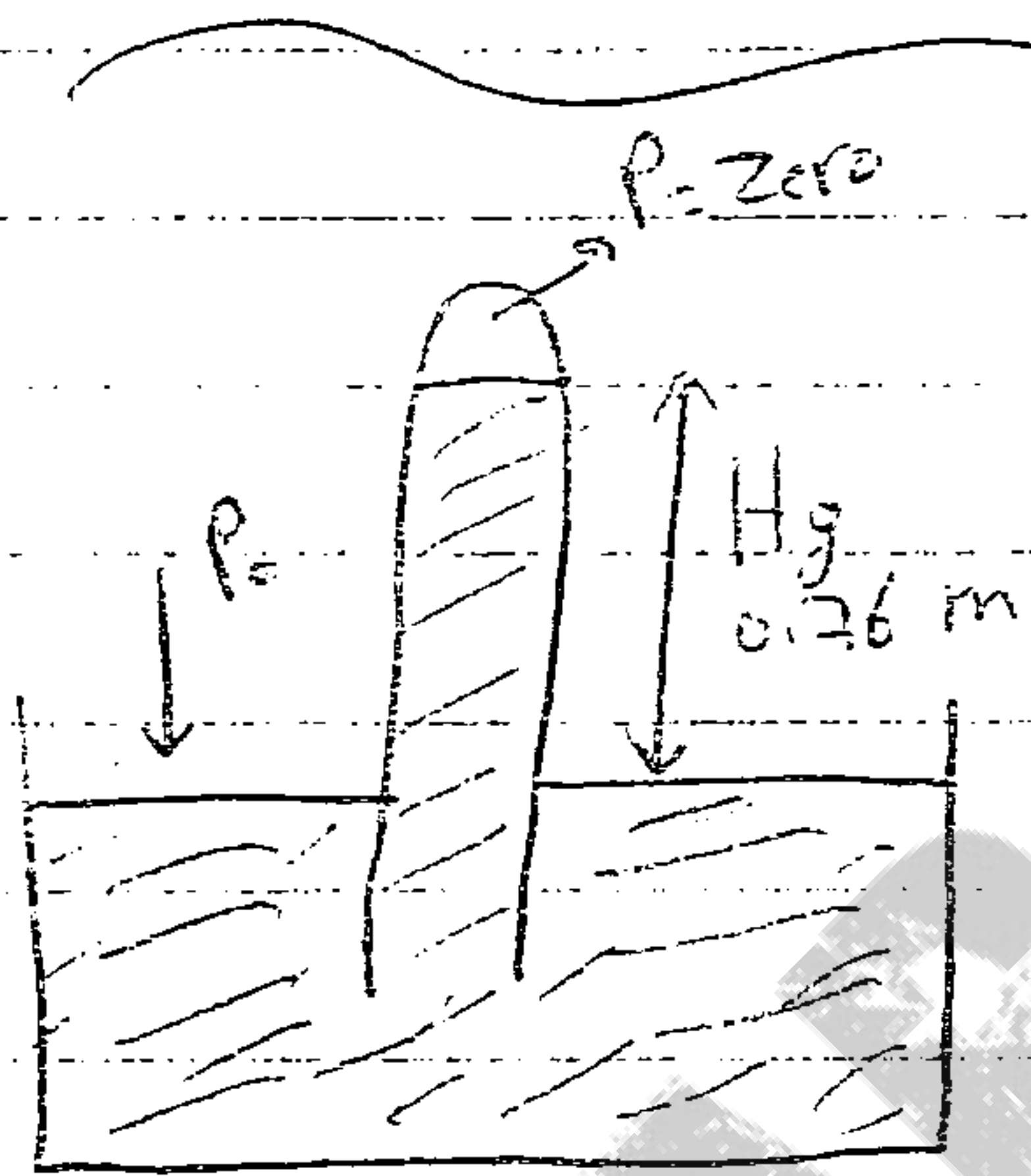
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gauge
 $K = 1000 \text{ N/m}$



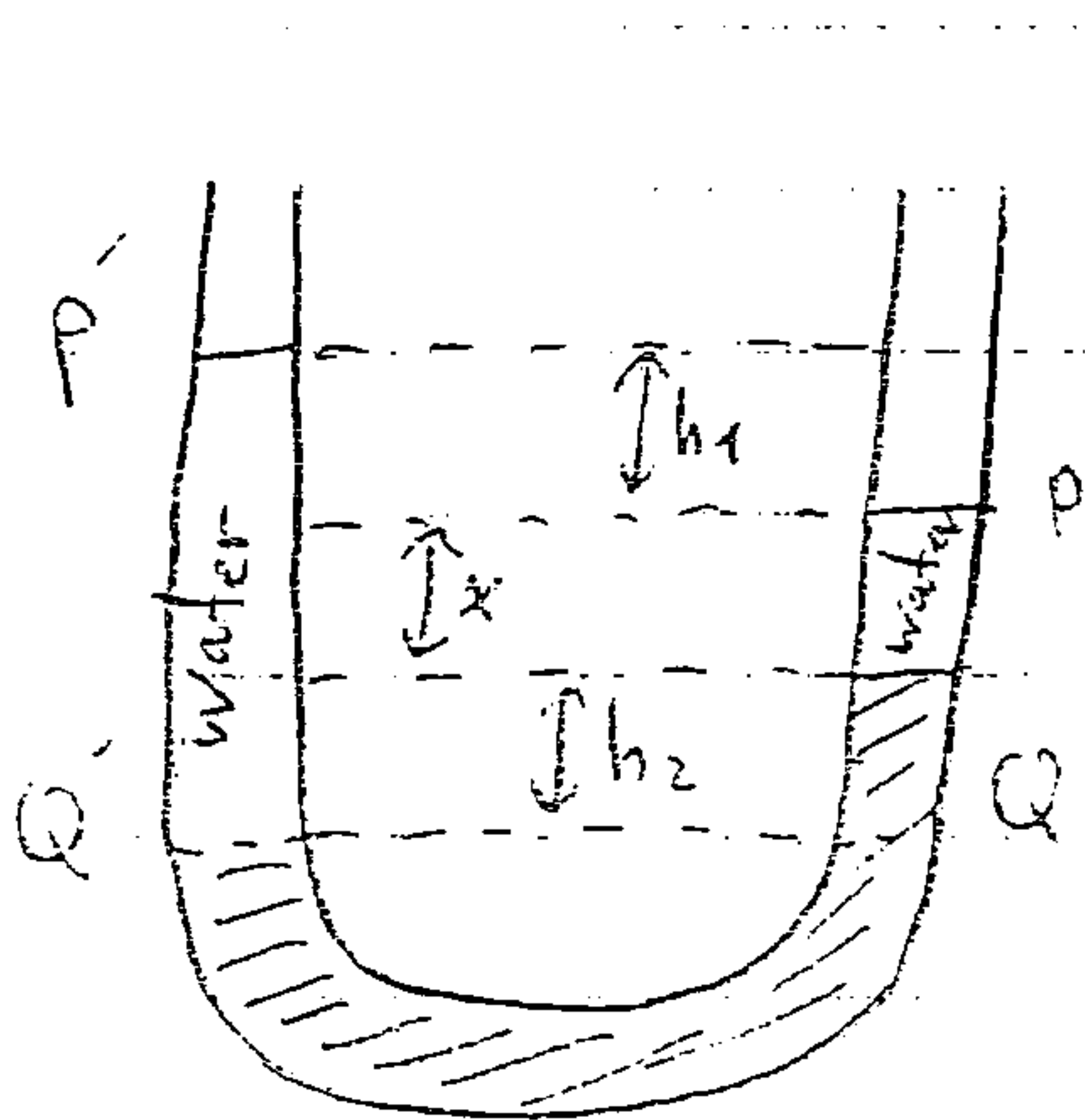
Find (h)

Force = Pressure $\times$ Area
 $Kx = (\rho g h) (10^{-4} \pi)$



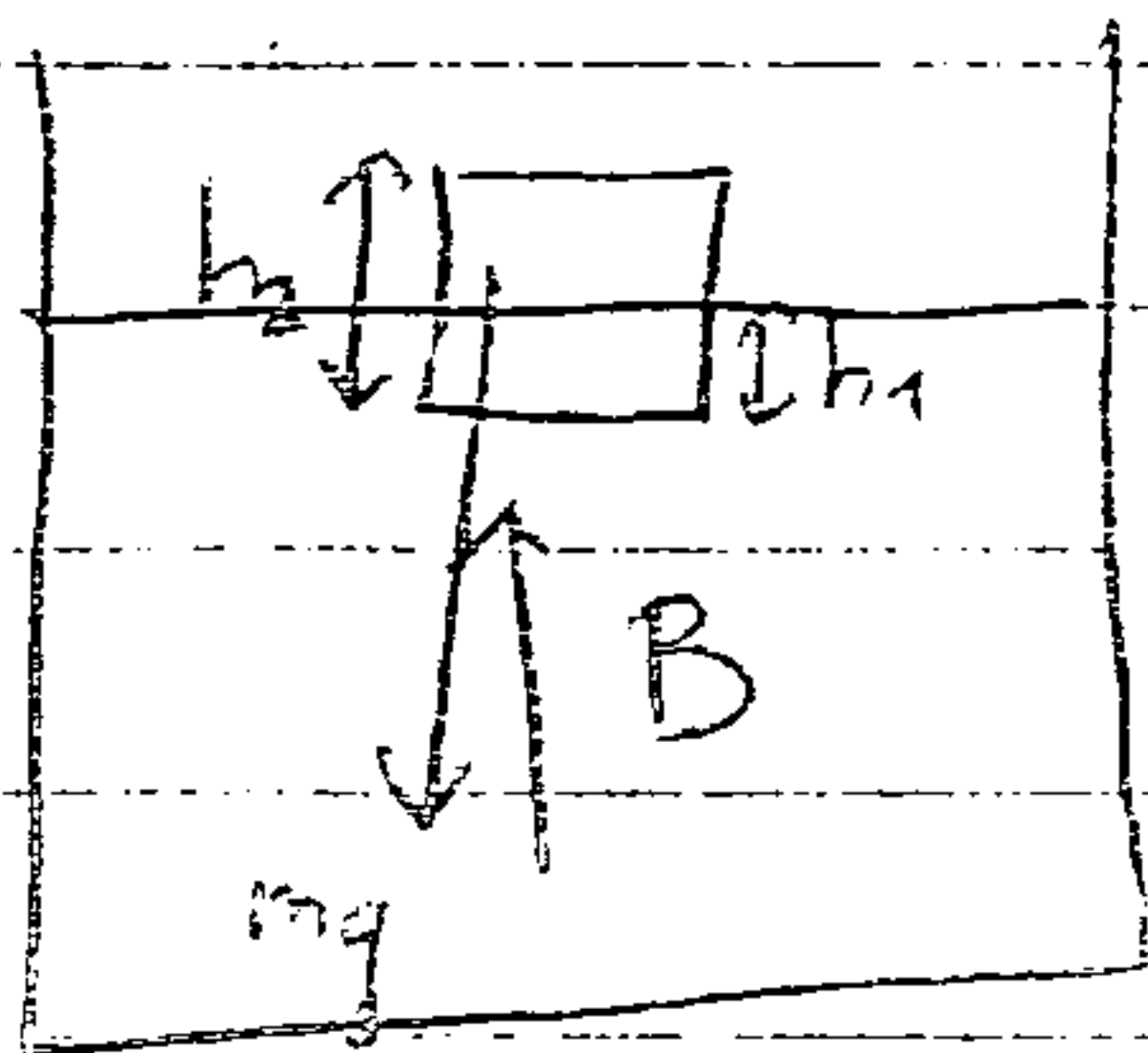
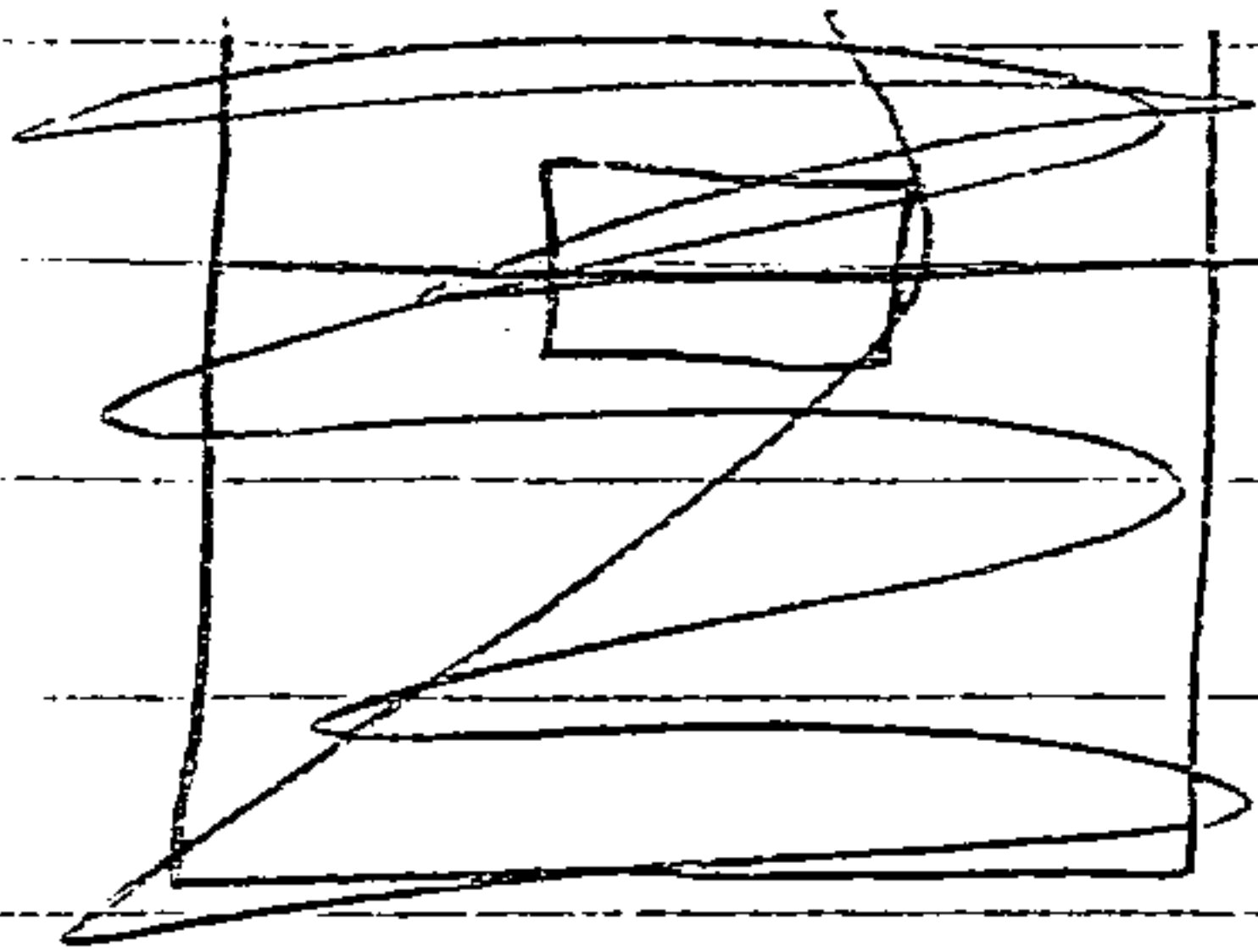
$\rho_1 g h_1 = \rho_2 g h_2$

$\frac{\rho_w}{\rho_{Hg}} = \frac{h_2 H_g}{h_w} \Rightarrow 13.6 = \frac{h_w}{0.76 \text{ m}}$



$h_2 = 1 \text{ cm}$
 $h_1 = ??$

Example:-



Find (h_1)

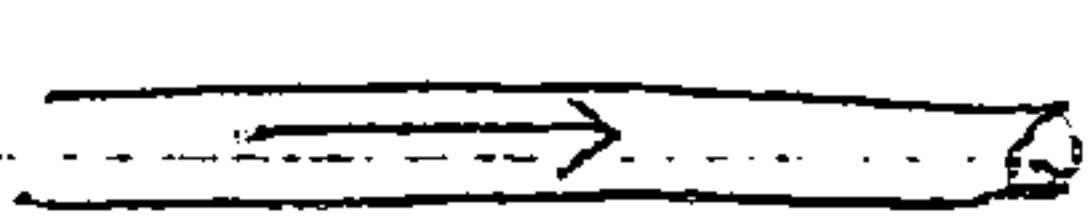
~~179~~

$$(Volume_{cube}) \rho_{ice} g = (h_1) (Area) \rho_{water} g$$

$$(h_2) (Area)$$

$$\frac{h_1}{h_2} = \frac{\rho_{ice}}{\rho_{water}} = \frac{917 \text{ kg/m}^3}{1030 \text{ kg/m}^3} \approx 0.9 \Rightarrow 90\%$$

$V = ?$



$2 \text{ cm} = 2r$

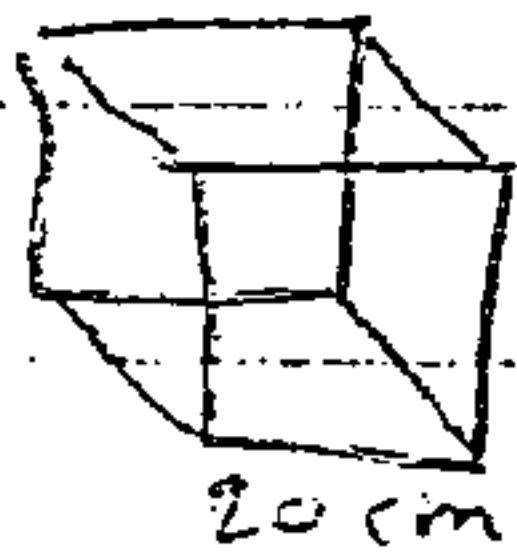
$r = 10^{-2} \text{ m}$

20 L

60 seconds
to fill
the bucket

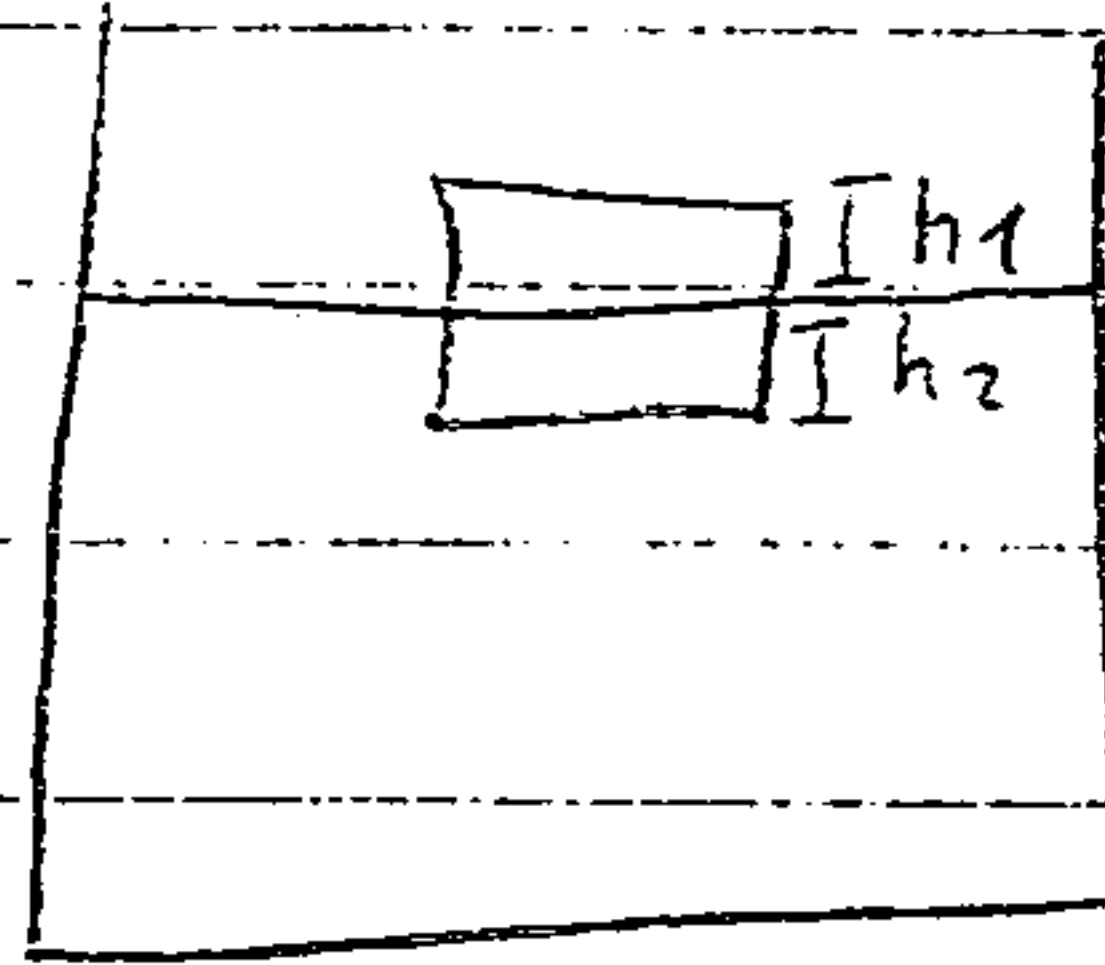
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cube of wood



20 cm

$$\rho_{\text{wood}} = 650 \frac{\text{kg}}{\text{m}^3}$$



$$M_g = B$$

$$(\text{Volume}) (\rho) g = B$$

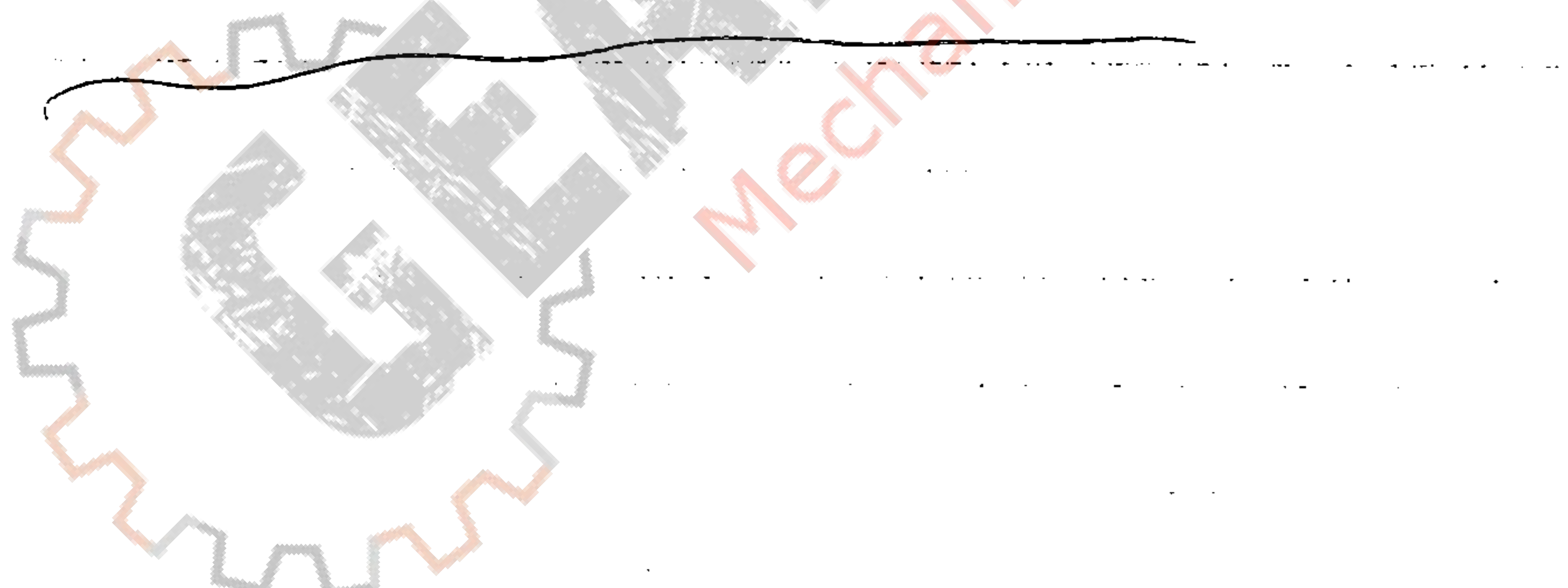
$$(\cancel{A_{\text{top}}}) (h_1 + h_2) (\rho_{\text{wood}}) g = (\cancel{A_{\text{bottom}}}) (h_2) \rho_{\text{water}} g$$

~~$$h_2 = 20 \times 10^{-2} \times 650$$~~

$$h_2 = \frac{20 \times 10^{-2} \times 650}{1000} = 13 \text{ cm}$$

$$h_1 = 20 - 13 = 7$$

$$\frac{7}{20}$$



Chap 4
SP 4.15.16
Post script 1

SP 4.15.16
Post script 1

(2016)